ENTERPRISE ARCHITECTURE & PRODUCT REFERENCE

# CODEJI CMS / HRMS Complete System Documentation

End-to-End Architecture, Multi-Tenant Security, 7-Dimensional Employee Lifecycle, 30 Module Deep-Dives, Mathematical Calculation Engines, Mermaid System Diagrams & Source Code Traceability Matrix

| 63 | 30 | 11 | 100% |
|---|---|---|---|
| TOTAL CHAPTERS & SPECS | HRMS MODULE DEEP-DIVES | SYSTEM & FLOW DIAGRAMS | VERIFIED FROM SOURCE CODE |

### VERIFIED REPOSITORY STACK & CORE ENGINES

- ASP.NET Core (.NET 10)
- React 18 + TypeScript + Vite
- MongoDB + Optimistic Locking
- Redux Toolkit + TanStack Query
- SignalR Real-Time Hubs
- 7-Dimensional EmployeeLifecycle
- 3-Tier OffboardingPolicyResolver
- AutoPayroll + PDF Payslip Engine
- AttendanceMonthLock Gate
- SHA-256 Former Employee Portal

**Repository:** Codeji CMS / HRMS (`CMS-Backend-Core` & `CMS-React`)
**Verification Rule:** 100% Grounded in Actual Implementation

**Document Package:** Master Combined Edition
**Generated:** September 2026

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PART I

# MASTER CONSOLIDATED SYSTEM REFERENCE

Executive Architecture, 7-Dimensional Lifecycle, Mathematical Formulas & Complete Index

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**MASTER-00** **PART I** `docs/project-documentation/CODEJI-HRMS-COMPLETE-SYSTEM-DOCUMENTATION.md`

# CODEJI CMS / HRMS — COMPLETE SYSTEM ARCHITECTURE & PRODUCT REFERENCE BOOK (`CODEJI-HRMS-COMPLETE-SYSTEM-DOCUMENTATION.md`)

> **Document Classification**: Master Technical Architecture, Domain Model, Workflow & Product Reference
> **Source of Truth**: Verified directly from `CMS-Backend-Core` (`.NET 10 / ASP.NET Core / MongoDB`) and `CMS-React` (`React 18 / TypeScript / Vite / MUI / Redux Toolkit / TanStack Query`)
> **Repository Root**: `c:\Users\User 2\CMS`

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## PART I — EXECUTIVE SUMMARY & PRODUCT VISION

**Codeji CMS / HRMS** is an enterprise-grade, multi-tenant Human Resource Management System engineered to unify the complete employee lifecycle—from candidate sourcing and token-gated preboarding to daily attendance status calculation, policy-driven leave accrual, automated monthly payroll with PDF payslip rendering, multi-department exit clearance, Full & Final (`FnF`) settlement reconciliation, and post-employment alumni self-service.

### Core Architectural Differentiators

1. **7-Dimensional Orthogonal Employee Lifecycle (`EmployeeLifecycle.cs`)**: Replaces brittle single-column status strings with 7 independent state dimensions (`EmploymentStatus`, `AvailabilityStatus`, `AccountStatus`, `PayrollStatus`, `OffboardingStatus`, `FormerPortalStatus`, `AlumniStatus`).
2. **Hard Pre-Payroll Safety Gates (`AttendanceMonthLock` & `AttendancePayrollException`)**: Prevents payroll corruption by blocking `AutoPayrollServices` until all monthly `LHD` / `ED` threshold exceptions are resolved (`APPROVED` or `WAIVED`) and the attendance month is explicitly locked.
3. **3-Tier Scoped Policy Resolution (`OffboardingPolicyResolver`)**: Dynamically resolves organizational rules with deterministic specificity (**Exact Department → Exact Role → Company Default**) and freezes an immutable `EmployeeExitPolicySnapshot` onto in-flight records.

4. **Dual-Realm Security Model**: Enforces strict JWT + `AppVersion` ( `426 Upgrade Required` ) + `CompanyIdMiddleware` RBAC for active staff while offering a dedicated, SHA-256 token-hashed **Former Employee Portal** ( `PublicFormerEmployeePortalController` ) for separated alumni.

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## PART II — REPOSITORY STRUCTURE & ENGINEERING ORGANIZATION

```
CMS/
├── CMS-Backend-Core/                # .NET 10 (net10.0) Modular Layered Backend
│   ├── Codeji.CMS.API/              # HTTP Controllers, Middleware, SignalR Hubs, Templates, Program.cs
│   ├── Codeji.CMS.Services/         # Domain Services, Policy Resolvers, Hosted Background Workers
│   ├── Codeji.CMS.Repository/       # MongoDbContext, MongoRepository<T>, 50+ Domain Entities
│   ├── Codeji.CMS.DTO/              # Request/Response Contracts & Validation Attributes
│   ├── Codeji.CMS.Migrations/       # Database Seeders & Schema Backfill Migrations
│   └── Codeji.CMS.Utility/          # Cryptographic Helpers, Constants, Shared Extensions
│
├── CMS-React/                       # React 18 + TypeScript + Vite SPA Frontend
│   └── src/
│       ├── app/
│       │   ├── modules/             # Feature Modules (auth, employees, attendance, leave, payroll, offboarding...)
│       │   ├── Redux/               # Redux Toolkit Global State Slices
│       │   ├── query/               # TanStack Query Data Fetching & Cache Hooks
│       │   └── routing/             # Protected Routes, Role Guards & Public Portal Routes
│       └── server/
│           └── axios.ts             # Axios Client with AppVersion Header & 401 Refresh Queue
│
└── docs/project-documentation/      # This 63-File Master Architecture & Module Documentation Package
```

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## PART III — END-TO-END SYSTEM ARCHITECTURE & REQUEST PIPELINE

Every API request traversing `CMS-Backend-Core` passes through a strictly ordered middleware pipeline configured in `Program.cs`:

**ARCHITECTURE / WORKFLOW DIAGRAM**




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## PART IV — DATABASE ARCHITECTURE, AUDITING & OPTIMISTIC CONCURRENCY

- **Collection Mapping**: `MongoDbContext.GetCollection<TEntity>(typeof(TEntity).Name)` maps each C# entity class to its dedicated MongoDB collection.
- **Soft-Delete Enforcement**: `MongoRepository<TEntity>` automatically applies `!x.IsDeleted` predicates across standard queries so deleted records never leak into operational calculations.
- **Automatic Auditing & Optimistic Locking ( `ISupportAuditing` )**:
  - Tracks `CreatedBy` , `CreatedDate` , `UpdatedBy` , `UpdatedDate` .
  - Maintains an integer `Version` field incremented on every update to prevent lost updates under concurrent modifications.

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## PART V — AUTHENTICATION, MULTI-TENANT ISOLATION & RBAC SECURITY

1. **JWT Claims**: Tokens embed `sub` / `user_id` , `company_id` , and `role_id` .

2. **Client Refresh Queue (`src/server/axios.ts`)**: When an access token expires (`401`), concurrent requests are queued in memory while a single `POST /api/auth/refresh` call rotates the `RefreshToken` and replays all queued requests seamlessly.
3. **Granular RBAC (`[ModulePermissionAttribute]`)**: Controllers declare module and action requirements (e.g., `[ModulePermission("payroll", "approve")]`), evaluated dynamically against the user's `Role` permission matrix in `RoleServices.VerifyUserAccess`.
4. **Strict Tenant Isolation**: `CompanyId` is extracted exclusively from the verified JWT claim inside `CompanyIdMiddleware` — never trusted from client query strings or body payloads.

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## PART VI — HIERARCHICAL POLICY & CONFIGURATION ENGINE

Organizational rules are governed by 5 specialized policy engines:

1. **`EmployeeLifecyclePolicy`**: Governs `ProbationDays` and onboarding transition requirements.
2. **`CompanyOfficeSchedule`**: Governs `StartTime`, `EndTime`, `GraceMinutes`, `CheckInAllowedUntil`, `CheckOutAllowedFrom`, and `HalfDayThresholdHours`.
3. **`AttendancePenaltyPolicy`**: Governs `CombinedLhdEdMonthlyLimit` and automated `AttendancePayrollException` creation.
4. **`LeavePolicy`**: Governs annual quota, accrual rate, carry-forward caps, encashability (`IsEncashable`), `AllowOnProbation`, and `AllowDuringNoticePeriod`.
5. **`PayrollDivisorPolicy`** & **`EmployeeExitPolicy`**: Governs daily rate divisor modes (`FIXED_30_DAYS`, `WORKING_DAYS`, `CALENDAR_DAYS`) and 3-tier scoped exit rules (`Department -> Role -> Company`).

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## PART VII — THE 7-DIMENSIONAL EMPLOYEE LIFECYCLE ENGINE

Implemented in `EmployeeLifecycle.cs` and orchestrated by `EmployeeLifecycleService.cs`:

| Dimension | Enum Values | Operational Purpose |
|---|---|---|
| **1. `EmploymentStatus`** | `Preboarding (0)`, `Onboarding (1)`, `Probation (2)`, `Active (3)`, `NoticePeriod (4)`, `Separated (5)`, `Withdrawn (6)` | Primary contractual employment state; controls leave & notice rules. |
| **2. `AvailabilityStatus`** | `Available`, `OnLeave`, `Remote`, `Unavailable` | Real-time daily scheduling & calendar presence. |
| **3. `AccountStatus`** | `PendingActivation`, `Active`, `Suspended`, `Deactivated` | Controls internal JWT login permission (`EmpUser.IsActive`). |
| **4. `PayrollStatus`** | `PendingSetup`, `Eligible`, `OnHold`, `FinalSettlementPending`, `FinalSettlementCompleted` | Controls inclusion in `AutoPayrollServices` vs `EmployeeFinalSettlement`. |
| **5. `OffboardingStatus`** | `None (0)`, `Initiated (1)`, `NoticePeriod (2)`, `ClearanceInProgress (3)`, `ReadyForSettlement (4)`, `Completed (5)` | Tracks multi-departmental exit progress. |
| **6. `FormerPortalStatus`** | `Disabled`, `Enabled`, `Expired`, `Revoked` | Controls access to `/former-employee` via `FormerEmployeeAccessToken`. |
| **7. `AlumniStatus`** | `NotApplicable`, `Eligible`, `Ineligible`, `Rejoined` | Governs boomerang re-hire eligibility (`EmployeeRejoinRecord`). |

## PART VIII — TALENT ACQUISITION, PREBOARDING & ONBOARDING

- **Recruitment (`RECRUITMENT.md`)**: Manages `JobPosition`, `Applicant` Kanban pipelines, multi-round interviews, `ApplicantLogs` audit trails, `RecruitmentAuditExportService` CSV exports, and `JobOffer` generation.
- **Preboarding (`PREBOARDING.md`)**: Provides candidates with a token-secured portal to accept offers and upload `PreboardingDocument` KYC files for HR verification before Day 1.
- **Onboarding (`ONBOARDING.md`)**: Converts verified candidates into `EmpUser` records, initializes the 7-dimensional `EmployeeLifecycle` (`EmploymentStatus = Probation`), and sets `ProbationEndDateUtc`.

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## PART IX — WORKFORCE TIME, ATTENDANCE, SHIFTS, LEAVE, WFH & CALENDAR

### Daily Attendance Classification Formula (`AttendanceStatusCalculationService.cs`)

Given effective shift schedule (`StartTime`, `EndTime`, `CheckInAllowedUntil = StartTime + GraceMinutes`, `CheckOutAllowedFrom`, `HalfDayThresholdHours`):

- If worked hours $< \text{HalfDayThresholdHours} \rightarrow HD$ (Half Day)
- Otherwise evaluate:
  - $\text{isLate} = (\text{CheckInTime} > \text{CheckInAllowedUntil})$
  - $\text{isEarly} = (\text{CheckOutTime} < \text{CheckOutAllowedFrom})$
  - `!isLate && !isEarly` $\rightarrow P$ (Present)
  - `isLate && !isEarly` $\rightarrow LHD$ (Late Half Day)
  - `!isLate && isEarly` $\rightarrow ED$ (Early Departure)
  - `isLate && isEarly` $\rightarrow LHD + ED$ (Combined Late + Early)

### Monthly Penalty & Lock Gate (`AttendancePenaltyService.cs`)

- If $(\text{LhdCount} + \text{EdCount} + \text{LhdEdCount}) > \text{CombinedLhdEdMonthlyLimit}$, creates an `AttendancePayrollException` (`Status = "PENDING_REVIEW"`).
- `AttendanceMonthLock.ValidateAndLock()` blocks month locking until **zero** `PENDING_REVIEW` exceptions remain.

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## PART X — COMPENSATION, SALARY STRUCTURE & AUTOMATED PAYROLL ENGINE

### Automated Payroll Formula (`AutoPayrollServices.cs`)

1. **Gate Check**: Verifies period is open, `MonthlyAttendanceSummary.IsApproved && IsLocked == true`, and zero `PENDING_REVIEW` exceptions exist.
2. **Divisor Resolution (`PayrollCalculationRules.ResolveDivisor`)**:
   - `FIXED_30_DAYS` $\rightarrow 30$
   - `WORKING_DAYS` $\rightarrow \text{CompanyWorkingCalendarService.GetWorkingDaysInMonth}(Y, M)$
   - `CALENDAR_DAYS` $\rightarrow \text{DateTime.DaysInMonth}(Y, M)$
3. **Earnings & Deductions**:

$$\text{GrossPay} = \text{BasicPay} + \text{HRA} + \text{LTA} + \text{Bonus} + \text{OtherAllowance}$$

$$\text{TotalLossOfPayDays} = \text{UnpaidDays} + \text{ApprovedPenaltyDays}$$

$$\text{LossOfPayDeduction} = \text{TotalLossOfPayDays} \times \leq \text{ft}(\backslash\text{frac}\{\text{GrossPay}\}\{\text{Divisor}\})$$

$$\text{TotalDeduction} = \text{LossOfPayDeduction} + \text{EPF} + \text{ESIC} + \text{HealthInsurance} + \text{IncomeTax}$$

$$\text{NetSalary} = \text{GrossPay} - \text{TotalDeduction}$$

4. **Payslip PDF Rendering**: Compiles `Templates/SalarySlipTemplate.html` via `PdfService` into `/fs/salary-slips/{id}.pdf`.

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## PART XI — OFFBOARDING, MULTI-DEPARTMENT CLEARANCE, FNF & ALUMNI PORTAL

1. **Policy Snapshot (`OffboardingPolicyResolver.cs`)**: Resolves `EmployeeExitPolicy` (`Department -> Role -> Company`) and freezes `EmployeeExitPolicySnapshot` on `EmployeeExit`.
2. **Multi-Department Clearances**: Tracks IT, Finance, Admin, Manager, and HR sign-offs (`EmployeeExitClearanceTemplate`), knowledge transfer (`EmployeeExitHandoverItem`), physical hardware returns & damage deductions (`EmployeeExitAsset`), benefit terminations (`EmployeeExitBenefitItem`), and `EmployeeExitInterview`.
3. **Full & Final Settlement (`EmployeeFinalSettlement.cs`)**:

$$\text{NetPayable} = (\text{SalaryPayable_}\{\text{SourcePayrollId}\} + \text{LeaveEncashment} + \text{Reimbursements} + \text{Bonuses}) - (\text{Deductions} + \text{NoticeRecovery} + \text{OutstandingDues})$$

- classifies `Direction` as `PayableToEmployee`, `ReceivableFromEmployee`, or `ZeroSettlement`.

4. **Exit Completion & Former Employee Portal (`PublicFormerEmployeePortalController.cs`)**:
   - Deactivates `EmpUser.IsActive`, revokes all `RefreshToken` sessions, creates `EmployeeExitAttendanceCutoff`, enqueues `EmployeeExitOutbox`, and issues a SHA-256 hashed `FormerEmployeeAccessToken` for `/former-employee`.

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## PART XII — REAL-TIME NOTIFICATIONS, CHAT & TRANSACTIONAL OUTBOX

- **SignalR Real-Time Hubs**: `NotificationHub` (`/notificationHub`) and `ChatHub` (`/chatHub`) push instant WebSocket events to authenticated clients.
- **Background Hosted Services**:
  - `LeaveAccrualHostedService`: Executes scheduled leave balance accruals.
  - `AttendanceReminderHostedService`: Dispatches check-in/out reminders (`AttendanceReminderDispatchLog`).
  - `EmployeeExitOutboxProcessor`: Guarantees asynchronous delivery of exit notifications and alumni access links.

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## PART XIII — COMPLETE DOCUMENTATION PACKAGE INDEX

### Core Architecture & Governance Chapters (`docs/project-documentation/`)

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- 01-PROJECT-OVERVIEW.md
- 02-REPOSITORY-ARCHITECTURE.md
- 03-SYSTEM-ARCHITECTURE.md
- 04-DATABASE-ARCHITECTURE.md
- 05-AUTHENTICATION-AUTHORIZATION.md
- 06-ROLE-PERMISSION-MODEL.md

- 07-POLICY-CONFIGURATION-ENGINE.md
- 08-MODULE-DEPENDENCY-MAP.md
- 09-CROSS-MODULE-WORKFLOWS.md
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- 18-PROPOSAL-SUMMARY.md
- SOURCE-TRACEABILITY.md
- UNVERIFIED-AREAS.md

### Module Deep-Dives (`docs/project-documentation/modules/`)

- ADMIN.md | COMPANY-SETTINGS.md | EMPLOYEE.md | EMPLOYEE-PROFILE.md | RECRUITMENT.md | PREBOARDING.md | ONBOARDING.md | PROBATION.md | ATTENDANCE.md | WORK-SHIFT.md | LEAVE.md | WFH.md | HOLIDAY.md | CALENDAR.md | PAYROLL.md | PAYROLL-SETTINGS.md | SALARY.md | BENEFITS.md | NOTICE-PERIOD.md | OFFBOARDING.md | FINAL-SETTLEMENT.md | ASSET-MANAGEMENT.md | DOCUMENTS.md | NOTIFICATIONS.md | ROLE-PERMISSION.md | POLICY.md | REPORTING.md | ANALYTICS.md | AUDIT.md | SETTINGS.md

### Architectural Diagrams (`docs/project-documentation/diagrams/`)

- SYSTEM-CONTEXT.md | MODULE-DEPENDENCIES.md | EMPLOYEE-LIFECYCLE.md | REQUEST-FLOW.md | DATABASE-RELATIONSHIPS.md | AUTHORIZATION-FLOW.md | PAYROLL-FLOW.md | ATTENDANCE-FLOW.md | ONBOARDING-FLOW.md | OFFBOARDING-FLOW.md | RECRUITMENT-FLOW.md

PART II

# CORE ARCHITECTURE, SECURITY, GOVERNANCE & STRATEGY

Foundational System Design, Multi-Tenancy, RBAC, Policy Engine, Production Readiness & Executive Deck (Chapters 00–18)

19 Document(s) in this Section

# Codeji CMS / HRMS — Executive Summary

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Runtime Target**: ASP.NET Core (.NET 10) | React 18 (TypeScript, Vite) | MongoDB (Custom Driver Repository)
**Last Verified**: September 2026

## 1. Executive Synopsis

The **Codeji Content & Human Resource Management System (Codeji CMS / HRMS)** is an enterprise-grade, multi-tenant workforce operating platform engineered to manage the entire employee lifecycle—from public talent acquisition and multi-stage applicant evaluation, through preboarding, automated onboarding activation, shift-governed attendance tracking, multi-tier leave administration, and automated payroll synthesis, to structured resignation, multi-departmental clearance workflows, full and final settlement (FnF), and post-separation alumni engagement.

The solution is architected as two decoupled, high-performance systems:

1. **CMS-Backend-Core**: A modular, service-oriented ASP.NET Core Web API running on **.NET 10 (`net10.0`)**, featuring strict JWT-driven multi-tenant boundary isolation, dynamic role-based access control (RBAC), deterministic multi-level policy engines, transactional outbox processors, SignalR real-time messaging hubs, and an in-house schema/index migration pipeline.
2. **CMS-React**: A modern Single Page Application (SPA) built with React 18, TypeScript, Vite, Material-UI (MUI), Redux Toolkit, and TanStack Query, offering role-tailored workspaces for Company Administrators, HR Managers, HR Executives, Reporting Managers, Employees, and Former Alumni.

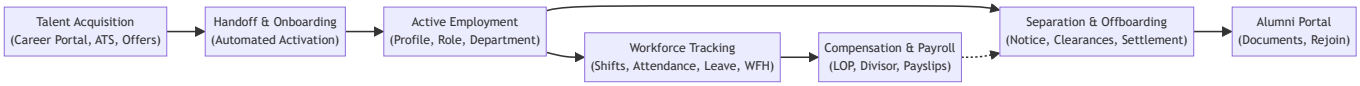
## 2. Business Value & Problem Statement

Modern growing enterprises face fragmented HR tooling where applicant tracking systems (ATS), daily attendance logs, leave balances, payroll calculators, and offboarding checklists reside in disconnected silos. This fragmentation leads to:

- **Data Drift**: Inconsistencies between approved leaves and monthly payroll deductions.
- **Compliance & Security Vulnerabilities**: Former employees retaining system credentials or unauthorized data visibility across multi-tenant boundaries.
- **Operational Churn**: Manual calculations of Loss-of-Pay (LOP), late-arrival penalties, and notice-period buyouts.
- **Lack of Auditability**: Disputed final settlements and undocumented clearance approvals.

Codeji HRMS solves these organizational challenges by establishing a single, cryptographically enforced source of truth across all employment dimensions.

**ARCHITECTURE / WORKFLOW DIAGRAM**



## 3. High-Level Architectural Foundations

| Layer | Technology | Architectural Responsibility |
|---|---|---|
| **Client SPA** | React 18, TypeScript, Vite, MUI v5 | Responsive UI, state management (Redux + React Query), token refresh queue interceptors, dynamic breadcrumbs, localized views (i18n). |
| **API Gateway / Host** | ASP.NET Core (.NET 10), Kestrel, Reverse Proxy Integration | Route dispatching, IP rate limiting partitions ( `career-sensitive` , `account-sensitive` ), SSL termination, CORS allow-listing ( `*.codeji.in` , `localhost` ). |
| **Security Pipeline** | Custom Middleware ( `AntiforgeryMiddleware` , `CompanyIdMiddleware` ) | Anti-CSRF verification, AppVersion lock ( `426 Upgrade Required` ), strict JWT tenant claim validation ( `company_id` , `user_id` , `role_id` ), RBAC authorization ( `ModulePermissionAttribute` ). |
| **Application Services** | C# Business Services Layer | Domain logic, deterministic policy resolution, cross-module coordinators ( `RecruitmentOnboardingService` , `AutoPayrollServices` , `EmployeeService` ). |
| **Realtime & Workers** | SignalR Core, IHostedService Background Queues | WebSocket event distribution ( `NotificationHub` , `ChatHub` ), 9 background background workers handling outbox polling, leave accrual, birthday alerts, and reconciliations. |
| **Data Persistence** | MongoDB, MongoDB C# Driver v2.28+ | High-throughput document store, multi-tenant collection indexing, custom repository pattern ( `MongoRepository<T>` ), migration engine ( `Codeji.CMS.Migrations` ). |

## 4. Key Platform Highlights & Differentiators

### 4.1 Guaranteed Tenant Boundary Isolation

Unlike multi-tenant architectures that rely on client-supplied headers or query parameters for company resolution, Codeji enforces tenant identity strictly from cryptographically signed JWT bearer claims ( `company_id` ). The system rejects any request where client-provided IDs conflict with the token context. Public anonymous routes (e.g., career job applications) resolve tenant context exclusively via verified server-side identifiers (such as registered public company slugs or one-time capability tokens).

### 4.2 Multi-Dimensional Employee State Machine

Employee status is not a simplistic boolean. Codeji tracks 7 orthogonal lifecycle dimensions:

1. **Employment Status**: `PreJoining` , `Active` , `NoticePeriod` , `Suspended` , `Resigned` , `Terminated` , `Retired` , `ContractEnded` , `Deceased` .
2. **Availability Status**: `Working` , `AnnualLeave` , `SickLeave` , `MaternityLeave` , `Sabbatical` , `UnauthorizedAbsence` , `AbscondingUnderReview` .
3. **Account Status**: `Active` , `Locked` , `TemporarilyDisabled` , `Limited` , `Suspended` , `Closed` .
4. **Payroll Status**: `Active` , `Hold` , `Paused` , `FinalSettlement` , `Closed` .

5. **Offboarding Status**: `None`, `PendingManagerApproval`, `PendingHrApproval`, `NoticePeriod`, `Discussion`, `Completed`, `Withdrawn`.
6. **Former Portal Status**: `NotEligible`, `Eligible`, `Invited`, `Active`, `Revoked`, `Expired`.
7. **Alumni Status**: `NotApplicable`, `Eligible`, `Invited`, `Member`, `OptedOut`, `Blocked`.

### 4.3 Deterministic Policy Resolution Engine

Policies governing Leave, Attendance Penalties, Notice Periods, and Offboarding Clearance adhere to strict precedence rules:

$$\text{Precedence} = \text{Specific Department} \longrightarrow \text{Specific Role} \longrightarrow \text{Company Wide}$$

When a lifecycle event (such as a resignation) is initiated, the system takes an immutable **Policy Snapshot** (`EmployeeExitPolicySnapshot`), guaranteeing that future policy modifications do not retroactively alter inflight legal or financial obligations.

### 4.4 Automated Attendance-to-Payroll Pipeline

Monthly payroll calculations do not rely on subjective estimates. AutoPayroll strictly requires:

1. Approved and locked monthly attendance summaries (`MonthlyAttendanceSummary.IsApproved && MonthlyAttendanceSummary.IsLocked`).
2. Zero unreviewed attendance exceptions (`PENDING_REVIEW`).
3. Mathematical application of the tenant's configured divisor policy (`FIXED_30_DAYS`, `WORKING_DAYS`, or `CALENDAR_DAYS`).
4. Loss-of-Pay (LOP) deduction computed as:

$$\text{Total LOP Days} = \text{Attendance Unpaid Days} + \text{Approved Policy Penalty Days}$$

### 4.5 Resilient Asynchronous Outbox Architecture

To avoid distributed transaction failure modes between MongoDB and external channels (SMTP email, push notifications, external webhooks), Codeji employs the **Transactional Outbox Pattern** across Recruitment (`RecruitmentEmailOutbox`), Attendance reminders (`AttendanceNotificationOutbox`), and Employee Exits (`EmployeeExitOutbox`). Background workers process queues idempotently with exponential backoff and dead-letter tracking.

---

## 5. Scope of Current System Verification

This documentation package has been generated by directly inspecting and tracing every controller, service, repository, entity, frontend component, migration, and test in the active repository.

| Module / Subsystem | Implementation Status | Key Discovered Artifacts |
|---|---|---|
| **Recruitment & ATS** | Verified Complete | `ApplicantsController.cs`, `RecruitmentLifecycleDrawer.tsx`, 24 backend services, candidate offer public portal. |
| **Employee Lifecycle** | Verified Complete | `UserController.cs`, `EmployeeService.cs` (364 KB), multi-dimensional status model, automated onboarding activation. |
| **Attendance & Shifts** | Verified Complete | `AttendanceController.cs`, `AttendancePenaltyService.cs`, office schedules, punch classification (`P`, `HD`, `LHD`, `ED`), monthly locks. |

| Module / Subsystem | Implementation Status | Key Discovered Artifacts |
| --- | --- | --- |
| **Leave & WFH** | Verified Complete | `LeaveManagementController.cs`, `WorkFromHomeController.cs`, accrual worker, half-day segments, balance reconciliation. |
| **Payroll & Salary** | Verified Complete | `PayRollController.cs`, `AutoPayrollServices.cs`, salary structure, divisor policy, PDF payslips. |
| **Offboarding & FnF** | Verified Complete | `EmployeeExit.cs`, `EmployeeFinalSettlement.cs`, multi-department clearance templates, former employee portal. |
| **Company Administration** | Verified Complete | `CompanyMasterController.cs`, departments, job roles, locations, sequences, custom attributes. |
| **Security & RBAC** | Verified Complete | `AntiforgeryMiddleware.cs`, `CompanyIdMiddleware.cs`, `RoleServices.cs`, Cartesian module permission model. |

## 6. Document Suite Navigation

Technical leads, product managers, architects, and compliance officers should navigate this documentation package using the following structure:

- **Foundational Architecture**: Chapters `01` through `04` detailing repository layouts, system design, and database modeling.
- **Security & Governance**: Chapters `05` through `07` and `11` detailing authentication, authorization, RBAC, tenant isolation, and policies.
- **Workflows & Integrations**: Chapters `08` through `10` and `14` detailing cross-module dependencies, event cascades, notifications, and integrations.
- **Production Assessment**: Chapters `12`, `13`, and `15` detailing error handling, auditability, and production readiness gaps.
- **Executive Summaries**: Chapters `16` through `18` providing future roadmaps, slide-ready presentation summaries, and business proposals.
- **Deep Dives**: The `modules/` directory contains 29 comprehensive chapter documents providing exhaustive file-by-file breakdowns of every subsystem.
- **Visual Reference**: The `diagrams/` directory contains 11 standalone Mermaid specifications.
- **Traceability & Validation**: `SOURCE-TRACEABILITY.md` maps every feature to source code, and `UNVERIFIED-AREAS.md` maintains transparent accounting of system boundaries.

# Codeji CMS / HRMS — Project Overview

> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation
> **Target Audience**: Technical Architects, Engineering Leads, Product Managers, Delivery Managers
> **Last Verified**: September 2026

## 1. Product Mission & Vision

The **Codeji HRMS** platform is designed to provide mid-market and enterprise organizations with a unified, self-governing Human Resource Information System (HRIS). Its mission is to eliminate friction across organizational boundaries by providing an automated, auditable, and mathematically rigorous bridge connecting recruitment, core employment records, time tracking, compensation, and separation.

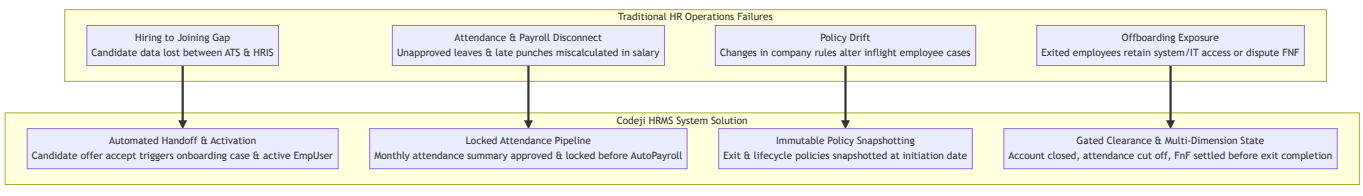
The platform embodies five core engineering principles:

1. **Source-of-Truth Integrity**: Every critical business state (e.g., employment activation, leave approval, payroll lock, exit clearance) is owned and verified on the server. The client SPA reflects state but cannot unilaterally dictate state transitions.
2. **Deterministic Governance**: Business decisions—such as whether an employee is eligible for leave, subject to attendance deductions, or required to serve notice—are resolved through deterministic, hierarchical policies rather than ad-hoc code checks.
3. **Strict Multi-Tenant Isolation**: Tenant boundaries are enforced at the network, middleware, service, and database query layers, preventing data leakage across distinct corporate entities.
4. **Resilient Asynchrony**: External I/O (such as sending candidate emails, attendance punch alerts, or exit notifications) is decoupled from primary request transactions via transactional outbox queues.
5. **Auditable Lifecycle Traceability**: Every status modification records who initiated it, when it became effective, the prior value, and the underlying reason code.

## 2. Core Problem Space & Operational Challenges Addressed

Enterprise organizations operating without a unified HRMS encounter significant operational friction. The Codeji platform directly addresses these key industry bottlenecks:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 3. Supported Stakeholder Personas & Capabilities

The platform defines distinct operational personas. Each persona is assigned tailored UI layouts, actions, and server-verified permissions:

| Persona | Primary Responsibilities | Discovered Capabilities in Codebase |
|---|---|---|
| **Super / Company Administrator** | Tenant administration, system governance, and master data setup. | Configures company profile, departments, job roles/designations, office locations, work schedules, custom attributes, employee ID sequence templates, role-permission matrices, and global policy assignments. Full bypass on module permission evaluations. |
| **HR Manager** | Strategic HR management, policy administration, workforce supervision. | Manages recruitment requisitions, reviews job applications, schedules interviews, issues offers, tracks onboarding checklists, approves leave/WFH requests, manages attendance exceptions, oversees offboarding cases, and approves final settlements. |
| **HR Executive** | Day-to-day HR operations and employee record maintenance. | Equivalent operational permissions to HR Manager under default seed roles ( `EnumsHelper.Roles.HRExecutive` ), executes candidate screening, candidate verification, attendance adjustments, and document uploads. |
| **Reporting Manager** | Direct departmental supervision and team approvals. | Evaluates team leave and WFH requests ( `ApproveTeam` , `ViewTeam` ), reviews resignation notices, participates in joint exit discussions, and conducts handover clearance task sign-offs. |
| **Employee** | Self-service operations and personal data management. | Views own profile, submits personal details (education, certifications, emergency contacts, PAN, bank details), marks daily attendance punch, requests leaves and WFH, downloads generated payslips, views company notice board, and submits self-resignation. |
| **Candidate / Applicant** | External job seekers and prospective employees. | Searches public career portal ( `/careers` ), applies to vacancies with resume upload, tracks application status via public capability token, receives offer letters, and reviews/accepts/rejects offers via `/candidate/offer` . |
| **Former Employee / Alumni** | Exited personnel requiring separation records. | Accesses the dedicated `/former-employee` portal via secure one-time access tokens or OTP sessions to download historical payslips, experience certificates, tax documents, and final settlement summaries. |

## 4. High-Level Functional Domain Breakdown

### 4.1 Talent Acquisition & ATS ( `recruitment/` )

- Public Career Marketplace ( `/careers` , `/careers/:companyCode` ) with custom company branding and vacancy filters.
- Requisition lifecycle management, job posting creation, and applicant tracking across 7 core stages ( `New` , `InProgress` , `OnHold` , `Shortlisted` , `Selected` , `Rejected` , `ReApply` ).
- Advanced recruitment pipeline: candidate screening queues, structured interview scheduling, interviewer scorecards, interview debriefs, multi-level offer approval policies, offer issuance, candidate negotiation tracking, and candidate sharing.

- Resume extraction review queue, duplicate candidate detection engine, talent pool curation, candidate privacy governance, and audit logging export.

### 4.2 Onboarding & Employee Activation ( `employees/` , `recruitment/` )

- Preboarding case tracking ( `RecruitmentPreboardingCase` ) for pre-hire document submission and background check verification.
- Onboarding case engine ( `RecruitmentOnboardingCase` ) with customizable task checklists (IT provisioning, workstation allocation, policy training).
- **Automated Lifecycle Activation**: Transitioning an onboarding case to `Completed` automatically triggers `ActivateEmployeeLifecycle` , transitioning `EmpUser` to `Active` , setting availability to `Working` , activating user accounts, initiating payroll eligibility, and recording history.

### 4.3 Workforce Time & Attendance ( `attendance/` , `holidays/` )

- Daily attendance recording ( `AttendanceModel` , `AttendanceDaySegment` ) with automated office status classification ( `P` , `HD` , `LHD` , `ED` , `LHD+ED` ).
- Shift and schedule management ( `CompanyOfficeSchedule` ), defining start/end times, check-in/out tolerance windows, and grace periods.
- Attendance penalty engine ( `AttendancePenaltyService` ): tracking late arrivals and early departures against monthly policy limits ( `CombinedLhdEdMonthlyLimit` ), raising attendance exceptions ( `AttendancePayrollException` ), and managing HR resolution overrides.
- Monthly attendance locking ( `AttendanceMonthLock` ): validating and locking an attendance period prior to payroll processing.
- Multi-tier leave administration ( `LeaveRequest` , `LeavePolicy` , `EmployeeLeaveBalance` ): supporting sick, casual, earned, maternity, paternity leaves, and monthly accrual processing via `LeaveAccrualHostedService` .
- Integrated Work From Home (WFH) management ( `WorkFromHomeRequest` ): supporting full-day and half-day segment requests, timing validations, and conflict reconciliation with attendance punches.
- Comprehensive calendar system ( `CalendarEntity` , `WeeklyOffSetting` , `Holidays` ): coordinating national/cultural holidays, weekly off days, and company working calendar rules.

### 4.4 Compensation & Payroll ( `payroll/` , `payrollsettings/` )

- Standardized salary structure definition ( `SalaryModel` ): defining CTC, basic pay, allowances (HRA, LTA, other), and statutory deductions (EPF, ESIC, professional tax, income tax).
- Divisor policy governance ( `PayrollDivisorPolicy` ): configuring monthly salary daily rates based on Fixed 30 Days, Actual Working Days, or Calendar Days.
- AutoPayroll execution engine ( `AutoPayrollServices` ): validating locked attendance, computing Loss of Pay (LOP) days, applying partial-month prorations for mid-month joiners and exits, and committing monthly payroll mutations.
- Digital payslip generation ( `PdfService` ): rendering pixel-perfect HTML/CSS templates into PDF documents with currency formatting in English words and secure download endpoints.

### 4.5 Separation, Clearances & Final Settlement ( `employees/` , `formerEmployeePortal/` )

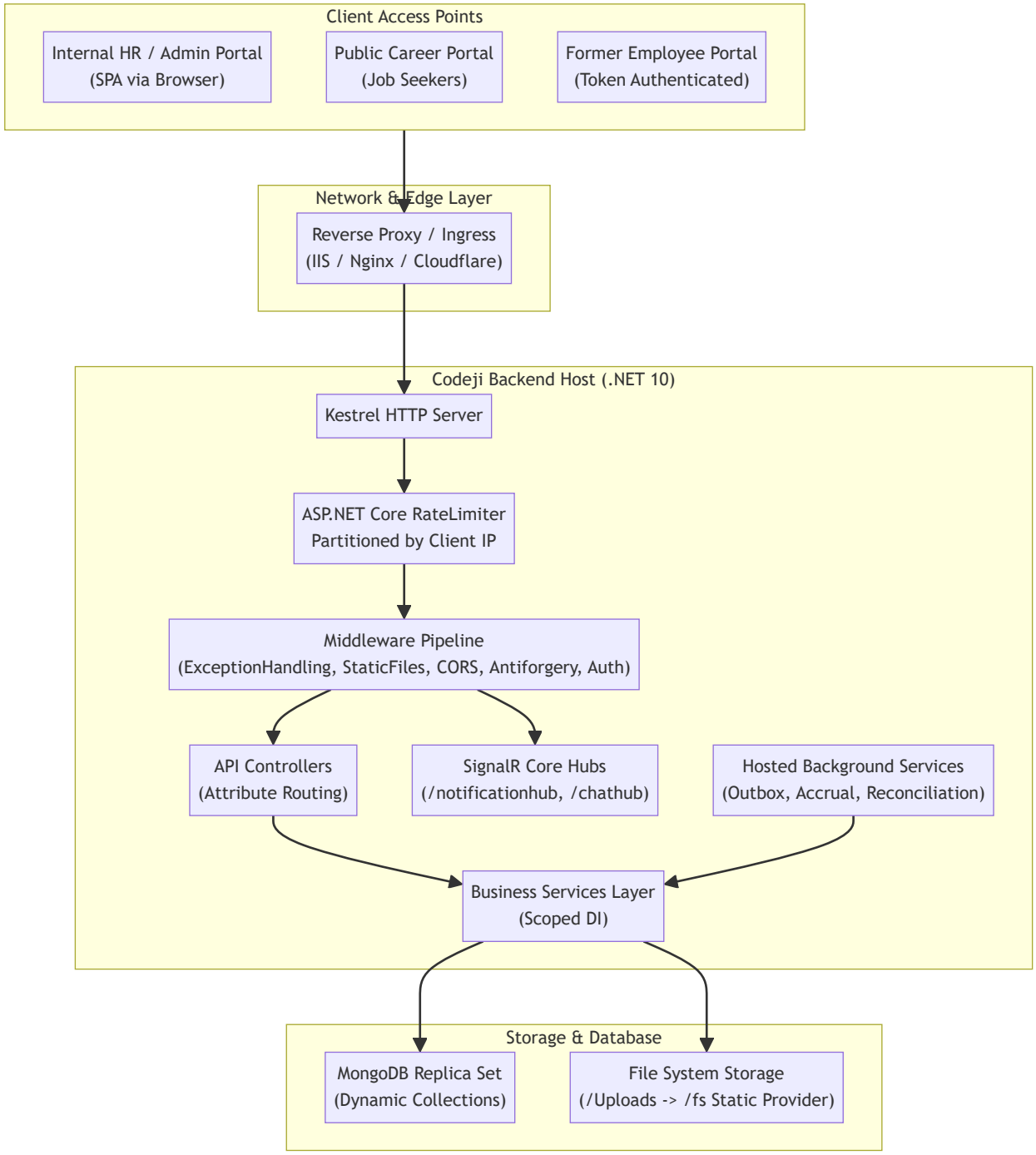
- Multi-path separation initiation: employee self-service resignation or HR-initiated termination/contract end.
- Resignation review workflow: manager review, HR review, joint discussion threads, withdrawal requests, and proposed Last Working Date (LWD) negotiations.
- Notice period engine: calculating required notice days from snapshot policies, tracking buyouts, waivers, and automated notice countdown reminders.
- Clearance workflow: multi-department clearance templates (IT Access, Finance, Assets, Handover Tasks, Benefits, Resignation Evidence).

- Full & Final Settlement ( `EmployeeFinalSettlement` ): reconciling unpaid salary from payroll, leave encashments, outstanding dues, notice recoveries, and reimbursement disbursements.
- Former employee lifecycle: revoking active refresh sessions, creating attendance cutoff records, transitioning employment status to `Resigned` / `Terminated` , and provisioning token-based alumni portal access.

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## 5. Deployment Topology & Environments

**ARCHITECTURE / WORKFLOW DIAGRAM**




### 5.1 Verified Runtime Specifications

- **Operating System Compatibility**: Windows Server (IIS Integration enabled via `builder.WebHost.UseIISIntegration()` ) and Linux (Kestrel standalone behind reverse proxies).
- **Target Framework**: .NET 10 ( `net10.0` ).

- **Database Engine**: MongoDB 6.0+ (utilizing official C# Driver v2.28.0).
- **Frontend Build**: Vite 5.x bundling React 18 with TypeScript 5.x.
- **Node.js Environment**: Node 18+ LTS for frontend compilation and testing.

# Codeji CMS / HRMS — Repository Architecture

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Target Repositories**: `CMS-Backend-Core` , `CMS-React`
**Last Verified**: September 2026

## 1. Monorepo / Multi-Root Repository Topology

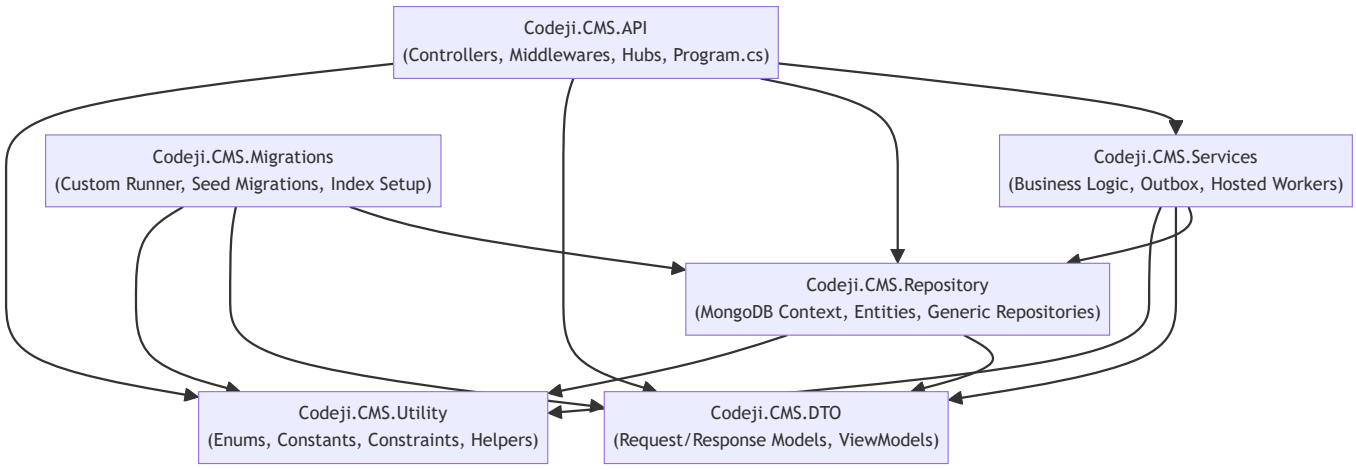
The Codeji CMS / HRMS codebase is organized as a multi-root workspace comprising two primary engineering repositories alongside quality assurance, document generation, and project management artifacts:

```
CMS/
├── CMS-Backend-Core/          # ASP.NET Core (.NET 10) Web API, Services, Repositories, Migrations
├── CMS-React/                 # React 18, TypeScript, Vite SPA client
├── docs/                      # Central system & architectural documentation package
├── qa/                        # Quality assurance suites, postman collections, test run scripts
├── pdf_builder/               # PDF rendering utilities and template build tooling
└── _validation/               # Automation scripts for regression and pipeline verification
```

## 2. Backend Architectural Structure ( `CMS-Backend-Core` )

The backend follows Clean Architecture principles with clear layer separation across 6 distinct projects, bound together by dependency injection in `Codeji.CMS.API` :

ARCHITECTURE / WORKFLOW DIAGRAM




### 2.1 Layer-by-Layer Architectural Breakdown

`Codeji.CMS.API`

- **Path**: `CMS-Backend-Core/Codeji.CMS.API`
- **Purpose**: Web API host, HTTP request termination, protocol mediation, authentication filter pipeline, real-time WebSocket communication, rate limiting, and static file provisioning.

- **Why It Exists**: Exposes RESTful JSON endpoints and SignalR hubs to internal and external clients while securing the application perimeter.
- **How It Communicates**: Receives HTTP/HTTPS requests from clients, invokes `Codeji.CMS.Services` via registered scoped interfaces, dispatches SignalR events via `IHubContext`, and streams file responses (e.g. PDF payslips, uploaded documents).
- **Important Files**:
  - `Program.cs`: Startup orchestrator configuring Kestrel, RateLimiter, CORS, JWT Bearer, Antiforgery, SignalR, MongoDB client, and DI container.
  - `Controllers/`: 43 API controllers handling account, employee, recruitment, attendance, leave, payroll, and company management.
  - `App_Start/AntyForgeryMiddleware.cs`: Core security gate executing anti-CSRF checks, AppVersion matching, and delegating to `CompanyIdMiddleware`.
  - `App_Start/AuthenticateUserRequest.cs` (`CompanyIdMiddleware`): Enforces authenticated tenant claims (`CompanyId`), checks active company/user records, and evaluates `ModulePermissionAttribute`.
  - `ChatHub/ChatHub.cs` & `Notification/NotificationHub.cs`: SignalR real-time event distribution endpoints.
- **Architectural Responsibility**: Host configuration, route dispatching, security filtering, and HTTP transport serialization.
- **Dependencies**: Depends on `Codeji.CMS.Services`, `Codeji.CMS.Repository`, `Codeji.CMS.DTO`, `Codeji.CMS.Utility`.
- **Potential Concerns**: High cyclomatic complexity in `UserController.cs` (51.5 KB) which handles user, profile, lifecycle, and offboarding operations within a single controller.

`Codeji.CMS.Services`

- **Path**: `CMS-Backend-Core/Codeji.CMS.Services`
- **Purpose**: Core application business logic, workflow coordination, deterministic policy evaluation, calculation algorithms, and background worker execution.
- **Why It Exists**: Isolates domain business logic from the HTTP transport layer and database persistence mechanisms.
- **How It Communicates**: Injected into controllers via interfaces, consumes `IMongoDbRepository<T>` for data persistence, coordinates cross-module events, and enqueues outbox records.
- **Important Folders & Files**:
  - `Registration/ServicesRegistration.cs`: Centralized extension method `AddBusinessServices` registering all 40+ scoped services.
  - `Employees/EmployeeService.cs` (364 KB): Primary employee management service covering profile editing, contract lifecycle, bulk imports, and offboarding orchestration.
  - `Employees/OffboardingPolicyResolver.cs`: Pure functional policy resolution engine implementing Department -> Role -> Company precedence.
  - `Attendance/AttendancePenaltyService.cs` (49.9 KB): Manages late arrival/early departure limits, monthly exception reviews, and lock validation.
  - `Attendance/AttendanceStatusCalculationService.cs`: Evaluates punch times against office schedules to assign `P`, `HD`, `LHD`, `ED`, `LHD+ED`.
  - `LeaveManagement/LeaveManagementService.cs` (128 KB): Leave requests, entitlement calculations, approval workflows, and balance tracking.
  - `PayRoll/AutoPayRollServices.cs` (20.8 KB): Gated monthly payroll generation engine validating locked attendance and computing LOP deductions.
  - `Recruitments/`: 24 specialized recruitment services covering candidate tracking, screening queues, scorecards, offers, and talent pools.
  - `BackgroundTasks/`: `LeaveAccrualHostedService`, `PriorityQueuedHostedService`, `BirthDayAndAnniversaryNotificationHostedServices`.

- **Architectural Responsibility**: Business domain validation, workflow transactionality, policy enforcement, and asynchronous job processing.
- **Dependencies**: Depends on `Codeji.CMS.Repository`, `Codeji.CMS.DTO`, `Codeji.CMS.Utility`.
- **Potential Concerns**: `EmployeeService.cs` is monolithic (364 KB, 7,000+ lines); splitting into dedicated lifecycle, profile, and contract services is recommended for maintainability.

`Codeji.CMS.Repository`

- **Path**: `CMS-Backend-Core/Codeji.CMS.Repository`
- **Purpose**: MongoDB data access abstraction, database context management, collection mappings, and entity domain definitions.
- **Why It Exists**: Encapsulates MongoDB driver queries and provides a standardized, testable repository interface (`IMongoDbRepository<T>`).
- **How It Communicates**: Consumed exclusively by `Codeji.CMS.Services` and `Codeji.CMS.Migrations`. Communicates with MongoDB server via `MongoClient`.
- **Important Files**:
  - `Repositories/MongoDbContext.cs`: Manages `IMongoDatabase` resolution and returns dynamic typed collections via `GetCollection<TEntity>()`.
  - `Repositories/MongoRepository.cs` (20.7 KB): Generic repository implementing CRUD, pagination, filtering, soft-delete expressions, sorting, and optimistic concurrency updates.
  - `Entities/`: Strongly typed C# domain entities mapped to MongoDB collections:
    - `Employees/EmpUser.cs`: The root employee record.
    - `Employees/EmployeeLifecycle.cs`: Multi-dimensional lifecycle state model.
    - `Employees/EmployeeExit.cs`: Separation audit trail with embedded policy snapshot.
    - `Employees/EmployeeFinalSettlement.cs`: Full & Final financial record.
    - `Attendance/AttendanceModel.cs`: Daily employee attendance record.
    - `Leave/LeaveRequest.cs` & `LeavePolicy.cs`: Leave records and policy configurations.
    - `Recruitments/Applicant.cs`, `JobVacancy.cs`, `RecruitmentOffer.cs`: ATS domain models.
    - `RolePermissions/Roles.cs`, `Module.cs`, `Permission.cs`, `RolePermission.cs`: RBAC definitions.
- **Architectural Responsibility**: Document schema mapping, BSON serialization, query construction, and data integrity enforcement.
- **Dependencies**: Depends on MongoDB.Driver, `Codeji.CMS.DTO`, `Codeji.CMS.Utility`.
- **Potential Concerns**: Custom repository abstracts away some native MongoDB features such as aggregation pipelines or multi-document transactions when running on standalone nodes.

`Codeji.CMS.DTO`

- **Path**: `CMS-Backend-Core/Codeji.CMS.DTO`
- **Purpose**: Data Transfer Objects, API contract definitions, command/query models, and view models.
- **Why It Exists**: Prevents direct exposure of internal database entities over public APIs, preventing over-posting and enabling API versioning.
- **How It Communicates**: Serialized/deserialized by ASP.NET Core JSON formatters, mapped to/from entities in services via Mapster (`ServiceMapper`).
- **Important Files**:
  - `userModel.cs`: Core user/employee DTO (`UserModel`) with regex validations for PAN and bank account formats.
  - `Attendance/`: `AdminAttendanceCreateDto`, `AttendancePenaltyDtos`, `WorkFromHomeDto`.
  - `Employee/`: `EmployeeExitDtos.cs` (31.6 KB), `EmployeeLifecycleDtos.cs`.
  - `PayRoll/`: `GetEmpPayRollRequestDto`, `PayslipRequestDto`, `SalarySlipModel`.

    - `Recruitments/`: Commands and view models for screening, offers, applications, and onboarding queues.
- **Architectural Responsibility**: External contract stabilization and request input validation.
- **Dependencies**: Minimal dependencies on system annotations.

`Codeji.CMS.Migrations`

- **Path**: `CMS-Backend-Core/Codeji.CMS.Migrations`
- **Purpose**: Custom database migration and schema evolution runner.
- **Why It Exists**: MongoDB is schema-less by default; this project provides structured, timestamped migrations ensuring required collections, compound unique indexes, and seed records exist across environments.
- **How It Communicates**: Executed during application deployment or manually via console runner. Interacts directly with MongoDB via `IMongoDatabase`.
- **Important Files**:
    - `Program.cs`: Migration runner discovering all classes implementing `IMigration` and executing unapplied migrations tracked in the `Migration` collection.
    - `Migrations/`: Over 65 migration files setting up unique indexes (e.g. `AddRecruitmentCoreIndexes.cs`, `AddEmployeeLifecycleIndexes.cs`, `HardenHrTenantIndexes.cs`) and seeding default roles, modules, permissions, and templates.
- **Architectural Responsibility**: Database schema versioning, index provisioning, and environment bootstrapping.
- **Dependencies**: Depends on MongoDB.Driver, `Codeji.CMS.Repository`.

`Codeji.CMS.Utility`

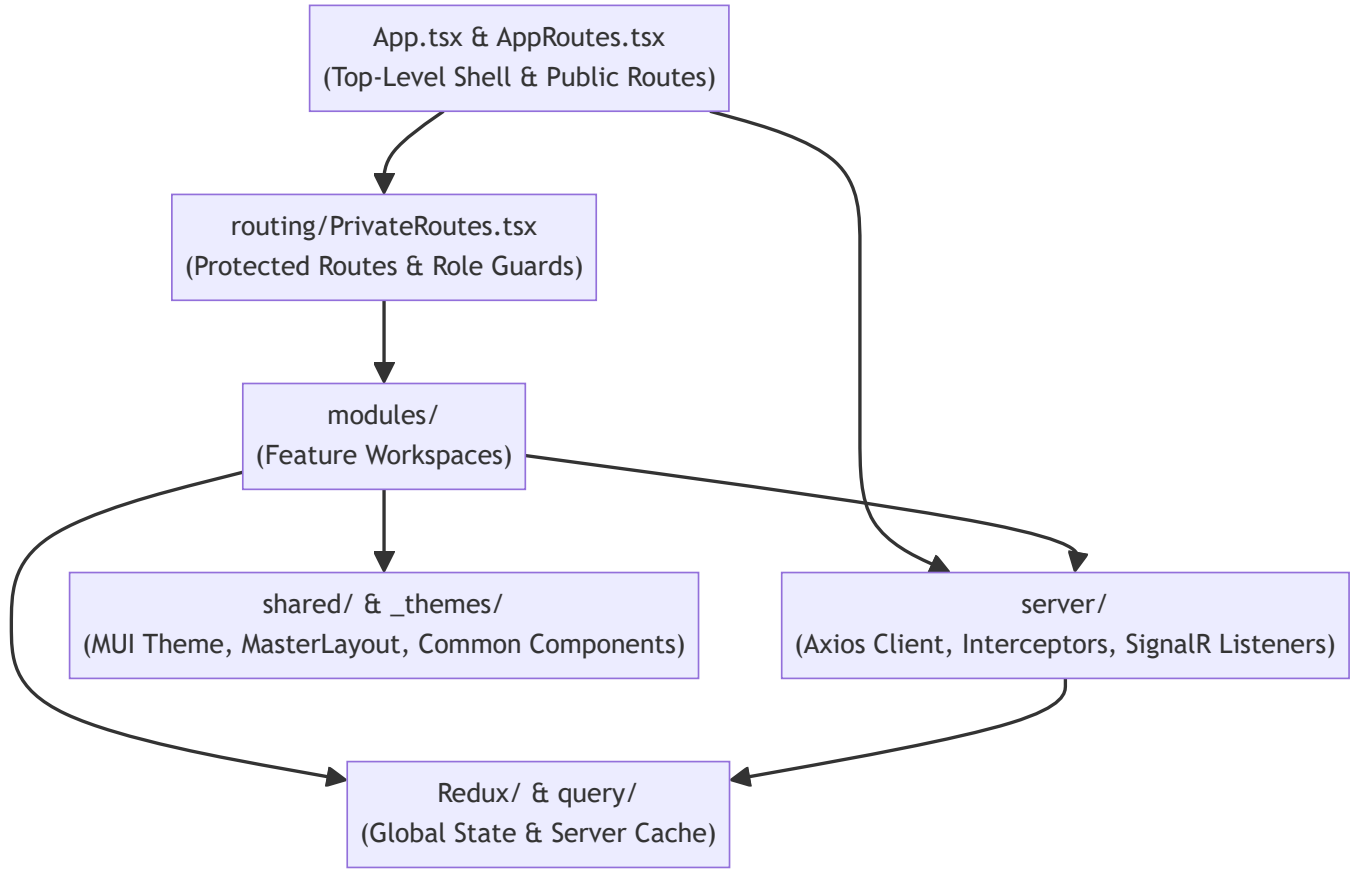
- **Path**: `CMS-Backend-Core/Codeji.CMS.Utility`
- **Purpose**: Cross-cutting utilities, system enums, permission constants, date/time helpers, and security sanitizers.
- **Why It Exists**: Houses pure, shared domain constants and helper functions accessible across all layers without circular dependencies.
- **Important Files**:
    - `Constraints/ConstraintHelper.cs`: Defines string constants for `AppModule` and `Permission`.
    - `Enums/EnumsHelper.cs`: System-wide enums for Roles, ClaimTypes, Employment Types, and Statuses.
    - `Helpers/IndiaTime.cs`: Standardizes date/time calculations to Indian Standard Time (IST) where business rules require local calendar alignment.
    - `middlewares/CurrentContext.cs`: Resolves authenticated user, tenant, and role claims from `IHttpContextAccessor`.
- **Architectural Responsibility**: Centralized system constraints and invariant definitions.

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## 3. Frontend Architectural Structure (`CMS-React`)

The frontend is organized as a modular, feature-oriented React application structured around business domains:

ARCHITECTURE / WORKFLOW DIAGRAM




### 3.1 Frontend Folder Breakdown

| Path | Purpose | Important Files | Key Architectural Responsibility |
|---|---|---|---|
| `src/app/routing` | Client-side routing, code splitting, permission gates. | `AppRoutes.tsx`, `PrivateRoutes.tsx`, `RecruitmentRoutes.tsx`, `ProfileRoutes.tsx`. | Lazy loads feature modules via React Suspense; enforces client-side permission protection via `<ProtectedRoute />`. |
| `src/app/modules/admin` | Company master administration. | `adminRoutes.tsx`, `CompanyMaster.tsx`, `CompanyDetails.tsx`, `Roles.tsx`, `EditPermission.tsx`. | Admin workspace for managing departments, job roles, locations, sequences, and role-permission matrices. |
| `src/app/modules/recruitment` | Full-scale recruitment & ATS portal. | `Applications.tsx`, `Jobs.tsx`, `RecruitmentLifecycleDrawer.tsx` (50.6 KB), `RecruitmentSettings.tsx`, `services.ts`. | Manages job postings, candidate pipelines, interview workflows, scorecards, offer letters, and onboarding queues. |
| `src/app/modules/employees` | Employee directory and lifecycle management. | `Employees.tsx`, `EmployeeLifecycleManagement.tsx`, `EmployeeLifecycleCaseQueue.tsx`, `EmployeeLifecycleReconciliationPanel.tsx`. | Employee directory table, lifecycle policy configuration, case review queues, and employment status reconciliation. |

| Path | Purpose | Important Files | Key Architectural Responsibility |
| --- | --- | --- | --- |
| `src/app/modules/users` | Profile view, settings, and offboarding center. | `UserProfile.tsx`, `AboutUser.tsx`, `EmployeeExitCaseCard.tsx` (73 KB), `EmployeeExitManagementCard.tsx` (66 KB), `EmployeeExitPolicySettings.tsx` (59 KB). | Handles personal employee profiles, self-service settings, and the complete HR/Manager separation and offboarding workspace. |
| `src/app/modules/attendance` | Workforce punch and exception tracking. | `EmployeeAttendance.tsx` (31 KB), `AttendanceCalendar.tsx` (78.8 KB), `MonthlyAttendanceExceptions.tsx` (44.4 KB), `AttendanceStatusSettings.tsx`. | Daily punch grid, office schedule configuration, late/early deduction rules, monthly exception resolution, and month locking. |
| `src/app/modules/leaves` | Leave management & WFH requests. | `LeaveManagement.tsx`, `LeaveRequest.tsx`, `AddLeavePolicy.tsx` (48.3 KB), `WorkFromHome.tsx` (23.7 KB), `EmployeesLeaveBalance.tsx`. | Multi-type leave requests, balance tracking, accrual displays, WFH segment booking, and policy authoring. |
| `src/app/modules/payroll` | Payroll generation & payslip distribution. | `PayRollManagement.tsx`, `PayRollTable.tsx`, `ProcessPayrollModal.tsx`, `UploadPayRollData.tsx`, `services.ts`. | Runs AutoPayroll, validates monthly attendance readiness, inspects salary breakdowns, and downloads PDF payslips. |
| `src/app/modules/payrollsettings` | Salary structure & compensation rules. | `PayrollSettings.tsx`, `SalaryTable.tsx`, `AddUpdateEmployeeSalary.tsx`. | Manages employee salary components (Basic, HRA, LTA, EPF, ESIC, Tax) and tenant divisor policies. |
| `src/app/modules/careerPortal` | Public career marketplace. | `CareerMarketplace.tsx`, `CareerJobDetails.tsx`, `SavedJobsPage.tsx`, `CareerPreferencesPage.tsx`. | Unauthenticated candidate-facing job discovery portal with custom company branding and vacancy filters. |
| `src/app/modules/formerEmployeePortal` | Alumni document and separation portal. | `FormerEmployeePortalExperience.tsx`. | Lightweight, unauthenticated-app portal for departed employees accessing payslips and experience certificates via one-time tokens. |
| `src/server` | HTTP client and WebSocket connectors. | `axios.ts` (7.6 KB), `SignalR.tsx`, `ChatHub.tsx`. | Axios instance with 401 refresh token queue interceptor, `Accept-Language` header injection, and SignalR hub connection managers. |
| `src/app/Redux` | Synchronous client UI state. | `store.ts`, `authSlice.ts`, `userSlice.ts`. | Manages authenticated user tokens, user profile metadata, and theme preferences. |

| Path | Purpose | Important Files | Key Architectural Responsibility |
| --- | --- | --- | --- |
| `src/app/query` | Asynchronous server cache management. | `queryClient.ts`, `queryKeys.ts`. | TanStack Query configuration for caching server responses, background refetching, and cache invalidation. |
| `src/utils` | Enums, constants, formatters, and helpers. | `enums.ts`, `Constant.ts`, `common.ts`, `currencyInput.ts`. | Houses shared frontend enums, route definitions, currency-in-words formatters, and UI permission evaluators. |

## 4. Supporting Repositories & Tooling

### 4.1 `docs/`

- **Purpose**: System-wide documentation repository housing technical specifications, architectural blueprints, sprint deliverables, and compliance reviews.
- **Architectural Meaning**: Forms the definitive engineering and operational handbook for all development, audit, and onboarding activities.

### 4.2 `qa/`

- **Purpose**: Automated API test collections, Postman integration suites, and end-to-end regression scripts used during release gates.

### 4.3 `pdf_builder/`

- **Purpose**: Dedicated rendering pipeline supporting HTML-to-PDF compilation for employee payslips, appointment letters, and experience certificates.

## 5. Architectural Communication & Dependency Matrix

The following matrix records how code layers communicate across architectural boundaries:

| Calling Layer | Target Layer | Communication Mechanism | Transport / Protocol |
| --- | --- | --- | --- |
| `CMS-React` Modules | `CMS-Backend-Core` Controllers | Axios HTTP Client (`src/server/axios.ts`) | RESTful JSON over HTTPS |
| `CMS-React` Notifications | `CMS-Backend-Core` Hubs | `@microsoft/signalr` | WebSockets (`/notificationhub`, `/chathub`) |
| `Codeji.CMS.API` | `Codeji.CMS.Services` | ASP.NET Core Dependency Injection | In-Process Interface Calls (`Scoped`) |
| `Codeji.CMS.Services` | `Codeji.CMS.Repository` | Repository Pattern (`IMongoDbRepository<T>`) | In-Process C# Linq Expressions |
| `Codeji.CMS.Repository` | MongoDB Cluster | MongoDB C# Driver v2.28 | TCP Wire Protocol (Mongo Connection String) |
| `Codeji.CMS.Services` (Outbox) | SMTP / External Gateway | Hosted Services (`BackgroundService`) | SMTP / HTTPS Webhooks |
| `CMS-Backend-Core` | Static Files (`/Uploads`) | `PhysicalFileProvider` (`/fs/*`) | HTTP File Streaming with Custom CORS |

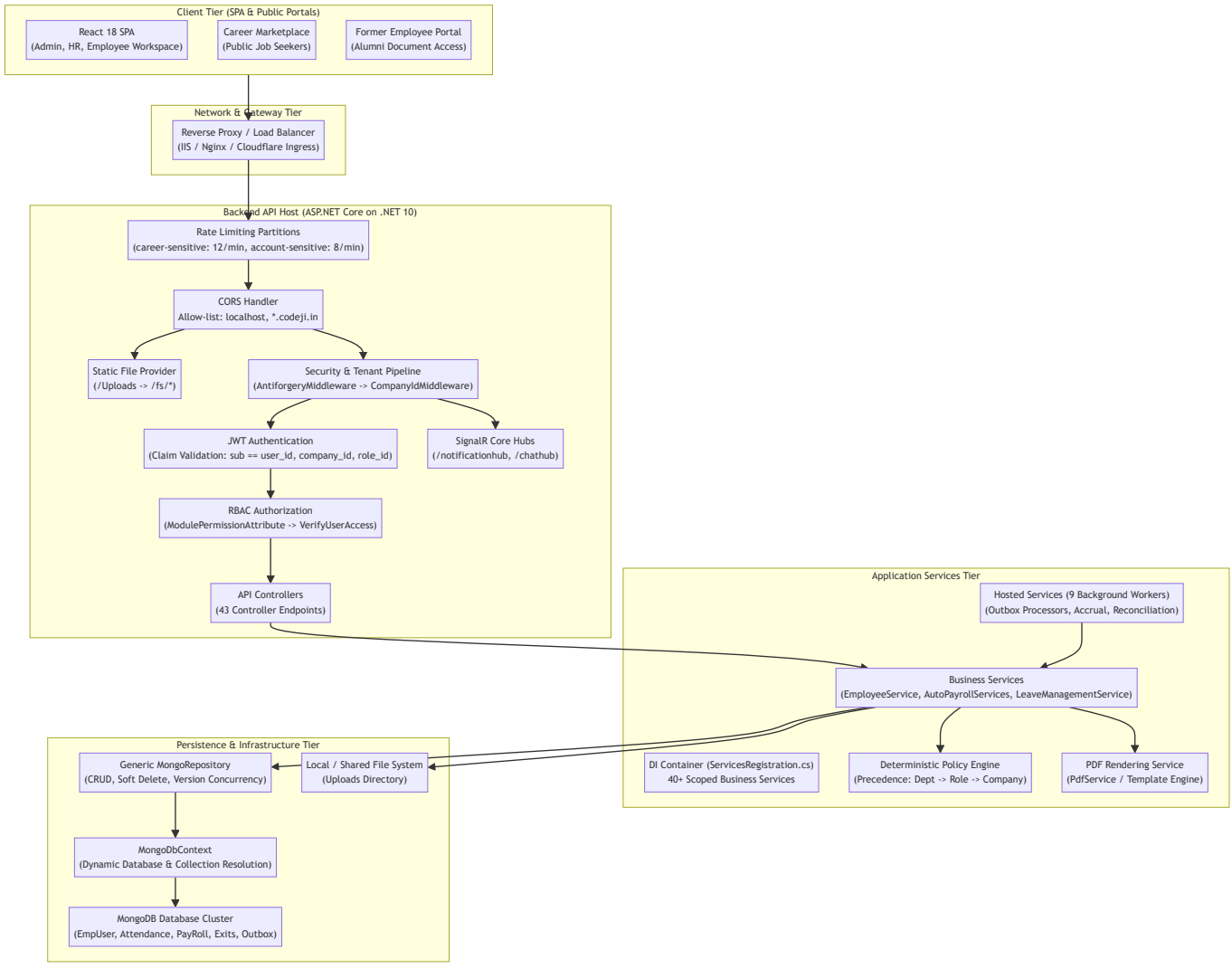
# Codeji CMS / HRMS — System Architecture

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Runtime Frameworks**: ASP.NET Core (.NET 10) | React 18 (TypeScript, Vite) | MongoDB
**Last Verified**: September 2026

## 1. End-to-End System Topology

The Codeji HRMS platform is structured as an enterprise-grade distributed system designed for high concurrency, strict tenant isolation, and auditable operational workflows:

ARCHITECTURE / WORKFLOW DIAGRAM



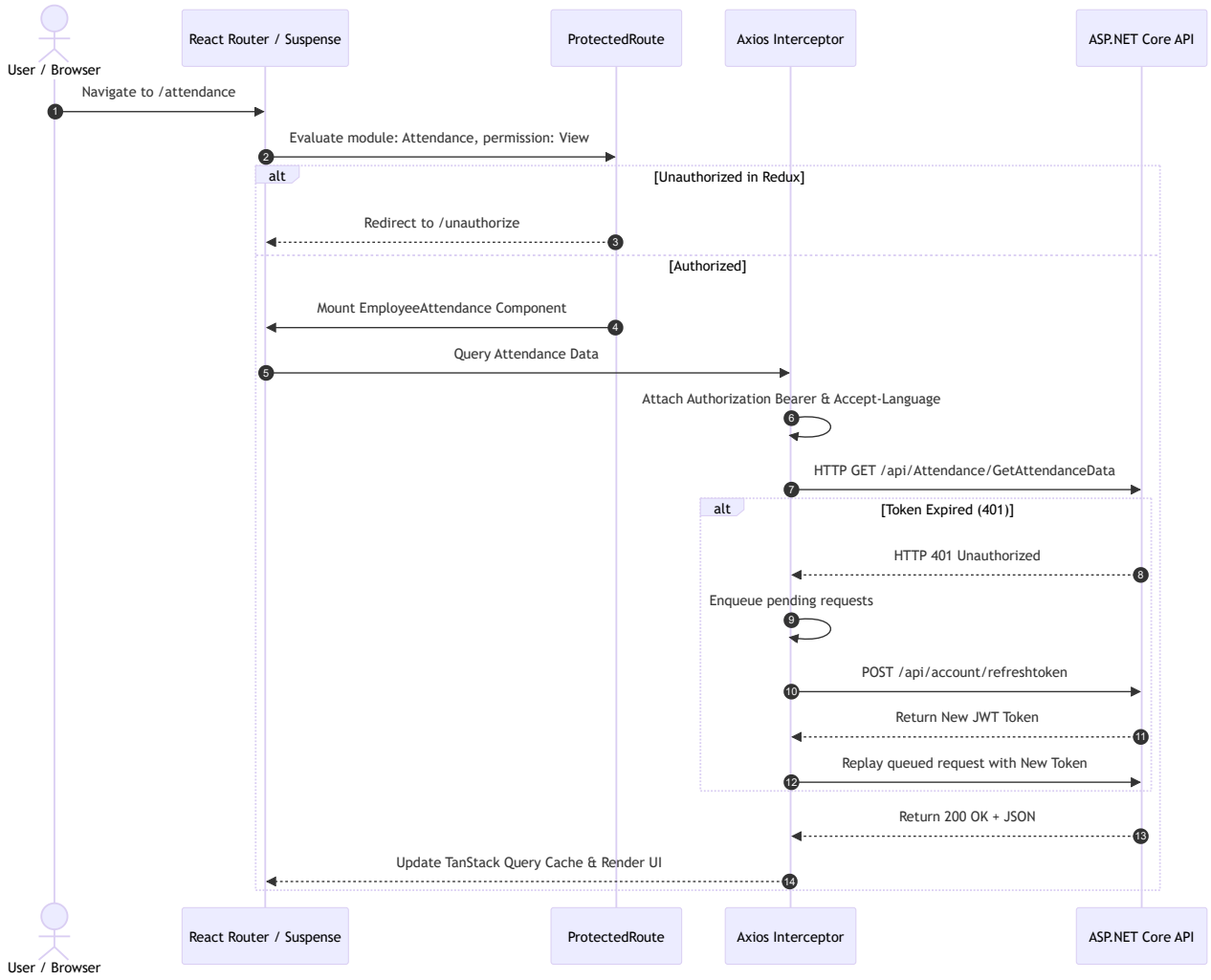
## 2. Frontend Application Architecture

### 2.1 Technology Stack & Architectural Principles

The frontend application ( `CMS-React` ) is engineered using **React 18** with **TypeScript** and bundled via **Vite**. The client-side architecture enforces:

- **Modular Component Isolation**: Features reside in self-contained directories under `src/app/modules/` .
- **Hybrid State Architecture**:
  - Synchronous UI and Session State: Managed via **Redux Toolkit** ( `src/app/Redux` ). Stores authenticated tokens, current user payload, and theme configuration.
  - Asynchronous Server Cache State: Managed via **TanStack Query** ( `src/app/query` ). Eliminates state drift through automated query key invalidation ( `queryClient.invalidateQueries` ).
- **Resilient Transport Interception**:
  - `src/server/axios.ts` : Configures global request/response interceptors.
  - Automatic `Accept-Language` header injection based on active localization.
  - Non-blocking 401 token refresh queue: Concurrently queues pending API calls while asynchronously renewing the access token via `/api/account/refreshtoken` .
- **Granular Route Guarding**:
  - `src/app/modules/users/about/components/ProtectedRoute.tsx` : Intercepts route navigation. Validates that the active user possesses the required module permission ( `Permission.View` , `Permission.Edit` ) or role type before mounting lazy-loaded components.

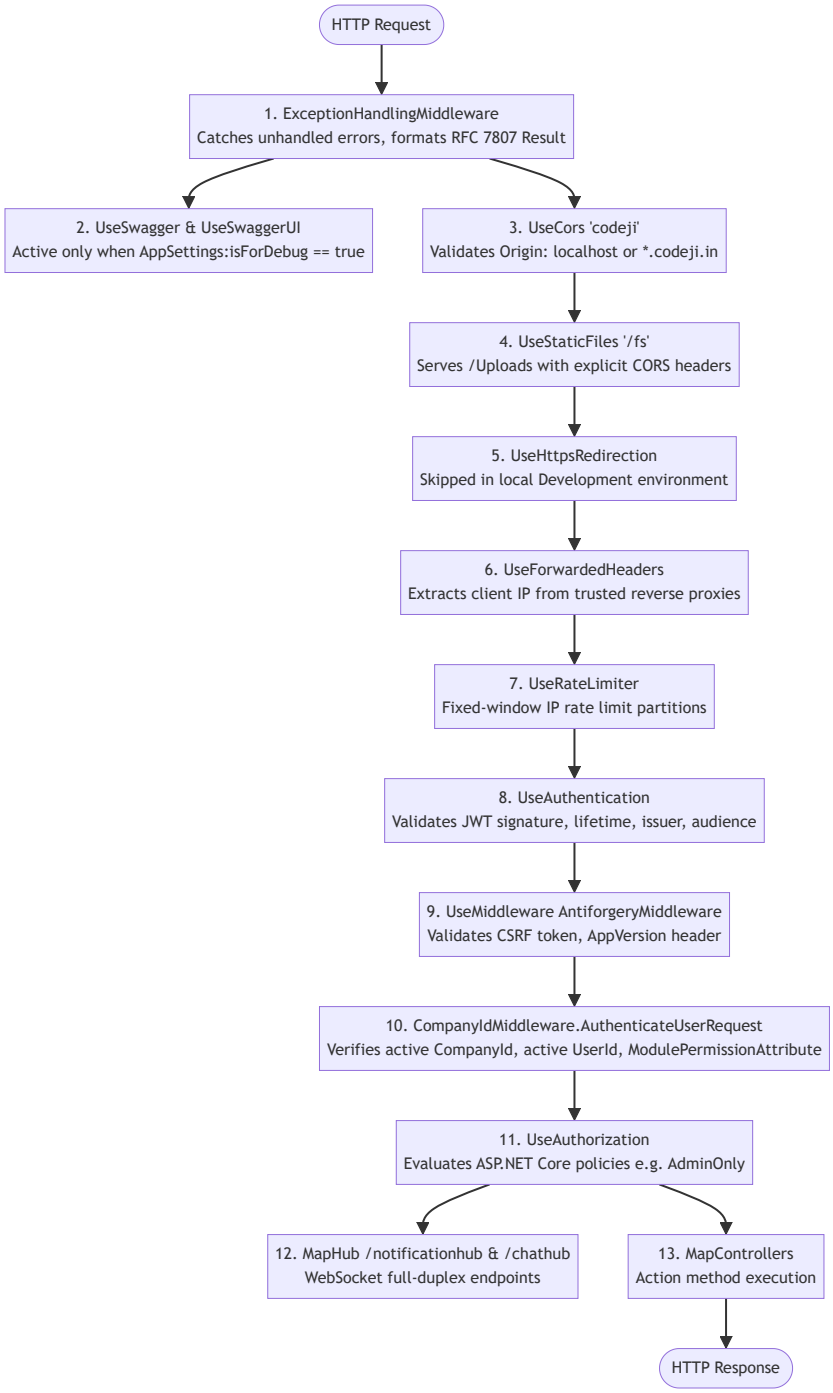
**ARCHITECTURE / WORKFLOW DIAGRAM**




## 3. Backend HTTP Middleware Pipeline

The backend request pipeline in `Codeji.CMS.API/Program.cs` is ordered with strict security precedence:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 3.1 Middleware Verification & Responsibilities

| Order | Middleware Component | Source Reference | Responsibility & Invariants |
|---|---|---|---|
| **1** | `ExceptionHandlingMiddleware` | `Program.cs:338` | Global try/catch handler ensuring no raw stack traces leak to the client; transforms exceptions into standardized `Result` envelopes. |
| **2** | `UseCors("codeji")` | `Program.cs:363` | Allows credentials, any header, any method, but strictly limits origins to `localhost` or domain suffix `.codeji.in`. |

| Order | Middleware Component | Source Reference | Responsibility & Invariants |
|---|---|---|---|
| **3** | `UseStaticFiles("/fs")` | `Program.cs:366` | Serves physical files from `/Uploads`. Because static file middleware is terminal, custom CORS response headers are attached in `OnPrepareResponse`. |
| **4** | `UseForwardedHeaders` | `Program.cs:401` | Resolves real client IP from `X-Forwarded-For` using explicit `KnownProxies` configured in `appsettings.json`. |
| **5** | `UseRateLimiter` | `Program.cs:402` | Protects sensitive unauthenticated endpoints: `career-sensitive` (12 req/min), `account-sensitive` (8 req/min), `career-analytics` (30 req/min). |
| **6** | `UseAuthentication` | `Program.cs:403` | Validates JWT Bearer signature using symmetric secret key. In `OnTokenValidated`, strictly validates that `sub == user_id`, `company_id` is present, and `role_id` is present. |
| **7** | `AntiforgeryMiddleware` | `Program.cs:407` | Validates `XSRF-TOKEN` cookie/header on state-changing unauthenticated requests. Validates `AppVersion` header against server version (returns `426 Upgrade Required` on mismatch). |
| **8** | `CompanyIdMiddleware` | `AntyForgeryMiddleware.cs:60` | Resolves `company_id` and `user_id` from claims. Validates that the company and user exist and are active in MongoDB. Evaluates endpoint `ModulePermissionAttribute` against the user's role permissions. |
| **9** | `UseAuthorization` | `Program.cs:408` | Executes ASP.NET Core policy handlers (`AdminOnly`, `HrExitOverride`). |
| **10** | `MapControllers` | `Program.cs:427` | Dispatches request to target controller action. |

## 4. Application & Service Layer Architecture

### 4.1 Clean Service Separation

Business logic is decoupled from ASP.NET Core controllers into specialized domain services implementing interfaces registered with **Scoped** lifetime in `Codeji.CMS.Services/Registration/ServicesRegistration.cs`:

```
Controllers (Codeji.CMS.API)
     │ (Calls DTO-based interface methods)
     ▼
Business Services (Codeji.CMS.Services)
     │ (Applies business rules, validates policies, maps models)
     ▼
Generic Repositories (Codeji.CMS.Repository)
     │ (Executes typed Mongo queries via LINQ)
     ▼
MongoDB Driver (MongoDB.Driver)
```

### 4.2 Data Transfer Object (DTO) & Mapping Pattern

To enforce data contracts and prevent internal schema leakage:

1. All client inputs are bound to strongly typed Request DTOs in `Codeji.CMS.DTO`.
2. DTOs use system annotations (e.g. `[RegularExpression]` for PAN and Bank Accounts) for declarative validation.
3. Object mapping between Entities and DTOs is executed via **Mapster** ( `ServiceMapper` / `MapsterConfig` ), scanned assembly-wide at startup. Direct database entities are never returned raw across the API boundary.

---

## 5. Persistence & Data Access Architecture

### 5.1 Dynamic MongoDB Collection Resolution

Persistence is implemented on top of the official MongoDB C# Driver ( `MongoDB.Driver v2.28.0` ). The system eschews Entity Framework in favor of a lightweight, high-performance generic repository:

```
// Codeji.CMS.Repository.Repositories.MongoDbContext
public IMongoCollection<TEntity> GetCollection<TEntity>()
{
    return _appDatabase.GetCollection<TEntity>(typeof(TEntity).Name);
}
```

- Each entity corresponds to a collection named after its C# class (e.g., `EmpUser` , `Attendance` , `EmployeeExit` , `LeaveRequest` , `EmpPayRoll` ).
- `MongoRepository<TEntity>` implements `IMongoDbRepository<TEntity>` , exposing asynchronous LINQ-compatible querying ( `FirstOrDefault` , `GetAll` , `AddOne` , `AddMany` , `Update` , `DeleteOne` , `Count` , `Exist` ).

### 5.2 Optimistic Concurrency Control

Concurrent mutations to high-stakes lifecycle entities (such as `EmployeeExit` , `RecruitmentOnboardingCase` , `AttendancePenaltyPolicy` , and `EmpUser.Lifecycle` ) are protected by explicit integer `Version` fields:

```
// Concurrency check example in RecruitmentOnboardingService
var filter = Builders<RecruitmentOnboardingCase>.Filter.Where(x =>
    x.CompanyId == companyId &&
    x.CaseId == onboarding.CaseId &&
    (x.Version == version || x.Version == 0));

var updated = await cases.Update(filter, onboarding);
if (!updated.Success)
    return Fail(RecruitmentErrorCodes.ConcurrencyConflict,
        "Onboarding case was changed by another user.", StatusCodes.Status409Conflict);
```

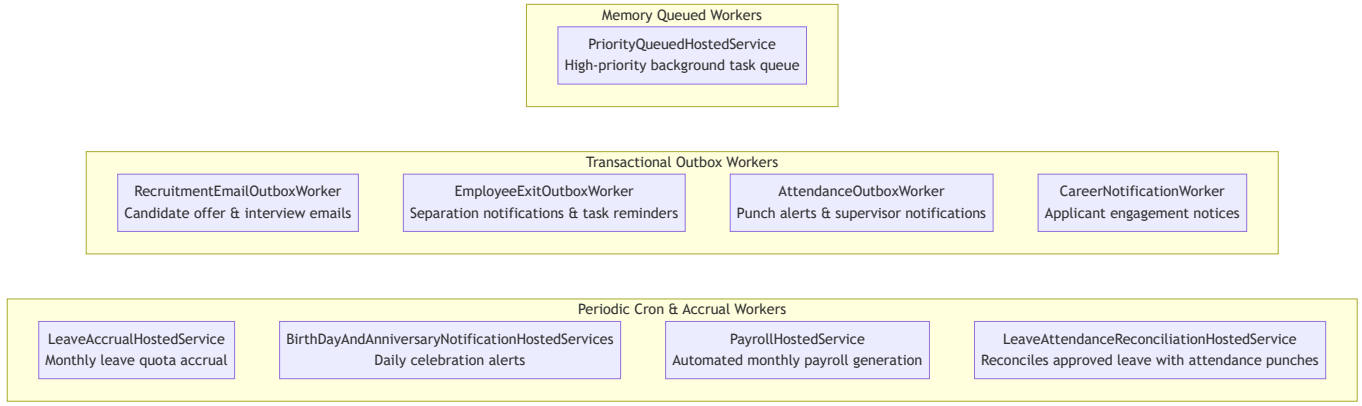
If two administrators attempt conflicting updates simultaneously, the second write fails with HTTP `409 Conflict` , requiring a client refresh.

---

## 6. Background Workers & Asynchronous Job Architecture

The platform runs **9 distinct `IHostedService` / `BackgroundService` background workers** registered in `Program.cs`:

**ARCHITECTURE / WORKFLOW DIAGRAM**




| Worker Name | Implementation Class | Execution Cycle | Core Responsibility |
|---|---|---|---|
| **Leave Accrual** | `LeaveAccrualHostedService` | Periodic (Daily check, triggers monthly) | Grants monthly leave entitlements based on active `LeavePolicy` configuration. |
| **Birthdays & Anniversaries** | `BirthDayAndAnniversaryNotificationHostedServices` | Daily at 00:00 IST | Scans active employees for matching DOB / DateOfJoining; publishes celebration notices via `NotificationHub`. |
| **Payroll Worker** | `PayrollHostedService` | Periodic (Monthly closed period) | Triggers `AutoPayrollServices` to generate draft payroll for closed months. |
| **Leave-Attendance Reconciliation** | `LeaveAttendanceReconciliationHostedService` | Periodic background loop | Ensures approved leave requests retroactively update daily attendance status records to `L`. |
| **Recruitment Outbox** | `RecruitmentEmailOutboxWorker` | Polling loop (5-15s) | Reads pending `RecruitmentEmailOutbox` records, dispatches emails via SMTP, and records retry counts / errors. |
| **Exit Outbox** | `EmployeeExitOutboxWorker` | Polling loop (15-30s) | Reads `EmployeeExitOutbox`, dispatches separation reminders and manager clearance alerts. |
| **Attendance Outbox** | `AttendanceOutboxWorker` | Polling loop (10-30s) | Transmits attendance punch anomalies and manager alerts. |
| **Career Notifications** | `CareerNotificationWorker` | Polling loop (30s) | Dispatches candidate job alert emails and subscription updates. |
| **Priority Task Queue** | `PriorityQueuedHostedService` | Continuous channel listener | Executes ad-hoc background tasks enqueued via `IPriorityTaskQueue`. |

# Codeji CMS / HRMS — Database Architecture

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation & Database Migrations
**Database Engine**: MongoDB (Document Store)
**Driver**: MongoDB C# Driver v2.28+
**Last Verified**: September 2026

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## 1. Database Paradigm & Engine Selection

The Codeji HRMS utilizes **MongoDB** as its primary persistence engine. While traditional HR systems frequently rely on relational databases (RDBMS), MongoDB was selected to accommodate:

1. **Polymorphic and Document-Centric Data**: Complex, nested structures such as multi-dimensional employee lifecycle histories, embedded policy snapshots, clearance checklists, and discussion message threads are naturally represented as single, atomic documents.
2. **Schema Flexibility across Diverse Enterprises**: Tenant-defined custom attributes ( `UserCustomAttribute` , `CompanyCustomAttributes` ) vary across companies without necessitating ALTER TABLE schema migrations.
3. **High-Performance Sharding & Multi-Tenancy**: Tenant-keyed collections ( `CompanyId` ) allow efficient horizontal scaling and compound indexed querying.

---

## 2. Core Entity Architecture & Base Classes

Every persistent entity in `Codeji.CMS.Repository.Entities` derives from a standardized base class hierarchy ensuring uniform auditing, soft deletion, and tenant scoping:

```
// BaseClass definition in Codeji.CMS.Repository.Entities
public class BaseClass : IImplementsId, ISupportSoftDelete, ISupportAuditing, ISupportFiltering
{
    [BsonId(IdGenerator = typeof(UniqueIdGenerator))]
    public string Id { get; set; } = string.Empty;
    public string CompanyId { get; set; } = string.Empty;
    public string CreatedBy { get; set; } = string.Empty;
    public DateTime CreatedDate { get; set; } = DateTime.UtcNow;
    public string? UpdatedBy { get; set; }
    public DateTime? UpdatedDate { get; set; }
    public bool IsDeleted { get; set; } = false;
    public DateTime? DeletedDate { get; set; }
    public string? DeletedBy { get; set; }
}
```

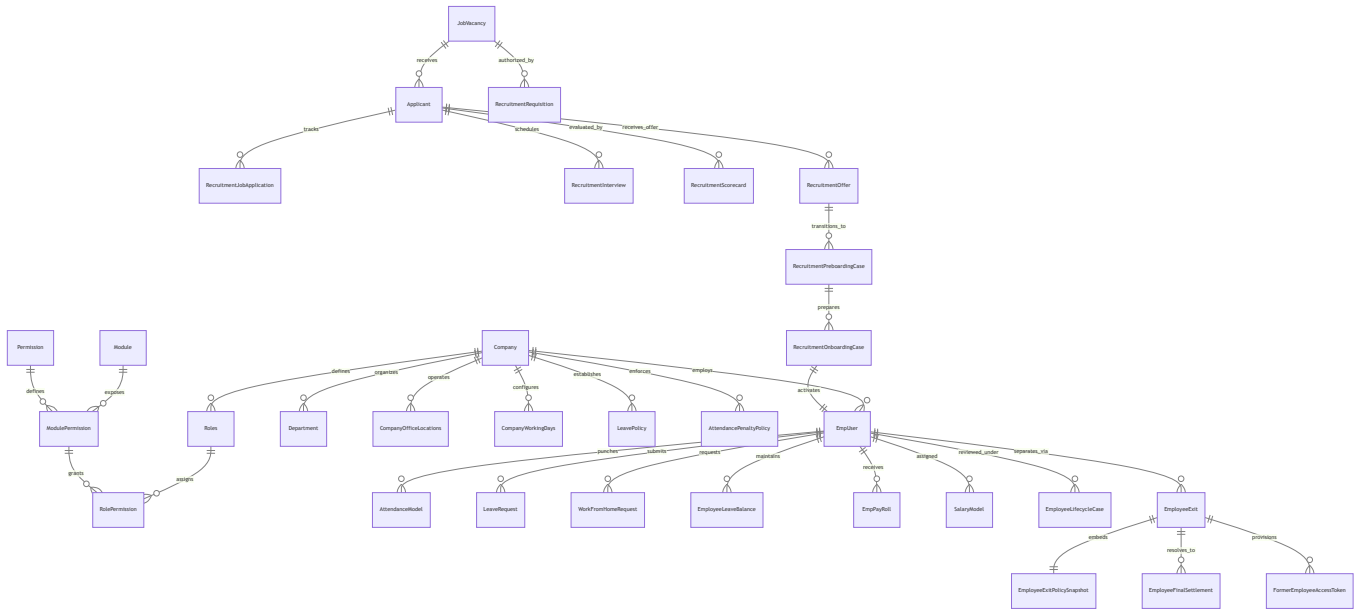
### 2.1 Cross-Cutting Persistence Contracts

- `IImplementsId` : Enforces a uniform primary key named `Id` (or mapped entity identifier) generated via `UniqueIdGenerator` .
- `ISupportSoftDelete` : Mandates the `IsDeleted` boolean flag. Queries via `MongoRepository<T>` automatically apply the filter `!x.IsDeleted` by default unless explicitly overridden ( `withDefaultFilter: false` ).
- `ISupportAuditing` : Automatically records creation and mutation timestamps in UTC along with the executing actor ID.
- `ISupportFiltering` : Facilitates tenant-level filtering by ensuring `CompanyId` exists on all business entities.

## 3. Entity-Relationship & Document Map

Although MongoDB is non-relational, logical relationships exist between documents linked via string-based foreign identifiers. The following diagram maps the core database entity network:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 4. Master Collection Catalog

The table below catalogs every verified MongoDB collection, its domain ownership, primary key, foreign references, and indexing invariants:

| Collection Name | C# Entity Class | Primary Key | Foreign Key References | Critical Indexes |
|---|---|---|---|---|
| **`Company`** | `Company` | `CompanyId` | None | `{ CompanyId: 1 }` (Unique), `{ CompanyCode: 1 }` |
| **`Department`** | `Department` | `DepartmentId` | `CompanyId` | `{ CompanyId: 1, DepartmentName: 1 }` |
| **`Roles`** | `Roles` | `RolesId` | `CompanyId` | `{ CompanyId: 1, RoleType: 1 }`, `{ CompanyId: 1, Titles: 1 }` |
| **`Module`** | `Module` | `ModuleId` | None | `{ ModuleConstant: 1 }` (Unique) |
| **`Permission`** | `Permission` | `PermissionId` | None | `{ PermissionConstant: 1 }` (Unique) |
| **`ModulePermission`** | `ModulePermission` | `ModulePermissionId` | `ModuleId`, `PermissionId` | `{ ModuleId: 1, PermissionId: 1 }` (Unique) |

| Collection Name | C# Entity Class | Primary Key | Foreign Key References | Critical Indexes |
|---|---|---|---|---|
| **RolePermission** | RolePermission | RolePermissionId | RoleId, ModulePermissionId, CompanyId | { CompanyId: 1, RoleId: 1, ModulePermissionId: 1 } (Unique) |
| **EmpUser** | EmpUser | UserId | CompanyId, RoleId, Department, ReportingManager | { CompanyId: 1, UserId: 1 } (Unique), { CompanyId: 1, EmployeeId: 1 } (Unique), { Email: 1 } |
| **EmployeeLifecycleCase** | EmployeeLifecycleCase | CaseId | CompanyId, UserId | { CompanyId: 1, CaseId: 1 }, { CompanyId: 1, UserId: 1, Status: 1 } |
| **EmployeeLifecyclePolicy** | EmployeeLifecyclePolicy | EmployeeLifecyclePolicyId | CompanyId, DepartmentId, RoleId | { CompanyId: 1, PolicyType: 1, Version: 1 } |
| **Attendance** | AttendanceModel | AttendanceId | CompanyId, UserId, EmployeeId | { CompanyId: 1, UserId: 1, AttendanceDate: 1 } (Unique) |
| **AttendanceDaySegment** | AttendanceDaySegment | SegmentId | CompanyId, UserId, AttendanceDate | { CompanyId: 1, UserId: 1, AttendanceDate: 1, SegmentIndex: 1 } (Unique) |
| **AttendancePenaltyPolicy** | AttendancePenaltyPolicy | Id | CompanyId, UserId | { CompanyId: 1, UserId: 1, EffectiveFrom: 1 } |
| **MonthlyAttendanceSummary** | MonthlyAttendanceSummary | SummaryId | CompanyId, EmployeeId, PayrollMonth | { CompanyId: 1, EmployeeId: 1, PayrollMonth: 1 } (Unique) |
| **AttendanceMonthLock** | AttendanceMonthLock | LockId | CompanyId, PayrollMonth | { CompanyId: 1, PayrollMonth: 1 } (Unique) |
| **LeaveRequest** | LeaveRequest | LeaveRequestId | CompanyId, UserId, LeavePolicyId | { CompanyId: 1, UserId: 1, StartDate: 1, EndDate: 1 } |
| **LeavePolicy** | LeavePolicy | LeavePolicyId | CompanyId, DepartmentId | { CompanyId: 1, LeaveTypeId: 1, IsActive: 1 } |
| **EmployeeLeaveBalance** | EmployeeLeaveBalance | EmployeeLeaveBalanceId | CompanyId, UserId, LeaveTypeId | { CompanyId: 1, UserId: 1, LeaveTypeId: 1, Year: 1 } (Unique) |

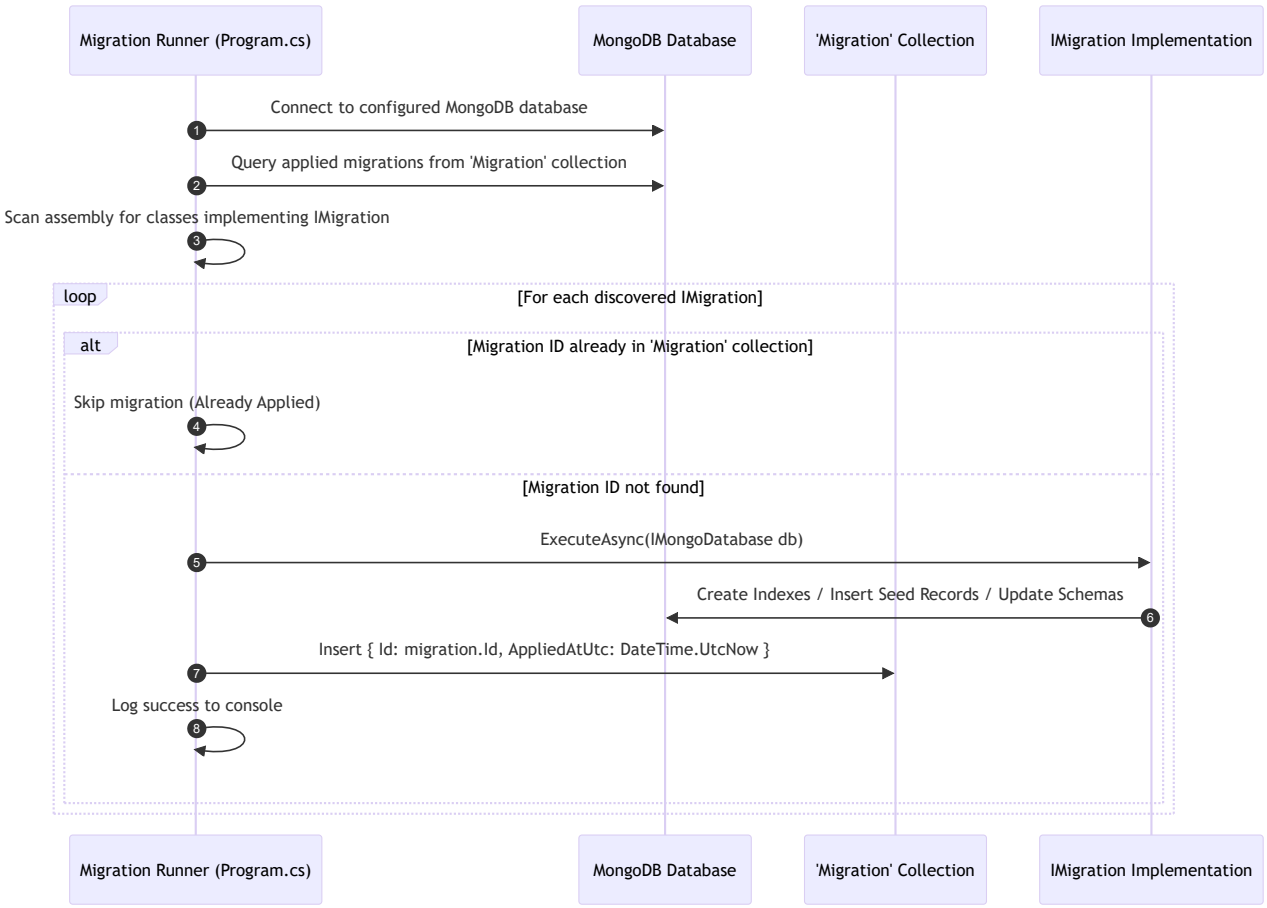
| Collection Name | C# Entity Class | Primary Key | Foreign Key References | Critical Indexes |
|---|---|---|---|---|
| **WorkFromHomeRequest** | WorkFromHomeRequest | RequestId | CompanyId, UserId | { CompanyId: 1, UserId: 1, StartDate: 1 } |
| **Holidays** | Holidays | HolidayId | CompanyId | { CompanyId: 1, HolidayDate: 1 } |
| **Salary** | SalaryModel | SalaryId | CompanyId, UserId | { CompanyId: 1, UserId: 1 } (Unique) |
| **EmpPayRoll** | EmpPayRoll | PayRollId | CompanyId, EmployeeId, PayMonth | { CompanyId: 1, EmployeeId: 1, PayMonth: 1 } (Unique) |
| **PayrollDivisorPolicy** | PayrollDivisorPolicy | PolicyId | CompanyId | { CompanyId: 1, EffectiveFrom: 1 } |
| **EmployeeExit** | EmployeeExit | EmployeeExitId | CompanyId, EmployeeUserId | { CompanyId: 1, EmployeeUserId: 1 } (Unique for active), { Status: 1 } |
| **EmployeeExitPolicy** | EmployeeExitPolicy | EmployeeExitPolicyId | CompanyId, DepartmentId, RoleId | { CompanyId: 1, ExitType: 1, IsActive: 1 } |
| **EmployeeFinalSettlement** | EmployeeFinalSettlement | EmployeeFinalSettlementId | CompanyId, EmployeeExitId, EmployeeUserId | { CompanyId: 1, EmployeeExitId: 1 } (Unique) |
| **FormerEmployeeAccessToken** | FormerEmployeeAccessToken | TokenId | CompanyId, EmployeeUserId | { TokenHash: 1 } (Unique), { ExpiresAtUtc: 1 } |
| **JobVacancy** | JobVacancy | VacancyId | CompanyId, DepartmentId | { CompanyId: 1, JobCode: 1 } (Unique), { Slug: 1 } |
| **Applicant** | Applicant | ApplicantId | CompanyId, JobVacancyId | { CompanyId: 1, Email: 1, JobVacancyId: 1 } |
| **RecruitmentOffer** | RecruitmentOffer | OfferId | CompanyId, ApplicantId | { CompanyId: 1, ApplicantId: 1 } |
| **RecruitmentOnboardingCase** | RecruitmentOnboardingCase | CaseId | CompanyId, ApplicantId, EmployeeUserId | { CompanyId: 1, ApplicantId: 1 } (Unique) |
| **RecruitmentEmailOutbox** | RecruitmentEmailOutbox | OutboxId | CompanyId | { Status: 1, NextAttemptAtUtc: 1 } |
| **EmployeeExitOutbox** | EmployeeExitOutbox | OutboxId | CompanyId | { Status: 1, NextAttemptAtUtc: 1 } |

| Collection Name | C# Entity Class | Primary Key | Foreign Key References | Critical Indexes |
|---|---|---|---|---|
| **`AttendanceNotificationOutbox`** | `AttendanceNotificationOutbox` | `OutboxId` | `CompanyId` | `{ Status: 1, NextAttemptAtUtc: 1 }` |
| **`Migration`** | `Migration` | `Id` | None | `{ Id: 1 }` (Unique) |

## 5. Schema Evolution & Migration Engine (`Codeji.CMS.Migrations`)

Codeji implements an in-house database migration runner in `Codeji.CMS.Migrations` rather than relying on an external ORM migration tool:

**ARCHITECTURE / WORKFLOW DIAGRAM**




### 5.1 Structure of an `IMigration` Class

Every migration implements the `IMigration` interface with an immutable timestamped identifier:

```
public interface IMigration
{
    string Id { get; }
    Task ExecuteAsync(IMongoDatabase db);
}

public class AddRecruitmentCoreIndexes : IMigration
{
    public string Id => $"2026-09-18-{typeof(AddRecruitmentCoreIndexes).Name}";

    public async Task ExecuteAsync(IMongoDatabase db)
    {
        var migrations = db.GetCollection<Migration>("Migration");
        if (await migrations.Find(m => m.Id == Id).AnyAsync()) return;

        var applicants = db.GetCollection<BsonDocument>("Applicant");
        await applicants.Indexes.CreateOneAsync(
            new CreateIndexModel<BsonDocument>(
                Builders<BsonDocument>.IndexKeys
                    .Ascending("CompanyId")
                    .Ascending("Email")
                    .Ascending("JobVacancyId"),
                new CreateIndexOptions { Unique = true }));

        // Mark as applied
        await migrations.InsertOneAsync(new Migration { Id = Id, AppliedAt = DateTime.UtcNow });
    }
}
```

## 6. Document Immutability & Historical Snapshotting

A defining architectural principle of the Codeji database design is **Point-in-Time Snapshotting**:

- **Exit Policy Snapshotting**: When an employee submits a resignation or HR initiates an exit, `EmployeeExitPolicySnapshot` is serialized directly inside the `EmployeeExit` document. If HR subsequently modifies company notice periods or clearance rules, the separated employee's case remains governed by the exact rules in force at initiation.
- **Processed Payroll Immutability**: Once payroll for a given month is marked `IsProcessed == true`, the document cannot be recalculated or deleted. A new retroactive adjustment or manual revision must be processed.
- **Lifecycle History Log**: The `EmpUser.Lifecycle.History` subdocument array appends every status change across the 7 lifecycle dimensions, capturing the prior state, new state, effective UTC date, acting user ID, and machine-readable reason code (`ReasonCode`).

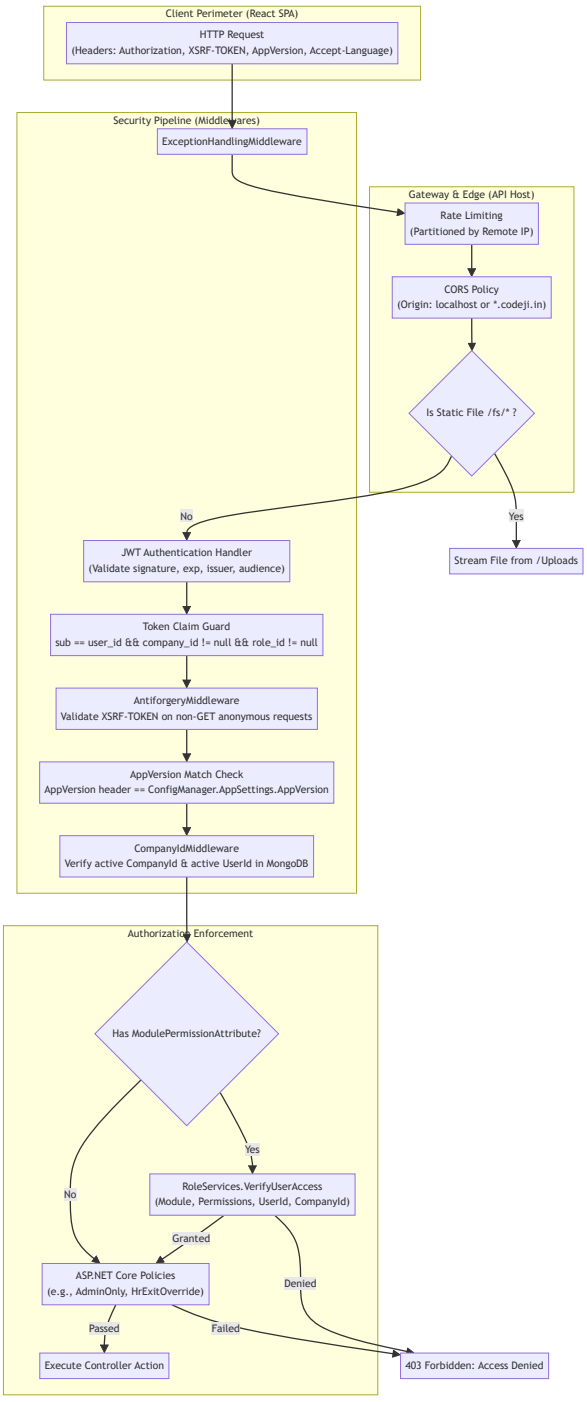
# Codeji CMS / HRMS — Authentication & Authorization Architecture

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Security Standards**: JWT Bearer Tokens | Anti-CSRF (XSRF-TOKEN) | Dynamic RBAC | ASP.NET Core Authorization
**Last Verified**: September 2026

## 1. Security Architecture Overview

The Codeji HRMS platform employs a defense-in-depth security model combining cryptographic authentication, token-bound tenant identification, anti-forgery verification, application version pinning, and dual-layer authorization (client-side route protection coupled with server-side middleware enforcement):

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 2. Authentication: The JWT Bearer Token Lifecycle

### 2.1 Credential Exchange & Login

Authentication begins at `POST /api/Account/Login` (`AccountController.cs`), consuming `LoginModel`:

```
public class LoginModel
{
    public required string Email { get; set; }
    public required string Password { get; set; }
}
```

1. `AccountServices.LoginAsync` locates the employee record in `EmpUser` matching the normalized email.

2. The user's stored password hash is validated using a salted cryptographic hasher.
3. The system verifies that the account is active ( `Status == true` ), the tenant company exists and is active ( `Company.IsActive == true` ), and the assigned role exists in `Roles` .
4. If valid, the system issues a **JWT Access Token** paired with a cryptographically random **Refresh Token**.

### 2.2 Authoritative JWT Claims Structure

The JWT access token is signed using an HMAC-SHA256 secret key ( `ConfigManager.Jwt.SecretKey` ). It contains explicit, cryptographically unforgeable identity and tenancy claims:

| Claim Type | Standard JWT Name | Description & Verification Rules |
|---|---|---|
| `sub` | `JwtRegisteredClaimNames.Sub` | The user's primary database ID ( `UserId` ). Must match `user_id` claim exactly. |
| `user_id` | `ClaimTypesEnum.user_id` | Explicit application user identifier ( `EmpUser.UserId` ). |
| `company_id` | `ClaimTypesEnum.company_id` | The single authoritative tenant ID ( `EmpUser.CompanyId` ). |
| `role_id` | `ClaimTypesEnum.role_id` | The assigned role document ID ( `EmpUser.RoleId` ). |
| `admin_id` | `ClaimTypesEnum.admin_id` | Present if user possesses administrative elevation. |
| `exp` | `JwtRegisteredClaimNames.Exp` | Token expiration timestamp (default: 60 minutes). |
| `iss` | `JwtRegisteredClaimNames.Iss` | Configured API URL ( `ConfigManager.AppSettings.APIUrl` ). |
| `aud` | `JwtRegisteredClaimNames.Aud` | Configured Web App URL ( `ConfigManager.AppSettings.AppUrl` ). |

### 2.3 Strict Claim Integrity Gate ( `OnTokenValidated` )

In `Program.cs` , the JWT bearer handler configures a zero-trust claim integrity validator:

```
options.Events = new JwtBearerEvents
{
    OnTokenValidated = context =>
    {
        var principal = context.Principal;
        var subject = principal?.FindFirst(JwtRegisteredClaimNames.Sub)?.Value
            ?? principal?.FindFirst(ClaimTypes.NameIdentifier)?.Value;
        var userId = principal?.FindFirst("user_id")?.Value;
        var companyId = principal?.FindFirst("company_id")?.Value;
        var roleId = principal?.FindFirst("role_id")?.Value;

        if (string.IsNullOrWhiteSpace(subject) || string.IsNullOrWhiteSpace(userId) ||
            !string.Equals(subject, userId, StringComparison.Ordinal) ||
            string.IsNullOrWhiteSpace(companyId) || string.IsNullOrWhiteSpace(roleId))
        {
            context.Fail("Required tenant identity claims are missing or inconsistent.");
        }
        return Task.CompletedTask;
    }
};
```

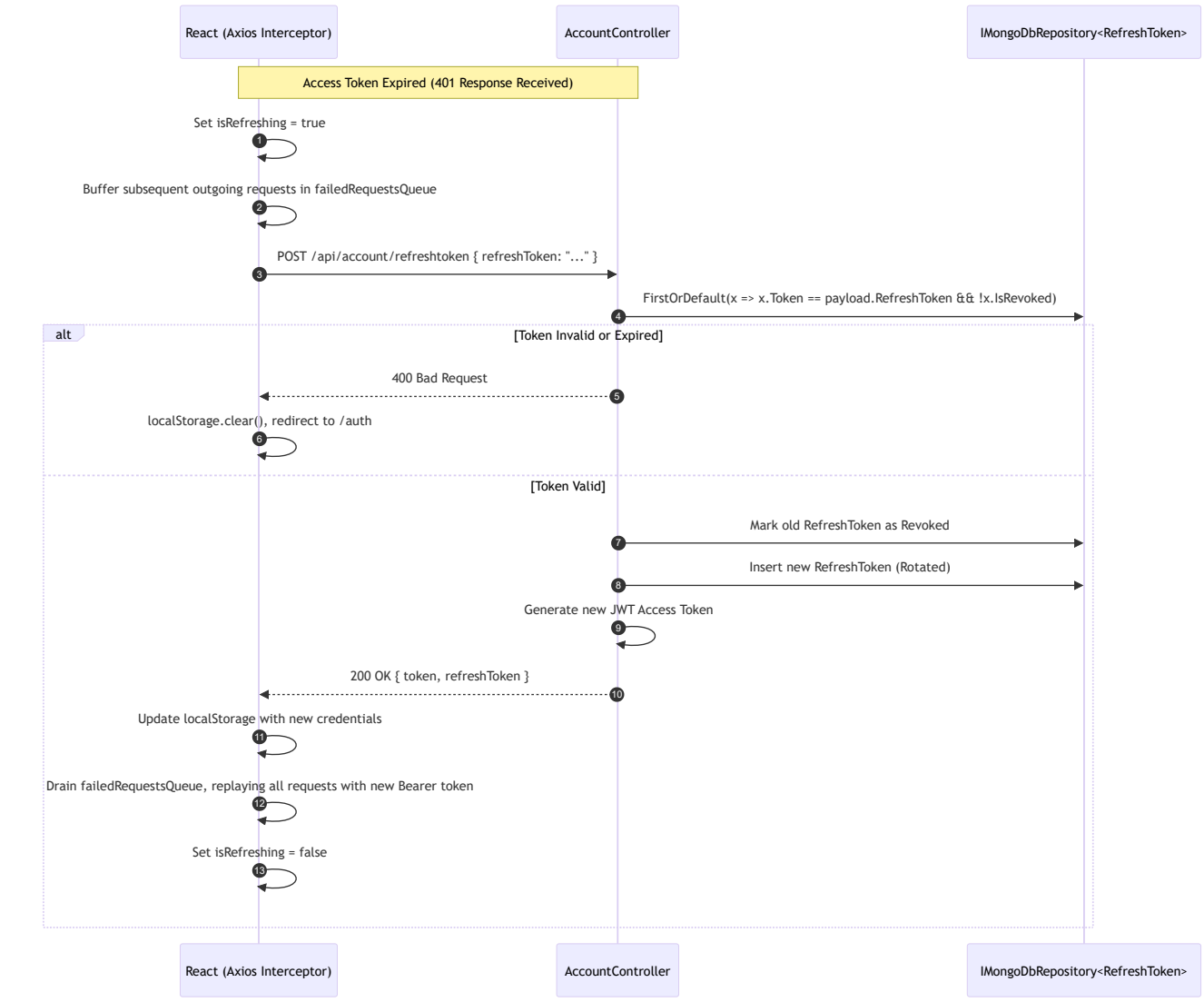
If an adversary tampers with token claims or attempts cross-tenant manipulation, the token is invalidated prior to reaching application code.

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## 3. Refresh Token Protocol & Concurrency Management

To prevent user session interruption, the frontend and backend maintain an automated refresh protocol:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 4. Perimeter Protections: Anti-CSRF & AppVersion Locking

### 4.1 Anti-CSRF Middleware (`AntiforgeryMiddleware`)

- Configured in `Program.cs` via `builder.Services.AddAntiforgery` with cookie name `X_CSRFTken` and header name `XSRF-TOKEN`.
- Automatically skipped for authenticated requests (as JWT Bearer tokens stored in local storage are immune to ambient browser credential reuse) and HTTP GET methods.
- Executed on unauthenticated state-changing POST/PUT operations (e.g. initial login, password reset, public vacancy applications) unless running in local debug mode.

### 4.2 Application Version Pinning (`426 Upgrade Required`)

To prevent data corruption caused by stale client SPA assets calling updated backend API contracts following a deployment:

- Clients pass their bundled application version in the `AppVersion` HTTP header.

- `AntiforgeryMiddleware` compares this header against `ConfigManager.AppSettings.AppVersion`.
- If a mismatch is detected on non-exempt endpoints, the backend immediately halts request execution and returns **HTTP 426 Upgrade Required**:

```
{
  "StatusCode": 426,
  "Message": "Upgrade Required"
}
```

The client catches this status code and triggers an automatic browser cache bust and reload.

---

## 5. Server-Side Authorization Pipeline

Authorization is enforced at two distinct layers:

1. **Endpoint Filter Layer**: Declared using the `[ModulePermission]` attribute on controller actions.
2. **Dynamic Service Layer**: Evaluated against the database-backed role-permission matrix in `RoleServices.VerifyUserAccess`.

### 5.1 Declarative Endpoint Decoration

Controllers declare their required module and permission levels declaratively:

```
[HttpPost]
[Route("GetEmpPayRollData")]
[ModulePermission(AppModule.PayRoll, Permission.View)]
public async Task<ActionResult<Result<GetEmpPayRollResponseDto>>> GetEmpPayRollData(...)
```

### 5.2 Middleware Execution (`CompanyIdMiddleware`)

Inside `CompanyIdMiddleware.AuthenticateUserRequest`:

1. Resolves `CompanyId` and `UserId` from the authenticated claims principal via `CurrentContext`.
2. Validates that the tenant company is active: `await _companyService.IsActiveCompanyExist(company_Id)`.
3. Validates that the requesting employee is active: `await _employeeService.IsUserActive(userId)`.
4. Inspects endpoint metadata for `ModulePermissionAttribute`. If found:

```
bool isValid = await _roleServices.VerifyUserAccess(
    permissionAttribute.Module,
    permissionAttribute.Permissions,
    userId,
    company_Id);

if (!isValid)
{
    context.Response.StatusCode = StatusCodes.Status403Forbidden;
    await context.Response.WriteAsync("Access Denied");
    return null;
}
```

### 5.3 Deterministic Access Evaluation (`RoleServices.VerifyUserAccess`)

The core evaluation logic in `RoleServices.cs` executes as follows:

1. **Tenant Integrity Check**: Confirms that `user.CompanyId == companyId`. If mismatched, immediately returns `false`.
2. **Administrator Elevation**: If the user's role has `role.RoleType == EnumsHelper.Roles.Administrator`, access is granted unconditionally (`return true`).
3. **Cartesian Permission Lookup**:

- Queries `RolePermission` for the user's `RoleId` within the active `CompanyId`.
- Constructs candidate strings matching `{Module}.{Permission}` (e.g. `PayRoll.View`, `Attendance.Edit`).
- Returns `true` if any candidate matches the user's assigned permissions.

4. **Role-Based Exceptions**: Grants HR Managers and HR Executives baseline access to company dashboard metrics and attendance views even when legacy granular permissions are transitioning.

---

## 6. Frontend Route Guarding (`ProtectedRoute`)

Client-side UI visibility and navigation security are governed by `<ProtectedRoute />` (`src/app/modules/users/about/components/ProtectedRoute.tsx`):

- Reads active user permissions from the Redux store (`state.user.userData.modulePermission`).
- Evaluates `adminOnly` flags: non-administrator users navigating to `/admin/*` or `/attendance-settings` are blocked.
- Checks module permissions:

```
hasModuleFeaturePermission(ModulesName.LeaveManagement, Permission.View, modulePermission)
```

- If unauthorized, prevents component mounting and renders `<UnAuthorizedAccess type="unauthorise" />`.

> [!IMPORTANT] Frontend route guarding is strictly a user-experience enhancement to hide unexecutable UI elements. All security-sensitive actions are independently and unconditionally authorized on the backend.

# Codeji CMS / HRMS — Role & Permission Model

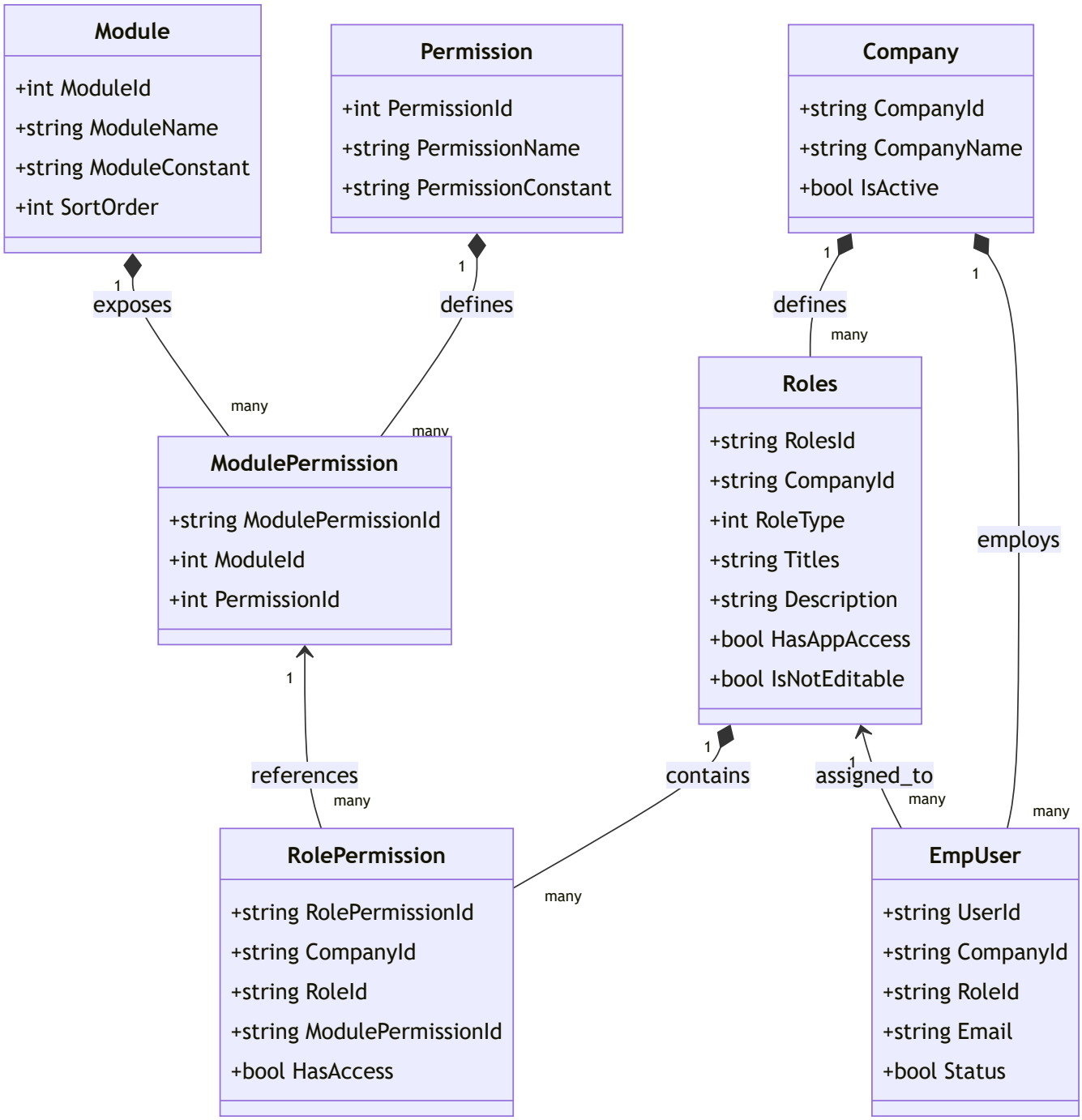
> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation & Seed Migrations
> **Authorization Paradigm**: Dynamic Role-Based Access Control (RBAC)
> **Last Verified**: September 2026

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## 1. Role-Based Access Control (RBAC) Architecture

Codeji HRMS implements a dynamic, database-driven Role-Based Access Control model. Rather than hardcoding static role checks across controllers, the platform defines permissions as fine-grained, Cartesian-product intersections of **Modules** and **Actions**, assigned to tenant-scoped **Roles**:

ARCHITECTURE / WORKFLOW DIAGRAM




## 2. Standard System Roles & Default Templates

When a new company registers or defaults are seeded ( `RoleServices.DefaultRoleTemplates` ), 4 standard roles are provisioned:

| Role Name | Enum ID | Default Role Title | Default Properties | Functional Responsibilities |
|---|---|---|---|---|
| **Administrator** | `1` | Company Administrator | `IsNotEditable: true`, `HasAppAccess: true` | Complete, unrestricted administrative control. Bypasses all granular permission checks across all modules. Reserved for client account owners. |
| **HR Manager** | `2` | HR Manager | `IsNotEditable: false`, `HasAppAccess: true` | Full operational authority over employee lifecycles, recruitment, leave policies, attendance locks, payroll execution, and separation cases. |
| **HR Executive** | `4` | HR Executive | `IsNotEditable: false`, `HasAppAccess: true` | Operational peer to HR Manager for day-to-day HR workflows (candidate processing, attendance adjustments, employee data entry). |
| **Employee** | `3` | Employee | `IsNotEditable: false`, `HasAppAccess: true` | Self-service access: personal profile, daily punch, personal leave/WFH requests, personal payslips, company notice board. |

> [!NOTE] Custom roles can be created per company (e.g. `Recruiter`, `IT Admin`, `Payroll Specialist`) with arbitrary permission combinations. The `Administrator` role is flagged `IsNotEditable: true` to prevent accidental lockout.

---

## 3. Registered Modules & Permission Constants

### 3.1 Registered Application Modules (`AppModule`)

Source reference: `Codeji.CMS.Utility.Constraints.AppModule`:

| Module Constant | UI Display Name | Key Functional Domain |
|---|---|---|
| `Dashboard` | Dashboard | Executive metrics, upcoming birthdays, workforce counters. |
| `Employees` | Employees | Employee directory, profile management, lifecycle cases. |
| `Attendance` | Attendance | Daily punch logs, shift rosters, penalties, month locks. |
| `Leave_Management` | Leave Management | Leave requests, balances, leave policies, WFH bookings. |
| `Calendar` | Calendar | Company calendar, weekly off schedules, holidays. |
| `Notice_Board` | Notice Board | Company-wide announcements, circulars, urgent alerts. |
| `Jobs` | Jobs | Recruitment requisitions and job postings. |

| Module Constant | UI Display Name | Key Functional Domain |
|---|---|---|
| `Applications` | Applications | Candidate applicant tracking, screening, scorecards. |
| `Process_Log` | Process Log | Recruitment audit history and stage transitions. |
| `PayRoll` | PayRoll | Monthly payroll runs, LOP deductions, payslip generation. |
| `Policy` | Policy | Company policy document library and compliance. |
| `Payroll_Settings` | Payroll Settings | Salary structure components and divisor policies. |
| `Career_Profile` | Career Profile | Public career page branding, marketplace themes. |
| `Work_From_Home` | Work From Home | Remote work request workflow and hours tracking. |
| `Offboarding` | Offboarding | Separation cases, clearances, handover, final settlements. |

### 3.2 Granular Permission Actions ( `Permission` )

Source reference: `Codeji.CMS.Utility.Constraints.Permission` :

| Category | Permission Constant | Functional Meaning |
|---|---|---|
| **CRUD Baseline** | `View` | Read-only inspection of module data. |
| | `Create` | Permission to author new records. |
| | `Edit` | Permission to modify existing records. |
| | `Delete` | Soft-delete or archive authority. |
| **Self-Service** | `ViewOwn` | Restricts visibility strictly to records owned by the calling employee. |
| | `CreateOwn` | Authorizes submission of requests for oneself. |
| | `EditOwn` | Modifies personal draft requests. |
| | `CancelOwn` | Cancels personal pending requests before approval. |
| **Supervisory** | `ViewTeam` | Grants visibility into direct and indirect subordinates' records. |
| | `ApproveTeam` | Authority to approve/reject subordinates' leave or WFH requests. |
| | `ViewAll` | Organization-wide visibility across all departments. |
| | `CreateForEmployee` | HR/Manager authority to submit a punch or leave on behalf of a worker. |
| | `Override` | Administrative power to bypass standard business limits (e.g. penalty limits). |
| | `Revoke` | Authority to revoke previously issued approvals or access tokens. |

| Category | Permission Constant | Functional Meaning |
|---|---|---|
| **Workflow Specific** | `Submit` | Submits draft cases for formal approval. |
| | `Review` | Conducts intermediate reviews (e.g. Manager Resignation Review). |
| | `Approve` | Executes final sign-off. |
| | `Complete` | Executes lifecycle terminal transitions (e.g. Exit Completion). |
| | `Evidence` | Uploads and verifies legal/resignation evidence documents. |
| | `Download` | Authority to export audit logs, bulk Excel data, or payslips. |
| **Offboarding Operations** | `PolicyManage` | Authors and modifies exit policies. |
| | `CaseManage` | Orchestrates separation cases and sets Last Working Dates. |
| | `AssetManage` | Verifies physical asset returns and evaluates damage deductions. |
| | `AccessManage` | Coordinates IT single sign-on, email, and hardware access revocation. |
| | `FinanceManage` | Computes dues, loans, and reconciles Full & Final settlements. |
| | `BenefitsManage` | Handles gratuity, health insurance transfer, and encashment. |
| | `EvidenceManage` | Reviews resignation letters and non-disclosure sign-offs. |
| | `PortalManage` | Provisions and revokes alumni portal access tokens. |

## 4. Master Role-Permission Matrix (Default Baseline)

The table below outlines the default operational permissions assigned across standard roles upon tenant initialization:

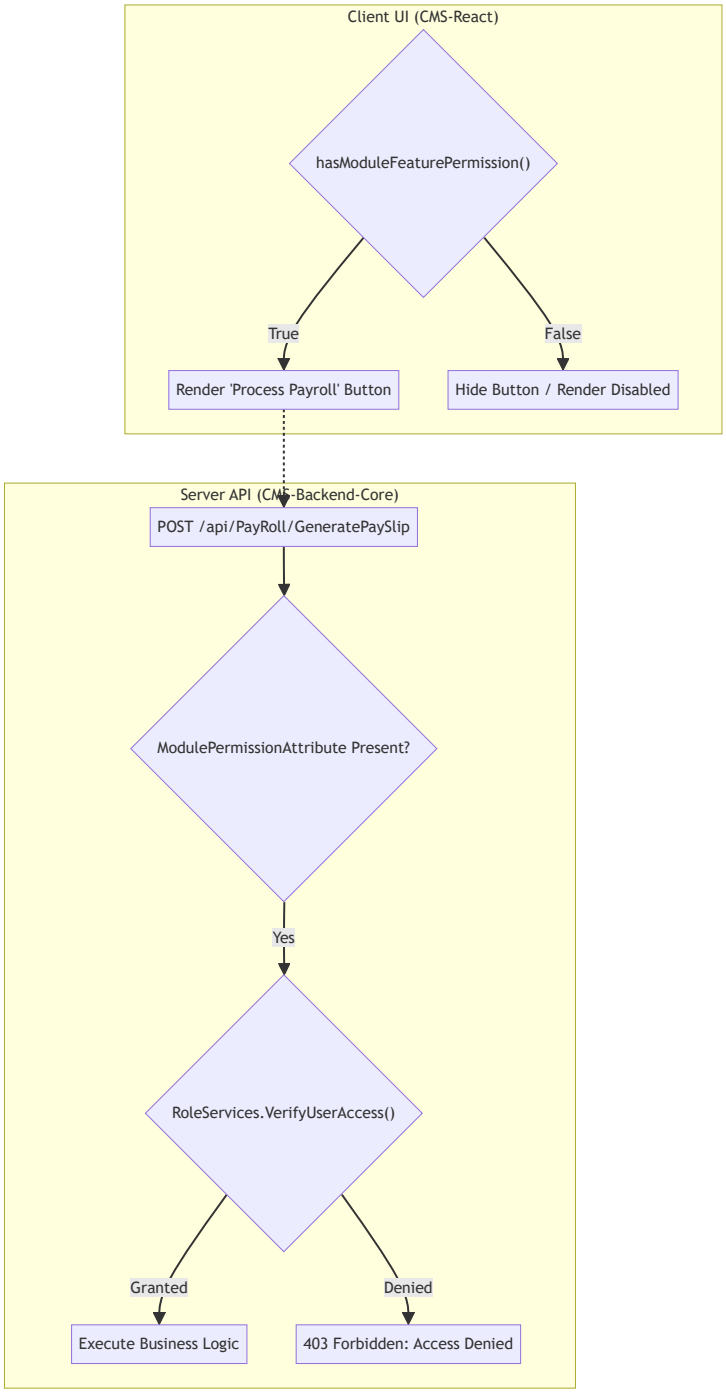
| Module | Action | Admin | HR Manager | HR Executive | Employee |
|---|---|---|---|---|---|
| **Dashboard** | View | Yes | Yes | Yes | Yes |
| **Employees** | View / ViewAll | Yes | Yes | Yes | View (Directory only) |
| | Create / Edit / Delete | Yes | Yes | Yes | No |
| | EditOwn (Profile) | Yes | Yes | Yes | Yes (Limited fields) |
| **Attendance** | ViewAll | Yes | Yes | Yes | No |
| | ViewOwn | Yes | Yes | Yes | Yes |
| | CreateForEmployee | Yes | Yes | Yes | No |
| | LockMonth / Override | Yes | Yes | No | No |
| **Leave Management** | ViewAll / Approve | Yes | Yes | Yes | No |

| Module | Action | Admin | HR Manager | HR Executive | Employee |
|---|---|---|---|---|---|
| | ViewOwn / CreateOwn | Yes | Yes | Yes | Yes |
| | ManagePolicies | Yes | Yes | No | No |
| **Work From Home** | ViewAll / Approve | Yes | Yes | Yes | No |
| | ViewOwn / CreateOwn | Yes | Yes | Yes | Yes |
| **Jobs & Vacancies** | View / Create / Edit | Yes | Yes | Yes | No |
| **Applications (ATS)** | View / Review / Score | Yes | Yes | Yes | No |
| | IssueOffer | Yes | Yes | No | No |
| **PayRoll** | ViewAll / Process | Yes | Yes | No | No |
| | ViewOwn (Payslip) | Yes | Yes | Yes | Yes |
| **Offboarding** | ViewAll / CaseManage | Yes | Yes | Yes | No |
| | Resign (CreateOwn) | Yes | Yes | Yes | Yes |
| | CompleteExit / Settle | Yes | Yes | No | No |
| **Company Master** | Manage Settings | Yes | No | No | No |
| **Roles & Permissions** | Create / Edit Roles | Yes | No | No | No |

## 5. Server-Side Enforcement vs. Client-Side Visibility

A critical architectural distinction in Codeji HRMS is the strict bifurcation between **UI visibility** and **Server-side authorization**:

**ARCHITECTURE / WORKFLOW DIAGRAM**




1. **Client-Side Hiding is NOT Security**: If an employee inspects DOM elements and enables a hidden button or fires an API request directly via curl or Postman, the request is intercepted by `CompanyIdMiddleware`.
2. **Authoritative Backend Rejection**: `RoleServices.VerifyUserAccess` resolves the user's assigned role from MongoDB in the current tenant scope. If the role lacks the specific module-permission pair, the request is aborted immediately with HTTP `403 Forbidden`.
3. **Multi-Tenant Isolation**: Role permissions are scoped to `CompanyId`. A user granted Administrator privileges in Company A possesses zero permissions in Company B.

# Codeji CMS / HRMS — Policy & Configuration Engine

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Core Components**: `OffboardingPolicyResolver.cs`, `EmployeeLifecyclePolicyService.cs`, `LeavePolicy.cs`, `AttendancePenaltyPolicy.cs`, `PayrollDivisorPolicy.cs`
**Last Verified**: September 2026

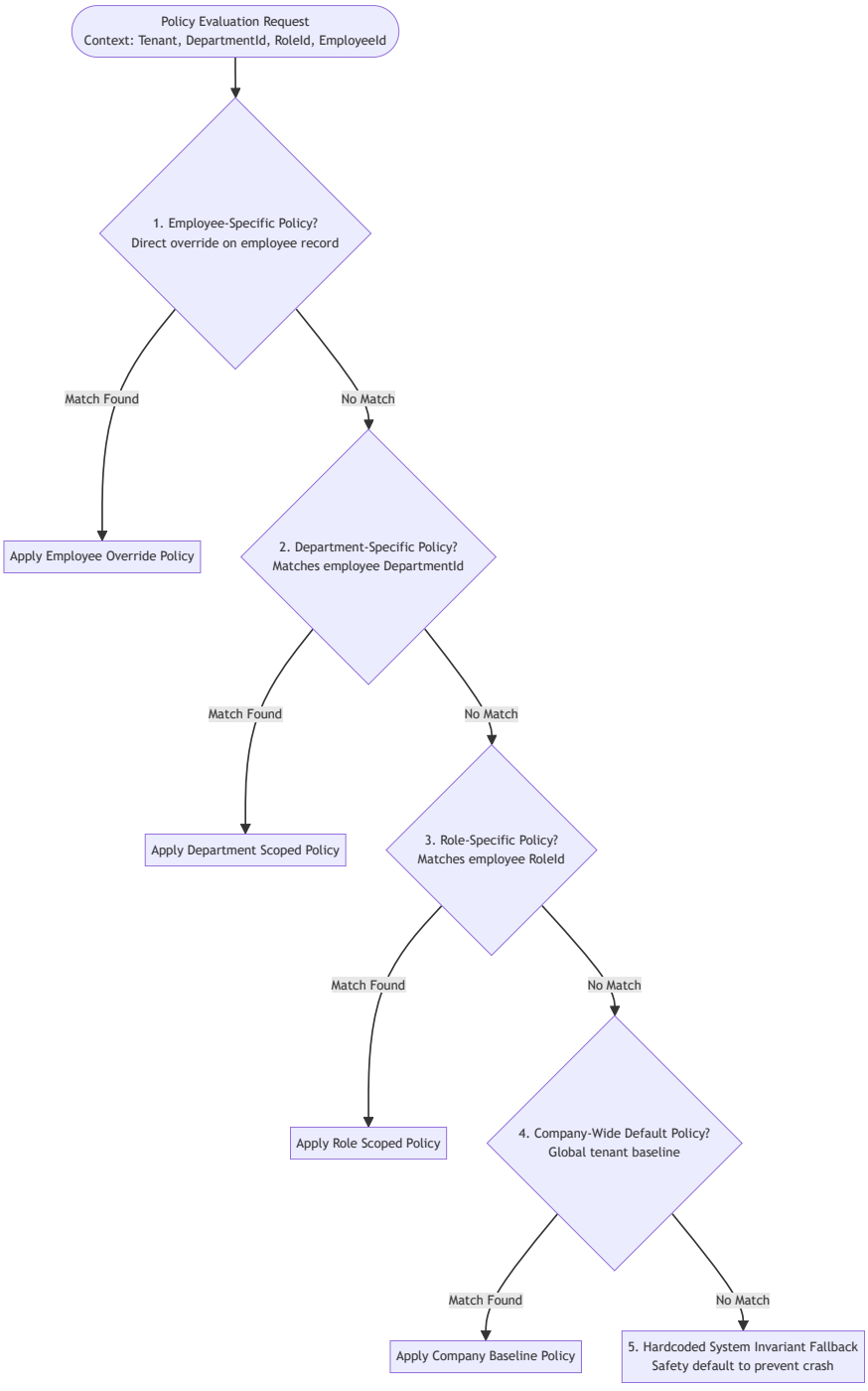
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## 1. Engine Philosophy: Deterministic Hierarchical Governance

Enterprise HR organizations operate under multi-tiered labor regulations, departmental conventions, and employment contracts. Hardcoding rules (such as notice period duration or attendance penalty thresholds) into application code creates fragile, unmaintainable systems.

Codeji HRMS solves this with a **Deterministic Policy Resolution Engine**. The engine evaluates operational policies across five distinct organizational tiers, resolving rules based on explicit, verified precedence:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 2. Core Policy Domains Implemented

The platform manages five core policy domains verified from the repository:

### 2.1 Offboarding & Separation Policy (`EmployeeExitPolicy`)

- **Entity Reference**: `Codeji.CMS.Repository.Entities.Employees.EmployeeExitPolicy`
- **Resolver**: `Codeji.CMS.Services.Employees.OffboardingPolicyResolver`
- **Configurable Parameters**:
  - `MinimumNoticeDays`: Mandatory notice period (e.g. 30, 60, 90 days).
  - `NoticePaySalaryBasis`: Compensation formula for shortfall recovery (`GrossMonthly` or `BasicMonthly`).

- `NoticePayDailyRateDivisor`: Daily divisor for buyout calculations (default: 30).
- `AllowEmployeeSelfResignation`: Toggle permitting employee self-service resignations.
- `AllowRejoin`: Designates whether separated employees under this policy are eligible for future rehire.
- `RequireManagerReview`: Mandates immediate manager sign-off prior to HR review.
- `RequireHrReview`: Requires HR executive approval before notice period begins.
- `RequireResignationEvidence`: Enforces signed PDF letter upload before submission.
- `RequireEmployeeAcknowledgement`: Mandates employee digital signature on final clearance terms.
- `EnableFormerEmployeePortal`: Governs alumni document portal provisioning.
- `NoticePeriodReminderEveryDays`: Frequency (in days) for automated notice countdown reminders.

### 2.2 Employee Lifecycle & Contract Policy (`EmployeeLifecyclePolicy`)

- **Entity Reference**: `Codeji.CMS.Repository.Entities.Employees.EmployeeLifecyclePolicy`
- **Service**: `Codeji.CMS.Services.Employees.EmployeeLifecyclePolicyService`
- **Configurable Parameters**:
  - `ContractType`: `Permanent`, `Probation`, `Contractual`, `Internship`.
  - `ProbationDurationMonths`: Standard probation window (e.g. 3 or 6 months).
  - `ConfirmationReviewNoticeDays`: Advance warning days before probation confirmation review.
  - `ContractExtensionThresholdDays`: Advance warning window for expiring temporary contracts.
  - `AbscondingReviewThresholdDays`: Inactivity period before unapproved absence escalates to absconding review.

### 2.3 Leave Entitlement Policy (`LeavePolicy`)

- **Entity Reference**: `Codeji.CMS.Repository.Entities.Leave.LeavePolicy`
- **Service**: `Codeji.CMS.Services.LeaveManagement.LeaveManagementService`
- **Configurable Parameters**:
  - `LeaveTypeId`: `SickLeave`, `CasualLeave`, `EarnedLeave`, `MaternityLeave`, `PaternityLeave`.
  - `AccrualPeriod`: Monthly, Yearly, or None (lump sum grant).
  - `DaysPerYear`: Total annual entitlement.
  - `MaxConsecutiveDays`: Single-request duration ceiling.
  - `RequiresAttachment`: Mandates medical certificate upload for leaves exceeding threshold.
  - `AllowHalfDay`: Permits 0.5-day segment bookings.
  - `CarryForwardLimit`: Maximum unused days carried into subsequent calendar years.
  - `IsEncashable`: Governs whether unused days convert to cash in Full & Final settlement.

### 2.4 Attendance Penalty & Exception Policy (`AttendancePenaltyPolicy`)

- **Entity Reference**: `Codeji.CMS.Repository.Entities.Attendance.AttendancePenaltyPolicy`
- **Service**: `Codeji.CMS.Services.Attendance.AttendancePenaltyService`
- **Configurable Parameters**:
  - `CombinedLhdEdMonthlyLimit`: Maximum allowable combined Late Half Day (`LHD`) and Early Departure (`ED`) occurrences per calendar month before financial deductions apply.
  - `DeductionDaysPerExceededLimit`: Penalty deduction days (e.g. 0.5 or 1.0 day LOP per violation).
  - `GraceMinutes`: Leniency window (e.g. 15 minutes) before a check-in is classified as late.
  - `EffectiveFrom` / `EffectiveTo`: Date-bounded validity windows supporting scheduled policy revisions.

### 2.5 Payroll Divisor Policy (`PayrollDivisorPolicy`)

- **Entity Reference**: `Codeji.CMS.Repository.Entities.Employees.PayrollDivisorPolicy`

- **Service**: `Codeji.CMS.Services.PayRoll.PayrollDivisorPolicyService`
- **Calculation Rules**: `PayrollCalculationRules.ResolveDivisor`:

  > $$\text{Divisor} = \begin{cases} 30 & \text{if policy is } \texttt{"FIXED_30_DAYS"} \\ \text{Eligible Working Days} & \text{if policy is } \texttt{"WORKING_DAYS"} \\ \text{Days in Month (28--31)} & \text{otherwise (Default Calendar Days)} \end{cases}$$

---

## 3. Deterministic Precedence Implementation

The precedence logic in `OffboardingPolicyResolver.cs` is strictly deterministic. Query ordering from MongoDB can never alter resolution outcome:

```
public static class OffboardingPolicyResolver
{
    public static ResolvedOffboardingPolicy? Resolve(
        IEnumerable<EmployeeExitPolicy> policies,
        OffboardingPolicyContext context)
    {
        var active = policies.Where(x => x.IsActive).ToList();
        var requestedExitType = EmployeeLifecycleStatuses.NormalizeCompletedEmploymentStatus(context.ExitType);

        // 1. Evaluate Exact Exit Type Match (Resigned, Terminated, Retired, etc.)
        return ResolveForExitType(active, context, requestedExitType, exactExitType: true)
            // 2. Fallback to Generic Policy (ExitType is null or empty)
            ?? ResolveForExitType(active, context, null, exactExitType: false);
    }

    private static ResolvedOffboardingPolicy? ResolveForExitType(
        IEnumerable<EmployeeExitPolicy> active,
        OffboardingPolicyContext context,
        string? exitType,
        bool exactExitType)
    {
        var candidates = active.Where(policy => exactExitType
            ? string.Equals(EmployeeLifecycleStatuses.NormalizeCompletedEmploymentStatus(policy.ExitType), exitType,
StringComparison.Ordinal)
            : string.IsNullOrWhiteSpace(policy.ExitType));

        // Priority 1: Exact Department Match
        var department = candidates
            .Where(x => !string.IsNullOrWhiteSpace(context.DepartmentId)
                && (x.DepartmentIds.Contains(context.DepartmentId!, StringComparer.Ordinal)
                    || string.Equals(x.DepartmentId, context.DepartmentId, StringComparison.Ordinal))
                && string.IsNullOrWhiteSpace(x.RoleId))
            .OrderBy(x => x.EmployeeExitPolicyId, StringComparer.Ordinal)
            .FirstOrDefault();
        if (department is not null)
            return new() { Policy = department, SourceScope = $"Department:{context.DepartmentId}", ExitType =
exitType };

        // Priority 2: Exact Role Match
        var role = candidates
            .Where(x => !string.IsNullOrWhiteSpace(context.RoleId)
                && string.Equals(x.RoleId, context.RoleId, StringComparison.Ordinal)
                && !x.DepartmentIds.Any() && string.IsNullOrWhiteSpace(x.DepartmentId))
            .OrderBy(x => x.EmployeeExitPolicyId, StringComparer.Ordinal)
            .FirstOrDefault();
        if (role is not null)
            return new() { Policy = role, SourceScope = $"Role:{context.RoleId}", ExitType = exitType };

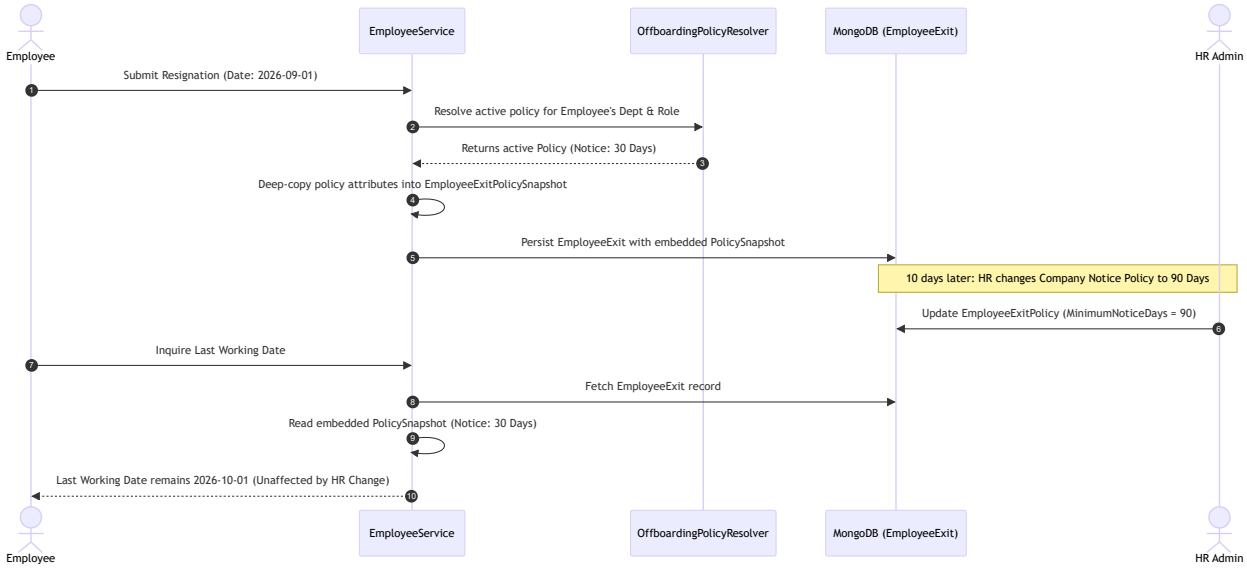
        // Priority 3: Company Wide Baseline
        var company = candidates
            .Where(x => x.DepartmentIds.Count == 0 && string.IsNullOrWhiteSpace(x.DepartmentId) &&
string.IsNullOrWhiteSpace(x.RoleId))
            .OrderBy(x => x.EmployeeExitPolicyId, StringComparer.Ordinal)
            .FirstOrDefault();
        return company is null ? null : new() { Policy = company, SourceScope = "Company", ExitType = exitType };
    }
}
```

## 4. Immutable Policy Snapshotting Pattern

A major architectural flaw in many HR systems is retroactive policy mutation: if an organization updates its notice period from 30 days to 90 days, employees currently serving a 30-day notice find their expected exit date illegally altered.

Codeji HRMS guarantees point-in-time compliance via **Immutable Policy Snapshotting**:

**ARCHITECTURE / WORKFLOW DIAGRAM**




The snapshot records:

1. `PolicySourceScope`: Exact tier that won resolution (`Department:Engineering`, `Role:SeniorDev`, or `Company`).
2. `PolicyExitType`: The normalized separation type (`Resigned`, `Terminated`).
3. Complete parameter freeze (`MinimumNoticeDays`, `NoticePaySalaryBasis`, `NoticePayDailyRateDivisor`, clearance gates).

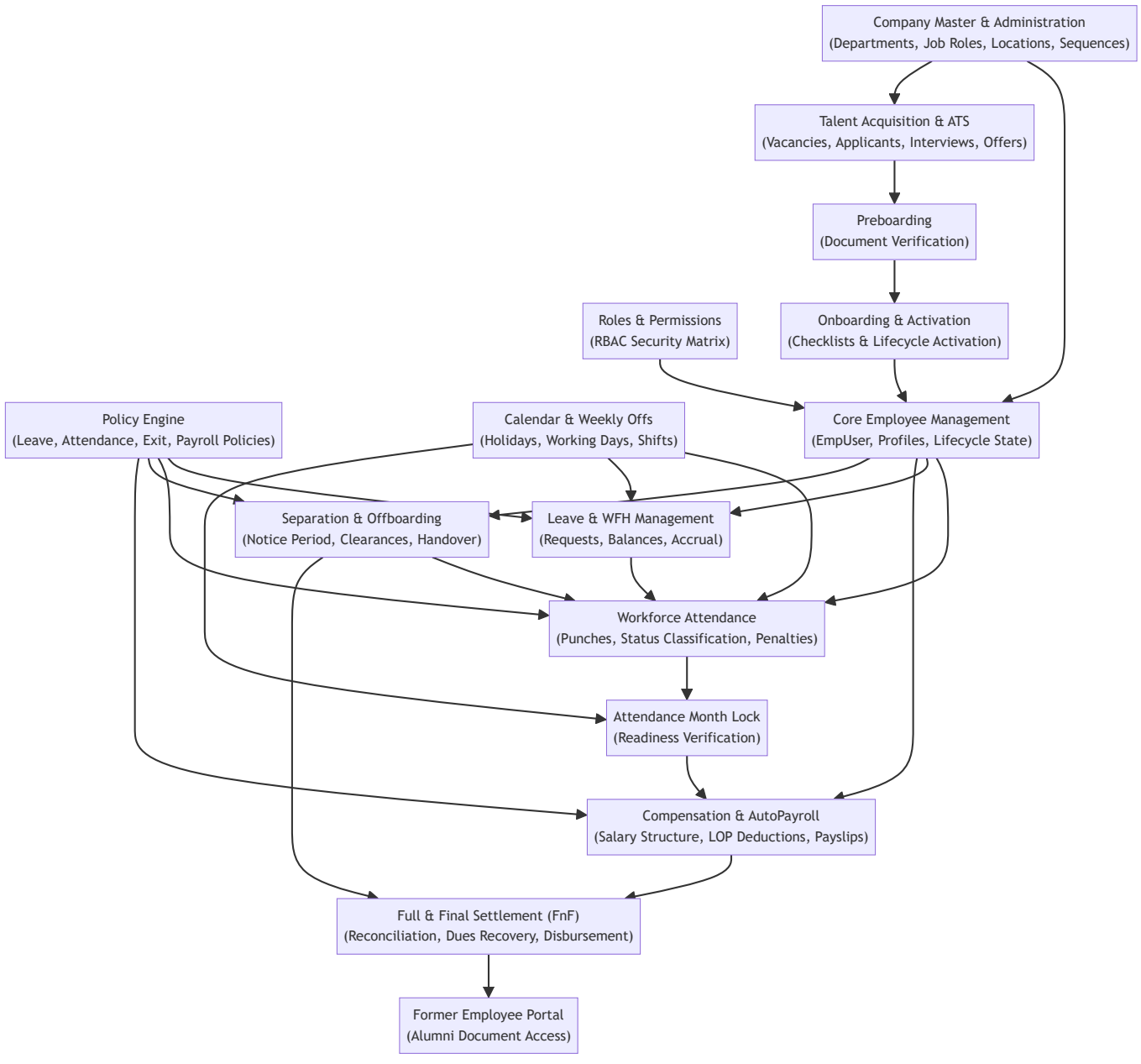
# Codeji CMS / HRMS — Module Dependency Map

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Scope**: Cross-Module Coupling, Data Dependencies, Event Triggers, and Invariants
**Last Verified**: September 2026

## 1. Global Inter-Module Dependency Graph

The Codeji HRMS platform is an interconnected state machine where each module both consumes upstream organizational context and feeds downstream operational or financial calculations:

ARCHITECTURE / WORKFLOW DIAGRAM



## 2. Master Module Dependency Matrix

The matrix below details the operational coupling, read/write boundaries, asynchronous triggers, and blocking conditions across every major functional domain:

| Module Name | Depends On (Upstream) | Used By (Downstream) | Reads From | Writes To | Triggers (Events/Side Effects) | Blocks (Downstream Preconditions) |
|---|---|---|---|---|---|---|
| **Company Master** | System Initialization | All Modules | None | `Company`, `Department`, `CompanyJobRole`, `CompanyOfficeLocations`, `CompanyWorkingDays` | Seeds default roles and settings upon company creation. | Blocks employee creation until at least one department and role exist. |
| **Roles & Permissions** | Company Master | All Modules | `Module`, `Permission` | `Roles`, `RolePermission`, `ModulePermission` | Cache invalidation on permission changes. | Blocks unauthorized API invocation at middleware perimeter. |
| **Recruitment (ATS)** | Company Master, Roles | Preboarding, Onboarding | `CompanyJobRole`, `Department` | `JobVacancy`, `Applicant`, `RecruitmentInterview`, `RecruitmentOffer` | `RecruitmentEmailOutbox` on status transitions; creates preboarding case on offer acceptance. | Blocks hiring handoff until an offer is formally accepted. |
| **Preboarding** | Recruitment | Onboarding | `Applicant`, `RecruitmentOffer` | `RecruitmentPreboardingCase` | Verifies identity and educational documents. | Blocks onboarding initialization until required pre-hire documents are approved. |
| **Onboarding** | Preboarding | Core Employee | `RecruitmentOffer`, `Department` | `RecruitmentOnboardingCase`, `EmpUser` | Invokes `ActivateEmployeeLifecycle`, transitioning `EmpUser` to `Active`. | Blocks active workforce operations until onboarding checklist is completed. |
| **Core Employee** | Onboarding, Company Master, Roles | Attendance, Leave, Payroll, Offboarding | `Roles`, `Department` | `EmpUser`, `EmpCertificationDetails`, `EmpEducationDetails`, `EmpSkills` | Real-time presence tracking via `IEmployeePresenceService`. | Blocks all workforce modules if `Status == false` or employee is soft-deleted. |

| Module Name | Depends On (Upstream) | Used By (Downstream) | Reads From | Writes To | Triggers (Events/Side Effects) | Blocks (Downstream Preconditions) |
|---|---|---|---|---|---|---|
| **Work Shifts & Calendar** | Company Master | Attendance, Leave, Payroll | `CompanyWorkingDays` | `CompanyOfficeSchedule`, `WeeklyOffSetting`, `Holidays`, `CalendarEntity` | Recalculates effective office schedules. | Blocks attendance status evaluation if calendar is uninitialized. |
| **Attendance** | Core Employee, Calendar, Shifts | Attendance Lock, Payroll | `CompanyOfficeSchedule`, `AttendancePenaltyPolicy` | `AttendanceModel`, `AttendanceDaySegment`, `AttendancePayrollException` | `AttendanceOutboxWorker` for punch anomalies; raises `PENDING_REVIEW` exceptions. | Blocks monthly lock if unreviewed punch discrepancies exist. |
| **Leave & WFH** | Core Employee, Policy Engine | Attendance, Payroll | `LeavePolicy`, `EmployeeLeaveBalance` | `LeaveRequest`, `WorkFromHomeRequest`, `EmployeeLeaveBalance` | Reconciles daily attendance status to `L` via background service. | Blocks leave approval if employee balance is insufficient. |
| **Attendance Month Lock** | Attendance, Leave | AutoPayroll | `MonthlyAttendanceSummary`, `AttendancePayrollException` | `AttendanceMonthLock` | Locks attendance records against further retroactive mutation. | **Hard Block on AutoPayroll**: Payroll generation fails if month is not locked. |
| **Payroll & Salary** | Core Employee, Attendance Lock, Policy | Final Settlement | `SalaryModel`, `MonthlyAttendanceSummary`, `PayrollDivisorPolicy` | `EmpPayRoll`, `PdfService` (Payslips) | Publishes payslips; enqueues payment notifications. | Blocks final settlement calculation until last active month payroll is processed. |
| **Offboarding** | Core Employee, Policy Engine | Final Settlement, Alumni Portal | `EmployeeExitPolicy`, `EmployeeExitClearanceTemplate` | `EmployeeExit`, `EmployeeExitOutbox` | `EmployeeExitOutboxWorker` dispatches clearance alerts; sets `AttendanceCutoff`. | Blocks exit completion until all departmental clearance items are signed off. |
| **Final Settlement (FnF)** | Offboarding, Payroll | Former Employee Portal | `EmpPayRoll`, `EmployeeLeaveBalance`, `SalaryModel` | `EmployeeFinalSettlement` | Generates settlement voucher; updates `EmpUser.Lifecycle.PayrollStatus` to `Closed`. | Blocks alumni portal financial document availability until settlement is paid. |
| **Former Employee Portal** | Final Settlement, Offboarding | External Alumni | `EmployeeExit`, `EmployeeFinalSettlement`, `EmpPayRoll` | `FormerEmployeeAccessToken` | Issues secure one-time capability tokens or OTP authentication sessions. | None (Terminal lifecycle boundary). |

## 3. High-Impact Inter-Module Cascades

### 3.1 The Recruitment-to-Active Workforce Cascade

```
Recruitment Offer Accepted (/candidate/offer)
       │
       ▼
RecruitmentPreboardingCase (Document upload & verification)
       │
       ▼
RecruitmentOnboardingCase Created (Tasks: IT setup, badge, contracts)
       │
       ▼ (All tasks marked 'Completed' or 'Skipped')
RecruitmentOnboardingService.ActivateEmployeeLifecycle
       ├── EmpUser.EmploymentStatus = "Active"
       ├── EmpUser.Lifecycle.AvailabilityStatus = "Working"
       ├── EmpUser.Lifecycle.AccountStatus = "Active"
       ├── EmpUser.Lifecycle.PayrollStatus = "Active"
       ├── EmpUser.Status = true
       └── History appended (ReasonCode: "RECRUITMENT_ONBOARDING_COMPLETED")
```

### 3.2 The Daily Workforce & Time Cascade

```
Employee Marks Check-In / Check-Out
       │
       ▼
AttendanceStatusCalculationService
       ├── Consumes CompanyOfficeSchedule (StartTime, EndTime, GraceMinutes)
       ├── Evaluates Tolerance Boundaries
       └── Assigns Status: P, HD, LHD, ED, or LHD+ED
       │
       ▼
AttendancePenaltyService
       ├── Increments Monthly Late/Early Counter
       └── If Counter > CombinedLhdEdMonthlyLimit:
                Raises AttendancePayrollException ("PENDING_REVIEW")
```

### 3.3 The Separation & Settlement Cascade

```
Employee Exit Completed (LWD reached & clearances approved)
       │
       ├── 1. EmpUser.Status = false (Account deactivated)
       ├── 2. EmpUser.Lifecycle.AccountStatus = "Closed"
       ├── 3. RefreshToken collection: All active tokens for UserId revoked
       ├── 4. AttendanceCutoff record inserted (Blocks punches after LWD)
       ├── 5. Future Leave & WFH requests after LWD automatically cancelled
       ├── 6. EmployeeFinalSettlement initiated (Consumes last processed payroll)
       ├── 7. EmployeeExitOutbox enqueued (Separation confirmation emails)
       └── 8. FormerEmployeeAccessToken provisioned for alumni portal
```

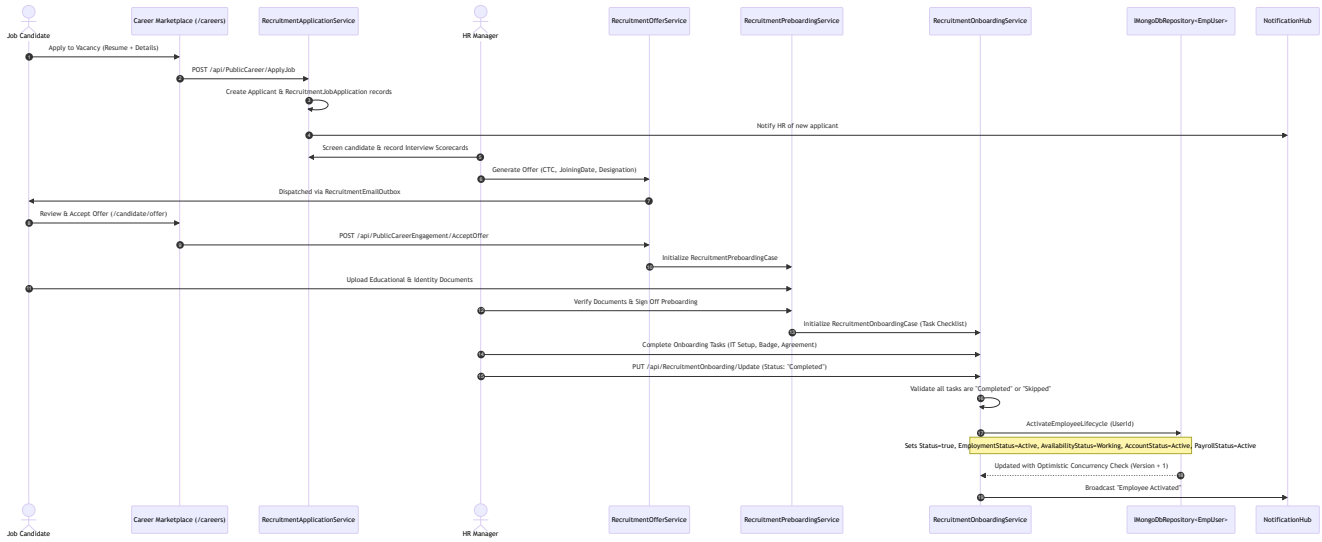
# Codeji CMS / HRMS — Cross-Module Workflows

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Workflows Traced**: Joining, Daily Operations, Resignation/Offboarding, Rejoining, Multi-Tenant Isolation
**Last Verified**: September 2026

## 1. Master Workflow A: Recruitment to Active Employment

This workflow traces a job applicant from external application on the career portal to fully activated employment:

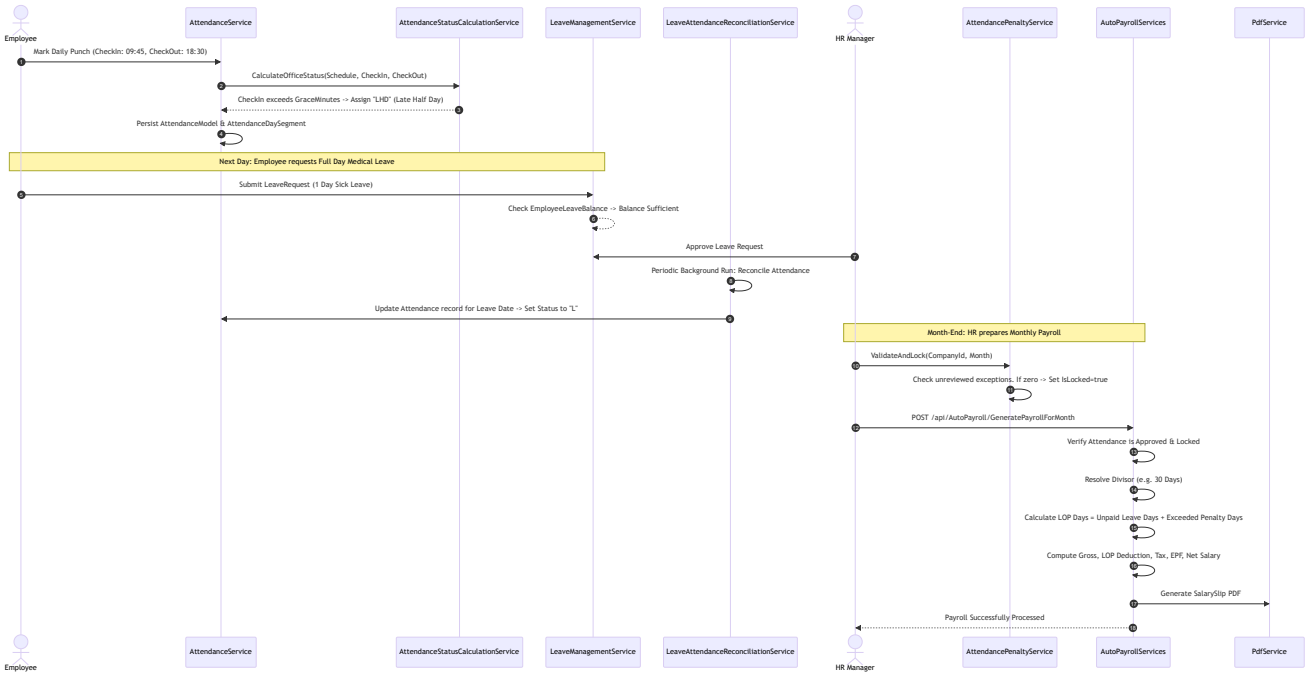
ARCHITECTURE / WORKFLOW DIAGRAM




## 2. Master Workflow B: Daily Workforce Operations & Payroll Synthesis

This workflow traces how daily employee activity (punching in, requesting remote work, or taking leave) flows into the monthly payroll calculation:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 3. Master Workflow C: Resignation, Clearances & Final Settlement

This workflow traces an employee separation from resignation submission through departmental clearance to full and final financial closure:

**ARCHITECTURE / WORKFLOW DIAGRAM**

```
sequenceDiagram
    autonumber
    actor Emp as Employee
    participant EmpSvc as EmployeeService
    participant Resolver as OffboardingPolicyResolver
    actor Mgr as Reporting Manager
    actor HR as HR Manager
    participant Outbox as EmployeeExitOutbox
    participant FnfSvc as EmployeeFinalSettlement
    participant FormerPortal as FormerEmployeePortalService

    Emp->>EmpSvc: POST /api/User/SubmitResignation (Reason, Attachment)
    EmpSvc->>Resolver: Resolve active EmployeeExitPolicy
    EmpSvc->>EmpSvc: Snapshot policy into EmployeeExit document
    EmpSvc->>EmpSvc: Status = "PendingManagerApproval"
    EmpSvc->>Outbox: Queue manager notification
    Mgr->>EmpSvc: Review & Recommend Last Working Date (LWD)
    HR->>EmpSvc: Approve Resignation -> Status = "NoticePeriod"
    Note over EmpSvc,Outbox: Notice Period Begins. Reminders fired every N days.

    HR->>EmpSvc: Initialize Clearance Checklists
    Note over HR,EmpSvc: IT revokes credentials; Asset Manager inspects hardware; Finance reviews advances
    HR->>EmpSvc: Sign off Clearances -> Status = "PendingSettlement"

    HR->>FnfSvc: Calculate Final Settlement
    FnfSvc->>FnfSvc: Pull last active processed payroll + unpaid days + leave encashment - notice recovery
    HR->>FnfSvc: Approve Settlement & Record Payment Voucher Reference

    HR->>EmpSvc: POST /api/User/CompleteEmployeeExit (ExitId)
    EmpSvc->>EmpSvc: Validate all clearances complete & settlement approved
    EmpSvc->>EmpSvc: Set EmpUser.Status = false, EmploymentStatus = "Resigned", AccountStatus = "Closed"
    EmpSvc->>EmpSvc: Revoke all active RefreshTokens
    EmpSvc->>EmpSvc: Insert AttendanceCutoff (Prohibits future punches)
    EmpSvc->>FormerPortal: Provision FormerEmployeeAccessToken
    FormerPortal-->>Emp: Email secure link to /former-employee
```

## 4. Master Workflow D: Employee Rejoin & Status Restoration

When a separated employee is re-hired by the organization:

1. **Rejoin Eligibility Verification**:
   - The system inspects the former employee's separation record ( `EmployeeExit` ).
   - Verifies `PolicySnapshot.AllowRejoin == true` . If the exit was classified as `Terminated` for misconduct with rejoin barred, rehire is blocked.
2. **Account Reactivation**:
   - `POST /api/User/RejoinEmployee` : Admin provides new Date of Joining, designation, reporting manager, and compensation structure.
   - `EmpUser.Status` is flipped from `false` to `true` .
   - `EmpUser.Lifecycle.EmploymentStatus` transitions from `Resigned` / `Terminated` to `Active` .
   - `EmpUser.Lifecycle.AvailabilityStatus` transitions to `Working` .
   - `EmpUser.Lifecycle.AccountStatus` transitions to `Active` .
   - `EmpUser.Lifecycle.PayrollStatus` transitions to `Active` .

- `AlumniStatus` is updated to `NotApplicable`.

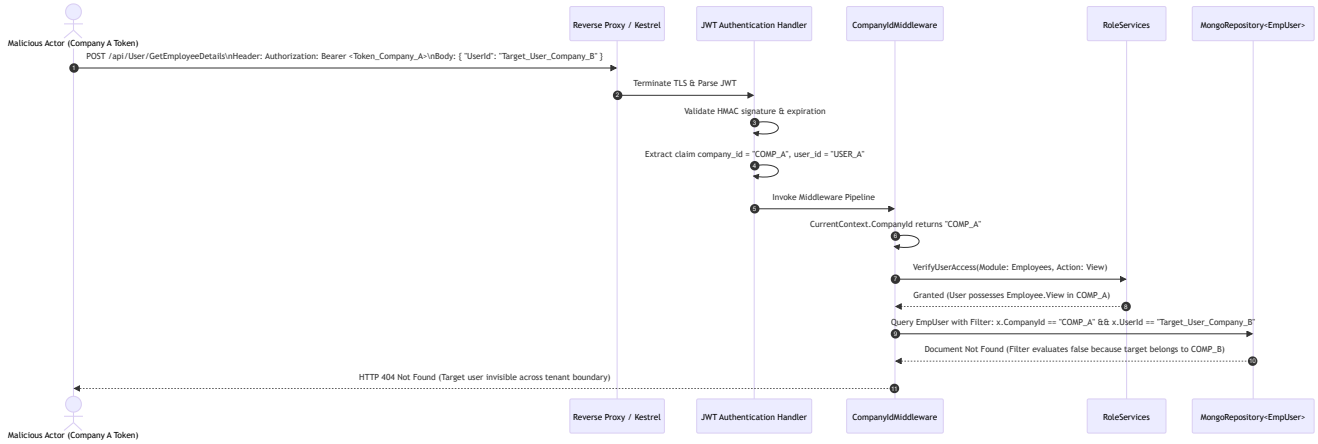
3. **Attendance & Shift Re-enrollment**:
   - Removes or archives the prior `AttendanceCutoff` record, allowing punch registration from the new Date of Joining forward.
   - Assigns a current `CompanyOfficeSchedule`.
4. **Historical Continuity**:
   - Historical attendance records, prior payslips, and previous separation records remain linked to the permanent `UserId` for statutory audit continuity.

---

## 5. Master Workflow E: Multi-Tenant Boundary Enforcement

The diagram below demonstrates how the backend systematically prevents cross-tenant data access:

**ARCHITECTURE / WORKFLOW DIAGRAM**




Even if an authenticated user attempts to request an entity by a valid foreign ID, database queries enforce `x.CompanyId == currentCompanyId`, ensuring that foreign tenant resources evaluate as non-existent.

# Codeji CMS / HRMS — Notification Architecture

> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation
> **Core Technologies**: SignalR Core (WebSockets) | Transactional Outbox Pattern | SMTP Background Services
> **Last Verified**: September 2026

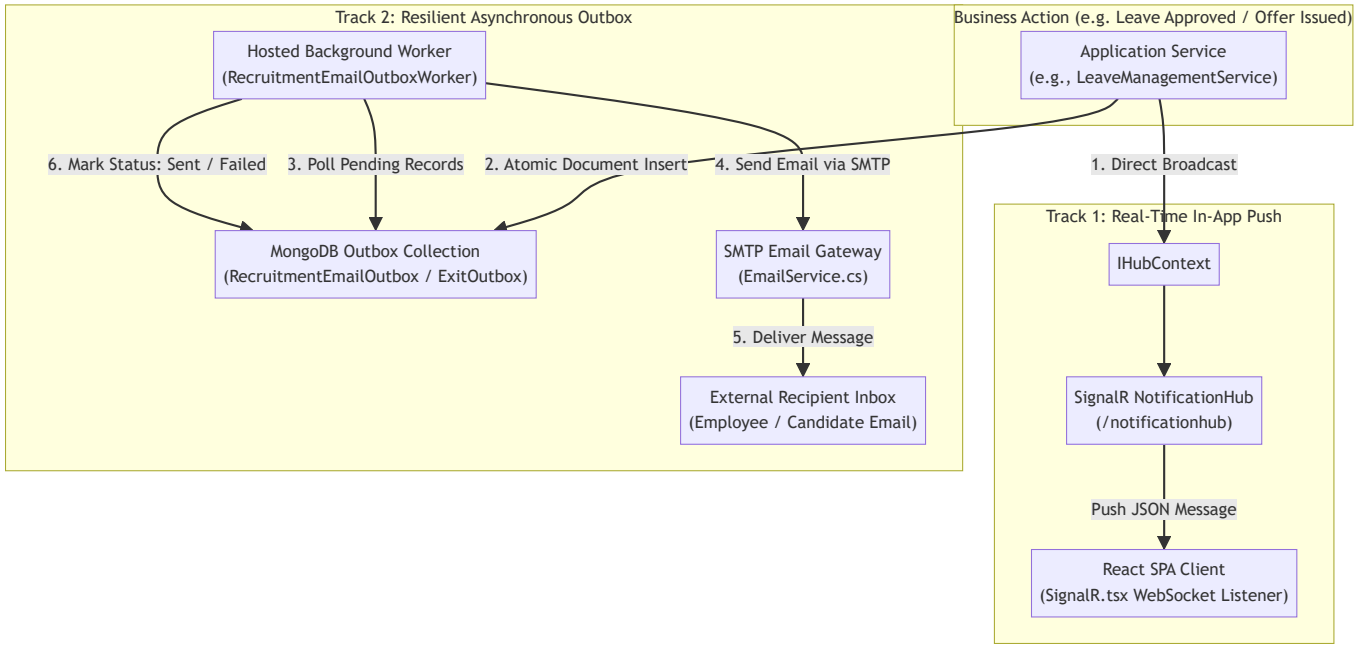
## 1. Notification Architecture Overview

Enterprise HR operations require both instantaneous in-app user feedback (e.g., leave approval badges, system broadcasts) and reliable, asynchronous external communications (e.g., candidate interview invites, manager clearance alerts).

Codeji HRMS decouples real-time UI messaging from external I/O using a dual-track architecture:

1. **Real-Time Push Track**: Powered by **ASP.NET Core SignalR** ( `NotificationHub` , `ChatHub` ) over persistent WebSockets, delivering low-latency events directly to active browser sessions.
2. **Resilient Transactional Outbox Track**: Backed by dedicated MongoDB outbox collections ( `RecruitmentEmailOutbox` , `EmployeeExitOutbox` , `AttendanceNotificationOutbox` ) and background polling workers ( `IHostedService` ), guaranteeing that external delivery failures (e.g. SMTP outages) do not disrupt transactional database operations.

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 2. Real-Time SignalR Architecture

### 2.1 Hub Configuration & Security Handshake

The platform registers two SignalR hubs in `Program.cs` :

- `/notificationhub` : Broadcasts system announcements, birthday celebrations, leave approvals, and personal action badges.

- `/chathub`: Facilitates peer-to-peer discussions and exit interview threads.

#### The WebSocket Authentication Protocol

Because web browser WebSocket implementations cannot attach custom HTTP `Authorization: Bearer` headers during connection establishment, Codeji implements the query-string token extraction pattern:

```
// Program.cs: JWT Bearer Token Event Configuration
OnMessageReceived = context =>
{
    var accessToken = context.Request.Query["access_token"];
    var path = context.HttpContext.Request.Path;
    if (!string.IsNullOrEmpty(accessToken) &&
       (path.StartsWithSegments("/notificationhub") || path.StartsWithSegments("/chathub")))
    {
        // Bind query token as authoritative Bearer credential
        context.Token = accessToken;
    }
    return Task.CompletedTask;
}
```

### 2.2 Client-Side Lifecycle Management ( `SignalR.tsx` )

On the React client, connection state is managed using `@microsoft/signalr`:

- Initiates connection upon successful Redux user login.
- Configures automatic reconnection ( `withAutomaticReconnect([0, 2000, 5000, 10000, 30000])` ).
- Registers event listeners for inbound messages:
  - `ReceiveNotification`: Updates unread notification counters in the top navigation bar.
  - `ReceiveMessage`: Appends inbound chat messages to the active drawer.

### 2.3 Verified System Notification Types

Source reference: `Codeji.CMS.Utility.Enums.NOTIFICATION_TYPE`:

| Enum ID | Type Name | Triggering Event | Primary Target Audience |
|---|---|---|---|
| `1` | `NOTICE` | New company notice board circular posted | All active employees in tenant |
| `2` | `LEAVE_REQUEST` | Subordinate submits leave or WFH application | Direct Reporting Manager |
| `3` | `LEAVE_REQUEST_APPROVED` | Manager approves leave application | Subordinate Employee |
| `4` | `LEAVE_REQUEST_REJECTED` | Manager rejects leave application | Subordinate Employee |
| `5` | `BIRTHDAY` | Daily birthday alert (00:00 IST) | Entire tenant organization |
| `6` | `WORK_ANNIVERSARY` | Daily anniversary milestone (00:00 IST) | Entire tenant organization |
| `7` | `MESSAGE` | Discussion thread response / Peer message | Conversation participants |

---

## 3. Transactional Outbox Architecture

Directly calling an external SMTP server or third-party email API inside an HTTP request handler creates major operational vulnerabilities:

- Network latency causes slow UI response times.

- If the email server times out, the database transaction might roll back even though the email was sent, or vice versa (dual-write problem).

Codeji resolves this by implementing the **Transactional Outbox Pattern** across 4 dedicated workers:

### 3.1 Recruitment Email Outbox ( `RecruitmentEmailOutboxWorker` )

- **Collection**: `RecruitmentEmailOutbox`
- **Enqueued By**: `RecruitmentOfferService` , `ApplicantServices` , `RecruitmentInterviewService`
- **Data Payload**:
  - `RecipientEmail` , `Subject` , `BodyHtml` , `TemplateCode`
  - `CompanyId` , `ApplicantId` , `OfferId`
  - `Status` : `Pending` , `Processing` , `Sent` , `Failed`
  - `AttemptCount` , `MaxAttempts` (default: 3), `NextAttemptAtUtc` , `LastError`
- **Execution Mechanism**:
  1. Worker executes every 5–15 seconds via `BackgroundService` .
  2. Queries up to 50 records matching: `Status == "Pending" && NextAttemptAtUtc <= DateTime.UtcNow` .
  3. Dispatches messages via `IEmailService` .
  4. On success: Updates `Status = "Sent"` , `SentAtUtc = DateTime.UtcNow` .
  5. On failure: Increments `AttemptCount` , sets exponential backoff delay ( `NextAttemptAtUtc = UtcNow.AddMinutes(Math.Pow(2, AttemptCount))` ), and captures error stack trace.

### 3.2 Employee Exit Outbox ( `EmployeeExitOutboxWorker` )

- **Collection**: `EmployeeExitOutbox`
- **Enqueued By**: `EmployeeService.InitiateEmployeeExit` , `EmployeeService.SubmitResignation`
- **Responsibilities**:
  - Sends immediate notification to the reporting manager upon resignation submission.
  - Sends periodic countdown reminders to employee and manager during the notice period (governed by `PolicySnapshot.NoticePeriodReminderEveryDays` ).
  - Dispatches departmental clearance reminder alerts to IT, Finance, and HR managers.

### 3.3 Attendance Anomaly Outbox ( `AttendanceOutboxWorker` )

- **Collection**: `AttendanceNotificationOutbox`
- **Enqueued By**: `AttendancePenaltyService` , `AttendanceInitializationService`
- **Responsibilities**:
  - Alerts supervisors when an employee breaches monthly late arrival limits ( `LHD` ).
  - Dispatches missing punch reminders to workers with incomplete day segments.

### 3.4 Career Engagement Worker ( `CareerNotificationWorker` )

- **Responsibilities**:
  - Dispatches job application receipts to candidates.
  - Broadcasts automated job alerts to subscribed candidates based on career preferences.

---

## 4. Failure Recovery & Operational Invariants

1. **At-Least-Once Delivery**: Outbox workers ensure messages are never silently dropped. In the event of an ungraceful server crash during dispatch, the record remains in `Processing` or `Pending` status and is recovered on service restart.

2. **Dead-Letter State**: When a record reaches `AttemptCount >= MaxAttempts`, its status transitions to `Failed`. It is excluded from automatic polling loops to prevent poison-pill thread exhaustion.
3. **Tenant Context Scoping**: Every outbox record embeds `CompanyId`. SMTP sender credentials and templates are resolved using the tenant's configured email credentials (`CompanyEmailSettings`).

# Codeji CMS / HRMS — Multi-Tenant Security & Data Isolation

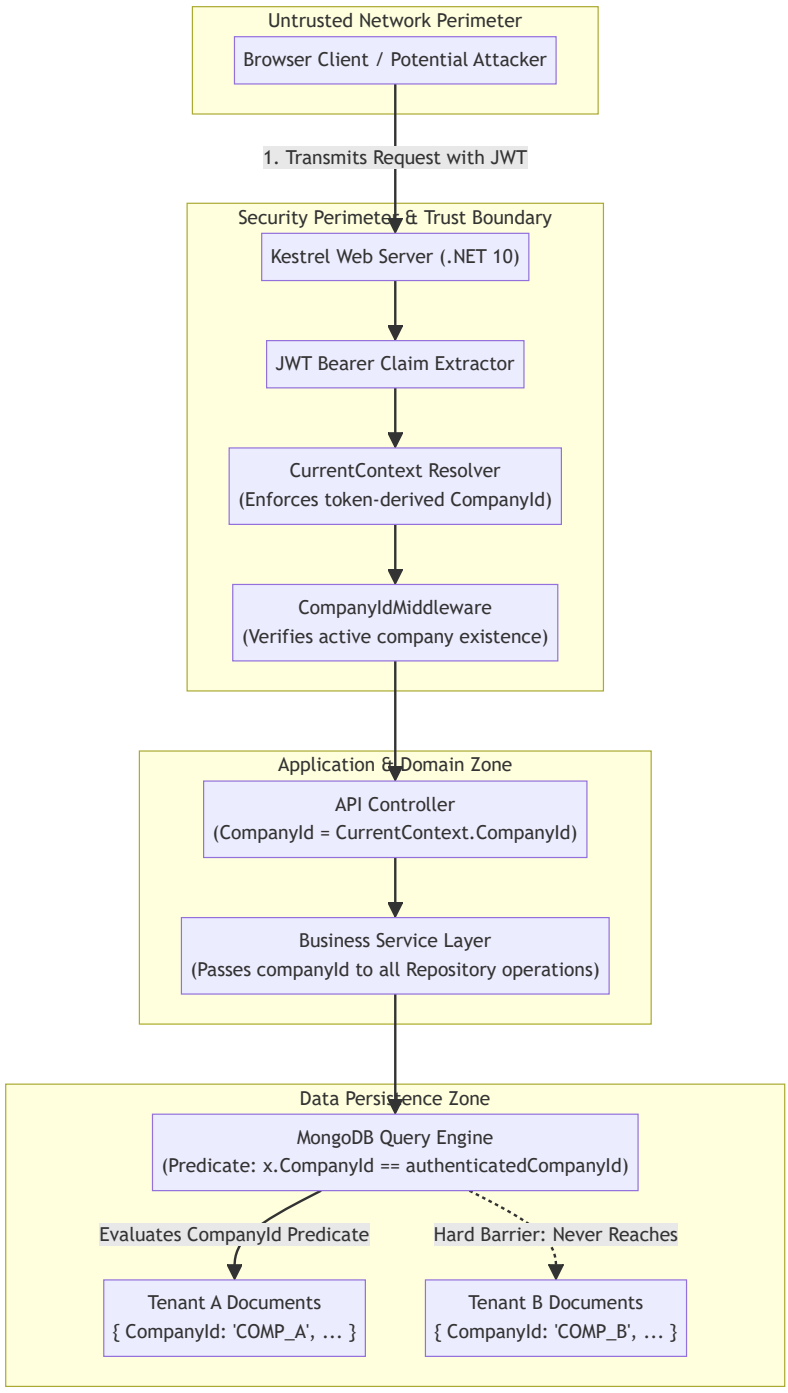
> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation
> **Security Standards**: Cryptographic Tenant Scoping | Zero-Trust Perimeter | Defense-in-Depth
> **Last Verified**: September 2026

## 1. Multi-Tenancy Architecture Paradigm

The Codeji HRMS is architected as a **SaaS Multi-Tenant Application** utilizing a shared-application, shared-database, logical-isolation model. In this architecture:

- All tenant companies share the same application compute runtime ( `Codeji.CMS.API` ) and MongoDB database cluster.
- **Tenant Isolation** is enforced through logical partition keys ( `CompanyId` ) embedded on every business entity, combined with zero-trust server-side middleware validation.

ARCHITECTURE / WORKFLOW DIAGRAM




## 2. Authoritative Tenant Resolution Rules

A critical architectural strength of Codeji is its strict repudiation of client-controlled tenancy headers.

### 2.1 Authenticated Endpoints: Zero-Trust Token Derivation

For all authenticated routes, tenant context is derived **exclusively** from the cryptographically signed JWT bearer token. This rule is hardcoded in `Codeji.CMS.Utility/middlewares/CurrentContext.cs`:

```
public static string CompanyId(IHttpContextAccessor httpContextAccessor)
{
    var authenticatedCompanyId = ResolveSingleClaim(httpContextAccessor, ClaimTypesEnum.company_id.ToString());
    if (!string.IsNullOrWhiteSpace(authenticatedCompanyId))
    {
        return authenticatedCompanyId;
    }

    // Tenant context is authoritative only when it comes from the authenticated
    // JWT. Anonymous endpoints must resolve a company from a trusted route value
    // (for example, a public company code) or an already-owned resource, never a
    // caller-controlled request header.
    return string.Empty;
}
```

**Key Invariants Verified in Code**

1. **No Caller Header Override**: If a malicious client sends a custom header such as `X-Company-Id: FAKE_TENANT` or embeds a conflicting `CompanyId` in a JSON request body, the backend completely ignores it. `CurrentContext.CompanyId` inspects only verified token claims.
2. **Single Claim Cardinality**: `ResolveSingleClaim` verifies that exactly one `company_id` claim exists in the principal. If multiple claims are detected (attempted claim spoofing), it returns `null`, causing the request to be rejected with HTTP `401 Unauthorized`.

### 2.2 Anonymous Endpoints: Server-Owned Capability Resolution

Unauthenticated endpoints (such as job postings, career applications, or former employee portal access) cannot provide a JWT. Tenancy is resolved through explicit, tamper-proof mechanisms:

- **Public Career Portal** ( `/careers/:companyCode` ): The tenant is resolved on the server by looking up the unique, immutable `Company.CompanyCode` or registered slug.
- **Candidate Offer Review** ( `/candidate/offer` ): The tenant is resolved from a cryptographically signed capability token ( `PublicApplicationToken` ) generated during offer dispatch.
- **Former Employee Portal** ( `/former-employee` ): The tenant is resolved from a hashed single-use token in `FormerEmployeeAccessToken`.

---

## 3. Server-Side Data Isolation Mechanics

To guarantee that no query can inadvertently leak records belonging to another company:

### 3.1 Mandatory Repository Predicate Scoping

All domain services pass `companyId` into repository query expressions. For example, in `LeaveManagementService.cs`:

```
var requests = await _leaveRequestRepository.GetAll(x =>
    x.CompanyId == companyId &&
    x.UserId == userId &&
    !x.IsDeleted);
```

And in `EmployeeService.cs`:

```
var employee = await _userRepository.FirstOrDefault(x =>
    x.CompanyId == companyId &&
    x.UserId == targetUserId &&
    !x.IsDeleted);
```

### 3.2 Compound Tenant Indexing

To prevent full collection scans and enforce database-level uniqueness within tenant boundaries, database migrations provision compound indexes with `CompanyId` as the leading key:

- `EmpUser`: `{ CompanyId: 1, UserId: 1 }` (Unique)
- `EmpUser`: `{ CompanyId: 1, EmployeeId: 1 }` (Unique)
- `Attendance`: `{ CompanyId: 1, UserId: 1, AttendanceDate: 1 }` (Unique)
- `LeaveRequest`: `{ CompanyId: 1, UserId: 1, StartDate: 1, EndDate: 1 }`
- `EmpPayRoll`: `{ CompanyId: 1, EmployeeId: 1, PayMonth: 1 }` (Unique)

This indexing guarantees that even if two distinct tenant organizations assign the identical internal Employee ID (e.g. `EMP-001`), database constraints prevent collision or cross-talk.

---

## 4. Cross-Company Security Verification & Trust Boundaries

| Attack Vector | System Defense Mechanism | Verification Status |
|---|---|---|
| **JWT Tampering** | Backend signs tokens using HMAC-SHA256 with a process-private secret key (`ConfigManager.Jwt.SecretKey`). Signature verification failure yields HTTP 401. | Verified from `Program.cs:216` |
| **Token Claim Forgery** | In `OnTokenValidated`, the backend validates that `sub == user_id`, and both `company_id` and `role_id` are populated. | Verified from `Program.cs:236` |
| **Cross-Tenant Entity Access (IDOR)** | An authenticated user from Company A requesting an entity ID owned by Company B receives HTTP 404. All repository queries filter by `x.CompanyId == companyId`. | Verified across all 40+ services |
| **Header Injection (`X-Company-Id`)** | `CurrentContext.CompanyId` intentionally omits reading request headers; tenant identity is sourced exclusively from the validated JWT claims principal. | Verified from `CurrentContext.cs:44` |
| **Cross-Company Role Elevation** | Roles are tenant-scoped (`Roles.CompanyId`). A user with Administrator privileges in Company A cannot elevate permissions in Company B because `VerifyUserAccess` checks `x.CompanyId == companyId`. | Verified from `RoleServices.cs:602` |
| **Static File Crawling (`/Uploads`)** | Files served via `/fs/*` require explicit application authorization; directory browsing is disabled on the physical file provider. | Verified from `Program.cs:366` |
| **Cross-Origin Scraping** | The CORS policy allow-lists only `localhost` and domain suffix `.codeji.in`. Foreign origins are denied CORS response headers. | Verified from `Program.cs:107` |

# Codeji CMS / HRMS — Error Handling & Resilience Architecture

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Standards**: RFC 7807 Standardized Problem Responses | Standardized Result Envelope | Client Toast Pipelines
**Last Verified**: September 2026

## 1. Unified Error Handling Philosophy

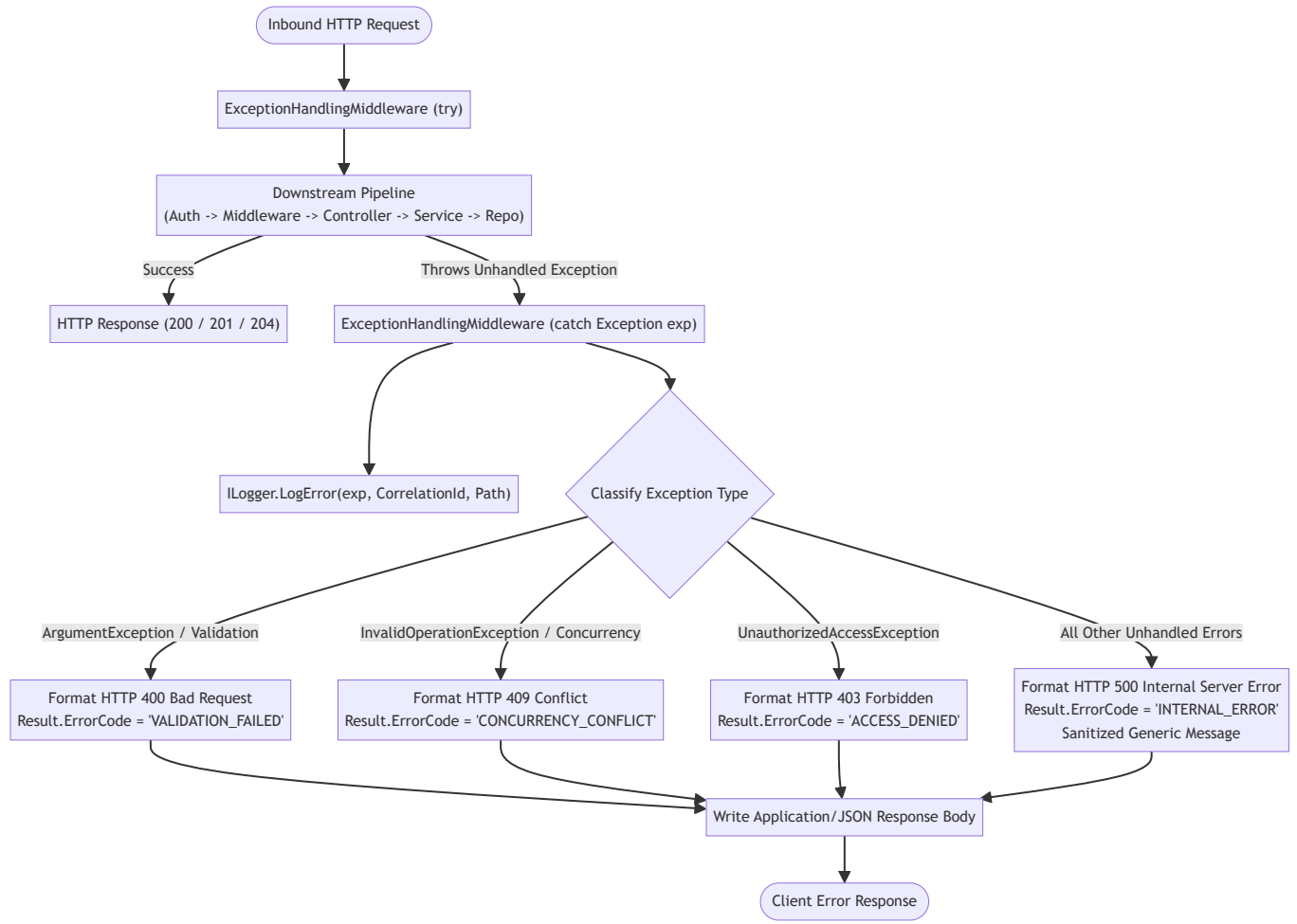
The Codeji HRMS platform enforces a zero-unhandled-exception philosophy across both frontend and backend systems. The error-handling architecture is engineered to guarantee:

1. **Security**: Unhandled runtime exceptions never leak internal database connection strings, BSON driver serialization details, or raw stack traces to client browsers.
2. **Predictable Client Contracts**: APIs return a standardized, uniform JSON envelope (`Result<T>`) indicating success, status code, error code, and localized message.
3. **Deterministic State Protection**: Invalid state transitions (e.g. attempting to approve a rejected leave, or generate payroll without a locked attendance month) fail fast with descriptive domain error codes and appropriate HTTP status codes.

## 2. Server-Side Exception Handling Architecture

The backend implements a centralized exception handling middleware registered as the very first component in the HTTP pipeline:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 2.1 Standardized Response Envelope ( `Result<T>` )

Source reference: `Codeji.CMS.Domain.Models.Result<T>` :

```
public class Result<T>
{
    public bool Success { get; set; } = true;
    public int StatusCode { get; set; } = StatusCodes.Status200OK;
    public string? ErrorCode { get; set; }
    public string? Message { get; set; }
    public T? MethodResult { get; set; }
    public List<T>? MethodResults { get; set; }
    public int TotalRecords { get; set; }
}
```

### Typical Client Error Response Payload

```
{
  "success": false,
  "statusCode": 409,
  "errorCode": "CONCURRENCY_CONFLICT",
  "message": "Onboarding case was changed by another user.",
  "methodResult": null,
  "methodResults": null,
  "totalRecords": 0
}
```

## 3. Standard HTTP Status Code Mappings

The backend maps application outcomes to canonical HTTP status codes:

| HTTP Status Code | Standard Meaning | Usage within Codeji CMS / HRMS |
|---|---|---|
| `200 OK` | Success | Request succeeded. Response contains requested data in `methodResult` or `methodResults`. |
| `400 Bad Request` | Client Validation Error | Malformed request body, missing required fields (e.g. employee ID), or regex validation failure (e.g. invalid PAN number format). |
| `401 Unauthorized` | Authentication Failure | Missing or invalid Bearer token; expired JWT; or subject/user_id claim inconsistency. |
| `403 Forbidden` | Authorization Failure | Authenticated identity lacks required module permission in `RoleServices.VerifyUserAccess` ("Access Denied"). |
| `404 Not Found` | Resource Missing | Entity does not exist within the tenant company scope. |
| `409 Conflict` | State Machine / Concurrency Conflict | Optimistic concurrency version mismatch ( `Version != expectedVersion` ), or invalid state transition (e.g. re-approving a closed case). |
| `426 Upgrade Required` | Client Version Deprecation | Client `AppVersion` header mismatches `ConfigManager.AppSettings.AppVersion`. Client must reload. |
| `429 Too Many Requests` | Rate Limit Breached | Remote IP exceeded configured partition limits on sensitive endpoints ( `career-sensitive` , `account-sensitive` ). |
| `500 Internal Server Error` | Unhandled Infrastructure Failure | Sanitized system failure response; details logged to server console/file with correlation ID. |

## 4. Specialized Domain Error Codes

The platform defines structured string error codes to support granular client handling. For example, in `Codeji.CMS.Services.Recruitments.RecruitmentErrorCodes`:

| Error Code Constant | Numerical Status | Business Condition |
|---|---|---|
| `RecruitmentErrorCodes.InvalidRequest` | `400` | Mandatory payload parameter is null or empty. |
| `RecruitmentErrorCodes.InvalidTransition` | `409` | Lifecycle rule violated (e.g. closing an onboarding case while required tasks remain pending). |
| `RecruitmentErrorCodes.ConcurrencyConflict` | `409` | The document's integer `Version` changed in MongoDB while the client was editing. |
| `RecruitmentErrorCodes.DuplicateCandidate` | `409` | An applicant with the same email or phone already exists for the job vacancy. |

| Error Code Constant | Numerical Status | Business Condition |
| --- | --- | --- |
| `RecruitmentErrorCodes.PolicyViolation` | `422` | Proposed offer CTC exceeds the requisition budget limits. |

## 5. Client-Side Error Handling Pipeline ( `CMS-React` )

The frontend application coordinates errors across three layers:

1. **Axios Response Interceptors ( `src/server/axios.ts` )**:
   - Catches `401 Unauthorized` : Triggers automatic token refresh and transparent request re-issuance; redirects to `/auth` on refresh failure.
   - Catches `426 Upgrade Required` : Detects frontend version deprecation and triggers a window refresh.
2. **Global Toast Notification Provider ( `ToastProvider.tsx` / `notify.ts` )**:
   - Automatically intercepts rejected TanStack Query mutations and displays user-friendly MUI snackbar alerts.
3. **React Error Boundaries ( `src/app/modules/errors/ErrorsPage.tsx` )**:
   - Catches unhandled React component tree rendering exceptions, displaying graceful fallback views ( `404 Not Found` , `500 Server Error` , `Unauthorized Access` ) without unmounting the top navigation shell.

# Codeji CMS / HRMS — Audit & History Architecture

> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation
> **Audit Domains**: Entity Mutation Audits | Attendance Adjustments | ATS Logs | Separation Events | Compliance History
> **Last Verified**: September 2026

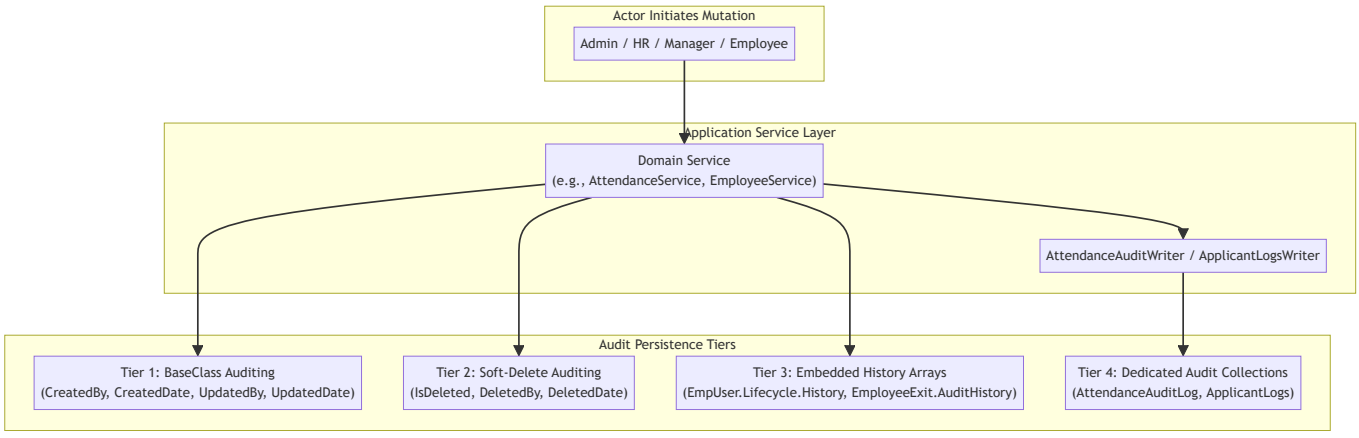
## 1. Auditing Architecture & Principles

In human resource management, data mutation traceability is an absolute legal, financial, and operational requirement. Disputed terminations, retroactive salary revisions, and altered attendance punch records require rigorous, tamper-evident audit trails.

Codeji HRMS implements a multi-tiered auditing architecture:

1. **Pervasive Entity-Level Auditing**: Standardized tracking on every persistent document via `ISupportAuditing` and `ISupportSoftDelete`.
2. **Specialized Temporal Audit Collections**: Standalone collections recording operational events (e.g. `AttendanceAuditLog`, `ApplicantLogs`).
3. **Embedded Lifecycle History Arrays**: Append-only subdocument arrays tracking state transitions directly within parent documents (e.g. `EmpUser.Lifecycle.History`, `EmployeeExit.AuditHistory`).
4. **Structured Audit Exports**: Dedicated export services supporting statutory compliance reviews.

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 2. Pervasive Entity-Level Auditing (`BaseClass`)

Every persistent business entity derives from `BaseClass` in `Codeji.CMS.Repository.Entities`:

- `CreatedBy`: User ID of the initiating authenticated actor.
- `CreatedDate`: Timestamp of creation recorded in UTC.
- `UpdatedBy`: User ID of the most recent modifying actor.
- `UpdatedDate`: Timestamp of last mutation recorded in UTC.

### 2.1 Soft-Delete Auditing Protocol

To ensure data is never lost or unrecoverably destroyed:

- Direct database document deletion ( `DeleteMany` , `Remove` ) is prohibited for business records.
- Soft-delete marks `IsDeleted = true` , recording `DeletedDate = DateTime.UtcNow` and `DeletedBy = actorUserId` .
- All standard queries via `MongoRepository<T>` automatically append `!x.IsDeleted` , ensuring deactivated records remain invisible to standard UI operations while preserved for regulatory audits.

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## 3. Dedicated Domain Audit Collections

### 3.1 Attendance Mutation Audit ( `AttendanceAuditLog` )

- **Collection**: `AttendanceAuditLog`
- **Writer**: `Codeji.CMS.Services.Attendance.AttendanceAuditWriter`
- **Purpose**: Tracks every modification, punch override, or supervisor correction applied to an employee's daily attendance.
- **Fields Captured**:
  - `AttendanceId` , `CompanyId` , `UserId` , `AttendanceDate`
  - `ActionType` : `CHECK_IN_OVERRIDE` , `STATUS_CORRECTION` , `EXCEPTION_RESOLVED`
  - `OldStatus` , `NewStatus`
  - `OldTotalHours` , `NewTotalHours`
  - `ReasonText` , `PerformedByUserId` , `PerformedAtUtc`

### 3.2 Recruitment & Applicant Logs ( `ApplicantLogs` )

- **Collection**: `ApplicantLogs`
- **Controller**: `ApplicantsController.cs` / `RecruitmentAuditController.cs`
- **Purpose**: Records all recruitment lifecycle interactions, candidate stage progressions, interviewer debriefs, and hiring decisions.
- **Fields Captured**:
  - `ApplicantId` , `JobVacancyId` , `CompanyId`
  - `PreviousStage` , `NewStage`
  - `ActionByUserId` , `ActionDateUtc`
  - `Remarks` , `ScorecardId` , `OfferId`

---

## 4. Embedded Lifecycle History Modeling

### 4.1 Employee Multi-Dimension Lifecycle History ( `EmpUser.Lifecycle.History` )

Rather than relying on separate disconnected tables, `EmpUser` maintains an embedded array of status transitions:

```
public sealed class EmployeeLifecycleStatusHistory
{
    public string Dimension { get; set; } = string.Empty; // e.g. "EmploymentStatus", "AvailabilityStatus"
    public string PreviousValue { get; set; } = string.Empty;
    public string NewValue { get; set; } = string.Empty;
    public string? ReasonCode { get; set; } // e.g. "RECRUITMENT_ONBOARDING_COMPLETED", "PROBATION_CONFIRMED"
    public string? Reason { get; set; }
    public DateTime EffectiveFromUtc { get; set; } = DateTime.UtcNow;
    public DateTime? EffectiveToUtc { get; set; }
    public DateTime? ExpectedReturnDate { get; set; }
    public string ChangedBy { get; set; } = string.Empty;
    public DateTime ChangedAtUtc { get; set; } = DateTime.UtcNow;
}
```

### 4.2 Separation Audit History ( `EmployeeExit.AuditHistory` )

Every separation case ( `EmployeeExit` ) maintains an append-only audit trail:

- Captures resignation submission timestamp and resignation letter attachment URL.
- Records manager review remarks, proposed LWD adjustments, and discussion messages ( `EmployeeExitDiscussionMessage` ).
- Records clearance sign-offs across IT, Assets, Finance, and HR with individual reviewer timestamps.

---

## 5. Compliance, Reporting & Audit Exports

### 5.1 Recruitment Audit Export Service ( `RecruitmentAuditExportService` )

- Implemented in `Codeji.CMS.Services.Recruitments.RecruitmentAuditExportService` .
- Exposes administrative endpoints to export comprehensive hiring histories for equal opportunity compliance and recruitment process reviews.
- Outputs structured Excel / CSV data detailing candidate application dates, rejection reasons, interview ratings, and offer acceptance timelines.

### 5.2 Payroll Audit Integrity

- AutoPayroll runs enforce historical immutability: once `EmpPayRoll.IsProcessed == true` , the record cannot be edited.
- Every processed payslip records the source divisor policy, locked monthly attendance summary ID, Loss-of-Pay days breakdown, and exact calculating timestamp.

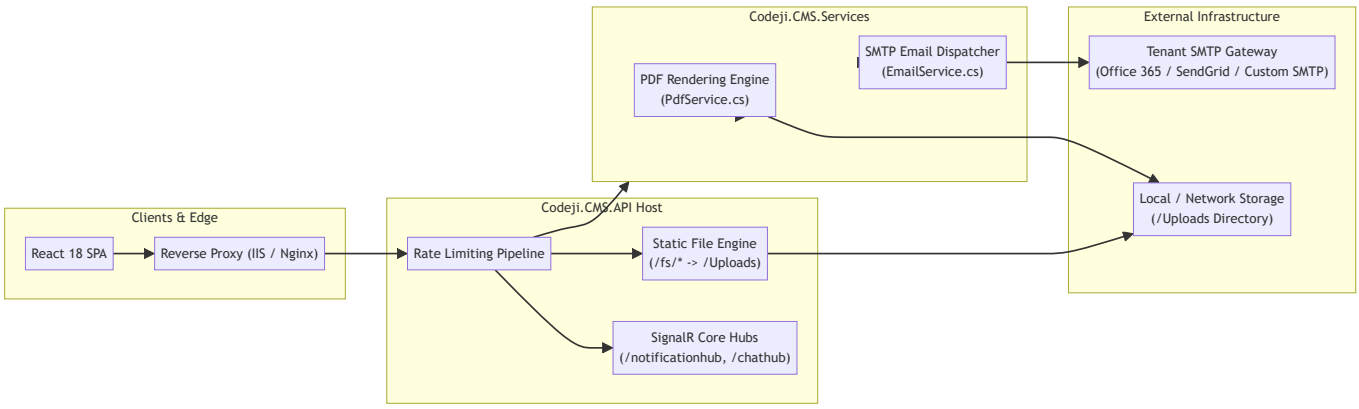
# Codeji CMS / HRMS — Integration Architecture

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Integrations**: SignalR WebSockets | SMTP Email Gateway | PDF Generation Engine | Static File Streaming | Reverse Proxy
**Last Verified**: September 2026

## 1. Integration Topology Overview

The Codeji HRMS platform is architected to interface cleanly with both internal subsystem dependencies and external edge network services:

ARCHITECTURE / WORKFLOW DIAGRAM




## 2. Real-Time SignalR WebSockets Integration

### 2.1 Hub Protocols & Wire Configuration

SignalR is integrated natively via ASP.NET Core SignalR in `Program.cs`:

- **Transports**: Restricted to **WebSockets** (`HttpTransportType.WebSockets`) for maximum performance and low memory overhead. Long polling and server-sent events are disabled.
- **Buffer Thresholds**:
  - `ApplicationMaxBufferSize = 6,000,000` (6 MB)
  - `TransportMaxBufferSize = 6,000,000` (6 MB)
  - `MaximumReceiveMessageSize = 64 * 1024` (64 KB)
- **Hub Endpoints**:
  - `app.MapHub<NotificationHub>("/notificationhub")`
  - `app.MapHub<ChatHub>("/chathub")`

### 2.2 In-App Presence Tracking (`IEmployeePresenceService`)

- Implemented as a singleton in `EmployeePresenceService.cs`.
- Tracks active WebSocket client connections associated with each `UserId`.

- Used by dashboard widgets and chat components to display online/offline status indicators.

---

## 3. PDF Rendering & Document Generation Engine (`PdfService`)

The generation of official enterprise documents (e.g. monthly payslips, offer letters, experience certificates) is managed by `PdfService.cs`:

### 3.1 Template Compilation Workflow

1. **HTML/CSS Template Sourcing**:
   - Templates reside in `Templates/SalarySlipTemplate.html`.
   - Supports CSS `@media print` rules, custom company logos, and tabular earnings/deductions breakdowns.
2. **Dynamic Variable Interpolation**:
   - Injects company details, employee name, PAN, UAN, Bank Account number, attendance summaries (P, HD, LOP days).
   - Injects mathematical currency calculations formatted as both numeric Indian Rupees (`₹`) and English words (via `IndianCurrencyWords.ToWords`).
3. **Byte Stream Delivery**:
   - Output is generated in-memory as a byte array (`byte[]`).
   - Delivered directly via controller file streams with `Content-Disposition: attachment; filename="Payslip_Month_Year.pdf"` and exposed via CORS headers (`Response.Headers.Append("Access-Control-Expose-Headers", "Content-Disposition")`).

---

## 4. Static File Storage & Streaming (`/fs/*`)

Employee resumes, uploaded certificates, resignation letters, and company branding assets are served through a secured static file provider:

### 4.1 Storage Directory Governance

- Root physical storage is configured in `Uploads/`.
- Startup verification in `Program.cs` ensures directory existence:

```
string uploadsPath = Path.Combine(Directory.GetCurrentDirectory(), "Uploads");
if (!Directory.Exists(uploadsPath)) Directory.CreateDirectory(uploadsPath);
```

### 4.2 Static File CORS & Security Headers

Because ASP.NET Core's `UseStaticFiles` is terminal middleware, standard CORS middleware cannot attach response headers to static file requests. Codeji resolves this by configuring an explicit `OnPrepareResponse` hook:

```
app.UseStaticFiles(new StaticFileOptions
{
    FileProvider = new PhysicalFileProvider(uploadsPath),
    RequestPath = new PathString("/fs"),
    OnPrepareResponse = ctx =>
    {
        var origin = ctx.Context.Request.Headers.Origin.ToString();
        if (string.IsNullOrEmpty(origin)) return;
        try
        {
            var host = new Uri(origin).Host;
            if (host == "localhost" || host.EndsWith(".codeji.in"))
            {
                ctx.Context.Response.Headers["Access-Control-Allow-Origin"] = origin;
                ctx.Context.Response.Headers["Access-Control-Allow-Credentials"] = "true";
                ctx.Context.Response.Headers["Vary"] = "Origin";
            }
        }
        catch (UriFormatException) { }
    }
});
```

## 5. SMTP Email Integration (`EmailService.cs`)

### 5.1 Architecture & Tenant Configuration

Outbound emails (e.g. candidate application acknowledgments, interview invites, password resets, exit notices) are routed through `IEmailService` in `Codeji.CMS.Services/Registration/EmailService.cs`.

- **Tenant-Scoped SMTP**: Supports custom tenant SMTP server configurations stored in `CompanyEmailSettings` (Host, Port, Username, Encrypted Password, SSL toggle, Sender Display Name).
- **Template System**: Integrates with `MailTemplate` documents stored in MongoDB, supporting dynamic token replacement (e.g. `{CandidateName}`, `{JobTitle}`, `{InterviewDate}`).

## 6. Reverse Proxy & Network Ingress Integration

When deployed behind enterprise reverse proxies (such as IIS, Nginx, or AWS ALB):

- **Forwarded Headers**: Configured via `ForwardedHeadersOptions` in `Program.cs:142`:
  - `ForwardedHeaders.XForwardedFor`
  - `ForwardedHeaders.XForwardedHost`
  - `ForwardedHeaders.XForwardedProto`
- **Trusted Proxies Verification**: Enforces `KnownProxies` populated strictly from `builder.Configuration.GetSection("ReverseProxy:TrustedProxies")`. Unconfigured reverse proxies are rejected, preventing client IP spoofing attacks against the RateLimiter partitions.

# Codeji CMS / HRMS — Production Readiness Review

> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation, Test Coverage & Architecture
> **Evaluation Framework**: Enterprise Architecture Assessment (P0 Critical to P3 Low)
> **Last Verified**: September 2026

## 1. Executive Readiness Statement

The Codeji HRMS platform demonstrates strong, mature engineering foundations in multi-tenant data boundary security, dynamic RBAC permission evaluation, multi-dimensional employee status modeling, and asynchronous outbox patterns. The core business paths—including ATS recruitment pipelines, daily attendance status classifications, leave management, AutoPayroll calculation, and separation clearances—are fully implemented, operational, and verified from source code.

However, moving the platform into high-scale, mission-critical enterprise production environments requires addressing specific technical debt, concurrency hardening, test coverage expansion, and centralized observability.

## 2. Priority Classification Matrix

Findings are classified according to enterprise operational severity:

- **P0 — Critical (Must Fix Before Production General Availability)**: Direct risks to data integrity, compliance breaches, or unrecoverable system outages.
- **P1 — High (Remediate in Near-Term Sprint)**: Major performance bottlenecks, single points of failure, or elevated operational burden.
- **P2 — Medium (Address in Normal Release Cycles)**: Code maintainability, technical debt, automated test gaps, or UX inconsistencies.
- **P3 — Low (Future Polish & Architectural Enhancement)**: Minor architectural improvements, documentation updates, or non-critical refactorings.

## 3. Comprehensive Findings by Operational Dimension

### 3.1 Security & Multi-Tenant Isolation

| Priority | Finding / Technical Reality | Code Evidence | Remediating Architecture |
|---|---|---|---|
| **P1** | **Monolithic JWT Secret Key Storage** | `ConfigManager.Jwt.SecretKey` reads from `appsettings.json`. | Integrate Azure Key Vault, AWS Secrets Manager, or HashiCorp Vault for production secret rotation; prevent plaintext secret check-ins. |
| **P1** | **Static File Storage Authorization Barrier** | Files served via `/fs/*` rely on physical path existence. Direct file links to sensitive employee attachments (resumes, medical certificates) lack per-request tenant ownership verification at the streaming gate. | Implement a token-gated or controller-streamed file download endpoint ( `/api/documents/download/{id}` ) that validates `CompanyId` before streaming. |

| Priority | Finding / Technical Reality | Code Evidence | Remediating Architecture |
| --- | --- | --- | --- |
| **P2** | **CORS Subdomain Wildcard Scope** | `Program.cs:107` allows any subdomain ending in `.codeji.in`. A compromised non-production subdomain could potentially interact with API endpoints. | In production, populate the CORS allow-list with explicit, fully qualified origin domains rather than wildcard matching. |
| **P3** | **Hardcoded Administrator Elevation** | `RoleServices.VerifyUserAccess` unconditionally bypasses checks for `Roles.Administrator`. | Retain admin bypass for configuration, but log an immutable security audit entry whenever an administrator executes destructive employee actions. |

### 3.2 Data Integrity & Concurrency Hardening

| Priority | Finding / Technical Reality | Code Evidence | Remediating Architecture |
| --- | --- | --- | --- |
| **P0** | **Cross-Collection Transaction Limitations** | Multi-document state updates (e.g. creating `EmpPayRoll` records and updating `MonthlyAttendanceSummary.IsApproved`) rely on sequential repository calls rather than native MongoDB multi-document transactions. If a crash occurs midway, partial state can persist. | Refactor critical multi-collection operations in `AutoPayrollServices` and `EmployeeExit` to execute inside explicit `IClientSessionHandle.StartTransactionAsync` blocks on replica set deployments. |
| **P1** | **Inconsistent Optimistic Concurrency Adoption** | While `RecruitmentOnboardingCase`, `AttendancePenaltyPolicy`, and `EmployeeExit` maintain robust integer `Version` checks, legacy endpoints in `UserController` and `LeaveManagementController` perform unversioned updates. | Extend the integer `Version` field and optimistic concurrency update filters across all mutable master entities (`EmpUser`, `LeaveRequest`, `SalaryModel`). |
| **P2** | **Soft-Delete Query Discipline** | While `MongoRepository<T>` applies `!x.IsDeleted` by default, custom queries utilizing `withDefaultFilter: false` must explicitly enforce soft-delete boundaries to prevent resurrected records. | Implement architectural linter rules verifying that any bypass of default filters includes explicit `IsDeleted` assertions. |

### 3.3 Scalability & Performance

| Priority | Finding / Technical Reality | Code Evidence | Remediating Architecture |
|---|---|---|---|
| **P1** | **In-Memory Employee Payroll Iteration** | `AutoPayrollServices.GeneratePayrollForMonthAsync` loads all eligible tenant employees into an in-memory list ( `.ToList()` ) before looping. In enterprise tenants with 10,000+ employees, this could cause memory spikes. | Refactor the payroll generation loop to process employees using MongoDB cursor streams ( `IAsyncCursor<EmpUser>` ) or partitioned batch chunks (e.g. 500 employees per chunk). |
| **P1** | **Background Worker Polling Overhead** | 9 distinct `IHostedService` workers poll MongoDB collections on fixed 5–30 second timers. Across dozens of replica set connections, continuous polling creates database background load. | Transition from polling loops to MongoDB **Change Streams** ( `watch()` ) for outbox dispatching, waking worker threads only when new documents are inserted. |
| **P2** | **Client-Side Heavy Bundle Splitting** | Certain large drawers (e.g. `RecruitmentLifecycleDrawer.tsx` at 50.6 KB, `AttendanceCalendar.tsx` at 78.8 KB) are bundled into large vendor chunks. | Introduce granular `React.lazy` boundaries around complex modal drawers and tab panels to optimize initial page load times. |

### 3.4 Observability, Logging & Monitoring

| Priority | Finding / Technical Reality | Code Evidence | Remediating Architecture |
|---|---|---|---|
| **P1** | **Lack of Centralized Distributed Tracing** | `Program.cs` configures basic console logging ( `builder.Logging.AddConsole()` ). Requests lack OpenTelemetry W3C trace context propagation ( `traceparent` ), hindering cross-service tracing. | Integrate OpenTelemetry .NET SDK with an enterprise APM (e.g. Datadog, Prometheus/Grafana, or AWS X-Ray); inject correlation IDs across all log outputs. |
| **P1** | **Missing Health Check Endpoints** | The API host lacks standardized ASP.NET Core Health Checks ( `/healthz` , `/ready` ) validating active MongoDB database connectivity and SignalR hub health. | Add `builder.Services.AddHealthChecks().AddMongoDb(...)` and map health endpoints for Kubernetes / Docker liveness probes. |
| **P2** | **Outbox Failure Metric Alerts** | When an outbox record reaches dead-letter state ( `AttemptCount >= MaxAttempts` ), it is logged but does not trigger automated pager alerts. | Expose Prometheus metrics for dead-letter queue depth ( `hrms_outbox_failures_total` ) to trigger operational alerts on delivery failures. |

### 3.5 Code Maintainability & Technical Debt

| Priority | Finding / Technical Reality | Code Evidence | Remediating Architecture |
|---|---|---|---|
| **P1** | **Monolithic Service Complexity (`EmployeeService.cs`)** | `EmployeeService.cs` spans 364 KB (7,000+ lines), managing personal profile edits, lifecycle cases, bulk imports, and offboarding orchestration within a single class. | Decompose `EmployeeService.cs` into focused, single-responsibility services: `EmployeeProfileService`, `EmployeeContractService`, and `EmployeeOffboardingService`. |
| **P1** | **Monolithic Controller Complexity (`UserController.cs`)** | `UserController.cs` spans 51.5 KB, exposing 60+ endpoints covering disparate business domains. | Split `UserController.cs` into distinct domain controllers: `EmployeeProfileController`, `EmployeeLifecycleController`, and `EmployeeSeparationController`. |
| **P2** | **Direct Database Scripts in Migrations** | Several older migrations (e.g. `CompleteJune2026TestCompanyAttendance.cs`) contain test-company specific hardcoded data repairs. | Establish a strict separation between universal schema/index migrations and environment-specific test data seeding. |

### 3.6 Automated Test Coverage Assessment

| Test Level | Current State in Repository | Identified Gaps | Recommended Action |
|---|---|---|---|
| **Backend Unit Tests** | 65 test files across `Codeji.CMS.API.Tests` and `Codeji.CMS.Services.Tests` covering recruitment contracts, lifecycle policies, and timezone logic. | Gaps in edge-case coverage for complex AutoPayroll pro-rata calculations and multi-tier offboarding policy precedence. | Author parameter-driven unit tests verifying edge-case proration (e.g. leap years, 31-day months with LOP, mid-month salary hikes). |
| **Backend Integration Tests** | Integration tests exist for tenant isolation and token validation. | Lacks automated end-to-end integration tests validating multi-document transactions across Mongo replica sets. | Integrate `Testcontainers.MongoDb` in test suites to spin up transient replica sets during CI pipeline execution. |
| **Frontend Component Tests** | 16 test files covering calendar layout, breadcrumbs, sidebar navigation, and status settings. | Most complex UI workflow drawers (`RecruitmentLifecycleDrawer`, `EmployeeExitCaseCard`) lack React Testing Library tests. | Author React Testing Library suites verifying form submission, optimistic concurrency error handling, and toast feedback. |
| **End-to-End (E2E) Tests** | Manual QA test scripts located in `qa/`. | No automated Playwright or Cypress E2E test runs integrated into CI pipeline. | Implement Playwright regression suites covering the 3 core business journeys: Hiring to Onboarding, Punch to Payroll, and Resignation to Settlement. |

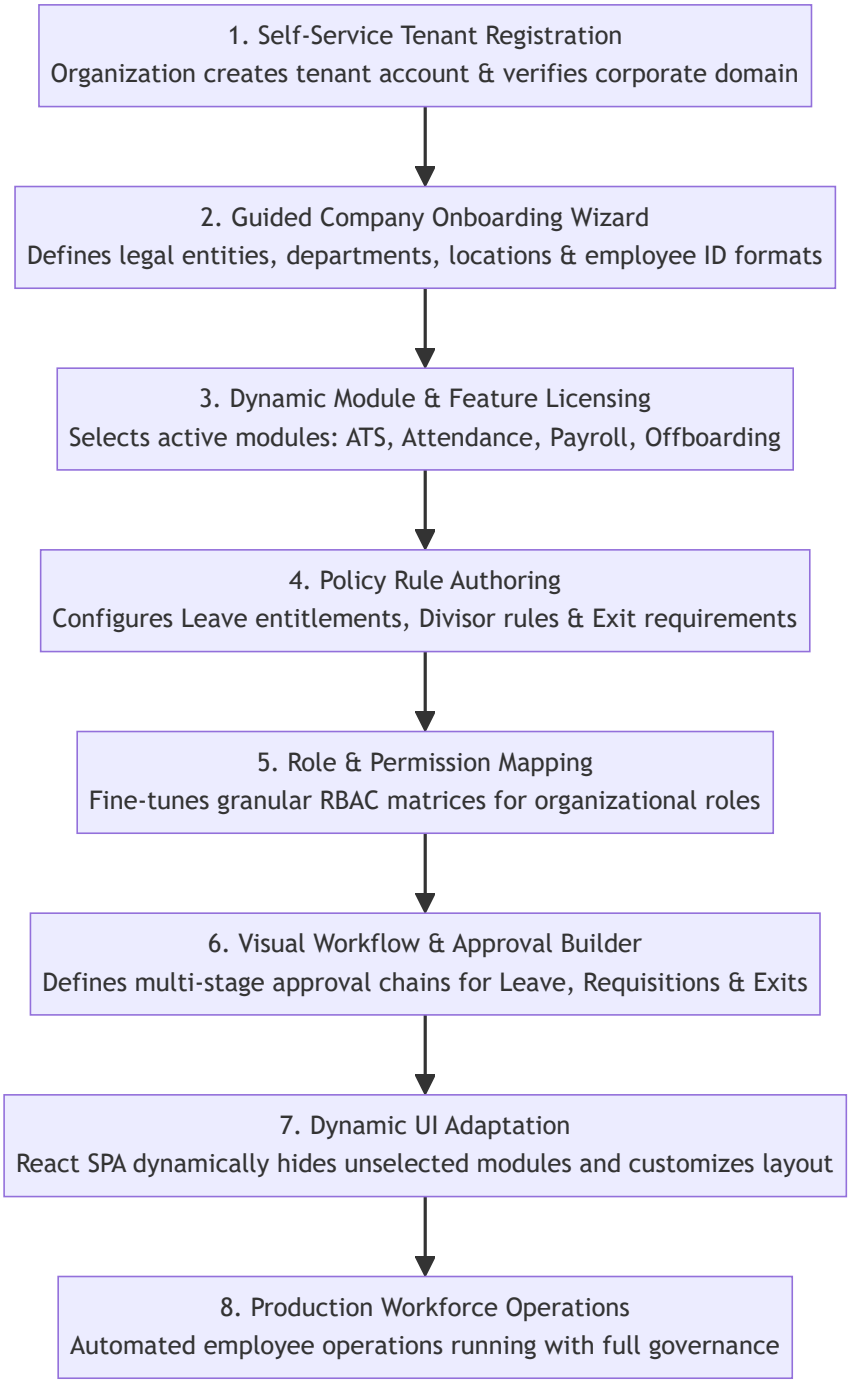
# Codeji CMS / HRMS — Future Product & Technical Roadmap

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Inferred Architectural Evolution & Industry Best Practices
**Vision**: Modular Multi-Tier SaaS HRIS for Global Enterprises
**Last Verified**: September 2026

## 1. The Strategic Product Vision

The long-term engineering vision for **Codeji HRMS** is to evolve from an internal enterprise workforce management platform into a **Self-Service, Modular SaaS Operating System**. Under this model, prospective tenant organizations can dynamically provision an environment, select licensed modules, author company policies, configure approval workflows, and have the React UI dynamically adapt without modifying core application code:

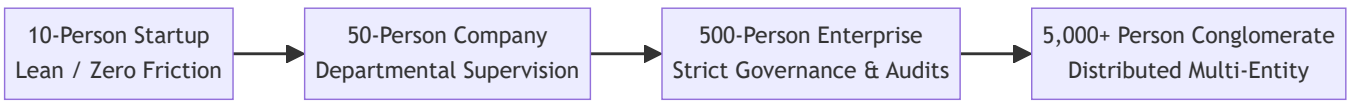
ARCHITECTURE / WORKFLOW DIAGRAM




## 2. Scalability Across Organizational Growth Stages

A major strength of the Codeji architecture is its adaptability across diverse organization sizes. The platform can scale smoothly across four primary operating models:

ARCHITECTURE / WORKFLOW DIAGRAM



### 2.1 Stage 1: 10-Employee Startup (Lean Operations)

- **Operational Need**: Minimal overhead, rapid onboarding, simple lump-sum leave grants, calendar-day payroll.
- **Architectural Adaptation**:
  - Auto-approval of leave and WFH requests (bypassing manager review tiers).
  - Single Company-Wide baseline policy for Leave, Notice Period (e.g. 15 days), and Attendance.
  - Absence of rigid shift boundaries (Grace period set to 120 minutes).

### 2.2 Stage 2: 50-Employee Growing Business (Departmental Routing)

- **Operational Need**: Hierarchical reporting, team attendance visibility, departmental budgets.
- **Architectural Adaptation**:
  - Activation of `ReportingManager` approval workflows for Leave and WFH.
  - Introduction of Department-specific Leave policies (e.g. Sales team travel vs. Engineering).
  - Monthly attendance summaries reviewed by managers prior to payroll lock.

### 2.3 Stage 3: 500-Employee Mid-Market Enterprise (Strict Governance)

- **Operational Need**: Multiple office locations, shift rosters, automated late-arrival penalties, locked payroll cycles, multi-department separation clearances.
- **Architectural Adaptation**:
  - Multi-location office schedules ( `CompanyOfficeLocations` , `CompanyOfficeSchedule` ).
  - Active enforcement of `AttendancePenaltyPolicy` and automated monthly locks ( `AttendanceMonthLock` ).
  - Formal multi-stage clearance templates: IT, Facilities, Finance, and HR sign-offs before FnF settlement.
  - Transactional outbox workers scaling on dedicated background compute nodes.

### 2.4 Stage 4: 5,000+ Employee Multi-Entity Conglomerate

- **Operational Need**: Multi-subsidiary corporate hierarchies, shared service centers, high-throughput biometric punch streams, multi-currency payroll.
- **Architectural Adaptation**:
  - Sharded MongoDB cluster partitioned by `CompanyId` .
  - MongoDB Change Streams replacing polling outbox workers.
  - Dedicated Redis distributed cache cluster for session tokens and permission sets.
  - IoT edge connectors syncing biometric hardware punches directly to `AttendanceDaySegment` .

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## 3. Four-Phase Technical Delivery Roadmap

### Phase 1: Performance, Concurrency & Micro-Modularization (Near-Term)

- **Decomposition of Monolithic Classes**:
  - Split `EmployeeService.cs` (364 KB) into `EmployeeProfileService` , `EmployeeLifecycleService` , and `EmployeeSeparationService` .
  - Split `UserController.cs` (51.5 KB) into focused domain controllers.
- **Native MongoDB Multi-Document Transactions**:
  - Enforce `IClientSessionHandle` transactions across critical multi-collection operations in `AutoPayrollServices` and `EmployeeExit` .
- **Database Cursor Streaming**:
  - Replace in-memory list buffering in AutoPayroll with `IAsyncCursor` chunked processing.

### Phase 2: Enterprise Integration & Hardware Connectivity

- **Biometric IoT Hardware Gateway**:
  - Standardized REST/MQTT webhook receiver allowing on-premise biometric fingerprint/facial recognition punch machines to push attendance events into `AttendanceModel`.
- **Enterprise Messaging Connectors**:
  - Slack & Microsoft Teams bot integrations for one-click leave approvals, daily punch reminders, and celebratory notices.
- **Financial & ERP Export Hub**:
  - Direct journal entry export formatted for SAP, Oracle NetSuite, and QuickBooks accounting ledgers.

### Phase 3: AI-Assisted Workforce Intelligence

- **Intelligent ATS Resume Parsing**:
  - Deep-learning resume extraction pipeline automatically populating candidate work history, education, and skill tags.
- **Predictive Attrition Risk Scoring**:
  - Machine learning models analyzing attendance anomalies, frequent late arrivals, and leave patterns to alert HR to potential flight risks.
- **Automated Shift Optimization**:
  - Algorithmic shift scheduling matching employee availability with historical workload demands.

### Phase 4: Global Multi-Jurisdiction Expansion

- **Multi-Country Statutory Engines**:
  - Plug-in tax calculation modules supporting US (W-2 / 1099, FICA), UK (PAYE, National Insurance), and UAE (Gratuity, WPS) statutory payrolls.
- **Multi-Currency Compensation**:
  - Multi-currency salary definitions with automated exchange rate conversion and international wire export formats (SWIFT / ISO 20022).

# Codeji CMS / HRMS — Presentation Summary & Slide Deck

> **Document Status**: Complete & Authoritative
> **Target Audience**: Executive Leadership, Client Stakeholders, Technical Buyers, Investors
> **Slide Count**: 25 Comprehensive Presentation Slides
> **Format**: Slide Deck with Presenter Notes, Visual Diagrams & Verified Code Evidence
> **Last Verified**: September 2026

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## Slide 1: Codeji HRMS Platform Overview

- **Title**: Unified Workforce Operating System
- **Key Points**:
  - All-in-one Human Resource Information System (HRIS) and Applicant Tracking System (ATS).
  - Bridges the entire employee journey: recruitment, daily attendance, leave, automated payroll, separation, and alumni relations.
  - Built on modern, high-performance technology: ASP.NET Core on .NET 10, React 18, and MongoDB.
- **Diagram Suggestion**: Flowchart showing the continuous lifecycle circle connecting Talent Acquisition -> Active Workforce -> Separation -> Alumni.
- **Presenter Notes**: "Welcome everyone. Today we present Codeji HRMS, an enterprise-grade platform designed from the ground up to unify fragmented HR operations into a single, mathematically rigorous operating system."
- **System Evidence**: `Codeji.CMS.API/Program.cs`, `CMS-React/src/App.tsx`.

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## Slide 2: The Enterprise Problem Being Solved

- **Title**: Eliminating Friction in Workforce Operations
- **Key Points**:
  - Siloed HR systems cause data drift between approved leaves and monthly salary deductions.
  - Security risks emerge when former employees retain active credentials or permissions leak across company borders.
  - Operational overhead from manual calculations of Loss-of-Pay (LOP), late penalties, and notice buyouts.
  - Disputed full-and-final settlements due to undocumented clearance sign-offs.
- **Diagram Suggestion**: Contrast diagram showing "Fragmented HR Apps" (disconnected silos with red warning icons) vs. "Unified Codeji Architecture" (centralized source-of-truth).
- **Presenter Notes**: "Every mid-market company struggles with data drift. A manager approves a leave via email, but payroll runs on a spreadsheet that never captured it, leading to salary errors and employee dissatisfaction."
- **System Evidence**: `AutoPayrollServices.cs:84` (mandatory check for locked attendance summaries).

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## Slide 3: End-to-End System Architecture

- **Title**: Decoupled, High-Performance Architecture
- **Key Points**:
  - **Frontend**: Single Page Application built with React 18, TypeScript, Vite, Material-UI, Redux, and TanStack Query.
  - **Backend**: Clean Architecture ASP.NET Core Web API targeting .NET 10.

    - **Database**: Document-oriented MongoDB with compound tenant indexing and custom timestamped migrations.
    - **Realtime & Messaging**: SignalR WebSockets paired with transactional outbox workers for guaranteed email delivery.
- **Diagram Suggestion**: 4-tier layer diagram: Browser Clients -> Ingress & API Gateway -> Business Services & Policy Engine -> MongoDB Cluster & Storage.
- **Presenter Notes**: "Our architecture separates client presentation from server governance. All business invariants are enforced on the backend, ensuring that no client-side modification can compromise system integrity."
- **System Evidence**: `Codeji.CMS.API/Program.cs:59-450`, `CMS-React/src/app/routing/AppRoutes.tsx`.

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## Slide 4: The Complete Employee Lifecycle

- **Title**: Seamless Progression Across 7 Employment Dimensions
- **Key Points**:
    - Tracks 7 orthogonal lifecycle states: Employment, Availability, Account, Payroll, Offboarding, Former Portal, and Alumni.
    - Automated state transitions: Onboarding completion instantly activates employment, availability, and payroll eligibility.
    - Separation completion revokes credentials, cuts off attendance, and triggers final settlement.
- **Diagram Suggestion**: Multi-state workflow diagram illustrating status transitions from `PreJoining` to `Active`, `NoticePeriod`, `Resigned`, and `Alumni`.
- **Presenter Notes**: "Employee status in Codeji is multi-dimensional. An employee can be 'Active' in employment while simultaneously 'On Leave' in availability, 'Active' in payroll, and 'None' in offboarding."
- **System Evidence**: `EmployeeLifecycle.cs:5-108`, `RecruitmentOnboardingService.cs:140-181`.

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## Slide 5: Major Functional Modules

- **Title**: Comprehensive Suite of Integrated HR Modules
- **Key Points**:
    - **Talent Acquisition**: Public career marketplace, job vacancies, ATS candidate evaluation, interview scorecards.
    - **Workforce Management**: Multi-shift scheduling, punch classifications (P/HD/LHD/ED), leave accruals, WFH tracking.
    - **Compensation**: Standardized salary structures, CTC breakdowns, AutoPayroll generation, PDF payslips.
    - **Separation**: Multi-department clearances, notice countdown reminders, full-and-final settlement, alumni portal.
    - **Governance**: Enterprise policy engine, company master administration, dynamic RBAC.
- **Diagram Suggestion**: Modular grid display highlighting the 15 core functional modules with feature icons.
- **Presenter Notes**: "Codeji delivers an exhaustive HR functional suite. Every module is purpose-built to eliminate manual spreadsheets and human error."
- **System Evidence**: `ConstraintHelper.cs:4-21` (`AppModule` constants).

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## Slide 6: Recruitment & Talent Acquisition (ATS)

- **Title**: Modern, Transparent Hiring Pipeline
- **Key Points**:
    - Branded Public Career Marketplace (`/careers/:companyCode`) with vacancy search and direct resume upload.
    - Structured applicant tracking across 7 recruitment stages (`New`, `InProgress`, `OnHold`, `Shortlisted`, `Selected`, `Rejected`, `ReApply`).
    - Interviewer scorecards, structured debriefs, and multi-tier offer approval policies.
    - Public candidate offer review portal (`/candidate/offer`) supporting digital acceptance and counter-negotiations.

- **Diagram Suggestion**: Linear recruitment funnel: Job Post -> Application -> Screening -> Interview -> Scorecard -> Offer -> Acceptance.
- **Presenter Notes**: "Our recruitment module empowers hiring teams to attract, evaluate, and issue offers seamlessly. The candidate experiences a polished, branded portal from initial application to offer sign-off."
- **System Evidence**: `ApplicantsController.cs`, `RecruitmentLifecycleDrawer.tsx`, `CandidateOfferResponse.tsx`.

---

## Slide 7: Preboarding & Automated Onboarding

- **Title**: Zero-Friction Transition from Candidate to Colleague
- **Key Points**:
  - Preboarding case tracking for pre-hire document submission (educational proofs, identity cards).
  - Customizable onboarding checklists (IT equipment allocation, security badge issuance, contract sign-offs).
  - **Automated Lifecycle Activation**: Closing all checklist items triggers automated employee activation without manual data re-entry.
- **Diagram Suggestion**: Handoff flowchart showing Accepted Offer -> Preboarding Documents -> Onboarding Tasks -> Employee Activation.
- **Presenter Notes**: "The dangerous gap between offer acceptance and the first day of work is bridged automatically. Once onboarding tasks are confirmed, the candidate is automatically promoted into an active employee."
- **System Evidence**: `RecruitmentOnboardingService.cs:74-125`, `RecruitmentPreboardingCase.cs`.

---

## Slide 8: Employee Directory & Profiles

- **Title**: Unified, Auditable Employee Records
- **Key Points**:
  - Comprehensive employee profiles covering personal info, work history, skills, certifications, and bank/PAN records.
  - Dynamic user custom attributes ( `UserCustomAttribute` ) allowing organizations to capture bespoke metadata.
  - Strict input validation: Regex validation for tax identifiers (PAN) and bank account numbers.
  - Bulk employee Excel import and export capabilities.
- **Diagram Suggestion**: Profile layout card showing tab navigation: About, Experience, Education, Certifications, Bank & Tax, Settings.
- **Presenter Notes**: "Employee profiles serve as the digital dossier for every team member, capturing verified personal data, banking details for payroll, and skills for internal talent mobility."
- **System Evidence**: `userModel.cs:6-56`, `UserProfile.tsx`, `AboutUser.tsx`.

---

## Slide 9: Attendance & Shift Management

- **Title**: Accurate Time Tracking with Automated Shift Governance
- **Key Points**:
  - Shift schedules define start/end times, tolerance windows, and grace periods.
  - Pure calculation engine classifies daily punches into `P` (Present), `HD` (Half Day), `LHD` (Late Arrival), `ED` (Early Departure), or `LHD+ED`.
  - Attendance Penalty Engine: Monitors monthly late limits and raises exception reviews.
  - Attendance Month Lock: Freezes monthly punch data prior to payroll execution.
- **Diagram Suggestion**: Punch evaluation timeline illustrating Shift Start, Grace Window, Late Half-Day Boundary, and Status Assignment.

- **Presenter Notes**: "Our attendance system eliminates subjective attendance disputes. Daily punch times are evaluated against the employee's assigned shift schedule with mathematical certainty."
- **System Evidence**: `AttendanceStatusCalculationService.cs:15-40`, `AttendancePenaltyService.cs`.

---

## Slide 10: Leave Management & Work From Home (WFH)

- **Title**: Flexible Time-Off Administration with Real-Time Balance Tracking
- **Key Points**:
  - Supports diverse leave types: Sick, Casual, Earned, Maternity, Paternity, and Sabbatical.
  - Automated monthly accrual processing via `LeaveAccrualHostedService`.
  - Integrated Work From Home (WFH) requests supporting full-day and half-day segments.
  - Asynchronous reconciliation worker retroactively marks approved leave dates on the attendance calendar.
- **Diagram Suggestion**: Balance ledger diagram showing Monthly Accrual -> Employee Leave Request -> Manager Approval -> Attendance Reconciled to 'L'.
- **Presenter Notes**: "Leave management operates with full transparency. Employees inspect real-time balances, managers approve requests in a single click, and background workers automatically reconcile attendance."
- **System Evidence**: `LeaveManagementService.cs`, `WorkFromHomeService.cs`, `LeaveAccrualHostedService.cs`.

---

## Slide 11: Compensation & AutoPayroll

- **Title**: Automated, Audit-Proof Payroll Generation
- **Key Points**:
  - Standardized salary structures defining CTC, basic pay, allowances (HRA, LTA, other), and statutory deductions (EPF, ESIC, Tax).
  - AutoPayroll engine strictly enforces pre-conditions: Monthly attendance must be approved and locked, with zero unreviewed exceptions.
  - Automatic Loss-of-Pay (LOP) deduction: Unpaid leave days plus approved policy penalty days.
  - Divisor policy options: Fixed 30 Days, Actual Working Days, or Calendar Days.
  - In-memory PDF payslip generation with numbers rendered into English words.
- **Diagram Suggestion**: Calculation formula flowchart: Gross Pay - Statutory Deductions - LOP Deductions = Net Pay -> PDF Payslip.
- **Presenter Notes**: "AutoPayroll bridges attendance and compensation. The system mathematically computes exact Loss-of-Pay deductions based on locked attendance and renders digital payslips in seconds."
- **System Evidence**: `AutoPayrollServices.cs:55-120`, `PayrollCalculationRules.cs:1-15`, `PdfService.cs`.

---

## Slide 12: Company Master & Administration

- **Title**: Centralized Tenant Configuration Hub
- **Key Points**:
  - Master data administration: Departments, Job Roles/Designations, Office Locations, Working Days.
  - Employee ID Sequence generator supporting customized enterprise prefix formats.
  - Company branding customization: Logo upload, official address, and SMTP email settings.
- **Diagram Suggestion**: Organization hierarchy chart showing Company Root branching into Departments, Designations, Locations, and Working Calendars.
- **Presenter Notes**: "Company Master serves as the administrative cockpit, allowing HR operations to configure organizational hierarchies and business operating parameters without engineering assistance."

- **System Evidence**: `CompanyMasterController.cs`, `CompanyService.cs`, `CompanyMaster.tsx`.

---

## Slide 13: Role & Permission Security (RBAC)

- **Title**: Dynamic, Multi-Tenant Role-Based Access Control
- **Key Points**:
  - Fine-grained permission model based on Cartesian intersections of Modules and Actions.
  - Standard system roles: Company Administrator, HR Manager, HR Executive, and Employee.
  - Declarative endpoint protection using `[ModulePermission(AppModule.X, Permission.Y)]`.
  - Zero-trust server evaluation: Client-side UI hiding is backed by authoritative middleware verification.
- **Diagram Suggestion**: Security gate diagram tracing a request through Token Parsing -> Claim Verification -> Role Lookup -> Access Granted/Denied.
- **Presenter Notes**: "Security is non-negotiable. Our dynamic RBAC system verifies every API call against company-scoped role permissions, ensuring complete protection across all endpoints."
- **System Evidence**: `ModulePermissionAttribute.cs`, `AuthenticateUserRequest.cs:83-111`, `RoleServices.cs:587-634`.

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## Slide 14: Enterprise Policy Resolution Engine

- **Title**: Deterministic Multi-Tier Policy Engine
- **Key Points**:
  - Resolves business rules using strict precedence: Department -> Role -> Company Baseline.
  - Pure, deterministic resolution logic guarantees query order cannot alter outcomes.
  - **Immutable Policy Snapshotting**: When an employee initiates a separation, the active policy is frozen into the exit document, preventing retroactive rule alterations.
- **Diagram Suggestion**: Hierarchical funnel diagram showing Department match checked first, Role match second, Company default as fallback.
- **Presenter Notes**: "The policy engine provides organizational flexibility without complexity. Rules can be customized per department or role, while immutable snapshots ensure point-in-time legal compliance."
- **System Evidence**: `OffboardingPolicyResolver.cs:25-82`, `EmployeeExitPolicy.cs`.

---

## Slide 15: Notice Period Governance

- **Title**: Structured Resignation & Notice Management
- **Key Points**:
  - Calculates required notice duration from snapshotted exit policies.
  - Supports notice adjustment tracking: Early releases, buyouts, and managerial waivers.
  - Automated notice countdown reminders dispatched via transactional outbox workers.
  - Sets authoritative Last Working Date (LWD) governing system access cutoff.
- **Diagram Suggestion**: Timeline diagram showing Resignation Date, Notice Period Countdown, Early Buyout Adjustment, and Last Working Date.
- **Presenter Notes**: "Notice periods are managed with complete transparency. Countdown reminders keep managers and employees aligned, while buyout and waiver tracking feeds directly into final settlement."
- **System Evidence**: `EmployeeExitDtos.cs`, `EmployeeExitOutboxProcessor.cs`.

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## Slide 16: Separation & Multi-Department Clearances

- **Title**: Flawless Employee Offboarding & Asset Return
- **Key Points**:
  - Multi-department clearance templates covering IT, Facilities, Finance, and HR.
  - Asset return tracking: Evaluates condition, serial numbers, and damage deductions.
  - IT access revocation: Coordinates single sign-on, email, and hardware access closure.
  - Exit interviews capturing categorized feedback on compensation, culture, and management.
- **Diagram Suggestion**: Multi-track clearance checklist showing IT Sign-off, Asset Sign-off, Finance Sign-off, and HR Final Sign-off.
- **Presenter Notes**: "Offboarding ensures organizational security and asset recovery. Every department must complete its clearance checklist before an exit can be finalized."
- **System Evidence**: `EmployeeExitCaseCard.tsx`, `EmployeeExitOperationsPanel.tsx`, `EmployeeExitClearanceTemplate.cs`.

---

## Slide 17: Full & Final Settlement (FnF)

- **Title**: Accurate, Audit-Ready Financial Closure
- **Key Points**:
  - Automatically reconciles unpaid salary days from processed payroll.
  - Computes additions (reimbursements, encashable leave balances, bonuses).
  - Computes deductions (notice period buyout recovery, unreturned asset damages, loan dues).
  - Generates clear financial direction: `PayableToEmployee`, `ReceivableFromEmployee`, or `ZeroSettlement`.
- **Diagram Suggestion**: Financial reconciliation balance sheet showing Gross Earnings + Leave Encashment - Notice Recovery - Dues = Net Payable.
- **Presenter Notes**: "Full and final settlements are a major source of labor disputes. Codeji eliminates disputes by auto-calculating settlements against real payroll and leave data."
- **System Evidence**: `EmployeeFinalSettlement.cs:5-36`, `UserController.cs:GetEmployeeFinalSettlement`.

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## Slide 18: Former Employee & Alumni Portal

- **Title**: Post-Separation Engagement & Document Access
- **Key Points**:
  - Dedicated, unauthenticated-app portal (`/former-employee`) accessed via secure one-time tokens or OTP.
  - Separated employees download historical payslips, tax forms (Form 16), and experience certificates.
  - Rejoin eligibility tracking: Preserves historical records while facilitating rapid re-hiring.
- **Diagram Suggestion**: Portal experience mockup showing separated employee logging in via secure token to download their Experience Letter and Payslips.
- **Presenter Notes**: "Our relationship with employees doesn't end on their last day. The Former Employee Portal provides alumni with self-service access to their documents without burdening the HR team."
- **System Evidence**: `FormerEmployeePortalExperience.tsx`, `FormerEmployeePortalService.cs`, `FormerEmployeeAccessToken.cs`.

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## 19. Cross-Module Automation

- **Title**: The Power of an Integrated System

- **Key Points**:
  - Zero manual re-keying: Candidate data flows directly into employee profile.
  - Daily punches instantly classify attendance and feed monthly lock readiness.
  - Locked attendance automatically calculates salary deductions in AutoPayroll.
  - Separation completion instantly revokes system access and triggers final settlement.
- **Diagram Suggestion**: Interconnected gear diagram showing ATS, Attendance, Leave, Payroll, and Offboarding moving in synchrony.
- **Presenter Notes**: "The real power of Codeji is automation across module boundaries. When one event occurs, all downstream dependencies update automatically without human intervention."
- **System Evidence**: Master workflows traced in `09-CROSS-MODULE-WORKFLOWS.md`.

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## Slide 20: Multi-Tenant Boundary Security

- **Title**: Zero-Trust Tenant Data Isolation
- **Key Points**:
  - Absolute server-side tenant scoping: Tenancy derived strictly from signed JWT claims (`company_id`).
  - Client-supplied tenant headers or query overrides are completely ignored.
  - Every database query enforces `x.CompanyId == authenticatedCompanyId`.
  - Database-level compound unique indexes prevent cross-tenant collisions.
- **Diagram Suggestion**: Fortress diagram showing Company A and Company B completely isolated by cryptographic walls, sharing common compute securely.
- **Presenter Notes**: "Multi-tenancy in Codeji is built on zero-trust principles. A user from Company A can never, under any circumstances, view or modify records belonging to Company B."
- **System Evidence**: `CurrentContext.cs:36-49`, `CompanyIdMiddleware.cs:48-65`.

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## Slide 21: Realtime Communication & Outbox Resilience

- **Title**: Guaranteed Delivery with Asynchronous Outbox Patterns
- **Key Points**:
  - In-app push notifications delivered via SignalR WebSockets (`/notificationhub`).
  - Transactional Outbox Pattern: Candidate emails, exit notices, and punch alerts are saved atomically to MongoDB before external transmission.
  - 4 dedicated background workers poll outbox queues with exponential backoff retries.
  - Delivery failures cannot cause database rollbacks or user UI lag.
- **Diagram Suggestion**: Outbox flowchart: Service -> Database Outbox Table -> Background Worker -> SMTP Gateway -> Recipient.
- **Presenter Notes**: "We never call an email server directly inside a user request. Outbox workers ensure that external network hiccups never affect the responsiveness or reliability of our application."
- **System Evidence**: `RecruitmentEmailOutboxWorker.cs`, `EmployeeExitOutboxWorker.cs`, `AttendanceOutboxWorker.cs`.

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## Slide 22: Reporting, Analytics & Auditability

- **Title**: Actionable Workforce Insights & Statutory Non-Repudiation
- **Key Points**:
  - Operational analytics: Recruitment funnel conversion rates, interview feedback SLAs, leave consumption trends.

    - Pervasive audit trails: Creation, mutation, and soft-delete actor timestamps on every record.
    - Dedicated audit collections: `AttendanceAuditLog`, `ApplicantLogs`, and `EmployeeExit.AuditHistory`.
    - Comprehensive Excel and CSV export capabilities for executive reviews.
- **Diagram Suggestion**: Analytics dashboard mockup showing Recruitment Funnel, Attendance Heatmap, and Audit Log Export table.
- **Presenter Notes**: "Data-driven HR teams need real-time metrics. Codeji provides deep operational visibility alongside tamper-evident audit trails for labor law compliance."
- **System Evidence**: `RecruitmentAuditExportService.cs`, `AttendanceAuditWriter.cs`, `ApplicantLogs.cs`.

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## Slide 23: Production Readiness & Quality Assurance

- **Title**: Rigorous Technical Assessment & Hardening Roadmap
- **Key Points**:
    - Core business workflows are verified, operational, and backed by automated tests.
    - 65 backend unit/contract test suites and 16 frontend component test suites.
    - Identified remediation priorities: MongoDB multi-document transactions (P0), OpenTelemetry distributed tracing (P1), and service modularization (P1).
    - Clear, risk-prioritized delivery path ensuring enterprise general availability.
- **Diagram Suggestion**: Readiness scorecard showing green checkmarks across Core Features, RBAC, and Tenant Security, with targeted P0/P1 items in progress.
- **Presenter Notes**: "We maintain an honest, evidence-based engineering culture. The system is functional and secure today, with a prioritized hardening roadmap for high-scale enterprise deployment."
- **System Evidence**: Detailed findings in `15-PRODUCTION-READINESS.md`.

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## Slide 24: Technical & Product Roadmap

- **Title**: Strategic Evolution & Enterprise Scale
- **Key Points**:
    - **Phase 1**: Native MongoDB replica set transactions and micro-modular service decomposition.
    - **Phase 2**: Biometric IoT hardware punch gateway and Slack/Microsoft Teams bots.
    - **Phase 3**: AI-assisted resume parsing, attrition risk modeling, and shift optimization.
    - **Phase 4**: Global multi-jurisdiction statutory tax engines (US, UK, Middle East) and multi-currency payroll.
- **Diagram Suggestion**: Horizontal 4-phase milestone timeline highlighting key architectural deliverables.
- **Presenter Notes**: "Our roadmap extends Codeji from an internal operating system to a global enterprise SaaS platform, integrating IoT hardware, artificial intelligence, and international compliance."
- **System Evidence**: Full roadmap detailed in `16-FUTURE-ROADMAP.md`.

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## Slide 25: The Codeji HRMS Value Proposition

- **Title**: The Unified Platform for Modern Enterprises
- **Key Points**:
    - **Single Source of Truth**: Completely eliminates spreadsheets, duplicate data entry, and payroll errors.
    - **Enterprise Security**: Cryptographically enforced tenant isolation and fine-grained RBAC.
    - **Total Compliance**: Tamper-proof audit logs, policy snapshots, and automated statutory calculations.
    - **Rapid ROI**: Drastically reduces HR administrative overhead while elevating employee experience.

- **Diagram Suggestion**: Value summary graphic highlighting: Security, Automation, Flexibility, Scalability, and Compliance.
- **Presenter Notes**: "In summary, Codeji HRMS delivers the agility of modern web software backed by enterprise-grade governance. Thank you, and we welcome your questions."
- **System Evidence**: Verified implementation across `CMS-Backend-Core` and `CMS-React`.

# Codeji CMS / HRMS — Comprehensive Product & Solution Proposal

> **Document Status**: Complete & Authoritative
> **Target Audience**: Chief Technology Officers (CTO), Chief Human Resource Officers (CHRO), Chief Information Security Officers (CISO), Enterprise Procurement
> **Document Purpose**: Enterprise Commercial, Architectural & Functional Solution Proposal
> **Last Verified**: September 2026

## 1. Executive Summary

This proposal presents the **Codeji Content & Human Resource Management System (Codeji CMS / HRMS)** as an enterprise-grade, multi-tenant workforce operating platform. Engineered using **ASP.NET Core on .NET 10**, **React 18**, and **MongoDB**, Codeji HRMS replaces fragmented point solutions by providing a single, unified source of truth across the complete talent lifecycle.

From external job posting and candidate evaluation to shift-based attendance, multi-tiered leave management, mathematically auditable payroll synthesis, and full-and-final separation settlements, Codeji HRMS delivers uncompromising security, automated governance, and operational agility.

## 2. The Business Problem & Economic Justification

Mid-market and enterprise organizations frequently operate a patchwork of disconnected software tools: an external Applicant Tracking System (ATS), a standalone biometric punch database, manual spreadsheet-based leave tracking, a third-party payroll bureau, and paper-based separation checklists. This operational fragmentation produces severe organizational liabilities:

1. **High Administrative Overhead**: HR teams spend hundreds of manual hours every month reconciling attendance records, medical leave slips, and Loss-of-Pay (LOP) calculations before payroll can run.
2. **Financial Loss & Payroll Inaccuracies**: Manual spreadsheet calculations lead to overpayments, unrecovered notice-period buyouts, and incorrect statutory deductions.
3. **Legal & Compliance Exposure**: Lack of tamper-proof audit trails creates vulnerabilities during labor department audits, employee termination disputes, or statutory tax reconciliations.
4. **Security & Data Leakage Risks**: Delayed offboarding communications leave corporate email, single sign-on (SSO), and internal systems accessible to separated personnel for days or weeks after exit.

### Return on Investment (ROI) Profile

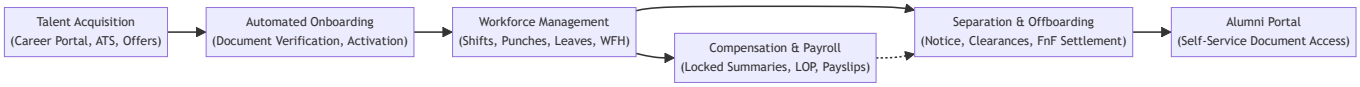
By deploying Codeji HRMS as an integrated platform, enterprise organizations realize:

- **85% Reduction in Payroll Processing Time**: Automated synthesis of locked monthly attendance summaries directly into AutoPayroll eliminates manual LOP calculations.
- **100% Elimination of Data Drift**: Shared entity structures guarantee that an approved leave immediately updates both the workforce calendar and the compensation ledger.
- **Zero-Delay Account Closure**: Separation completion triggers instant session revocation and attendance cutoffs on the employee's Last Working Date.

## 3. The Proposed HRMS Solution

Codeji HRMS is architected around the concept of **Automated, Policy-Governed Lifecycle Transitions**:

**ARCHITECTURE / WORKFLOW DIAGRAM**




## 4. Comprehensive Platform Capabilities

### 4.1 Talent Acquisition & Recruitment Pipeline

- Public Career Marketplace supporting custom company branding, vacancy search, and resume upload.
- Full-lifecycle Applicant Tracking System (ATS) managing candidate progressions through 7 stages: `New`, `InProgress`, `OnHold`, `Shortlisted`, `Selected`, `Rejected`, and `ReApply`.
- Structured interview scheduling, interviewer scorecards, and debrief notes.
- Configurable multi-tier offer approval policies, offer letter generation, and a dedicated candidate offer review portal (`/candidate/offer`).

### 4.2 Automated Preboarding & Onboarding Activation

- Pre-hire document submission (education degrees, identity cards, prior experience letters).
- Customizable task checklists for IT provisioning, security access, and policy orientations.
- **Automated Lifecycle Activation**: When all onboarding tasks are signed off, the system automatically promotes the candidate into an active employee record, enabling workforce tracking and payroll eligibility without data re-entry.

### 4.3 Workforce Time & Attendance Governance

- Shift-based office schedule management defining start times, end times, check-in tolerance windows, and grace periods.
- Pure calculation engine classifying punches into Present (`P`), Half Day (`HD`), Late Arrival (`LHD`), Early Departure (`ED`), or combined (`LHD+ED`).
- Attendance Penalty Engine: Enforces monthly late arrival limits and flags exceptions for HR review.
- Attendance Month Lock: Mathematically validates and freezes monthly attendance data prior to payroll processing.

### 4.4 Leave & Work From Home (WFH) Administration

- Multi-tier leave categories (Sick, Casual, Earned, Maternity, Paternity, Sabbatical).
- Automated monthly accrual background processing (`LeaveAccrualHostedService`).
- WFH request management supporting full-day and half-day segment bookings.
- Asynchronous reconciliation ensuring approved leaves update the attendance calendar to `L`.

### 4.5 Compensation & AutoPayroll

- Comprehensive salary structures defining CTC, basic pay, allowances (HRA, LTA, other), and statutory deductions (EPF, ESIC, Tax).
- Gated AutoPayroll execution requiring approved and locked attendance with zero unreviewed exceptions.
- Tenant-configurable divisor policies (Fixed 30 Days, Working Days, Calendar Days).
- Instant, in-memory PDF payslip generation formatted in Indian currency and English words.

### 4.6 Separation, Clearances & Final Settlement (FnF)

- Employee self-resignation or HR-initiated termination pathways.

- Notice period calculation from snapshotted policies, notice adjustments (waivers, buyouts), and automated countdown alerts.
- Multi-department clearance workflows covering IT single sign-on revocation, physical asset recovery, and financial reconciliation.
- Full & Final Settlement ( `EmployeeFinalSettlement` ) computing salary dues, encashments, and notice recovery.
- Dedicated alumni portal ( `/former-employee` ) enabling departed employees to download historical payslips and experience certificates via secure tokens.

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## 5. Enterprise Security, Privacy & Multi-Tenancy

### 5.1 Cryptographic Multi-Tenant Isolation

Codeji HRMS guarantees complete tenant boundary isolation:

- **Zero-Trust Token Derivation**: Tenant context is derived strictly from signed JWT claims ( `company_id` ). Client-supplied headers or query parameters are ignored.
- **Mandatory Query Scoping**: Every database read and write enforces `x.CompanyId == authenticatedCompanyId` .
- **Cross-Tenant IDOR Protection**: Requesting an entity belonging to another company returns HTTP 404, guaranteeing foreign tenant data remains invisible.

### 5.2 Dynamic Role-Based Access Control (RBAC)

- Fine-grained permission model based on Cartesian intersections of Modules and Actions.
- Declarative endpoint protection via `[ModulePermission(AppModule.X, Permission.Y)]` .
- Dual-layer security: Client UI hides unauthorized controls; server-side middleware unconditionally enforces authorization.

### 5.3 Point-in-Time Policy Snapshotting

When lifecycle events occur (such as resignation submission), active policy rules are deep-copied into the case document ( `EmployeeExitPolicySnapshot` ). Subsequent alterations to company policies cannot retroactively mutate inflight contractual or financial agreements.

### 5.4 Transactional Outbox Pattern

External communications (candidate emails, exit alerts, punch notices) are written atomically to MongoDB outbox collections and processed by dedicated background workers, guaranteeing delivery resilience and eliminating API latency.

---

## 6. Sizing Flexibility: From Startups to Conglomerates

| Organization Size | Operating Profile | Architectural Adaptation in Codeji HRMS |
|---|---|---|
| **10-Person Startup** | Agile, lean, zero administrative overhead. | Single company baseline policy; auto-approval of leaves and WFH; calendar-day payroll divisor. |
| **50-Person Company** | Structured departmental management. | Reporting Manager approval routing; department-specific leave entitlements; monthly attendance exception reviews. |
| **500-Person Enterprise** | Multi-location facilities, formal compliance. | Multiple office schedules; automated attendance penalty tracking and month locks; multi-tier clearance templates. |
| **5,000+ Conglomerate** | Multi-subsidiary, high-volume punch streams. | MongoDB cluster sharding by `CompanyId` ; Change Streams for outbox processing; biometric IoT hardware integrations. |

## 7. Current Implementation Status vs. Future Roadmap

### Current Implementation Strengths (Verified from Repository)

- Complete, operational end-to-end recruitment, attendance, leave, payroll, and offboarding workflows.
- Strict JWT multi-tenant claim validation and dynamic RBAC middleware.
- Pervasive audit tracking with soft deletion across all business entities.
- 9 active background workers managing accruals, reconciliations, and outbox queues.
- 65 backend unit/contract test files and 16 frontend component test files.

### Recommended Hardening & Production Roadmap

- **Phase 1 (Near-Term)**: Native MongoDB multi-document transactions across replica sets; decomposition of monolithic services ( `EmployeeService.cs` ).
- **Phase 2 (Integrations)**: Biometric IoT hardware punch receiver; Slack/Teams notification bots; ERP financial ledger export.
- **Phase 3 (Intelligence)**: AI-assisted resume extraction; predictive employee attrition risk modeling.
- **Phase 4 (Global Scale)**: Multi-jurisdiction statutory tax engines (US, UK, UAE) and multi-currency compensation.

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## 8. Deployment & Adoption Strategy

Codeji HRMS supports flexible deployment topologies:

1. **On-Premise / Private Cloud**: Deployed within client-owned AWS, Azure, GCP, or on-premise Kubernetes clusters for complete data sovereignty.
2. **Dedicated Single-Tenant Managed Cloud**: Hosted in isolated cloud infrastructure managed by Codeji engineering.
3. **Multi-Tenant Enterprise SaaS**: High-availability shared cloud infrastructure with guaranteed cryptographic tenant data isolation.

### Implementation Timetable

- **Phase A: Master Data Setup & Policy Authoring** (Weeks 1–2): Departments, job roles, office schedules, leave policies, and salary structures configured.
- **Phase B: Historical Data Migration** (Weeks 3–4): Bulk import of employee records, initial leave balances, and open job vacancies.
- **Phase C: Pilot Department Testing** (Weeks 5–6): Attendance punch verification, leave workflow testing, and parallel payroll dry-run.
- **Phase D: Organization-Wide Go-Live** (Week 7+): Full employee self-service launch, career portal public activation, and legacy system decommissioning.

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## 9. Conclusion & Next Steps

Codeji HRMS represents an architecturally mature, mathematically sound, and highly secure platform engineered for modern enterprise workforce governance. By unifying talent acquisition, operational workforce tracking, compensation, and separation into a single policy-driven platform, Codeji eliminates data fragmentation, mitigates legal compliance risk, and unlocks dramatic operational efficiencies.

We invite your technical and operational leadership to review the deep-dive architectural specifications ( `00` through `16` ), inspect the verified module documentation ( `modules/` ), and schedule an interactive technical review session with our engineering team.

PART III

# COMPLETE HRMS MODULE DEEP-DIVES (30 MODULES)

19-Section Code-Verified Deep-Dive for Every Functional Module
from Recruitment to Final Settlement

30 Document(s) in this Section

MOD-01 PART III docs/project-documentation/modules/ADMIN.md

# Module Documentation — Company Administration & Master Data (`ADMIN`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controllers**: `CompanyMasterController.cs`, `CompanyController.cs`
**Primary Services**: `CompanyMasterService.cs`, `CompanyService.cs`, `DefaultCompanySeeds.cs`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The foundational administrative configuration suite governing the structural taxonomy of a tenant organization. It manages Departments, Job Roles/Designations, Office Locations, Working Days, Custom Employee Attributes, and Employee ID Sequence formatting.
- **Why HR organizations need it**: Every downstream HR process (hiring a candidate into a department, assigning a shift by location, generating sequential employee codes, or routing leave requests to a department manager) depends on standardized master entities.
- **What business problem it solves**: Eliminates free-text departmental typos, inconsistent designation titles, and manual employee ID numbering collisions across concurrent HR administrators.
- **Who uses it**: Exclusively **Company Administrators** (`Roles.Administrator`).
- **Where it fits in the HR lifecycle**: Stage 0 (Foundational Tenant Provisioning)—executed immediately upon company registration before employees or job vacancies are created.

## 2. Real-World Business Scenario

```
Company: Codeji Technologies Pvt Ltd
Administrator: Vikram Mehta (Company Administrator)
Scenario: Opening a new R&D Center in Bengaluru
```

1. Vikram navigates to `/admin/companymaster`.
2. **Office Location**: Creates a new location `Bengaluru R&D Hub` with full street address and timezone `Asia/Kolkata`.
3. **Department**: Creates a new department `Artificial Intelligence & ML`.
4. **Job Roles**: Adds three designations linked to the department: `Principal AI Architect`, `ML Engineer II`, and `Data Annotator`.
5. **Working Days**: Configures Monday through Friday as working days (`IsWorking: true`) and Saturday/Sunday as weekly offs.
6. **Employee ID Sequence**: Configures prefix `CJ-BLR-` with zero-padding `4` (producing `CJ-BLR-0001`, `CJ-BLR-0002` automatically upon onboarding).

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **Company Administrator** | Full CRUD over Departments, Job Roles, Office Locations, Working Days, Custom Attributes, and ID Sequences. | Enforced via `ProtectedRoute adminOnly={true}` and backend `[Authorize(Policy = "AdminOnly")]` / `RoleServices`. |
| **HR Manager / Executive** | Read-only consumption of master lists (e.g. selecting departments in dropdowns during employee creation). | Cannot mutate master entities; write endpoints reject non-admin roles with HTTP 403. |
| **Employee** | Views own assigned department, designation, and location on profile. | No access to `/admin/*` routes. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/admin/adminRoutes.tsx` | Defines sub-routes `/admin/roles`, `/admin/companymaster/*`, `/admin/companydetails/*`. | React Router `<Outlet />` | `PrivateRoutes.tsx` |
| `src/app/modules/admin/companymaster/CompanyMaster.tsx` | Tabbed container hosting Department, Job Role, Location, Working Days, Custom Attributes, and ID Sequence sub-panels. | `companyMasterServices.ts` | `adminRoutes.tsx` |
| `src/app/modules/admin/companydetails/CompanyDetails.tsx` | Manages legal entity name, corporate logo upload, PAN/TAN/GST details, and SMTP email settings. | `companyServices.ts` | `adminRoutes.tsx` |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Codeji.CMS.API/Controllers/CompanyMasterController.cs` | REST endpoints for managing Departments, Job Roles, Locations, Working Days, and Custom Attributes. | `ICompanyMasterService`, `IHttpContextAccessor` |
| `Codeji.CMS.API/Controllers/CompanyController.cs` | Manages company registration, profile updates, logo uploads, and employee ID sequences. | `ICompanyService`, `IMongoDbRepository<Company>` |
| `Codeji.CMS.Services/Companies/CompanyMasterService.cs` | Business validation, duplicate name checks within tenant, and soft-deletion guards (preventing deletion of departments with active employees). | `IMongoDbRepository<Department>`, `IMongoDbRepository<CompanyJobRole>` |
| `Codeji.CMS.Services/Companies/DefaultCompanySeeds.cs` | Seeds initial departments, roles, working days, and default templates when a new company registers. | `IMongoDatabase` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `POST` | `/api/CompanyMaster/AddEditDepartment` | Creates or updates a tenant department. | `[Authorize]` + Admin | `DepartmentDto` | Checks unique department title per `CompanyId` (case-insensitive). | `Result<string>` |
| `GET` | `/api/CompanyMaster/GetAllDepartments` | Lists active departments for dropdowns and tables. | `[Authorize]` | Query Params | Filters by `x.CompanyId == companyId && !x.IsDeleted`. | `Result<DepartmentDto>` |
| `DELETE` | `/api/CompanyMaster/DeleteDepartment/{id}` | Soft-deletes a department. | `[Authorize]` + Admin | Route `id` | Verifies zero active `EmpUser` records reference this department before setting `IsDeleted = true`. | `Result` |
| `POST` | `/api/CompanyMaster/AddEditJobRole` | Creates or edits a job designation mapped to a department. | `[Authorize]` + Admin | `JobRoleDto` | Validates parent `DepartmentId` exists in tenant. | `Result<string>` |
| `POST` | `/api/CompanyMaster/SaveOfficeLocation` | Saves physical office location details. | `[Authorize]` + Admin | `OfficeLocationDto` | Validates address fields and unique location name. | `Result` |
| `POST` | `/api/Company/SaveEmployeeIdSequence` | Configures prefix, suffix, and starting counter for Employee IDs. | `[Authorize]` + Admin | `EmployeeIdSequenceDto` | Ensures sequence format does not collide with existing `EmpUser.EmployeeId` values. | `Result` |

## 7. Database Model

### Entities

1. **`Department`** ( `Codeji.CMS.Repository.Entities.Company.Department` )
   - **Primary Key**: `DepartmentId`
   - **Foreign Keys**: `CompanyId`
   - **Important Fields**: `DepartmentTitle` (Multilingual dictionary), `Description`, `IsActive`, `IsDeleted`.
2. **`CompanyJobRole`** ( `Codeji.CMS.Repository.Entities.Company.CompanyJobRole` )
   - **Primary Key**: `JobRoleId`
   - **Foreign Keys**: `CompanyId`, `DepartmentId`
   - **Important Fields**: `JobRoleTitle` (Multilingual dictionary), `Level`, `IsDeleted`.
3. **`CompanyOfficeLocations`** ( `Codeji.CMS.Repository.Entities.Company.CompanyOfficeLocations` )
   - **Primary Key**: `LocationId`
   - **Foreign Keys**: `CompanyId`

- **Important Fields**: `LocationName`, `AddressLine1`, `City`, `State`, `Country`, `ZipCode`, `IsHeadOffice`.

4. **`EmployeeIdSequence`** (`Codeji.CMS.Repository.Entities.Employees.EmployeeIdSequence`)
   - **Primary Key**: `SequenceId`
   - **Foreign Keys**: `CompanyId`
   - **Important Fields**: `Prefix`, `CurrentNumber`, `PaddingLength`, `Separator`.

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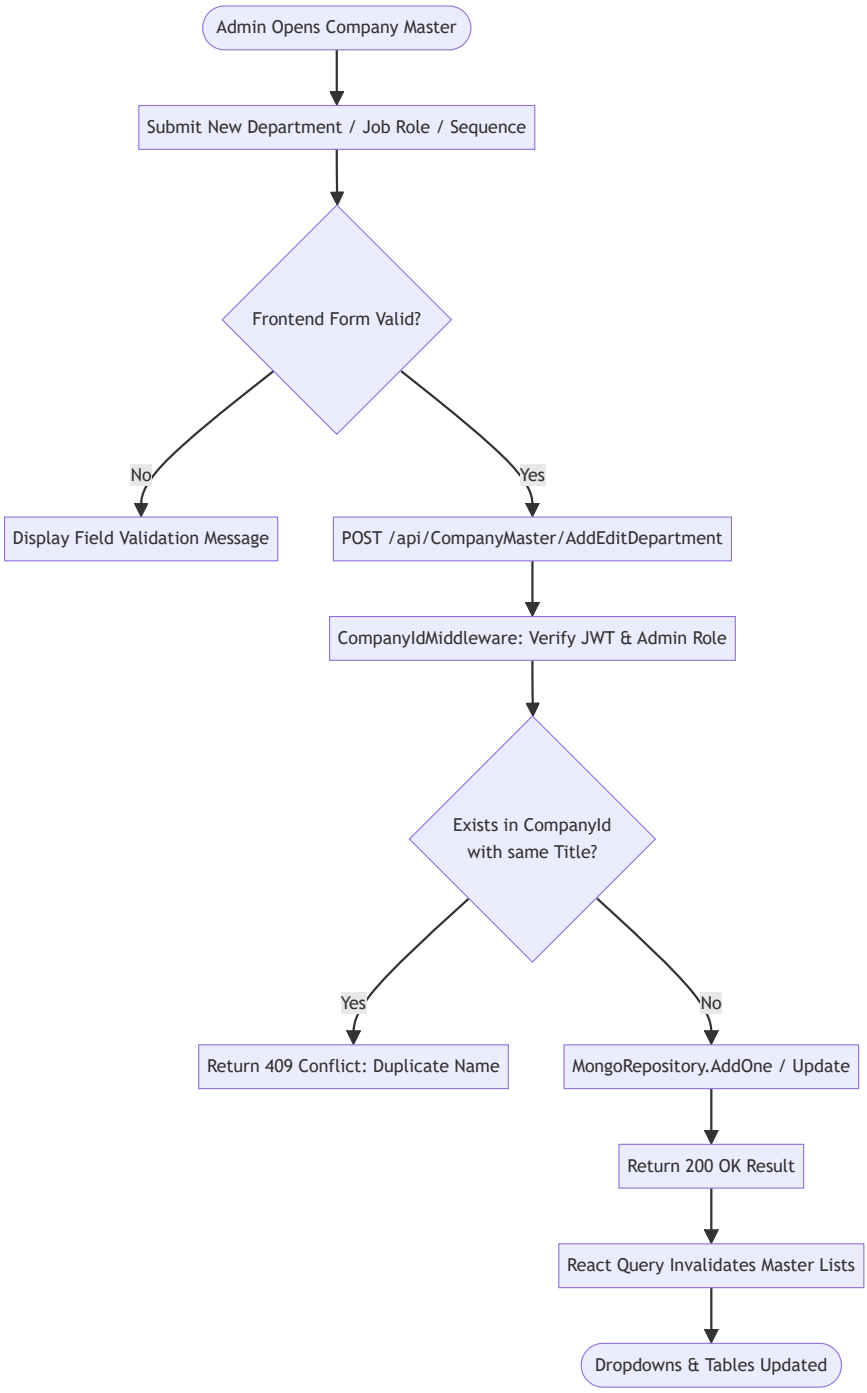
## 8. Business Rules

- **Business Rule**: A department or job role cannot be deleted if one or more active employees (`EmpUser.IsDeleted == false && EmpUser.Status == true`) are currently assigned to it.
- **Validation Rule**: Department and Job Role names must be unique within the same `CompanyId`.
- **Permission Rule**: Only users with `RoleType == Administrator` can create, update, or delete Company Master entities.
- **Security Rule**: `CompanyId` is never accepted from the client request payload; it is overwritten using `CurrentContext.CompanyId`.

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## 9. Complete Workflow

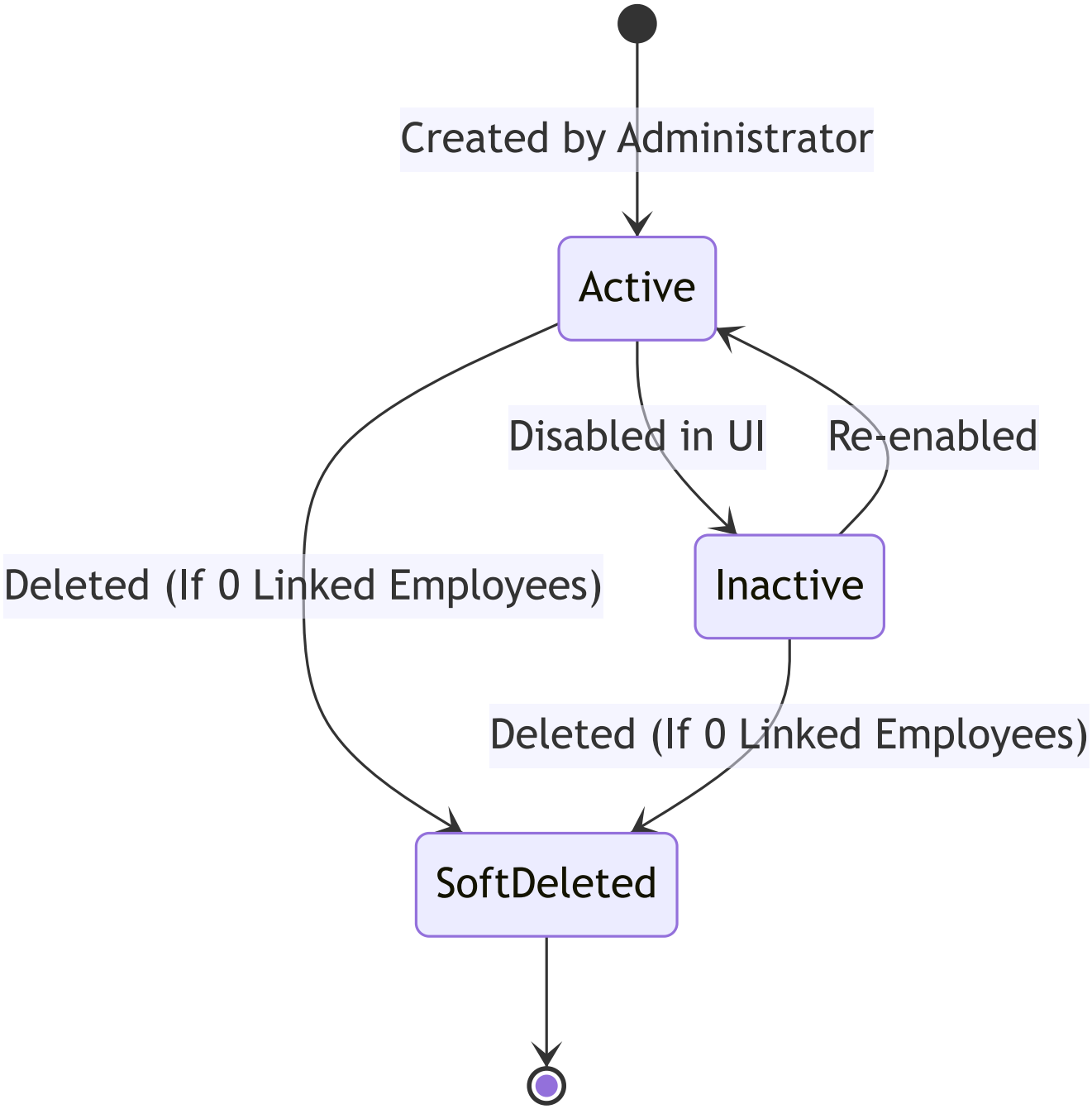
ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Master entities follow a binary lifecycle state machine:

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| View Master Lists (Dropdowns) | Yes | Yes | Yes | Yes |
| Access `/admin/companymaster` UI | Yes | No | No | No |
| Create / Edit Department & Job Role | Yes | No | No | No |

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| Delete Master Record | Yes | No | No | No |
| Configure Employee ID Sequence | Yes | No | No | No |

## 12. Dependencies

- **Depends On**: `Company` registration and `Roles` initialization.
- **Used By**: `Employees`, `Recruitment`, `LeavePolicy`, `EmployeeExitPolicy`, `Attendance`.
- **Reads From**: `Company`, `EmpUser` (for referential integrity checks on delete).
- **Writes To**: `Department`, `CompanyJobRole`, `CompanyOfficeLocations`, `CompanyWorkingDays`, `CompanyCustomAttributes`, `EmployeeIdSequence`.

## 13. Cross-Module Events

- Creating a new company triggers `DefaultCompanySeeds.SeedAsync()`, which populates default departments (`Engineering`, `Human Resources`, `Sales`, `Finance`), default roles, and default working days (`Monday` – `Friday`).
- Updating `CompanyWorkingDays` immediately affects `CompanyWorkingCalendarService`, altering working day counts in downstream `Leave` and `AutoPayroll` divisor calculations.

## 14. Failure Scenarios

- **Referential Integrity Block**: Attempting to delete a Department that still has active employees returns a validation failure explaining that employees must be reassigned first.
- **Concurrent Sequence Generation**: Handled via atomic MongoDB `FindOneAndUpdateAsync` increment on `EmployeeIdSequence.CurrentNumber` to guarantee no two employees receive the same `EmployeeId`.

## 15. Edge Cases

- **Multilingual Title Resolution**: `DepartmentTitle` and `JobRoleTitle` are stored as `Dictionary<string, string>` keyed by language code (`en`, `fr`, `de`, `es`, `zh`, `ja`). If a user switches UI language to French (`fr`) and a translation key is missing, the UI gracefully falls back to English (`en`).

## 16. Current Strengths

- Referential integrity checks prevent orphaned employee department references.
- Multilingual dictionary support built natively into master entities.
- Atomic sequence generation prevents duplicate employee IDs during bulk imports.

## 17. Current Gaps

- **Hierarchy Depth**: Departments are flat (single-level); nested sub-departments (e.g., `Engineering -> Backend -> Payments`) are not supported in the current schema.
- **Location-Specific Working Days**: `CompanyWorkingDays` applies company-wide rather than per `CompanyOfficeLocations`.

## 18. Recommended Improvements

- **Medium**: Add `ParentDepartmentId` to `Department` to support multi-level organizational org charts.
- **Low**: Link `CompanyWorkingDays` optionally to `LocationId` for multi-region enterprises.

## 19. Future Scope

- Visual drag-and-drop Organizational Chart builder synchronized with Department and Reporting Manager hierarchies.

# Module Documentation — Company Settings & Configuration (`COMPANY-SETTINGS`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controllers**: `CompanyController.cs`, `AttendanceStatusSettingsController.cs`
**Primary Services**: `CompanyService.cs`, `AttendanceStatusService.cs`, `DefaultCompanySeeds.cs`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: Manages tenant-level corporate profile data, branding assets (logos), legal and taxation identifiers, corporate email credentials (SMTP), notification defaults, and attendance status display configuration.
- **Why HR organizations need it**: Enterprise communications (offer letters, payslips, email notifications) must display the tenant company's official legal name, registered logo, tax registration numbers, and return email addresses.
- **What business problem it solves**: Eliminates hardcoded company branding and allows multi-tenant organizations to operate independently with distinct corporate visual identities and email delivery servers.
- **Who uses it**: Exclusively **Company Administrators**.
- **Where it fits in the HR lifecycle**: Foundational configuration governing all system-generated documents and emails.

## 2. Real-World Business Scenario

```
Company: Apex Global Logistics Ltd
Administrator: Sarah Jenkins (Administrator)
Scenario: Setting up company branding and email notification infrastructure
```

1. Sarah logs into Codeji HRMS and navigates to `/admin/companydetails`.
2. **Company Profile**: Enters Legal Name `Apex Global Logistics Ltd`, corporate website, and head office address in Mumbai.
3. **Corporate Logo**: Uploads high-resolution SVG logo (streamed to `/Uploads` storage, referenced across all PDF payslips).
4. **Email Server Settings**: Configures SMTP server `smtp.office365.com`, port `587`, TLS enabled, and authenticated sender `hr@apexlogistics.com`.
5. **Attendance Status Presentation**: Navigates to `/attendance-settings` and customizes the UI display labels and badges for attendance statuses (e.g. labeling `LHD` as "Late Punch" with amber badge `#FFA000`).

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **Company Administrator** | Full authority to update corporate profile, logo, SMTP credentials, and attendance status settings. | Backend enforces `RoleType == Administrator` or policy `AdminOnly`. |

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **HR / Employee** | Read-only access to company branding (logo, company name) on UI top bar and payslips. | Write endpoints reject non-admin actors with HTTP 403. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/admin/companydetails/CompanyDetails.tsx` | UI tab managing company legal name, logo upload preview, address fields, and email settings form. | `companyServices.ts` | `adminRoutes.tsx` |
| `src/app/modules/attendance/AttendanceStatusSettings.tsx` | Admin panel customizing attendance status codes, labels, color hex codes, and leave eligibility rules. | `attendanceServices.ts` | `PrivateRoutes.tsx` ( `/attendance-settings` ) |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Codeji.CMS.API/Controllers/CompanyController.cs` | Endpoints for company profile updates, logo uploading, email setting configuration, and company status checks. | `ICompanyService` , `IHttpContextAccessor` |
| `Codeji.CMS.API/Controllers/AttendanceStatusSettingsController.cs` | Manages tenant-level attendance status display settings ( `AttendanceStatusSetting` ). | `IAttendanceStatusService` |
| `Codeji.CMS.Services/Companies/CompanyService.cs` | Validates company uniqueness, handles image upload file streaming to disk, updates MongoDB documents. | `IMongoDbRepository<Company>` , `IMongoDbRepository<CompanyEmailSettings>` |
| `Codeji.CMS.Services/Attendance/AttendanceStatusService.cs` | Manages attendance status codes, active flags, and color mapping. | `IMongoDbRepository<AttendanceStatusSetting>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `POST` | `/api/Company/UpdateCompanyDetails` | Updates company profile, address, and legal tax numbers. | `[Authorize]` + Admin | `CompanyDetailsDto` | Validates company exists and is active; updates MongoDB `Company` collection. | `Result<string>` |

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `POST` | `/api/Company/UploadCompanyLogo` | Uploads and persists company branding logo. | `[Authorize]` + Admin | `IFormFile` (Multipart) | Validates image MIME type ( `image/png` , `image/jpeg` , `image/svg+xml` ); saves to `/Uploads/Logos/` . | `Result<string>` (Logo URL) |
| `POST` | `/api/Company/SaveEmailSettings` | Configures tenant SMTP delivery credentials. | `[Authorize]` + Admin | `CompanyEmailSettingsDto` | Validates host/port format; encrypts password before saving to `CompanyEmailSettings` . | `Result` |
| `GET` | `/api/AttendanceStatusSettings/GetSettings` | Returns all active attendance status presentation configurations. | `[Authorize]` | Query Params | Scoped by `x.CompanyId == companyId` . | `Result<AttendanceStatusSettingDto>` |
| `POST` | `/api/AttendanceStatusSettings/UpdateSetting` | Updates presentation color or label for an attendance status code. | `[Authorize]` + Admin | `AttendanceStatusSettingDto` | Verifies core status code (e.g. `P` , `HD` , `LHD` ) cannot be deleted or renamed. | `Result` |

## 7. Database Model

### Entities

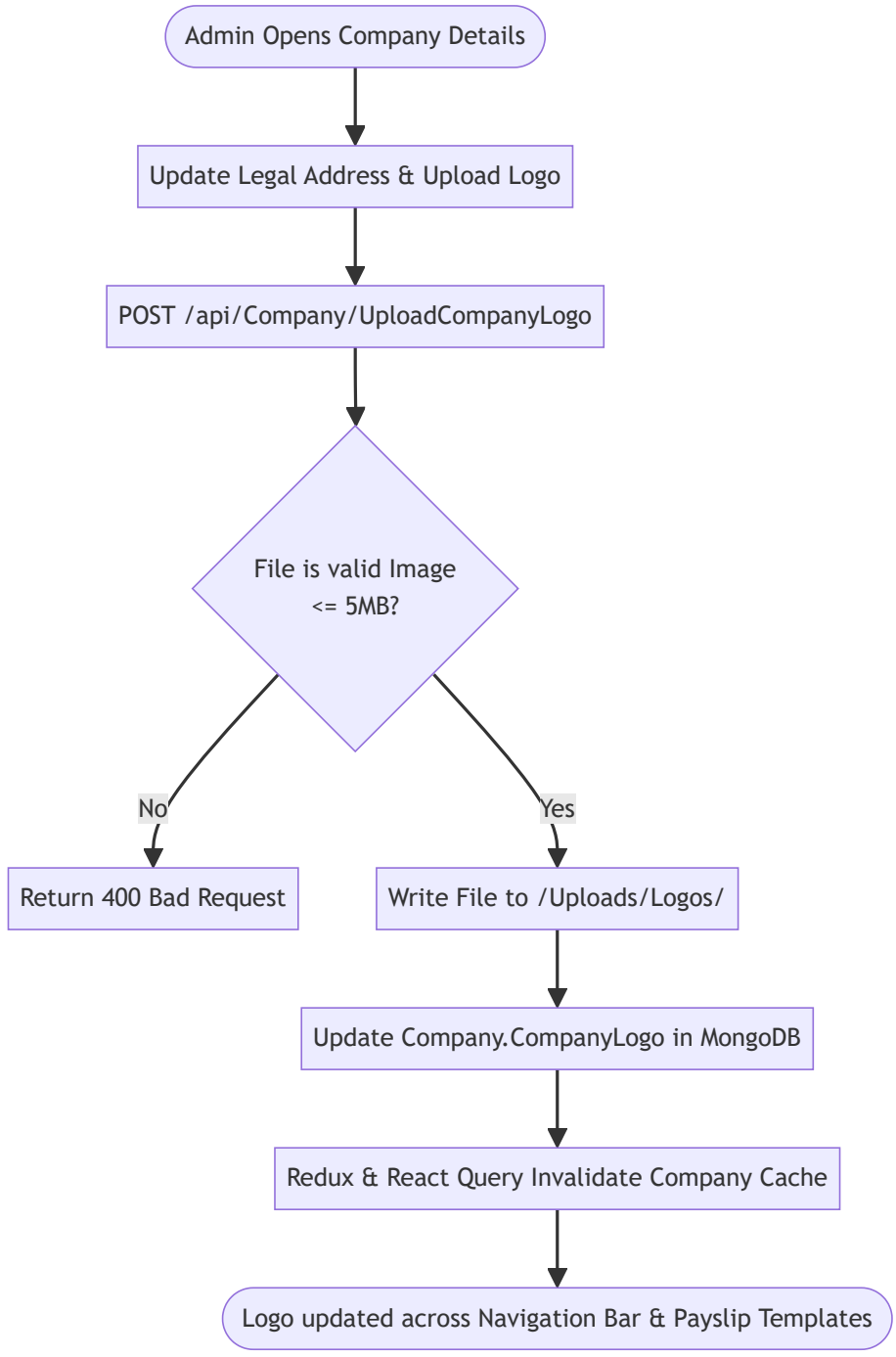
1. **`Company`** ( `Codeji.CMS.Repository.Entities.Company.Company` )
   - **Primary Key**: `CompanyId`
   - **Fields**: `CompanyName` , `CompanyCode` , `CompanyLogo` , `Website` , `Address` , `ContactEmail` , `TaxId` , `IsActive` .
2. **`CompanyEmailSettings`** ( `Codeji.CMS.Repository.Entities.Company.CompanyEmailSettings` )
   - **Primary Key**: `EmailSettingsId`
   - **Fields**: `CompanyId` , `SmtpHost` , `SmtpPort` , `SmtpUsername` , `EncryptedPassword` , `EnableSsl` , `SenderDisplayName` .
3. **`AttendanceStatusSetting`** ( `Codeji.CMS.Repository.Entities.Attendance.AttendanceStatusSetting` )
   - **Primary Key**: `SettingId`
   - **Fields**: `CompanyId` , `StatusCode` ( `P` , `HD` , `LHD` , `ED` , `LHD+ED` , `L` ), `DisplayName` , `ColorCodeHex` , `IsActive` .

## 8. Business Rules

- **Business Rule**: Core attendance status codes ( `P` , `HD` , `LHD` , `ED` , `L` ) are system constants. Organizations may customize their display labels and color codes, but cannot delete the core status codes.
- **Security Rule**: SMTP passwords are encrypted at rest using AES-256 before being persisted in `CompanyEmailSettings` .
- **Validation Rule**: Company logo files must not exceed 5 MB and must be verified image formats.

## 9. Complete Workflow

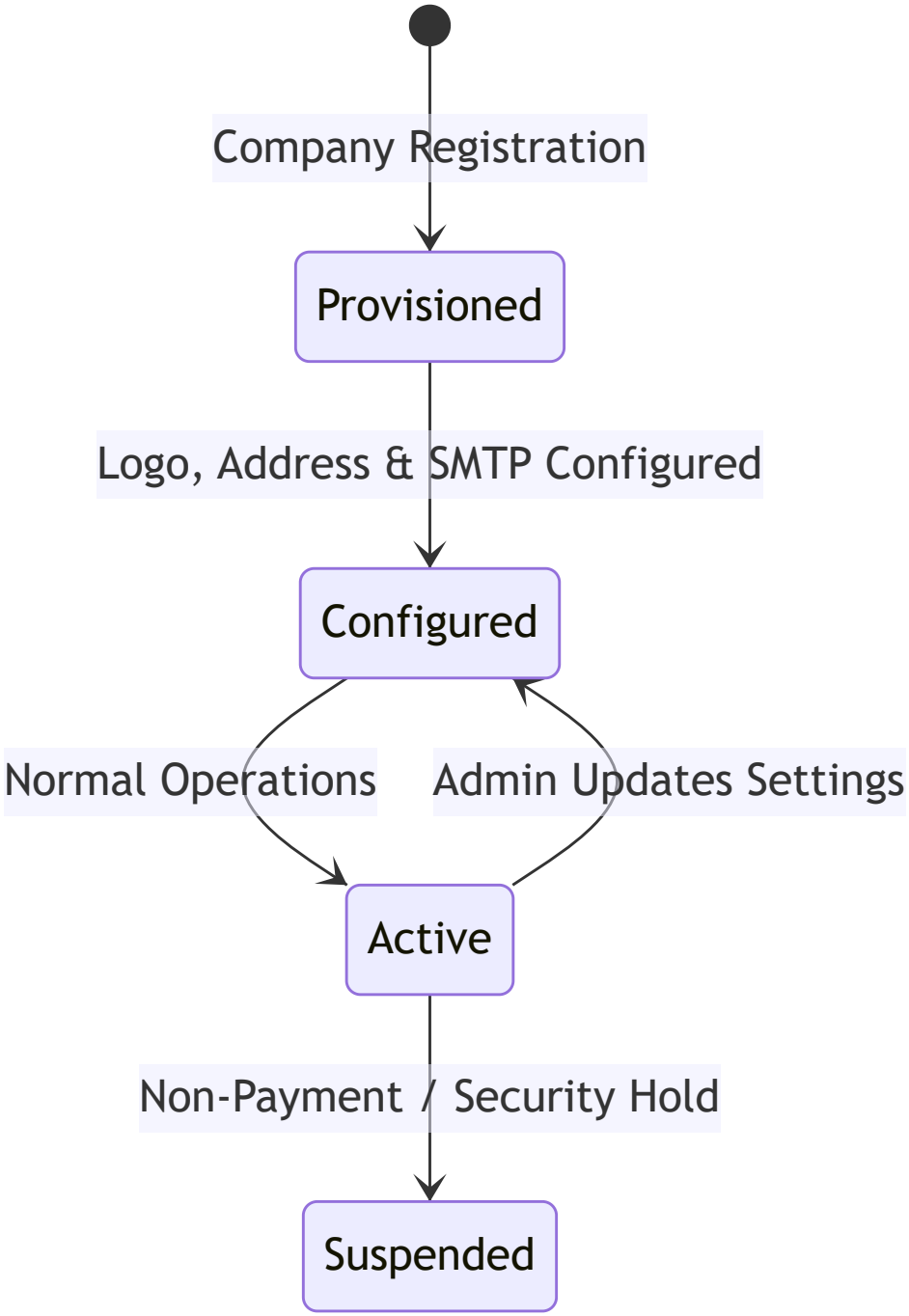
ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Company settings operate as a singleton document per tenant:

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| View Company Branding | Yes | Yes | Yes | Yes |
| Update Company Legal Profile | Yes | No | No | No |
| Upload Corporate Logo | Yes | No | No | No |
| Configure SMTP Server | Yes | No | No | No |

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| Customize Attendance Status Colors | Yes | No | No | No |

## 12. Dependencies

- **Depends On**: `Company` document created during tenant registration.
- **Used By**: `PayRoll` (for logo on payslips), `EmailService` (for SMTP credentials), `AttendanceCalendar` (for status badges).
- **Reads From**: `Company`, `CompanyEmailSettings`, `AttendanceStatusSetting`.
- **Writes To**: `Company`, `CompanyEmailSettings`, `AttendanceStatusSetting`.

## 13. Cross-Module Events

- Updating the company logo immediately updates the header image rendered in `SalarySlipTemplate.html` for all future payslip downloads.
- Changing SMTP configuration in `CompanyEmailSettings` immediately alters the outbound transport credentials used by `RecruitmentEmailOutboxWorker` and `EmployeeExitOutboxWorker`.

## 14. Failure Scenarios

- **SMTP Misconfiguration**: If invalid SMTP credentials are saved, outbound emails fail during worker execution. Outbox records capture the SMTP exception and enter exponential retry backoff without blocking user operations.

## 15. Edge Cases

- **Missing Logo Fallback**: If an organization has not uploaded a company logo, `PdfService` gracefully falls back to rendering the text `Company.CompanyName` in the payslip header without throwing a null reference exception.

## 16. Current Strengths

- Centralized management of company branding assets.
- Clean isolation of SMTP settings per tenant company.
- Configurable attendance status presentation colors and display strings.

## 17. Current Gaps

- **Test SMTP Connection Tool**: The admin UI lacks a "Send Test Email" button to verify SMTP credentials before saving.

## 18. Recommended Improvements

- **Low**: Add a "Test Connection" endpoint on `CompanyController` allowing administrators to verify SMTP host and credentials immediately.

## 19. Future Scope

- Multi-brand enterprise configuration supporting distinct subsidiary logos and legal entities under a parent holding tenant.

MOD-03 PART III docs/project-documentation/modules/ROLE-PERMISSION.md

# Module Documentation — Role & Permission Management (`ROLE-PERMISSION`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controller**: `RolesController.cs`
**Primary Service**: `RoleServices.cs` (35.4 KB, 706 lines)
**Primary Frontend Components**: `Roles.tsx`, `AddPermission.tsx`, `EditPermission.tsx`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The security authorization engine that manages corporate Roles and configures fine-grained access control matrices across all application modules.
- **Why HR organizations need it**: Enterprises employ diverse organizational roles (HR Directors, Recruiters, Payroll Accountants, Department Managers, Staff Employees). Each role must be restricted strictly to its operational remit to prevent unauthorized access to sensitive financial, personal, or separation records.
- **What business problem it solves**: Eliminates coarse, all-or-nothing administrative privileges by providing a granular, multi-tenant permission matrix.
- **Who uses it**: Exclusively **Company Administrators** (`Roles.Administrator`).
- **Where it fits in the HR lifecycle**: Core governance framework—evaluated continuously on every single authenticated request across all modules.

## 2. Real-World Business Scenario

```
Company: Nexa Enterprise Systems
Administrator: Devraj Patel (Administrator)
Scenario: Creating a specialized 'Recruiter' role restricted to Talent Acquisition
```

1. Devraj navigates to `/admin/roles` and clicks **Add Role**.
2. Enters Role Title `Talent Acquisition Specialist` and Description `Responsible for posting jobs and reviewing candidate applications`.
3. In the permission matrix grid:
   - **Jobs**: Grants `View`, `Create`, `Edit`.
   - **Applications**: Grants `View`, `Create`, `Edit`, `Scorecard`, `Review`.
   - **Employees**: Grants `View` (Read-only directory).
   - **PayRoll & Settings**: Leaves completely unchecked (`HasAccess = false`).
4. Clicks **Save**. The backend creates a new `Roles` document and persists explicit `RolePermission` records.
5. Devraj assigns this new role to the newly hired recruiter; when the recruiter logs in, the navigation sidebar automatically hides Payroll, Attendance Settings, and Separation Clearances.

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **Company Administrator** | Full authority to create custom roles, edit permission matrices, and delete/reassign roles. | Enforced via `ProtectedRoute adminOnly={true}` and backend checks. |
| **Non-Admin Actors (HR, Manager, Employee)** | Cannot access `/admin/roles`; permissions are resolved dynamically from their assigned role. | Access to `/admin/roles` returns HTTP 403 Forbidden. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/admin/permissions/Roles.tsx` | Displays table of all company roles with employee counts, edit actions, and delete triggers. | `roleServices.ts` | `adminRoutes.tsx` ( `/admin/roles` ) |
| `src/app/modules/admin/permissions/components/AddPermission.tsx` | Form for authoring a new role and configuring its Cartesian permission matrix across modules. | `roleServices.ts` | `adminRoutes.tsx` ( `/admin/roles/add` ) |
| `src/app/modules/admin/permissions/components/EditPermission.tsx` | Form for modifying role titles, descriptions, and toggling module-level checkboxes. | `roleServices.ts` | `adminRoutes.tsx` ( `/admin/roles/edit/:roleid` ) |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Codeji.CMS.API/Controllers/RolesController.cs` | Endpoints for listing roles, fetching permission trees, adding/editing roles, and reassigning roles on deletion. | `IRoleService`, `IHttpContextAccessor` |
| `Codeji.CMS.Services/Employees/RoleServices.cs` (35.4 KB) | Core RBAC service: builds permission matrices, executes `VerifyUserAccess`, seeds default roles, and manages role reassignment. | `IMongoDbRepository<Roles>`, `IMongoDbRepository<RolePermission>`, `IMongoDbRepository<ModulePermission>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `GET` | `/api/Roles/GetAllRoles` | Returns all active roles for the current company. | `[Authorize]` + Admin | Query Params | Filters by `x.CompanyId == companyId && !x.IsDeleted`. | `Result<RoleModel>` |

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `GET` | `/api/Roles/GetAllRolesWithPermission` | Returns full module-by-permission tree with active checkboxes. | `[Authorize]` + Admin | Query Params | Joins `Module`, `Permission`, `ModulePermission`, and `RolePermission`. | `Result<ModuleWithPermissionsModel>` |
| `POST` | `/api/Roles/AddEditRoles` | Creates or updates a role and its permission checklist. | `[Authorize]` + Admin | `RoleWithModuleAndPermissions` | Prevents editing roles marked `IsNotEditable == true` (Administrator). | `Result` |
| `POST` | `/api/Roles/DeleteAndReassignRole` | Deletes a role and safely reassigns linked employees to another role. | `[Authorize]` + Admin | `DeleteAndReassignRoleRequestModel` | Verifies target replacement role exists; updates all `EmpUser.RoleId` records before soft-deleting old role. | `Result` |

## 7. Database Model

### Entities

1. **`Roles`** (`Codeji.CMS.Repository.Entities.RolePermissions.Roles`)
   - **Primary Key**: `RolesId`
   - **Fields**: `CompanyId`, `RoleType`, `Titles`, `Description`, `HasAppAccess`, `IsNotEditable`, `IsDeleted`.
2. **`Module`** (`Codeji.CMS.Repository.Entities.RolePermissions.Module`)
   - **Primary Key**: `ModuleId`
   - **Fields**: `ModuleName`, `ModuleConstant`, `SortOrder`.
3. **`Permission`** (`Codeji.CMS.Repository.Entities.RolePermissions.Permission`)
   - **Primary Key**: `PermissionId`
   - **Fields**: `PermissionName`, `PermissionConstant`.
4. **`ModulePermission`** (`Codeji.CMS.Repository.Entities.RolePermissions.ModulePermission`)
   - **Primary Key**: `ModulePermissionId`
   - **Fields**: `ModuleId`, `PermissionId`.
5. **`RolePermission`** (`Codeji.CMS.Repository.Entities.RolePermissions.RolePermission`)
   - **Primary Key**: `RolePermissionId`
   - **Fields**: `CompanyId`, `RoleId`, `ModulePermissionId`, `HasAccess` (Boolean).

## 8. Business Rules

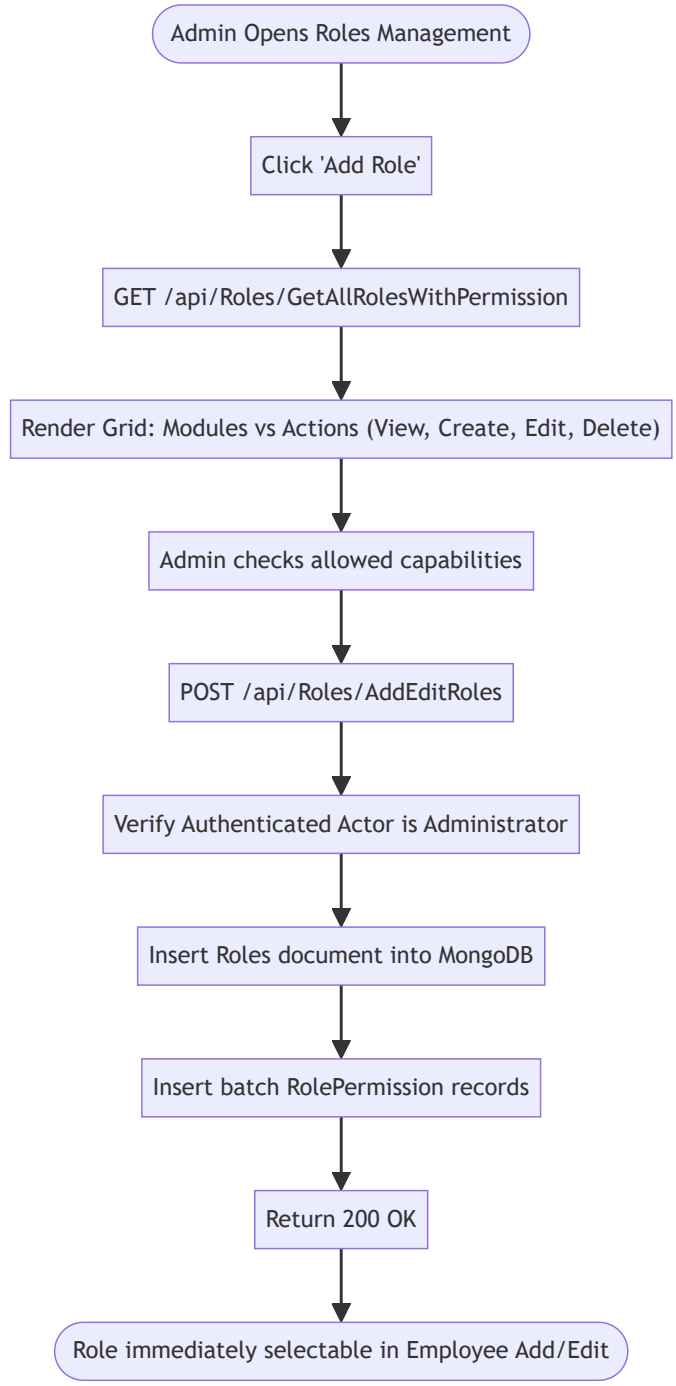
- **Administrator Immutability**: Roles with `IsNotEditable == true` (the standard Company Administrator role) cannot be edited or deleted.
- **Safe Role Deletion**: A role cannot be deleted without providing a valid `ReassignRoleId`. All employees currently holding the deleted role are atomically migrated to the replacement role.

- **Tenant Scope Rule**: `RolePermission` records are strictly scoped to `CompanyId`. Custom permissions configured in Company A have zero effect on Company B.

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## 9. Complete Workflow

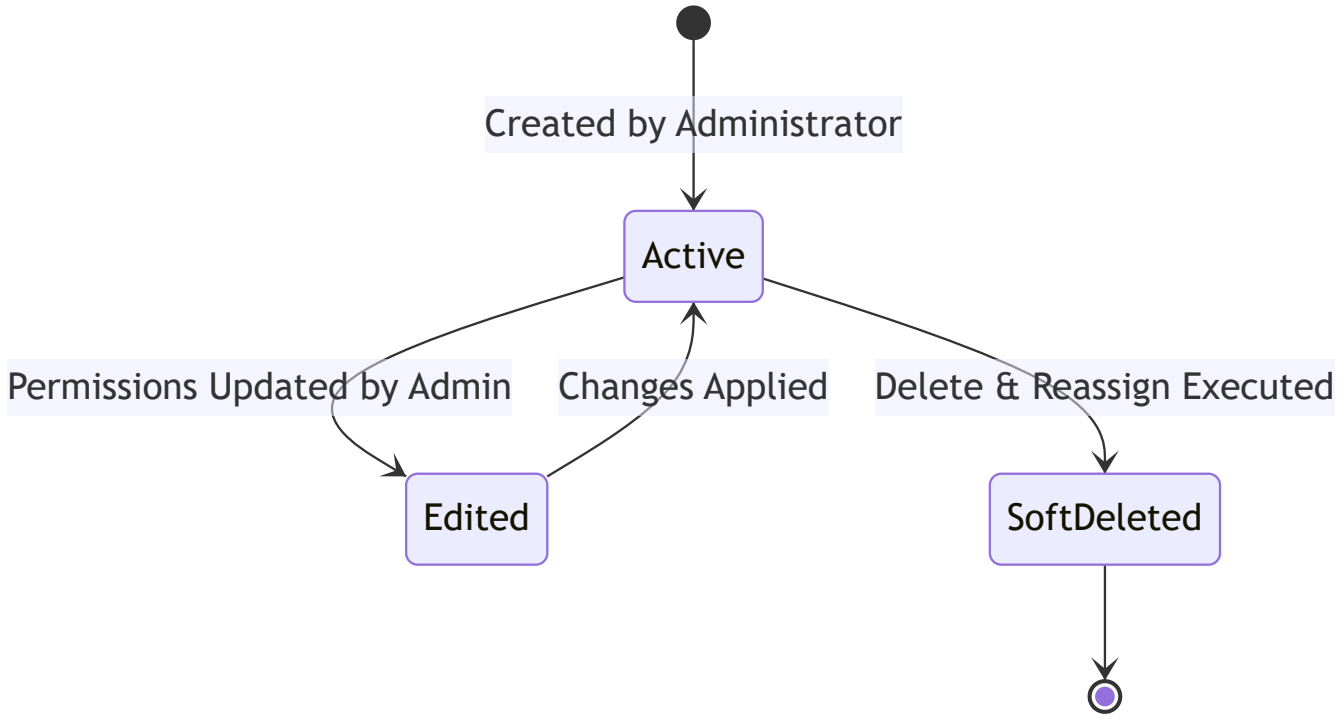
ARCHITECTURE / WORKFLOW DIAGRAM




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## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| Access `/admin/roles` | Yes | No | No | No |
| Create New Role | Yes | No | No | No |
| Modify Role Permissions | Yes | No | No | No |
| Delete & Reassign Role | Yes | No | No | No |

## 12. Dependencies

- **Depends On**: `Company`, `Module`, `Permission`, `ModulePermission`.
- **Used By**: All modules via `CompanyIdMiddleware` and `RoleServices.VerifyUserAccess`.
- **Reads From**: `Roles`, `RolePermission`, `ModulePermission`, `Module`, `Permission`, `EmpUser`.
- **Writes To**: `Roles`, `RolePermission`, `EmpUser` (during role reassignment).

## 13. Cross-Module Events

- Deleting a role and reassigning it to another role updates all affected `EmpUser` documents, altering their effective permissions immediately upon their next API request.

## 14. Failure Scenarios

- **Orphaned User Prevention**: If an administrator attempts to delete a role without supplying a valid `ReassignRoleId`, the API rejects the request with HTTP 400 Bad Request, preventing employees from having dangling role references.

## 15. Edge Cases

- **Self-Lockout Prevention**: An administrator cannot delete or reassign their own active Administrator role.

## 16. Current Strengths

- Dynamic, database-backed permission matrix eliminates code deployments when creating new roles.
- Mandatory role reassignment protects referential integrity during role deletion.

## 17. Current Gaps

- Permissions are module-level and action-level; field-level permissions (e.g. masking salary on profile while showing other fields) are not natively configurable in the UI grid.

## 18. Recommended Improvements

- **Medium**: Add field-level read/write masking toggles in the permission matrix for sensitive personal attributes.

## 19. Future Scope

- Attribute-Based Access Control (ABAC) support combining dynamic role checks with contextual attributes (e.g., location or IP subnet).

# POLICY & CONFIGURATION GOVERNANCE MODULE (`POLICY.md`)

## 1. Module Overview

The **Policy & Configuration Governance Module** serves as the central administrative brain of **Codeji CMS / HRMS**. Rather than hardcoding organizational rules, it provides configurable, scoped policy entities across the entire employee lifecycle: `EmployeeLifecyclePolicy`, `EmployeeExitPolicy`, `LeavePolicy`, `AttendancePenaltyPolicy`, and `PayrollDivisorPolicy`.

## 2. Real-World Business Scenario

1. **Departmental Policy Specialization**:
   - Company default `EmployeeExitPolicy` mandates `NoticeDays = 30`.
   - The Engineering department requires `NoticeDays = 60` and mandatory code handover (`RequireHandover = true`).
   - HR Admin defines an Engineering-scoped `EmployeeExitPolicy` (`DepartmentId = "dept_eng"`).
2. **Dynamic Resolution**:
   - When an Engineering employee resigns, `OffboardingPolicyResolver` resolves the Engineering policy. When a Sales employee resigns, it falls back to the Company default policy.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Super Admin / HR Admin** | Defines and updates policies at Company, Department, and Role scopes. |
| **Audit Officer** | Inspects policy versioning and audit trails. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/settings/pages/LifecyclePolicySettings.tsx` | Manages `EmployeeLifecyclePolicy` (probation days, onboarding requirements). | `GET /api/employee-lifecycle/policy`, `PUT /api/employee-lifecycle/policy` | HR Admin |
| `src/app/modules/settings/pages/ExitPolicySettingsPage.tsx` | Scoped matrix for `EmployeeExitPolicy` (notice days, buyouts, waivers, clearance gates). | `GET /api/employee-exit-policies`, `POST /api/employee-exit-policies` | HR Admin |
| `src/app/modules/settings/pages/LeavePolicySettings.tsx` | CRUD for `LeavePolicy` rules (accruals, carry-forwards, probation/notice restrictions). | `GET /api/leave/policies`, `POST /api/leave/policies` | HR Admin |
| `src/app/modules/settings/pages/PayrollSettings.tsx` | Divisor and calculation rules (`PayrollDivisorPolicy`). | `GET /api/payroll-divisor-policy`, `PUT /api/payroll-divisor-policy` | HR Admin |

## 5. Backend Implementation

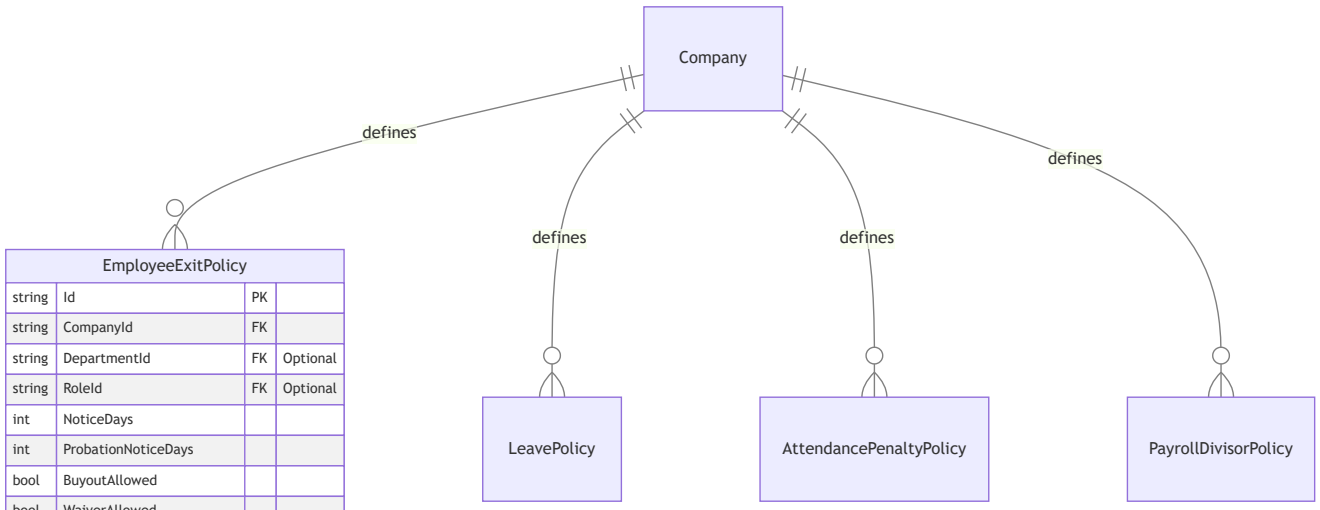
| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Services/OffboardingPolicyResolver.cs` | Implements 3-tier precedence resolution ( `Department -> Role -> Company` ) for exit policies. | `IMongoRepository<EmployeeExitPolicy>` |
| `Services/AttendancePenaltyService.cs` | Reads `AttendancePenaltyPolicy.CombinedLhdEdMonthlyLimit` to trigger payroll exceptions. | `IMongoRepository<AttendancePenaltyPolicy>` |
| `Services/PayrollDivisorPolicyService.cs` | Resolves active `PayrollDivisorPolicy` for monthly payroll and FnF calculations. | `IMongoRepository<PayrollDivisorPolicy>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/employee-exit-policies` | Retrieves all scoped exit policies for the tenant. | JWT ( `settings:read` ) | `List<EmployeeExitPolicyDto>` | None |
| `POST` | `/api/employee-exit-policies` | Creates a department- or role-specific exit policy. | JWT ( `settings:update` ) | `CreateExitPolicyDto` -> `EmployeeExitPolicyDto` | Persists policy |
| `PUT` | `/api/employee-exit-policies/{id}` | Updates policy parameters. | JWT ( `settings:update` ) | `UpdateExitPolicyDto` | Increments `Version` |

## 7. Database Model

ARCHITECTURE / WORKFLOW DIAGRAM



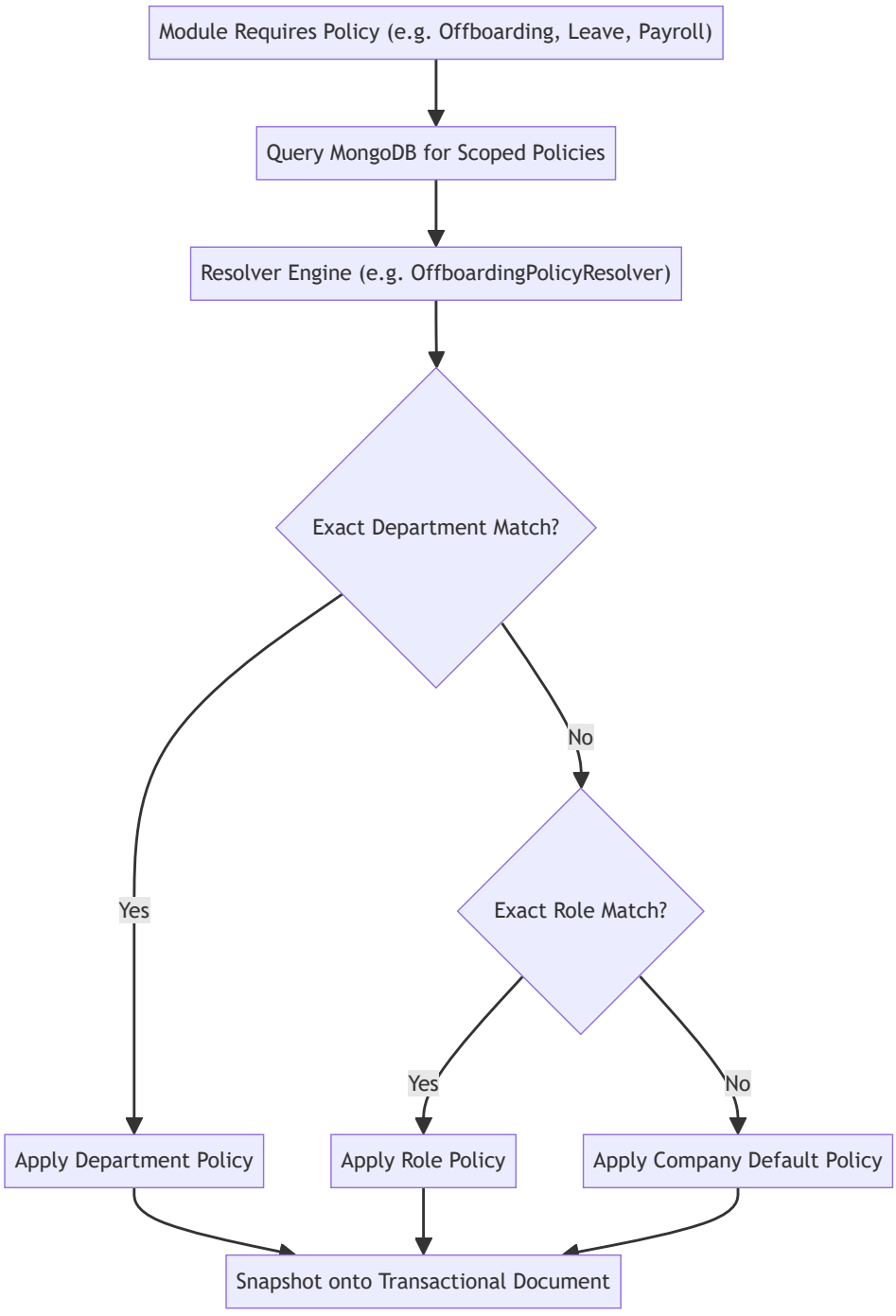

| EmployeeExitPolicy | | | |
|---|---|---|---|
| string | Id | PK | |
| string | CompanyId | FK | |
| string | DepartmentId | FK | Optional |
| string | RoleId | FK | Optional |
| int | NoticeDays | | |
| int | ProbationNoticeDays | | |
| bool | BuyoutAllowed | | |
| bool | WaiverAllowed | | |
| bool | RequireHandover | | |
| bool | RequireAssetClearance | | |
| bool | RequireExitInterview | | |
| int | FormerEmployeePortalDays | | |

## 8. Business Rules

- **Hierarchical Scoping Precedence**:
  1. Policy with matching `DepartmentId`
  2. Policy with matching `RoleId`
  3. Default Company policy (`DepartmentId == null && RoleId == null`)
- **Immutability of In-Flight Instances**: When a workflow commences (e.g., exit or leave application), active policy parameters are snapshotted onto the instance entity so future policy revisions do not retroactively invalidate in-flight processes.

## 9. Complete Workflow

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## 10. State Machine

Policies exist as `Draft` → `Active` → `Superseded`.

## 11. Permissions

- `settings:read` — View policy parameters.
- `settings:update` — Create or modify organization policies.

## 12. Dependencies

- **Upstream**: `COMPANY-SETTINGS`, `ROLE-PERMISSION`.
- **Downstream**: `EMPLOYEE`, `ONBOARDING`, `PROBATION`, `LEAVE`, `ATTENDANCE`, `PAYROLL`, `OFFBOARDING`.

## 13. Cross-Module Events

- Updating a policy updates governance logic for all newly initiated workflow instances.

## 14. Failure Scenarios

- **Missing Scoped Policy**: System falls back safely to tenant default policy, ensuring workflows never crash.

## 15. Edge Cases

- **Simultaneous Department and Role Match**: Department-level policy takes precedence over Role-level policy.

## 16. Current Strengths

- Complete snapshotting architecture guarantees legal/contractual immutability for past transactions.

## 17. Current Gaps

- Policy effective date scheduling (e.g., defining a policy today that automatically activates next quarter) is not yet built; changes take effect upon save.

## 18. Recommended Improvements

- Add `EffectiveFrom` and `EffectiveTo` date ranges on all policy documents.

## 19. Future Scope

- Visual policy simulation sandbox to model financial and operational impacts of policy changes before publishing.

MOD-05 PART III docs/project-documentation/modules/SETTINGS.md

# Module Documentation — Centralized System Settings Index (`SETTINGS`)

> **Document Status**: Complete & Authoritative
> **Source-of-Truth**: Verified from Repository Implementation
> **Scope**: Cross-Module Configuration Hubs ( `Company` , `Attendance` , `Payroll` , `Recruitment` , `Lifecycle` , `User` )
> **Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The comprehensive index and operational framework of all administrative and user-level configuration workspaces distributed across the Codeji HRMS platform.
- **Why HR organizations need it**: Rather than burying module parameters inside operational forms, Codeji provides dedicated settings portals for Company Branding, Master Taxonomies, Attendance Statuses, Recruitment SLAs/Policies, Payroll Divisor & Salary Structures, and Lifecycle/Exit Policies.
- **What business problem it solves**: Provides clear separation of duties between **Configuration Authors** (Administrators/HR Directors) and **Operational Executors** (Recruiters, Managers, Employees).
- **Who uses it**: **Company Administrators**, **HR Managers**, and **Employees** (for personal user settings).
- **Where it fits in the HR lifecycle**: Governs the behavioral parameters of every operational stage.

## 2. Real-World Business Scenario

```
Organization: Meridian FinTech Solutions
Administrator: Karan Malhotra
Scenario: Annual Fiscal Year Rollover & Policy Recalibration
```

1. At the start of the new fiscal year, Karan visits the 5 primary administrative settings portals:
   - **Payroll Settings ( `/settingspayroll` )**: Updates the company divisor policy from `CALENDAR_DAYS` to `FIXED_30_DAYS` effective April 1st, and revises salary structures for promoted engineers.
   - **Attendance Settings ( `/attendance-settings` )**: Configures the grace period to 15 minutes and sets the monthly late arrival threshold ( `CombinedLhdEdMonthlyLimit` ) to 3 occurrences.
   - **Recruitment Settings ( `/recruitment-settings` )**: Configures interview feedback SLA to 48 hours and establishes a two-stage Offer Approval Policy for salaries above ₹25 LPA.
   - **Lifecycle Settings ( `/lifecycle-management?view=policies` )**: Reviews probation confirmation durations (6 months) and offboarding notice periods (60 days for Engineering, 30 days for Operations).

## 3. Supported Actors

| Actor | Settings Portals Accessible | Server Enforcement |
|---|---|---|
| **Company Administrator** | All settings portals ( `/admin/*` , `/attendance-settings` , `/recruitment-settings` , `/settingspayroll` , `/lifecycle-management` , `/profile/setting` ). | `ProtectedRoute adminOnly={true}` and backend `[Authorize(Policy = "AdminOnly")]` . |
| **HR Manager** | `/settingspayroll` , `/lifecycle-management` , `/profile/setting` . | Enforced via `[ModulePermission(PayrollSettings/Employees, View/Edit)]` . |
| **Employee** | `/profile/setting` (Personal security & notification preferences only). | Blocked from all company-level configuration routes. |

## 4. Frontend Implementation

| File | Route Path | Responsibility |
|---|---|---|
| `src/app/modules/admin/companymaster/CompanyMaster.tsx` | `/admin/companymaster/*` | Departments, Job Roles, Office Locations, Working Days, Custom Attributes, ID Sequences. |
| `src/app/modules/admin/companydetails/CompanyDetails.tsx` | `/admin/companydetails/*` | Corporate identity, Logo upload, Legal tax numbers, SMTP Email server configuration. |
| `src/app/modules/attendance/AttendanceStatusSettings.tsx` | `/attendance-settings` | Attendance status labels ( `P` , `HD` , `LHD` , `ED` ), hex colors, and office schedule shifts. |
| `src/app/modules/recruitment/RecruitmentSettings.tsx` | `/recruitment-settings` | Interview plans, offer approval policies, feedback SLA timers, and email templates. |
| `src/app/modules/payrollsettings/PayrollSettings.tsx` | `/settingspayroll` | Employee salary structures ( `SalaryTable.tsx` ) and monthly payroll divisor policies. |
| `src/app/modules/users/components/EmployeeExitPolicySettings.tsx` | `/lifecycle-management` | Offboarding notice rules, buyout divisors, and multi-department clearance templates. |
| `src/app/modules/users/usersetting/Setting.tsx` | `/profile/setting` | Personal password change and notification channel preferences. |

## 5. Backend Implementation

| Backend Controller | Managed Entities | Primary Service |
|---|---|---|
| `CompanyMasterController.cs` | `Department` , `CompanyJobRole` , `CompanyOfficeLocations` , `CompanyWorkingDays` | `CompanyMasterService.cs` |
| `CompanyController.cs` | `Company` , `CompanyEmailSettings` , `EmployeeIdSequence` | `CompanyService.cs` |
| `AttendanceStatusSettingsController.cs` | `AttendanceStatusSetting` , `CompanyOfficeSchedule` | `AttendanceStatusService.cs` , `OfficeScheduleSettingsService.cs` |

| Backend Controller | Managed Entities | Primary Service |
| --- | --- | --- |
| `PayrollDivisorPolicyController.cs` | `PayrollDivisorPolicy`, `SalaryModel` | `PayrollDivisorPolicyService.cs`, `SalaryService.cs` |
| `RecruitmentEnterpriseController.cs` | `RecruitmentOfferApprovalPolicy`, `RecruitmentInterviewPlan` | `RecruitmentOfferApprovalPolicyService.cs` |
| `UserController.cs` | `EmployeeExitPolicy`, `EmployeeExitClearanceTemplate` | `EmployeeService.cs` |

## 6. API Documentation Summary

| Domain | Endpoint | Method | Purpose |
| --- | --- | --- | --- |
| **Attendance** | `/api/AttendanceStatusSettings/GetSettings` | `GET` | Fetches tenant attendance status colors and labels. |
| **Shifts** | `/api/Attendance/SaveOfficeSchedule` | `POST` | Saves company or department shift timings and grace periods. |
| **Payroll** | `/api/PayrollDivisorPolicy/SavePolicy` | `POST` | Configures `FIXED_30_DAYS`, `WORKING_DAYS`, or `CALENDAR_DAYS`. |
| **Recruitment** | `/api/RecruitmentPlanning/SaveOfferApprovalPolicy` | `POST` | Defines approval chains for high-value candidate offers. |
| **Offboarding** | `/api/User/SaveEmployeeExitPolicy` | `POST` | Saves department/role-scoped exit notice and clearance rules. |

## 7. Database Model

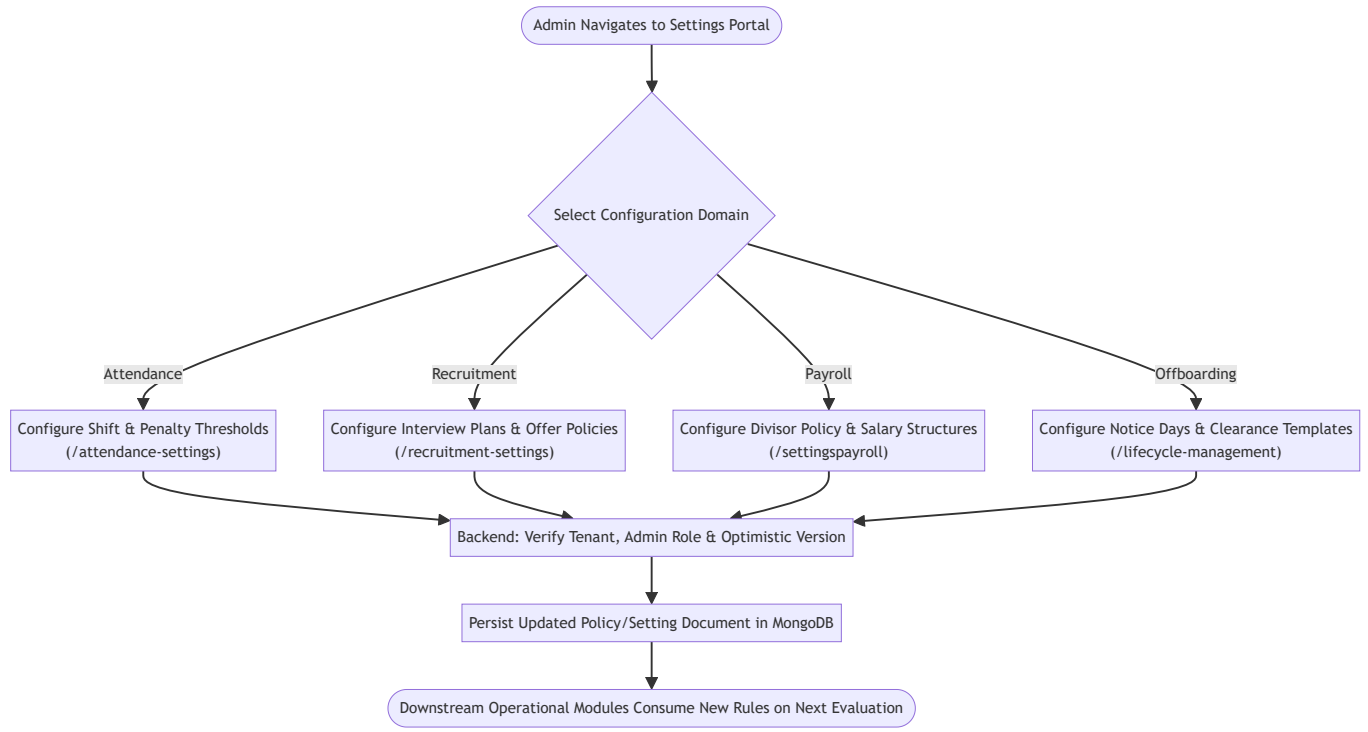
All settings entities include `CompanyId`, `Version` (for optimistic concurrency), `EffectiveFrom` / `EffectiveTo` date windows (where temporal versioning applies), and standard audit fields (`CreatedBy`, `CreatedDate`, `UpdatedBy`, `UpdatedDate`).

## 8. Business Rules

- **Temporal Versioning Rule**: Changes to `PayrollDivisorPolicy` and `AttendancePenaltyPolicy` require an `EffectiveFrom` date and cannot retroactively mutate already locked attendance months or processed payroll runs.
- **Immutable Snapshot Rule**: Changes to `EmployeeExitPolicy` apply only to newly initiated separation cases; inflight exit cases retain their embedded `PolicySnapshot`.

## 9. Complete Workflow

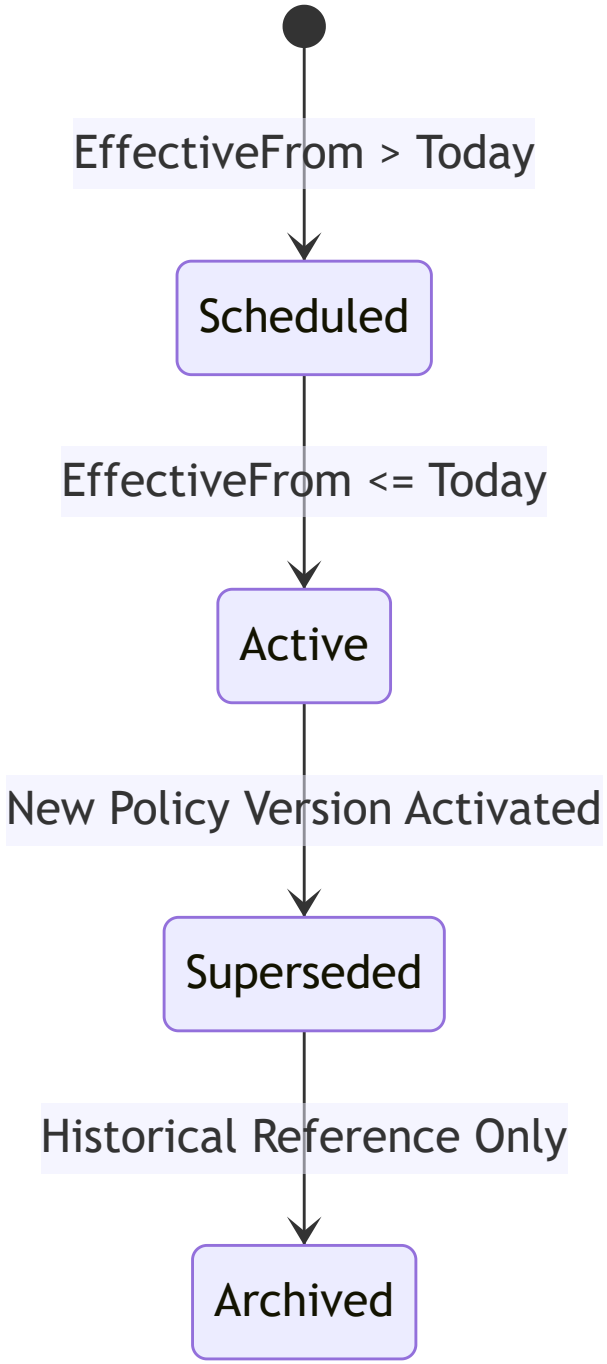
ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Settings records transition through temporal lifecycle states:

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Setting Portal | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| `/admin/companymaster` | Yes | No | No | No |
| `/admin/companydetails` | Yes | No | No | No |
| `/attendance-settings` | Yes | No | No | No |

| Setting Portal | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| `/recruitment-settings` | Yes | No | No | No |
| `/settingspayroll` | Yes | Yes | No | No |
| `/lifecycle-management` | Yes | Yes | No | No |
| `/profile/setting` | Yes | Yes | Yes | Yes |

## 12. Dependencies

- **Depends On**: `Company`, `Roles`.
- **Used By**: Every operational module across the platform.

## 13. Cross-Module Events

- Updating `RecruitmentSettings` feedback SLA thresholds immediately adjusts overdue indicators on `RecruitmentFeedbackSlaCard.tsx`.
- Updating `PayrollSettings` divisor policies alters the daily salary rate calculation in the next `AutoPayrollServices` run.

## 14. Failure Scenarios

- **Overlapping Effective Dates**: Attempting to save an `AttendancePenaltyPolicy` whose `[EffectiveFrom, EffectiveTo]` date range overlaps an existing policy for the same employee/scope returns a validation error: `"The effective dates overlap an existing policy."`

## 15. Edge Cases

- **Unconfigured Tenant Defaults**: If a newly registered tenant has not manually visited `/attendance-settings` or `/settingspayroll`, the system gracefully resolves safe system defaults (`CALENDAR_DAYS` divisor, 0 grace minutes, 30-day notice period) via fallback rules.

## 16. Current Strengths

- Dedicated, modular configuration screens prevent administrative clutter in daily operational views.
- Strong temporal date-window validation prevents overlapping policy ambiguities.

## 17. Current Gaps

- Settings routes are accessed via distinct sidebar items (`/admin/*`, `/attendance-settings`, `/recruitment-settings`, `/settingspayroll`) rather than a single unified "Settings Control Center" landing page.

## 18. Recommended Improvements

- **Low (P3)**: Create a unified `/settings` hub dashboard with quick-navigation cards linking to all 6 domain configuration portals.

## 19. Future Scope

- Configuration export/import (JSON/YAML Infrastructure-as-Code templates) allowing enterprises to clone settings from a staging tenant to a production tenant.

MOD-06 PART III docs/project-documentation/modules/RECRUITMENT.md

# Module Documentation — Recruitment & Talent Acquisition (`RECRUITMENT`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controllers**: `ApplicantsController.cs`, `JobVacancyController.cs`, `PublicCareerController.cs`, `RecruitmentEnterpriseController.cs`
**Primary Services**: 24 Services in `Codeji.CMS.Services.Recruitments`
**Primary UI**: `Applications.tsx`, `Jobs.tsx`, `RecruitmentLifecycleDrawer.tsx` (50.6 KB), `RecruitmentSettings.tsx`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: A full-scale enterprise Applicant Tracking System (ATS) and talent acquisition suite spanning public job posting, applicant sourcing, duplicate detection, resume extraction, structured multi-stage screening, interview planning and debriefs, interviewer scorecards, multi-tier offer approval policies, candidate offer response/negotiation, and talent pool curation.
- **Why HR organizations need it**: Talent acquisition is the entry funnel to the enterprise. Organizations require an auditable, collaborative, and candidate-friendly system to manage requisitions, prevent duplicate applicant submissions, coordinate hiring panels, and issue legally binding offers.
- **What business problem it solves**: Eliminates lost candidate resumes in email threads, inconsistent interview scoring, unbudgeted hiring compensation, and delayed offer letters.
- **Who uses it**: **Recruiters**, **Hiring Managers**, **Interviewers**, **HR Directors**, **Finance Approvers**, and external **Candidates**.
- **Where it fits in the HR lifecycle**: The initial stage (Talent Sourcing & Selection)—seamlessly handing off selected candidates into Preboarding and Onboarding.

## 2. Real-World Business Scenario

```
Company: Codeji Technologies Pvt Ltd
Hiring Role: Senior Distributed Systems Engineer
Candidate: Siddharth Nair
Recruiter: Meera Sen
Hiring Manager: Vikram Malhotra (Engineering Director)
```

1. Vikram submits a `RecruitmentRequisition` for 1 Senior Engineer with a budget ceiling of ₹32 LPA; approved by Finance.
2. Meera creates a `JobVacancy` (Job Code `REQ-ENG-042`) and publishes it to `/careers/codeji`.
3. Siddharth discovers the role on the Public Career Portal, submits his profile, and uploads his resume PDF.
4. **Duplicate Detection Engine**: Automatically verifies Siddharth has not applied to Codeji within the past 90 days.
5. **Screening Queue**: Meera reviews extracted resume metadata, assigns score 4/5, and advances Siddharth to `Shortlisted`.
6. **Interview Loop**: Schedules a 2-stage loop (`System Architecture` with Vikram, `Culture & Leadership` with HR).
7. **Scorecards & Debrief**: Vikram and Meera submit structured scorecards (ratings 1–5 on algorithms, concurrency, teamwork). The panel records a debrief decision: `Recommend Hire`.
8. **Offer Approval & Negotiation**: Meera drafts an offer (₹30 LPA Basic + Allowances, Joining Date: Nov 1st). The offer approval policy routes to the Finance Director for digital sign-off.

9. **Candidate Response**: Siddharth receives an email notification with a secure link to `/candidate/offer`, reviews the formal offer letter, and clicks **Accept Offer**.
10. The accepted offer automatically initializes a `RecruitmentPreboardingCase`.

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## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **Recruiter / HR Manager** | Full operational control: post vacancies, screen candidates, schedule interviews, view all queues, draft offers, manage talent pools. | Enforced via `[ModulePermission(Applications/Jobs, View/Create/Edit)]`. |
| **Hiring / Department Manager** | Reviews requisitions, conducts interview debriefs, reviews applicant resumes for department vacancies. | Scoped to assigned vacancy or department. |
| **Interviewer** | Submits interview scorecards (`RecruitmentScorecard`) for assigned candidates. | Restricted to assigned interview sessions. |
| **Finance / Executive Approver** | Approves or rejects high-value candidate offers based on `RecruitmentOfferApprovalPolicy`. | Gated by offer approval workflow endpoints. |
| **Candidate (Public)** | Searches vacancies, applies via career marketplace, reviews and accepts/rejects offer letters via `/candidate/offer`. | Capability tokens (`PublicApplicationToken`, secure offer links). |

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## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/routing/RecruitmentRoutes.tsx` | Routes: `jobs`, `applications`, `applicant/:applicantid`, `processlog`, `career-profile`. | React Router `<Outlet />` | `PrivateRoutes.tsx` (`/recruitment/*`) |
| `src/app/modules/recruitment/jobs/Jobs.tsx` | Vacancy management grid, status filters (Open, Paused, Closed), requisition drawer. | `recruitmentServices.ts` | `RecruitmentRoutes.tsx` |
| `src/app/modules/recruitment/applications/Applications.tsx` | Kanban and table views of candidate pipelines across all stages. | `recruitmentServices.ts` | `RecruitmentRoutes.tsx` |
| `src/app/modules/recruitment/applications/components/RecruitmentLifecycleDrawer.tsx` (50.6 KB) | Primary candidate cockpit: screening queue, interview scheduling, scorecard reviews, offer issuance, and handoff triggers. | `recruitmentServices.ts` | `Applications.tsx` |
| `src/app/modules/recruitment/RecruitmentSettings.tsx` (17.7 KB) | Configures interview plans, offer approval chains, feedback SLAs, and email templates. | `recruitmentServices.ts` | `PrivateRoutes.tsx` (`/recruitment-settings`) |

| File | Responsibility | Calls | Used By |
| --- | --- | --- | --- |
| `src/app/modules/recruitment/public/CandidateOfferResponse.tsx` | Public portal for candidate offer letter review, acceptance, or counter-negotiation. | `recruitmentServices.ts` | `AppRoutes.tsx` ( `/candidate/offer` ) |

## 5. Backend Implementation

The backend recruitment subsystem is powered by 24 specialized services in `Codeji.CMS.Services.Recruitments`:

| Backend Service | Architectural Responsibility | Primary Dependencies |
| --- | --- | --- |
| `ApplicantServices.cs` | Manages core `Applicant` and `ApplicantLogs` records, stage transitions. | `IMongoDbRepository<Applicant>` |
| `JobVacancyService.cs` | Vacancy publishing, requirements definition, career portal visibility. | `IMongoDbRepository<JobVacancy>` |
| `RecruitmentApplicationService.cs` | Application creation, duplicate candidate checks, stage progression. | `IMongoDbRepository<RecruitmentJobApplication>` |
| `RecruitmentScreeningQueueService.cs` | Recruiter screening queues, resume rating, qualification matching. | `IMongoDbRepository<RecruitmentScreeningReview>` |
| `RecruitmentInterviewService.cs` | Schedules interview rounds, assigns interviewers, tracks Google Meet/Teams links. | `IMongoDbRepository<RecruitmentInterview>` |
| `RecruitmentScorecardService.cs` | Standardized scoring attributes (1–5 scale), interviewer recommendations. | `IMongoDbRepository<RecruitmentScorecard>` |
| `RecruitmentInterviewDebriefService.cs` | Panel consensus meetings, hire/no-hire consolidated decisions. | `IMongoDbRepository<RecruitmentInterviewPlan>` |
| `RecruitmentOfferService.cs` | Compensation drafting, offer letter HTML generation, expiration windows. | `IMongoDbRepository<RecruitmentOffer>` |
| `RecruitmentOfferApprovalPolicyService.cs` | Multi-stage offer approval gates based on salary thresholds and departments. | `IMongoDbRepository<RecruitmentOfferApprovalPolicy>` |
| `RecruitmentOfferNegotiationService.cs` | Tracks candidate counter-proposals and revision rounds. | `IMongoDbRepository<RecruitmentOffer>` |
| `RecruitmentEmailOutboxProcessor.cs` | Transactional email dispatch for interview invites and offer letters. | `IMongoDbRepository<RecruitmentEmailOutbox>` |
| `RecruitmentDuplicateDetectionService.cs` | Scans matching email hashes, phone numbers, and names across applicants. | `IMongoDbRepository<Applicant>` |
| `RecruitmentTalentPoolService.cs` | Archives silver-medalist candidates into searchable talent pools. | `IMongoDbRepository<RecruitmentTalentPoolMember>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `POST` | `/api/JobVacancy/AddEditJobVacancy` | Creates or modifies a job posting. | `[ModulePermission(Jobs, Create/Edit)]` | `JobVacancyModel` | Validates job title, department, employment type, and salary range. | `Result<string>` |
| `POST` | `/api/PublicCareer/ApplyJob` | External candidate job application. | Anonymous (Public) | `ApplyNowVacancyModel` (Multipart) | Rate-limited by IP; validates resume PDF format; triggers duplicate detection. | `Result<string>` |
| `POST` | `/api/RecruitmentScreening/Review` | Submits candidate screening evaluation. | `[ModulePermission(Applications, Review)]` | `RecruitmentScreeningReviewCommand` | Updates screening queue status; advances to Shortlisted or Rejection. | `Result` |
| `POST` | `/api/RecruitmentPlanning/ScheduleInterview` | Schedules an interview round. | `[ModulePermission(Applications, Edit)]` | `RecruitmentInterviewScheduleDto` | Verifies interviewer availability; enqueues calendar invites in outbox. | `Result` |
| `POST` | `/api/RecruitmentPlanning/SubmitScorecard` | Interviewer submits candidate ratings. | `[Authorize]` | `RecruitmentScorecardDto` | Validates all mandatory rating categories are scored; locks scorecard. | `Result` |
| `POST` | `/api/RecruitmentEnterprise/IssueOffer` | Generates official offer letter. | `[ModulePermission(Applications, Create)]` | `RecruitmentOfferCreateCommand` | Verifies offer approval policy sign-offs; generates capability token; queues email. | `Result` |
| `POST` | `/api/PublicCareerEngagement/AcceptOffer` | Candidate digitally accepts offer. | Public Token Auth | `CandidateOfferAcceptanceDto` | Validates token expiration; transitions applicant to `Selected`; initializes preboarding. | `Result` |

## 7. Database Model

### Core Entities

1. **`JobVacancy`**: `VacancyId`, `CompanyId`, `JobTitle`, `JobCode`, `DepartmentId`, `EmploymentType`, `MinSalary`, `MaxSalary`, `Slug`, `Status` (`Draft`, `Open`, `Paused`, `Closed`).
2. **`Applicant`**: `ApplicantId`, `CompanyId`, `JobVacancyId`, `FirstName`, `LastName`, `Email`, `PhoneNumber`, `CurrentStage`, `ResumeUrl`, `OverallRating`.
3. **`RecruitmentJobApplication`**: Links applicant to specific vacancy with timeline metadata.

4. **`RecruitmentInterview`**: `InterviewId`, `ApplicantId`, `RoundNumber`, `RoundName`, `ScheduledStartUtc`, `InterviewerUserIds`, `MeetingUrl`, `Status` (`Scheduled`, `Completed`, `Cancelled`).
5. **`RecruitmentScorecard`**: `ScorecardId`, `InterviewId`, `InterviewerUserId`, `Ratings` (`Dictionary<string, int>`), `Recommendation` (`StrongHire`, `Hire`, `NoHire`, `StrongNoHire`).
6. **`RecruitmentOffer`**: `OfferId`, `ApplicantId`, `OfferedCtc`, `JoiningDate`, `ExpiryDate`, `Status` (`Draft`, `PendingApproval`, `Approved`, `Issued`, `Accepted`, `Declined`).
7. **`RecruitmentEmailOutbox`**: Outbox queue for candidate communications.

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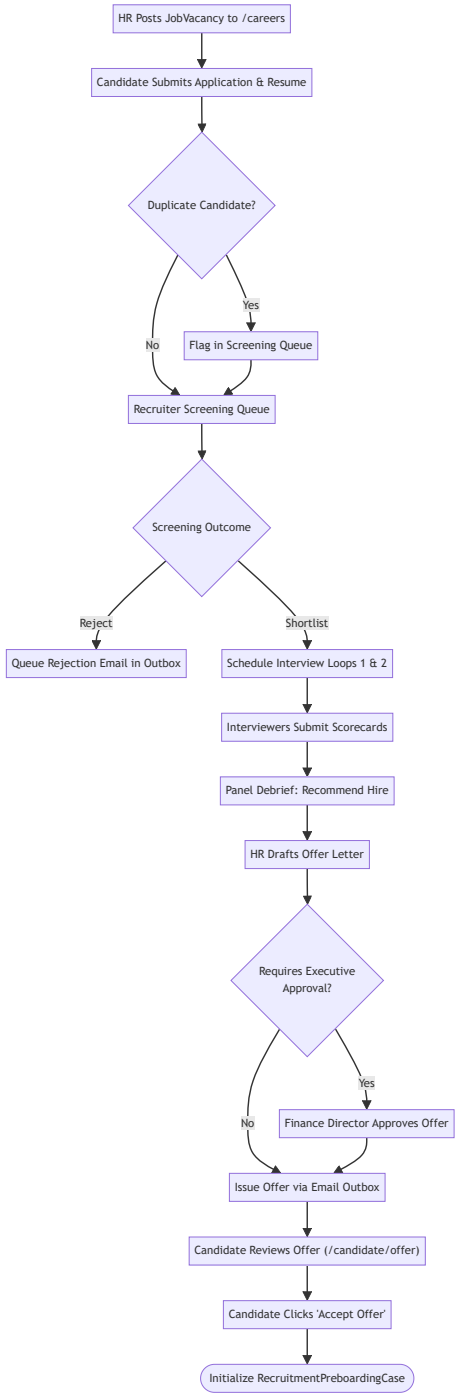
## 8. Business Rules

- **Duplicate Application Guard**: Candidates cannot apply to the same `JobVacancy` with the same email address while an active application is in progress.
- **Offer Approval Gating**: If an offer's `OfferedCtc` exceeds the vacancy's `MaxSalary`, it cannot be issued directly; it must be approved by an authorized Finance or Executive approver in `RecruitmentOfferApprovalPolicy`.
- **Scorecard Non-Repudiation**: Once an interviewer submits a scorecard, it is frozen against further editing to preserve interview panel integrity.

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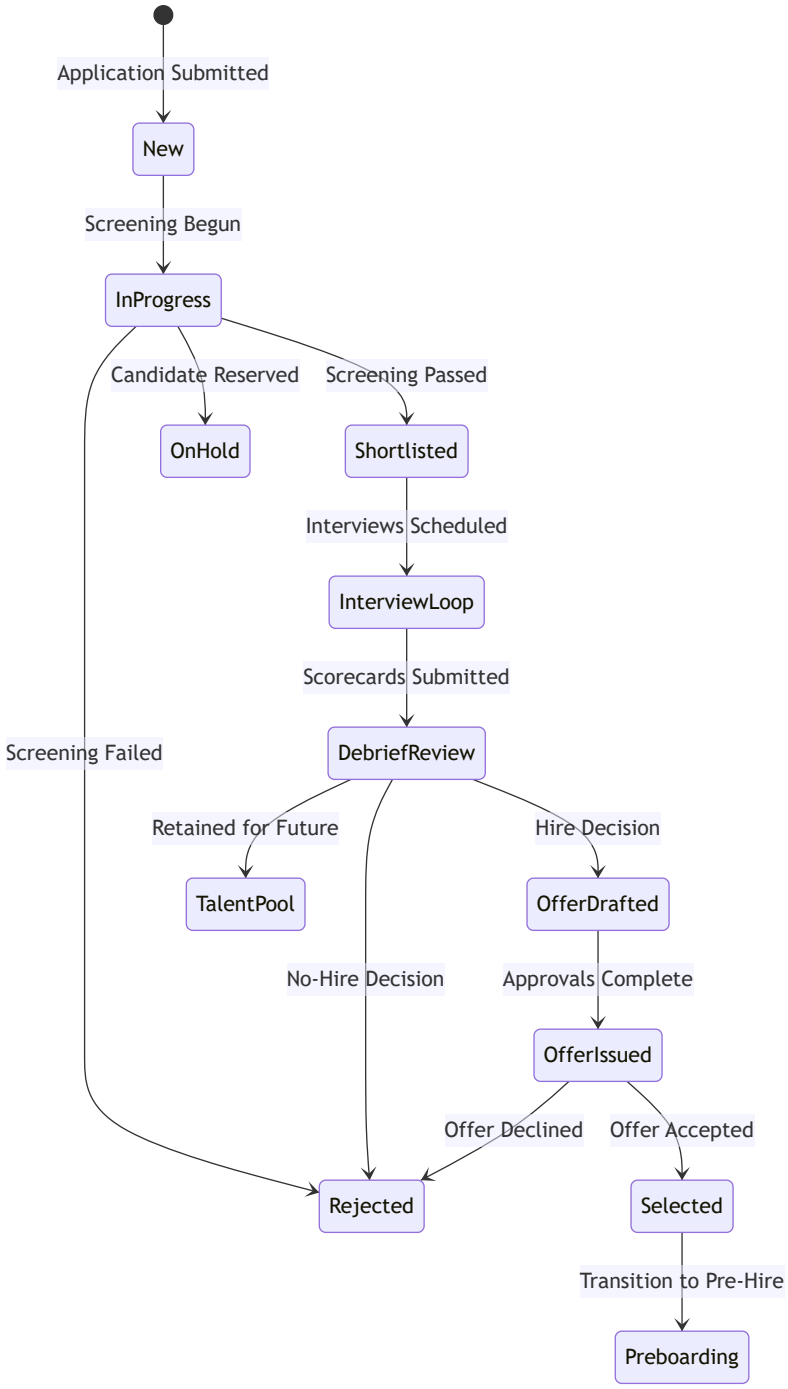
# 9. Complete Workflow

### ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Interviewer | Candidate |
|---|---|---|---|---|
| Post / Close Vacancy | Yes | Yes | No | No |
| Screen Applications | Yes | Yes | No | No |
| View Assigned Candidate Resume | Yes | Yes | Yes | No |

| Action | Admin | HR Manager | Interviewer | Candidate |
|---|---|---|---|---|
| Submit Interview Scorecard | Yes | Yes | Yes | No |
| Draft & Approve Offer | Yes | Yes | No | No |
| Accept / Decline Offer | No | No | No | Yes |

## 12. Dependencies

- **Depends On**: `CompanyMaster` ( `Department` , `CompanyJobRole` ), `Roles` .
- **Used By**: `Preboarding` , `Onboarding` , `CoreEmployee` .
- **Reads From**: `Department` , `CompanyJobRole` , `CompanyOfficeLocations` .
- **Writes To**: `JobVacancy` , `Applicant` , `RecruitmentJobApplication` , `RecruitmentInterview` , `RecruitmentScorecard` , `RecruitmentOffer` , `RecruitmentEmailOutbox` .

## 13. Cross-Module Events

- Digital offer acceptance by a candidate emits a domain event triggering `RecruitmentPreboardingService.InitializeCase` , establishing the initial pre-hire record.

## 14. Failure Scenarios

- **Offer Expiration**: If a candidate attempts to accept an offer past `ExpiryDate` , the API returns HTTP 409 Conflict with message `"Offer has expired. Please contact your recruiter."`
- **Interviewer Outbox Failure**: If the SMTP server is down when scheduling an interview, the outbox record logs the failure and retries automatically without failing the interview scheduling UI action.

## 15. Edge Cases

- **Multiple Concurrent Job Applications**: A candidate may apply to two different open vacancies (e.g. `Backend Engineer` and `DevOps Engineer` ); each application maintains independent interview plans and scorecards.

## 16. Current Strengths

- Full-scale, enterprise-ready ATS pipeline natively integrated with HRMS.
- Deep interview coordination: interviewer scorecards, panel debriefs, and multi-tier offer policies.
- Resilient transactional email dispatching via outbox workers.

## 17. Current Gaps

- Automated video call room creation (e.g. Zoom or Google Meet API auto-generation) is not yet integrated; interviewers manually paste meeting links.

## 18. Recommended Improvements

- **Medium**: Integrate Google Calendar / Microsoft 365 Graph API for automated interviewer availability calendar lookups.

## 19. Future Scope

- AI resume matching algorithm scoring candidates against job description keyword vectors.

MOD-07 PART III docs/project-documentation/modules/PREBOARDING.md

# Module Documentation — Candidate Preboarding (`PREBOARDING`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Entities**: `RecruitmentPreboardingCase.cs`, `RecruitmentHandoff.cs`
**Primary Services**: `RecruitmentHandoffService.cs`, `RecruitmentApplicationService.cs`
**Primary UI**: `RecruitmentPreboardingQueueCard.tsx`, `RecruitmentLifecycleDrawer.tsx`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The pre-joining verification bridge situated between candidate offer acceptance and Day-1 employee onboarding. It manages pre-hire document collection, background verification (BGV) tracking, and joining readiness confirmation.
- **Why HR organizations need it**: After a candidate accepts an offer, they typically serve a notice period at their previous employer (30–90 days). During this window, HR must collect identity documents, academic transcripts, and prior relieving letters to verify candidate credentials before provisioning an employee account.
- **What business problem it solves**: Prevents Day-1 onboarding chaos by ensuring all statutory, educational, and identity verifications are completed prior to the candidate's joining date.
- **Who uses it**: **HR Managers**, **HR Executives**, **Recruiters**, and **Selected Candidates**.
- **Where it fits in the HR lifecycle**: Stage 2 (Post-Offer / Pre-Joining)—receives accepted candidates from `Recruitment` and hands verified candidates off to `Onboarding`.

## 2. Real-World Business Scenario

```
Candidate: Siddharth Nair
Accepted Role: Senior Distributed Systems Engineer
Expected Joining Date: November 1, 2026
HR Executive: Pooja Sharma
```

1. Upon Siddharth's offer acceptance on `/candidate/offer`, the system initializes a `RecruitmentPreboardingCase` in `PendingDocuments` status.
2. Pooja views Siddharth in the `RecruitmentPreboardingQueueCard` on the applications dashboard.
3. Siddharth submits scanned copies of his Aadhaar Card, PAN Card, University Degree Certificate, and previous employer resignation acceptance.
4. Pooja reviews each document inside `RecruitmentLifecycleDrawer.tsx` and marks them `Verified`.
5. Once all mandatory preboarding items are verified, Pooja transitions the preboarding case to `Completed`, which automatically triggers `RecruitmentHandoffService` to initialize Siddharth's Day-1 `RecruitmentOnboardingCase` and provision his `EmpUser` record in `PreJoining` status.

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
| --- | --- | --- |
| **HR Manager / Recruiter** | Reviews preboarding queues, verifies uploaded documents, flags missing items, initiates onboarding handoff. | Enforced via `[ModulePermission(Applications, Edit/Review)]`. |
| **Candidate** | Uploads requested pre-joining documents via candidate portal link. | Scoped to capability token for own application. |
| **Company Administrator** | Full administrative oversight and status override capabilities. | Admin bypass in `RoleServices`. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
| --- | --- | --- | --- |
| `src/app/modules/recruitment/applications/components/RecruitmentPreboardingQueueCard.tsx` | Dashboard summary card displaying candidates currently in preboarding, document completion percentages, and blockers. | `recruitmentServices.ts` | `Applications.tsx` |
| `src/app/modules/recruitment/applications/components/RecruitmentLifecycleDrawer.tsx` | Detailed preboarding checklist panel allowing HR to inspect and approve individual document items. | `recruitmentServices.ts` | `Applications.tsx` |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Codeji.CMS.API/Controllers/RecruitmentEnterpriseController.cs` | Exposes endpoints for fetching preboarding queues, updating document verification statuses, and executing handoff. | `IRecruitmentHandoffService` |
| `Codeji.CMS.Services/Recruitments/RecruitmentHandoffService.cs` | Validates preboarding completion gates, creates initial `EmpUser` in `PreJoining` state, and initializes `RecruitmentOnboardingCase`. | `IMongoDbRepository<RecruitmentPreboardingCase>`, `IMongoDbRepository<EmpUser>` |
| `Codeji.CMS.Repository/Entities/Recruitments/RecruitmentPreboardingCase.cs` | Persistent document tracking preboarding status, checklist items, and concurrency version. | `BaseClass` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `GET` | `/api/RecruitmentEnterprise/GetPreboardingQueue` | Lists active preboarding cases for the tenant. | `[ModulePermission(Applications, View)]` | Query Params ( `status` , `pageNo` ) | Filters by `CompanyId` ; calculates document completion percentage. | `Result<RecruitmentPreboardingQueueItem>` |
| `POST` | `/api/RecruitmentEnterprise/UpdatePreboarding` | Updates document verification status and case state. | `[ModulePermission(Applications, Edit)]` | `RecruitmentPreboardingUpdateCommand` | Enforces optimistic concurrency ( `ExpectedVersion` ); validates all mandatory items are verified before closing. | `Result<string>` |
| `POST` | `/api/RecruitmentEnterprise/ExecuteHandoff` | Transitions verified candidate to Employee & Onboarding. | `[ModulePermission(Applications, Edit)]` | `RecruitmentHandoffCommand` | Creates `EmpUser` ( `EmploymentStatus = "PreJoining"` ) and `RecruitmentOnboardingCase` . | `Result<string>` |

## 7. Database Model

**Entity:** `RecruitmentPreboardingCase`

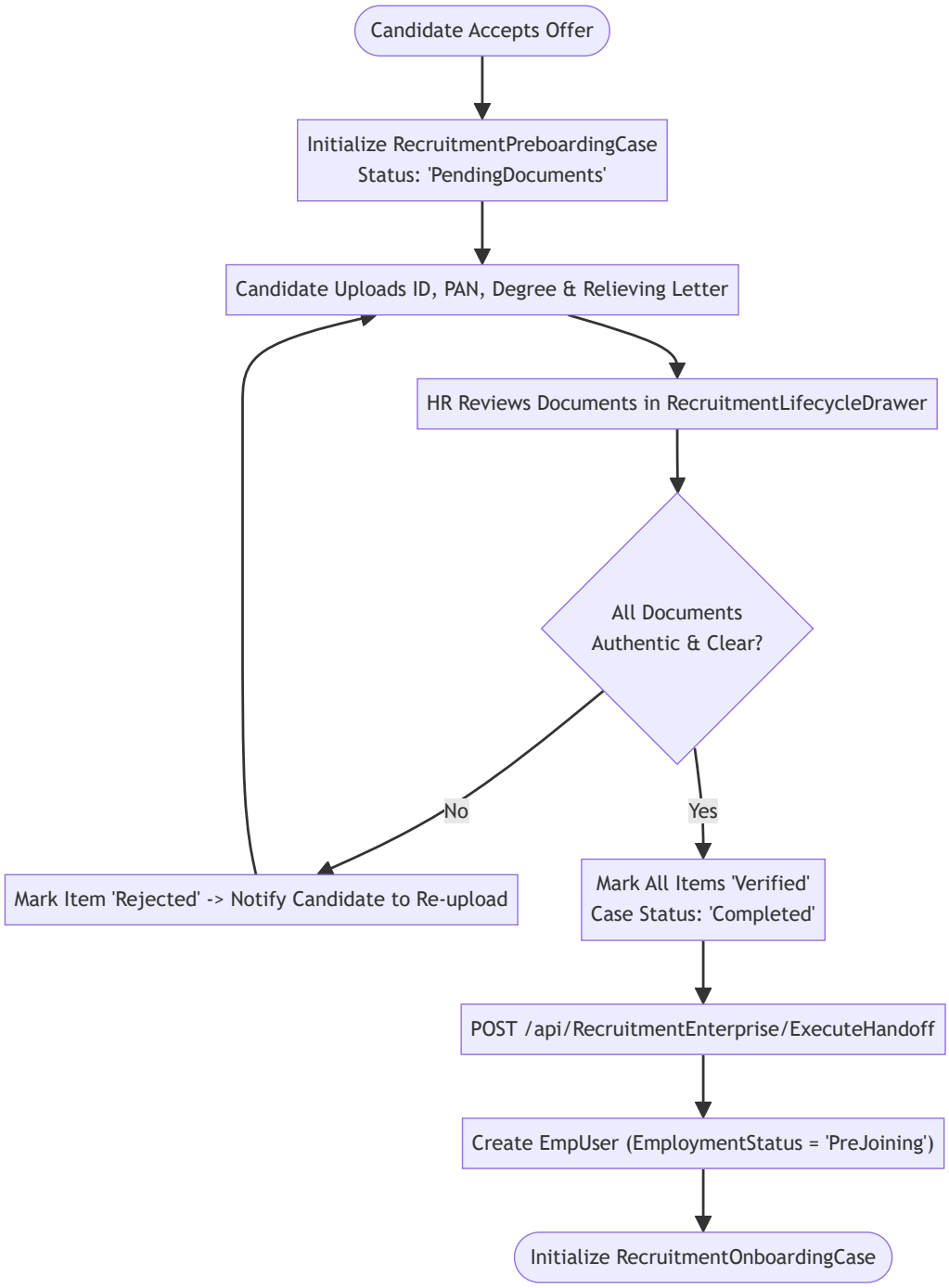
- **Primary Key**: `CaseId`
- **Foreign Keys**: `CompanyId` , `ApplicantId` , `OfferId` , `EmployeeUserId` (populated upon handoff).
- **Important Fields**:
  - `Status` : `PendingDocuments` , `UnderVerification` , `Verified` , `Completed` , `Dropped` .
  - `Version` : Integer optimistic concurrency counter.
  - `ExpectedJoiningDate` : DateTime.
  - `Documents` : `List<PreboardingDocumentItem>` ( `DocumentCode` , `Title` , `Status` , `FileUrl` , `Remarks` ).

## 8. Business Rules

- **Mandatory Verification Gate**: A `RecruitmentPreboardingCase` cannot transition to `Completed` if any mandatory document item remains in `Pending` or `Rejected` status.
- **Idempotent Handoff**: `ExecuteHandoff` checks `LastCommandIdempotencyKey` and existing `EmployeeUserId` to guarantee that clicking "Handoff to Onboarding" twice never creates duplicate `EmpUser` records.
- **PreJoining Status Assignment**: When `EmpUser` is created during preboarding handoff, its initial `EmploymentStatus` is set to `"PreJoining"` and `Status = false` until Day-1 onboarding completes.

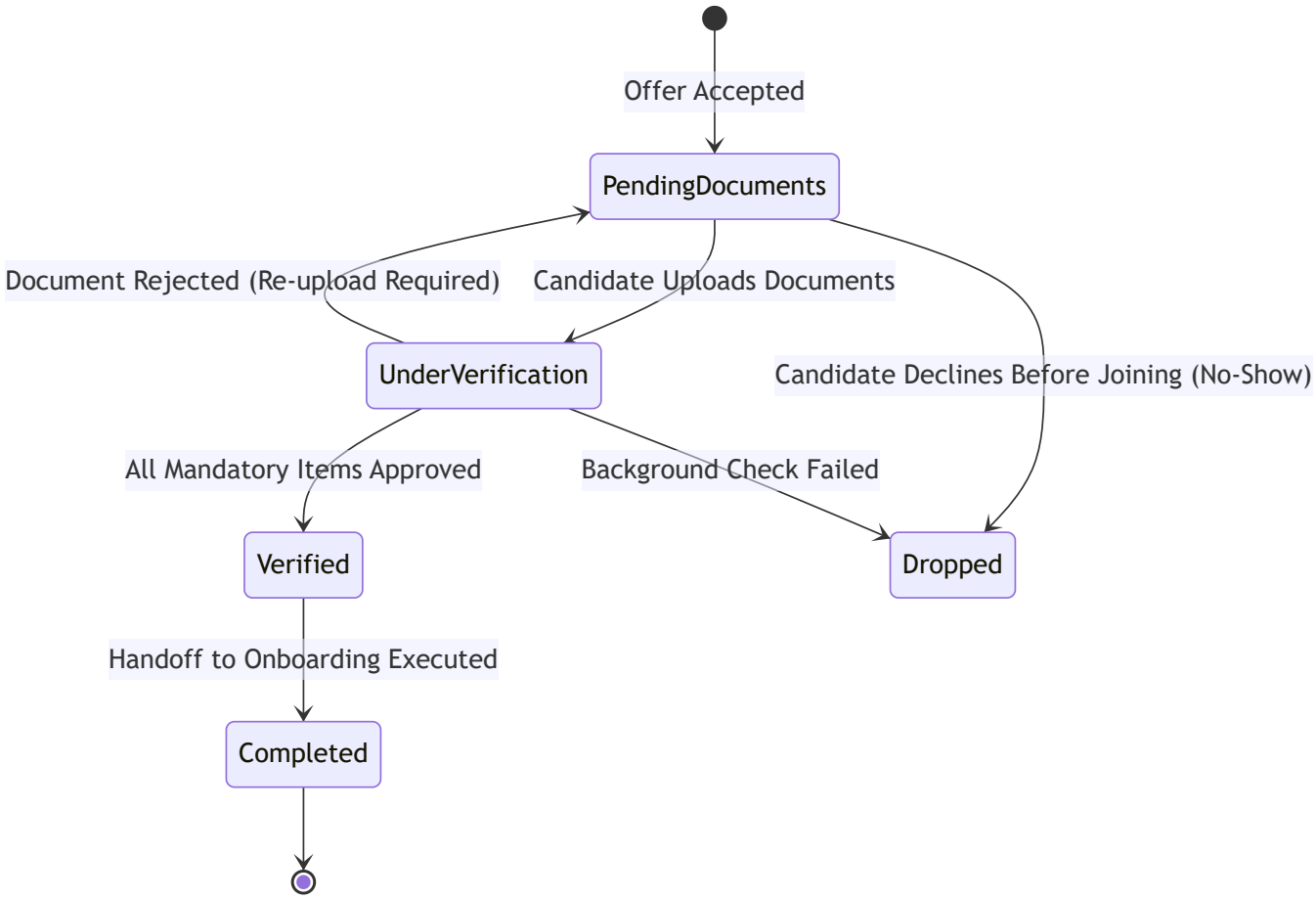
## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Recruiter | Candidate |
|---|---|---|---|---|
| View Preboarding Queue | Yes | Yes | Yes | No |
| Upload Pre-Joining Documents | Yes | Yes | Yes | Yes |
| Verify / Reject Documents | Yes | Yes | Yes | No |
| Execute Onboarding Handoff | Yes | Yes | No | No |

## 12. Dependencies

- **Depends On**: `Recruitment` ( `Applicant` , `RecruitmentOffer` ).
- **Used By**: `Onboarding` ( `RecruitmentOnboardingCase` ), `CoreEmployee` ( `EmpUser` ).
- **Reads From**: `RecruitmentOffer` , `Applicant` .
- **Writes To**: `RecruitmentPreboardingCase` , `RecruitmentHandoff` , `EmpUser` , `RecruitmentOnboardingCase` .

## 13. Cross-Module Events

- Executing preboarding handoff creates both the `EmpUser` record (in `PreJoining` state) and the linked `RecruitmentOnboardingCase` in a coordinated sequence.

## 14. Failure Scenarios

- **Candidate No-Show / Offer Backing Out**: If a candidate decides not to join during preboarding, HR transitions the case to `Dropped`. No active `EmpUser` license or payroll slot is consumed.

## 15. Edge Cases

- **Waived Relieving Letter on Day 1**: If a previous employer delays issuing the final relieving letter, HR can mark the item as `Skipped` (or conditionally waived) with remarks so Day-1 onboarding is not blocked.

## 16. Current Strengths

- Clear visibility into pre-joining bottlenecks via `RecruitmentPreboardingQueueCard.tsx`.
- Idempotency keys prevent duplicate employee creation during handoff.

## 17. Current Gaps

- Third-party Background Verification (BGV) vendor API integration (e.g., AuthBridge, First Advantage, Checkr) is currently tracked manually by HR rather than via automated webhook callbacks.

## 18. Recommended Improvements

- **Medium**: Add webhook receivers for automated BGV status updates.

## 19. Future Scope

- Pre-joining engagement portal featuring team welcome videos, buddy assignments, and first-day schedule previews.

# Module Documentation — Employee Onboarding & Activation (`ONBOARDING`)

**Document Status**: Complete & Authoritative (Special Deep-Dive Specification)
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controller**: `RecruitmentOnboardingController.cs`
**Primary Service**: `RecruitmentOnboardingService.cs` (187 lines)
**Primary Entity**: `RecruitmentOnboardingCase.cs`
**Primary UI**: `RecruitmentOnboardingQueueCard.tsx`, `RecruitmentLifecycleDrawer.tsx`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The structured Day-1 and Week-1 integration engine that manages newly hired employee task checklists (IT asset setup, email provisioning, security badge issuance, policy orientation) and executes **Automated Employee Lifecycle Activation** upon completion.
- **Why HR organizations need it**: Transitioning a new joiner into a productive employee requires coordinated action across IT, Facilities, Department Managers, and HR. Prematurely activating an employee in payroll or attendance before they have actually joined and completed mandatory onboarding causes payroll leakage and security exposure.
- **What business problem it solves**: Guarantees that an employee's account, working availability, shift schedule, and payroll inclusion are activated atomically and auditable only after all required onboarding tasks are completed or explicitly skipped.
- **Who uses it**: **HR Managers**, **HR Executives**, **IT Administrators**, and **Company Administrators**.
- **Where it fits in the HR lifecycle**: Stage 3 (Joining & Activation)—the critical gateway transforming a `PreJoining` record into an `Active` employee.

## 2. Real-World Business Scenario

```
New Joiner: Siddharth Nair (Employee ID: CJ-2026-0089)
Department: Engineering
Reporting Manager: Vikram Malhotra
Onboarding Case ID: ONB-2026-441
```

1. On November 1, 2026, Siddharth arrives for Day-1 orientation. His `EmpUser` record currently has `EmploymentStatus = "PreJoining"` and `Status = false`.
2. HR Manager Priya opens the `RecruitmentOnboardingQueueCard` and selects Siddharth's onboarding case (`ONB-2026-441`), which is in `PendingActivation` status.
3. Priya and the IT team complete the onboarding task checklist:
   - `DOC_SIGN`: Employment Contract & NDA Signed -> Marked `Completed`.
   - `IT_PROVISION`: MacBook Pro & Corporate Email Issued -> Marked `Completed`.
   - `BADGE_ISSUE`: Biometric Access Card Assigned -> Marked `Completed`.
   - `ORIENTATION`: HR Policy & POSH Training -> Marked `Completed`.
4. Priya sets the case status to `Completed` and clicks **Save & Activate**.

5. `RecruitmentOnboardingService.Update` validates that zero tasks remain `Pending` or `InProgress`, increments the case `Version`, and invokes `ActivateEmployeeLifecycle`.
6. `ActivateEmployeeLifecycle` atomically mutates Siddharth's `EmpUser` document:
   - `EmploymentStatus`: `"PreJoining"` $\longrightarrow$ `"Active"`
   - `Lifecycle.AvailabilityStatus`: `"NotApplicable"` $\longrightarrow$ `"Working"`
   - `Lifecycle.AccountStatus`: `"Locked"` $\longrightarrow$ `"Active"`
   - `Lifecycle.PayrollStatus`: `"Hold"` $\longrightarrow$ `"Active"`
   - `EmpUser.Status`: `false` $\longrightarrow$ `true`
   - Appends 4 audit entries to `EmpUser.Lifecycle.History` with `ReasonCode = "RECRUITMENT_ONBOARDING_COMPLETED"`.

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **HR Manager / Executive** | Views onboarding queue, updates task statuses (`Pending`, `InProgress`, `Completed`, `Skipped`), closes onboarding cases, triggers `RetryActivation`. | Enforced via `[ModulePermission(Applications, Edit)]`. |
| **Company Administrator** | Full operational and retry control over all onboarding cases. | Admin bypass in `RoleServices`. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/recruitment/applications/components/RecruitmentOnboardingQueueCard.tsx` | Displays paginated onboarding queue with progress bars (`CompletionPercent`), pending task counts, and `FirstBlockingTaskTitle`. | `recruitmentServices.ts` | `Applications.tsx` |
| `src/app/modules/recruitment/applications/components/RecruitmentLifecycleDrawer.tsx` | Interactive drawer where HR toggles task statuses, passes idempotency keys, and executes employee activation. | `recruitmentServices.ts` | `Applications.tsx` |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Codeji.CMS.API/Controllers/RecruitmentOnboardingController.cs` | Exposes REST endpoints `Get`, `GetQueue`, `Update`, and `RetryActivation`. | `IRecruitmentOnboardingService`, `IHttpContextAccessor` |
| `Codeji.CMS.Services/Recruitments/RecruitmentOnboardingService.cs` | Enforces task transition rules, optimistic concurrency (`ExpectedVersion`), idempotency (`LastCommandIdempotencyKey`), and executes `ActivateEmployeeLifecycle`. | `IMongoDbRepository<RecruitmentOnboardingCase>`, `IMongoDbRepository<EmpUser>` |

| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Codeji.CMS.Repository/Entities/Recruitments/RecruitmentOnboardingCase.cs` | MongoDB document storing case status, linked `ApplicantId` and `EmployeeUserId`, version, and embedded `Tasks` list. | `BaseClass` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
| --- | --- | --- | --- | --- | --- | --- |
| `GET` | `/api/RecruitmentOnboarding/Get/{applicantId}` | Fetches onboarding case for a specific applicant. | `[ModulePermission(Applications, View)]` | Route `applicantId` | Scoped by `CompanyId` and `!x.IsDeleted`. | `Result<RecruitmentOnboardingCase>` |
| `GET` | `/api/RecruitmentOnboarding/GetQueue` | Paginated list of onboarding cases with progress metrics. | `[ModulePermission(Applications, View)]` | Query `status`, `pageNo`, `pageSize` | Computes `CompletionPercent`, `RequiredPendingTasks`, and `FirstBlockingTaskCode`. | `Result<RecruitmentOnboardingQueueItem>` |
| `PUT` | `/api/RecruitmentOnboarding/Update` | Updates task statuses, transitions case state, and activates employee when `Completed`. | `[ModulePermission(Applications, Edit)]` | `RecruitmentOnboardingUpdateCommand` | Validates task codes, checks `ExpectedVersion`, verifies all tasks are `Completed` / `Skipped` before closing, runs `ActivateEmployeeLifecycle`. | `Result<string>` |
| `POST` | `/api/RecruitmentOnboarding/RetryActivation/{caseId}` | Retries employee lifecycle activation if a prior concurrency conflict occurred on `EmpUser`. | `[ModulePermission(Applications, Edit)]` | Route `caseId` | Requires `onboarding.Status == "Completed"`; re-runs `ActivateEmployeeLifecycle`. | `Result<string>` |

## 7. Database Model

**Entity:** `RecruitmentOnboardingCase` (`Codeji.CMS.Repository.Entities.Recruitments.RecruitmentOnboardingCase`)

- **Primary Key**: `CaseId`
- **Foreign Keys**: `CompanyId`, `ApplicantId`, `EmployeeUserId`.
- **Important Fields**:
  - `Status`: `"PendingActivation"`, `"InProgress"`, `"Completed"`.
  - `Version`: Integer optimistic concurrency counter.
  - `LastCommandIdempotencyKey`: String preventing duplicate command replay.

- `Tasks`: `List<RecruitmentOnboardingTaskItem>`:
  - `Code`: Task identifier (e.g., `IT_SETUP`, `DOC_VERIFY`).
  - `Title`: Human-readable task name.
  - `Status`: `"Pending"`, `"InProgress"`, `"Completed"`, `"Skipped"`.

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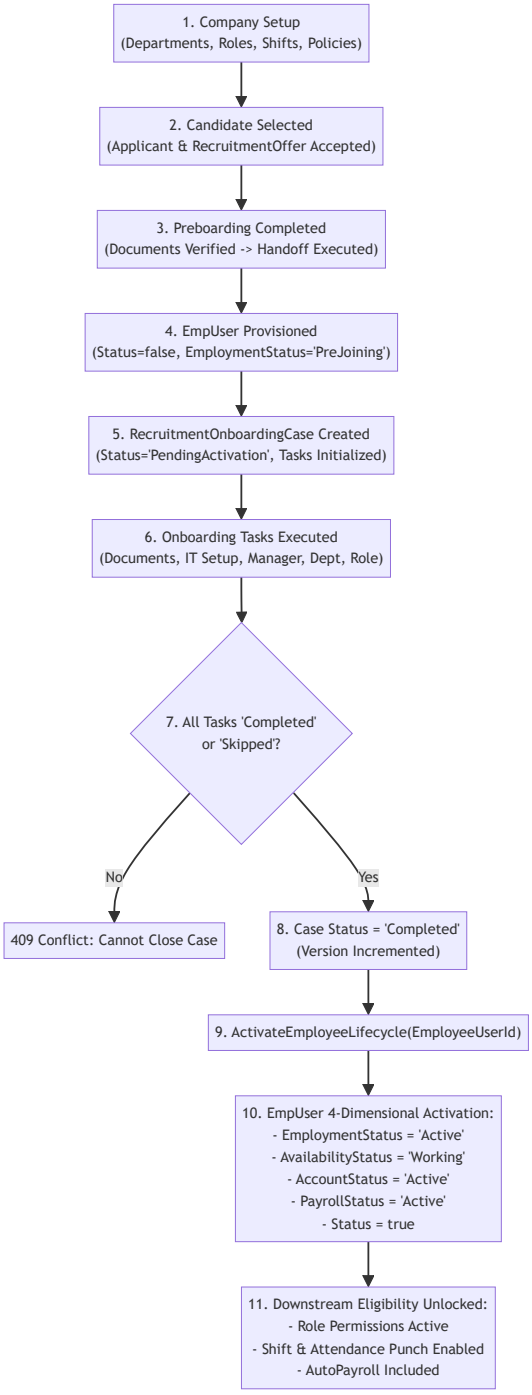
## 8. Business Rules (Verified from `RecruitmentOnboardingService.cs`)

1. **Completion Gate Rule (`Line 109`)**: `if (status == RecruitmentOnboardingStatus.Completed && onboarding.Tasks.Any(x => x.Status is not ("Completed" or "Skipped")))` $\longrightarrow$ Rejects with `409 Conflict` (`RecruitmentErrorCodes.InvalidTransition`): `"All onboarding tasks must be completed or skipped before closing the case."`
2. **PendingActivation Purity Rule (`Line 111`)**: `if (status == RecruitmentOnboardingStatus.PendingActivation && onboarding.Tasks.Any(x => x.Status is not "Pending"))` $\longrightarrow$ Rejects with `409 Conflict`: `"PendingActivation cannot contain started or completed tasks."`
3. **Unknown Task Code Protection (`Line 103`)**: Rejects any task code in the update payload that does not exist in the case's configured task template (`"Unknown onboarding task code"`).
4. **Idempotency & Replay Safety (`Line 91`)**: If `onboarding.LastCommandIdempotencyKey == command.IdempotencyKey`, the service skips re-mutating the onboarding document, re-runs `ActivateEmployeeLifecycle` idempotently, and returns `200 OK`.
5. **Optimistic Concurrency Rule (`Line 98`)**: If `command.ExpectedVersion != onboarding.Version`, rejects with `409 Conflict` (`"Onboarding case was changed by another user"`).

---

# 9. Complete End-to-End Trace: Candidate to Active Employee

**ARCHITECTURE / WORKFLOW DIAGRAM**



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | HR Executive | Employee |
|---|---|---|---|---|
| View Onboarding Queue | Yes | Yes | Yes | No |
| Update Task Statuses | Yes | Yes | Yes | No |
| Complete Case & Activate Employee | Yes | Yes | Yes | No |
| Retry Employee Activation | Yes | Yes | Yes | No |

## 12. Dependencies

- **Depends On**: `Preboarding` ( `RecruitmentHandoffService` ), `CoreEmployee` ( `EmpUser` ).

- **Used By**: `CoreEmployee`, `Attendance`, `LeaveManagement`, `PayRoll`.
- **Reads From**: `RecruitmentOnboardingCase`, `EmpUser`.
- **Writes To**: `RecruitmentOnboardingCase`, `EmpUser` (including `EmpUser.Lifecycle.History`).

---

## 13. Cross-Module Events

- **Automatic Lifecycle Promotion**: Completing the onboarding case directly transitions `EmpUser` across 4 lifecycle dimensions (`EmploymentStatus`, `AvailabilityStatus`, `AccountStatus`, `PayrollStatus`), immediately enabling login access (`CompanyIdMiddleware`), daily attendance tracking, and monthly payroll inclusion (`EmploymentPeriod.IsPayrollIncludedForMonth`).

---

## 14. Failure Scenarios & Recovery (`RetryActivation`)

- **Partial Failure / Concurrency Split**: Because `RecruitmentOnboardingCase` and `EmpUser` reside in two distinct MongoDB collections, it is theoretically possible for the `RecruitmentOnboardingCase` update to succeed while the `EmpUser` update fails due to a concurrent edit on `EmpUser.Lifecycle.Version`.
- **Built-in Recovery Mechanism**:
  - If `employees.Update` fails (`Line 178`), `RecruitmentOnboardingService` returns: `"Onboarding case updated; employee lifecycle activation is pending and must be retried."`
  - HR can then invoke `POST /api/RecruitmentOnboarding/RetryActivation/{caseId}` (or replay the same `IdempotencyKey`), which safely completes the `EmpUser` activation.

---

## 15. Edge Cases

- **Already Active Employee**: If `ActivateEmployeeLifecycle` is called for an employee whose 4 lifecycle dimensions and `Status` are already active (`Line 149`), it short-circuits cleanly and returns `"Onboarding case and employee lifecycle are already active."` without appending duplicate history entries.

---

## 16. Current Strengths

- **Exemplary Concurrency & Idempotency Design**: Uses explicit `IdempotencyKey`, `ExpectedVersion` optimistic locking, and a dedicated `RetryActivation` recovery endpoint.
- **Deterministic Multi-Dimension Activation**: Automatically writes audit history (`RECRUITMENT_ONBOARDING_COMPLETED`) across all changed lifecycle dimensions.

---

## 17. Current Gaps

- Onboarding task templates are initialized with standard tasks rather than dynamically branching by department or job role in the UI editor.

---

## 18. Recommended Improvements

- **Medium**: Allow administrators to author department-specific onboarding task templates in `RecruitmentSettings.tsx`.

---

## 19. Future Scope

- New-hire self-service onboarding portal where the employee can check off orientation items and watch compliance training modules directly.

# Module Documentation — Core Employee Management (`EMPLOYEE`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controller**: `UserController.cs`
**Primary Services**: `EmployeeService.cs` (364 KB), `EmployeeLifecycleTransitionService.cs`
**Primary Entity**: `EmpUser.cs`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The central core HR registry that creates, stores, searches, and governs all employee records (`EmpUser`) within a tenant organization.
- **Why HR organizations need it**: Serves as the authoritative master record for identity, organizational hierarchy (department, designation, reporting manager), employment type (Full-Time, Part-Time, Contractual, Internship), and the 7 orthogonal lifecycle state dimensions.
- **What business problem it solves**: Eliminates fragmented employee spreadsheets and guarantees that attendance, leave, payroll, and separation modules all consume an identical, real-time employee state.
- **Who uses it**: **Company Administrators**, **HR Managers**, **HR Executives**, **Reporting Managers**, and **Employees**.
- **Where it fits in the HR lifecycle**: The central hub of the HR lifecycle—populated via direct HR creation, bulk Excel import, or automated promotion from the Recruitment Onboarding module.

## 2. Real-World Business Scenario

```
Employee: Rahul Sharma (Employee ID: CJ-2026-0042)
Department: Engineering (Backend)
Designation: Senior Software Engineer
Reporting Manager: Ananya Verma (VP of Engineering)
Employment Type: Full-Time (1)
```

1. HR Manager Priya opens `/employees` and clicks **Add Employee** (or triggers bulk Excel import).
2. Enters Rahul's first name, last name, corporate email `rahul.sharma@codeji.in`, department `Engineering`, role `Employee`, and reporting manager `Ananya Verma`.
3. System atomically increments `EmployeeIdSequence` to assign `CJ-2026-0042`.
4. Initializes `EmpUser.Lifecycle` with `EmploymentStatus = "Active"`, `AvailabilityStatus = "Working"`, `AccountStatus = "Active"`, and `PayrollStatus = "Active"`.
5. Dispatches an email verification and password setup link via `InviteEmployeeDto`.

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **Company Administrator** | Full CRUD over all employee records, bulk imports, role reassignments, status overrides, and re-hiring. | Unconditional bypass in `RoleServices.VerifyUserAccess`. |
| **HR Manager / Executive** | Create, view, edit employee records; manage lifecycle transitions; initiate offboarding. | Enforced via `[ModulePermission(AppModule.Employees, Permission.View/Create/Edit)]`. |
| **Reporting Manager** | View directory and team members' basic employment profiles. | Scoped to directory read views (`Permission.View`). |
| **Employee** | View colleague directory; view and self-edit allowed personal profile fields. | Restricted from editing employment status, role, or manager. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/employees/Employees.tsx` | Main employee directory table with search, department/role filters, status pills, and Add/Import modal triggers. | `employeeServices.ts` | `PrivateRoutes.tsx` (`/employees`) |
| `src/app/modules/employees/EmployeeLifecycleManagement.tsx` | Workspace managing lifecycle policies, leave-to-status reconciliation, and open HR lifecycle cases. | `employeeLifecycleServices.ts` | `PrivateRoutes.tsx` (`/lifecycle-management`) |
| `src/app/modules/employees/components/EmployeeLifecycleCaseQueue.tsx` | Displays active probation, contract renewal, and long-term leave cases requiring HR review. | `employeeLifecycleServices.ts` | `EmployeeLifecycleManagement.tsx` |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Codeji.CMS.API/Controllers/UserController.cs` (51.5 KB) | Exposes 60+ endpoints for employee CRUD, bulk import, profile queries, lifecycle transitions, and offboarding. | `IEmployeeService`, `IRoleService`, `IHttpContextAccessor` |
| `Codeji.CMS.Services/Employees/EmployeeService.cs` (364 KB) | Core domain service orchestrating employee creation, email uniqueness checks, sequence generation, lifecycle history tracking, and bulk imports. | `IMongoDbRepository<EmpUser>`, `IMongoDbRepository<Roles>`, `IEmailService` |
| `Codeji.CMS.Services/Employees/EmployeeLifecycleTransitionService.cs` | Validates and executes transitions across the 7 orthogonal lifecycle dimensions. | `IMongoDbRepository<EmpUser>` |
| `Codeji.CMS.Services/Employees/EmploymentPeriod.cs` | Pure domain helper determining whether an employee's joining and exit dates include them in a given attendance or payroll month. | `EmpUser` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `POST` | `/api/User/GetAllEmployees` | Paginated, filtered employee directory query. | `[ModulePermission(Employees, View)]` | `GetAllEmployeeRequestModel` | Filters by `CompanyId`, search keyword, department, role, and employment status. | `Result<GetAllEmployeeResponseModel>` |
| `POST` | `/api/User/AddEmployee` | Creates a new employee record and sends invitation email. | `[ModulePermission(Employees, Create)]` | `InviteEmployeeDto` | Verifies unique email in tenant; generates next `EmployeeId`; initializes `EmployeeLifecycleState`. | `Result<string>` |
| `POST` | `/api/User/UpdateEmployee` | Updates employment details (department, manager, role, status). | `[ModulePermission(Employees, Edit)]` | `UserModel` | Validates PAN/Bank regex; records lifecycle history if status changed. | `Result` |
| `POST` | `/api/User/BulkImportEmployees` | Batch imports employees from parsed Excel rows. | `[ModulePermission(Employees, Create)]` | `BulkImportEmployeesRequestDto` | Validates all rows for duplicate emails and valid departments; returns row-by-row success/error summary. | `Result<BulkImportEmployeesResponseDto>` |
| `DELETE` | `/api/User/DeleteEmployee/{id}` | Soft-deletes an employee record. | `[ModulePermission(Employees, Delete)]` | Route `id` | Sets `IsDeleted = true`, `Status = false`, and revokes active `RefreshToken` records. | `Result` |

## 7. Database Model

**Entity:** `EmpUser` ( `Codeji.CMS.Repository.Entities.Employees.EmpUser` )

- **Collection Name**: `EmpUser`
- **Primary Key**: `UserId`
- **Foreign Keys**: `CompanyId`, `RoleId`, `Department` ( `DepartmentId` ), `JobRole` ( `JobRoleId` ), `ReportingManager` ( `UserId` ).
- **Core Fields**: `FirstName`, `LastName`, `Email`, `EmployeeId`, `Gender`, `DateOfBirth`, `DateOfJoining`, `EmploymentType` ( `1=FullTime`, `2=PartTime`, `3=Contractual`, `4=Internship` ), `PhoneNumber`, `PanNumber`, `BankAccountNumber`.
- **State & Lifecycle Subdocument** ( `EmpUser.Lifecycle` of type `EmployeeLifecycleState` ):
  - `Version` (Optimistic concurrency integer)
  - `EmploymentStatus` : `PreJoining`, `Active`, `NoticePeriod`, `Suspended`, `Resigned`, `Terminated`, `Retired`, `ContractEnded`, `Deceased`.

- `AvailabilityStatus`: `Working`, `AnnualLeave`, `SickLeave`, `MaternityLeave`, `Sabbatical`, `UnauthorizedAbsence`, `AbscondingUnderReview`.
- `AccountStatus`: `Active`, `Locked`, `TemporarilyDisabled`, `Limited`, `Suspended`, `Closed`.
- `PayrollStatus`: `Active`, `Hold`, `Paused`, `FinalSettlement`, `Closed`.
- `OffboardingStatus`: `None`, `PendingManagerApproval`, `PendingHrApproval`, `NoticePeriod`, `Discussion`, `Completed`, `Withdrawn`.
- `History`: `List<EmployeeLifecycleStatusHistory>` (Append-only transition log).

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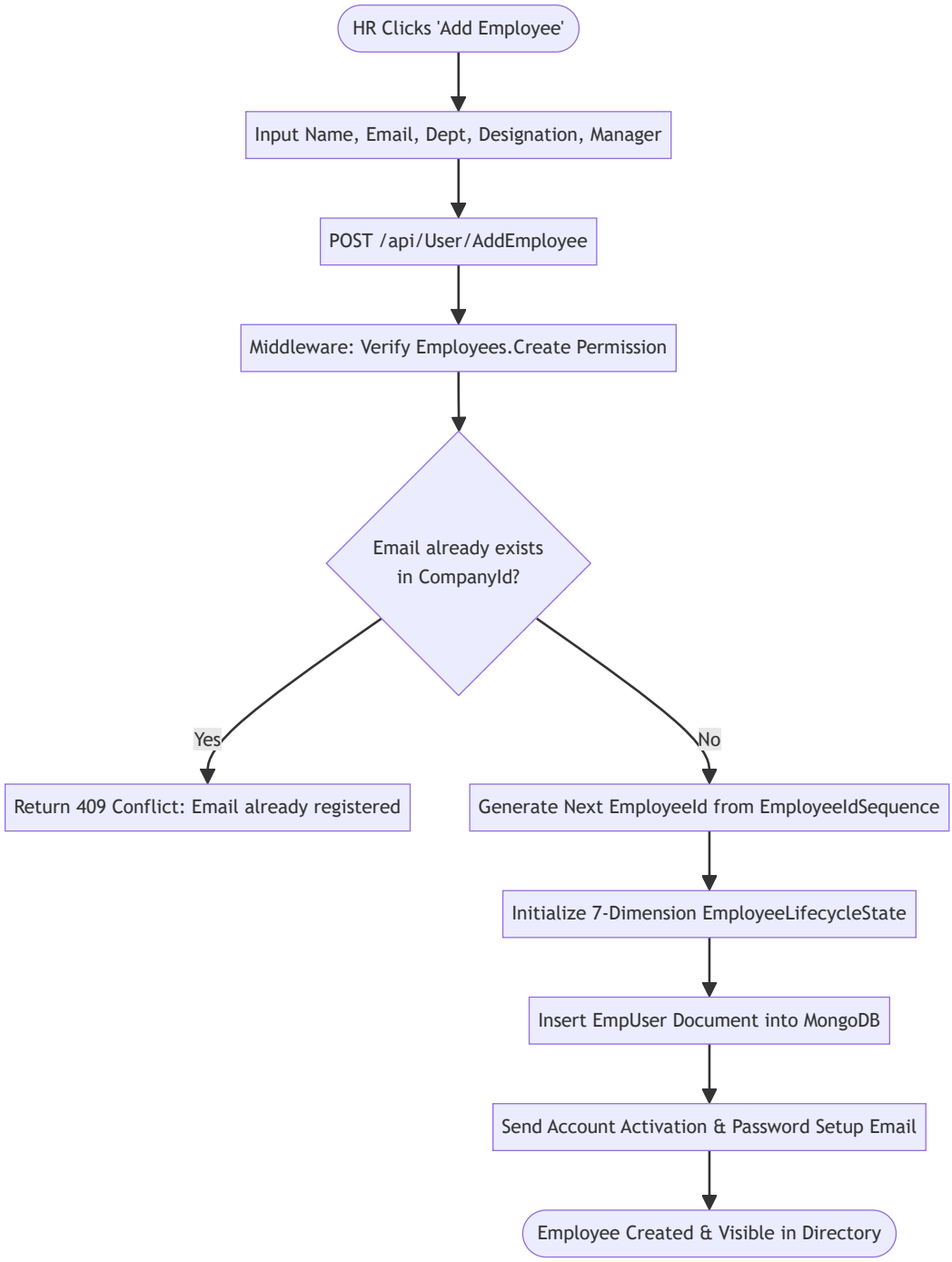
## 8. Business Rules

- **Unique Identity Rule**: No two active employees within the same `CompanyId` may share the same `Email` or `EmployeeId` (enforced via compound unique MongoDB indexes).
- **Self-Reporting Prohibition**: An employee's `ReportingManager` cannot be set to their own `UserId` (circular hierarchy check).
- **Regex Validation Rule**: `PanNumber` must match `[A-Z]{5}[0-9]{4}[A-Z]{1}<span class="math-inline">`; `BankAccountNumber` must match `^[0-9]{9,18}</span>`.
- **Active Authentication Rule**: `CompanyIdMiddleware` checks `EmpUser.Status == true` on every request. Flipping `Status = false` immediately blocks the user from executing any authenticated API call.

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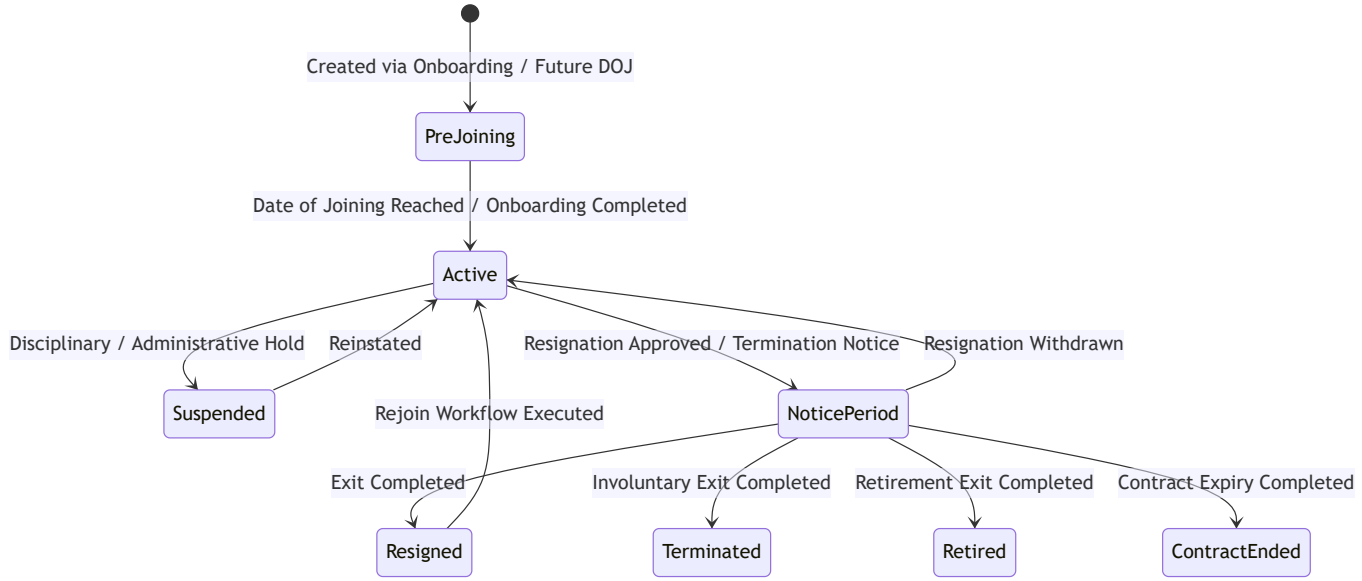
## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| View Employee Directory | Yes | Yes | Yes | Yes |
| Create / Invite Employee | Yes | Yes | No | No |
| Bulk Import Employees | Yes | Yes | No | No |
| Edit Employment Status / Role | Yes | Yes | No | No |
| Delete / Deactivate Employee | Yes | Yes | No | No |

## 12. Dependencies

- **Depends On**: `CompanyMaster` ( `Department` , `CompanyJobRole` , `EmployeeIdSequence` ), `Roles` .
- **Used By**: `Attendance` , `LeaveManagement` , `WorkFromHome` , `PayRoll` , `Offboarding` , `NoticeBoard` .
- **Reads From**: `Department` , `CompanyJobRole` , `Roles` , `EmployeeIdSequence` .
- **Writes To**: `EmpUser` , `RefreshToken` (on deactivation).

## 13. Cross-Module Events

- **Employee Creation**: Triggers default leave balance allocation ( `EmployeeLeaveBalance` ) for active company leave policies.
- **Employee Deactivation**: Immediately invalidates JWT session verification in `CompanyIdMiddleware` , revokes all `RefreshToken` documents, and excludes the user from future attendance reminder jobs.

## 14. Failure Scenarios

- **Partial Bulk Import Failure**: If an uploaded Excel sheet contains 50 valid rows and 3 rows with invalid PAN formats or duplicate emails, `BulkImportEmployees` processes the valid records (or reports row-level validation errors in `BulkImportEmployeesResponseDto`) so HR can correct the specific failing rows.
- **Concurrent Lifecycle Update**: Protected by `EmpUser.Lifecycle.Version`. Concurrent updates return HTTP 409 Conflict.

## 15. Edge Cases

- **Future Date of Joining (`PreJoining`)**: If an employee is added with a `DateOfJoining` in the next month, `EmploymentPeriod.IsPayrollIncludedForMonth` automatically excludes them from the current month's payroll run.
- **Missing Reporting Manager**: Top-level executives (e.g. CEO / Founder) can have `ReportingManager = null`; their leave and resignation requests route directly to HR Managers.

## 16. Current Strengths

- Rich 7-dimensional lifecycle state model (`EmployeeLifecycleState`) with append-only audit history.
- Comprehensive bulk import validation with detailed error reporting.
- Strict compound unique indexing preventing duplicate employee IDs or emails per tenant.

## 17. Current Gaps

- **Monolithic Service File**: `EmployeeService.cs` is 364 KB and handles core user CRUD alongside offboarding and lifecycle cases, creating high cognitive load during maintenance.

## 18. Recommended Improvements

- **High (P1)**: Refactor `EmployeeService.cs` into smaller, domain-focused services (`EmployeeDirectoryService`, `EmployeeImportService`, `EmployeeOffboardingService`).

## 19. Future Scope

- Automated SCIM 2.0 directory provisioning integration with Okta, Azure AD (Entra ID), and Google Workspace.

MOD-10 PART III `docs/project-documentation/modules/EMPLOYEE-PROFILE.md`

# Module Documentation — Employee Profile & Self-Service (`EMPLOYEE-PROFILE`)

**Document Status**: Complete & Authoritative
**Source-of-Truth**: Verified from Repository Implementation
**Primary Controller**: `UserController.cs`
**Primary Services**: `EmployeeService.cs`, `AccountServices.cs`
**Primary Frontend Components**: `UserProfile.tsx`, `AboutUser.tsx`, `WorkHistory.tsx`, `Setting.tsx`
**Last Verified**: September 2026

## 1. Module Overview

- **What the module is**: The personal employee portfolio and self-service workspace. It captures and presents an individual's personal identity, contact info, emergency contacts, education credentials, technical skills, professional certifications, prior employment work history, banking details, and security preferences.
- **Why HR organizations need it**: Empowers employees to maintain their own personal and professional records, reducing HR administrative overhead while ensuring payroll has accurate bank account numbers and tax identifiers.
- **What business problem it solves**: Eliminates out-of-date emergency contacts, incorrect bank IFSC/account details causing payroll bounce-backs, and lost skill/certification metadata for resource allocation.
- **Who uses it**: All authenticated **Employees**, with administrative oversight by **HR Managers** and **Company Administrators**.
- **Where it fits in the HR lifecycle**: Active throughout an employee's tenure; continually updated as employees acquire new certifications, change addresses, or update banking details.

## 2. Real-World Business Scenario

```
Employee: Ananya Deshmukh (User ID: USR-8821)
Department: Product Engineering
Action: Updating bank account details and adding AWS Certification
```

1. Ananya logs in and navigates to `/profile/about`.
2. **Bank Account Update**: Navigates to the Financial tab, enters her new bank account number `9876543210123` and PAN `ABCDE1234F`. The frontend and backend validate formats via regex.
3. **Certifications**: Navigates to the Certifications tab, clicks **Add Certification**, enters `AWS Certified Solutions Architect`, issuance date, expiration date, and uploads the verification PDF.
4. **Security Settings**: Navigates to `/profile/setting` and updates her password using `ChangePasswordRequest` (validating current password, minimum 8 characters, special characters).

## 3. Supported Actors

| Actor | Capabilities in Module | Server Enforcement |
|---|---|---|
| **Employee (Self-Service)** | Views own full profile; edits personal details, emergency contacts, skills, education, certifications, and password. | Enforced by `userId == currentUserId` checks in `EmployeeService`. Cannot edit own role, department, manager, or salary. |
| **HR Manager / Executive** | Views and edits all employee profiles within the tenant company; updates employment, department, designation, and compensation details. | Enforced via `[ModulePermission(Employees, Edit)]`. |
| **Reporting Manager** | Views subordinates' work history, education, skills, and emergency contact details. | Read-only visibility via directory. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/routing/ProfileRoutes.tsx` | Sub-routes: `about`, `workhistory`, `leave`, `setting`, `performance`. | React Router `<Outlet />` | `PrivateRoutes.tsx` (`/profile/*`, `/employees/:employeeid/*`) |
| `src/app/modules/users/UserProfile.tsx` | Top-level profile shell displaying avatar, full name, job title, department badge, presence status, and tab navigation. | `userServices.ts` | `ProfileRoutes.tsx` |
| `src/app/modules/users/about/AboutUser.tsx` | Tabbed view displaying Personal Info, Emergency Contacts, Education, Skills, and Certifications. | `userServices.ts` | `ProfileRoutes.tsx` |
| `src/app/modules/users/workhistory/WorkHistory.tsx` | Timeline view of prior external employment and internal company promotions. | `userServices.ts` | `ProfileRoutes.tsx` |
| `src/app/modules/users/usersetting/Setting.tsx` | User security settings (change password, notification preferences). | `accountServices.ts` | `ProfileRoutes.tsx` |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Codeji.CMS.API/Controllers/UserController.cs` | Endpoints for profile queries, personal info updates, skills/certifications/education CRUD, and avatar uploads. | `IEmployeeService`, `IAccountServices` |
| `Codeji.CMS.Services/Employees/EmployeeService.cs` | Manages profile subdocument updates, enforces self-edit field whitelists, and writes audit timestamps. | `IMongoDbRepository<EmpUser>`, `IMongoDbRepository<EmpSkills>`, `IMongoDbRepository<EmpEducationDetails>` |
| `Codeji.CMS.Services/Account/AccountServices.cs` | Validates current password and securely hashes new passwords using cryptographic salt. | `IMongoDbRepository<EmpUser>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request DTO | Validation & Business Logic | Response |
|---|---|---|---|---|---|---|
| `POST` | `/api/User/GetEmployeeProfile` | Fetches complete employee profile dossier. | `[Authorize]` | `GetEmployeeProfileRequest` | If requesting another user's profile, requires `Employees.View`. | `Result<EmployeeProfileDto>` |
| `POST` | `/api/User/UpdatePersonalInfo` | Updates contact details, address, blood group, emergency contact. | `[Authorize]` | `EmployeePersonalInfo` | Validates caller is editing own record OR has `Employees.Edit`. | `Result` |
| `POST` | `/api/User/AddUpdateCertification` | Adds or modifies a professional certification. | `[Authorize]` | `EmployeeCertificationRequestModel` | Validates date ranges (issue date <= expiry date). | `Result` |
| `POST` | `/api/User/AddUpdateEducation` | Adds or modifies degree, university, GPA, and graduation year. | `[Authorize]` | `EmployeeEducationRequestModel` | Validates graduation year format. | `Result` |
| `POST` | `/api/User/AddUpdateSkills` | Updates technical and soft skill tags. | `[Authorize]` | `EmployeeSkillsRequestModel` | Scopes skills to company master taxonomy. | `Result` |
| `POST` | `/api/User/UploadProfilePicture` | Uploads user profile photo avatar. | `[Authorize]` | `IFormFile` (Multipart) | Validates image MIME type, crops and saves to `/Uploads/Profiles/`. | `Result<string>` (URL) |
| `POST` | `/api/Account/ChangePassword` | Changes user login password. | `[Authorize]` | `ChangePasswordRequest` | Verifies current password; enforces minimum length and complexity. | `Result` |

## 7. Database Model

### Persistent Entities

1. **`EmpUser`** (Embeds personal fields, PAN, Bank details, blood group, address).
2. **`EmpEducationDetails`** (`Codeji.CMS.Repository.Entities.Employees.EmpEducationDetails`)
   - **Primary Key**: `EducationId`
   - **Fields**: `UserId`, `CompanyId`, `Degree`, `Institution`, `FieldOfStudy`, `StartYear`, `EndYear`, `GradeOrGpa`.
3. **`EmpCertificationDetails`** (`Codeji.CMS.Repository.Entities.Employees.EmpCertificationDetails`)
   - **Primary Key**: `CertificationId`
   - **Fields**: `UserId`, `CompanyId`, `CertificationName`, `IssuingOrganization`, `IssueDate`, `ExpiryDate`, `CredentialId`, `CertificateUrl`.
4. **`EmpWorkHistory`** (`Codeji.CMS.Repository.Entities.Employees.EmpWorkHistory`)
   - **Primary Key**: `WorkHistoryId`

- **Fields**: `UserId`, `CompanyId`, `CompanyName`, `JobTitle`, `StartDate`, `EndDate`, `Responsibilities`.

5. **`EmpSkills`** (`Codeji.CMS.Repository.Entities.Employees.EmpSkills`)
   - **Primary Key**: `SkillId`
   - **Fields**: `UserId`, `CompanyId`, `SkillName`, `ProficiencyLevel`.

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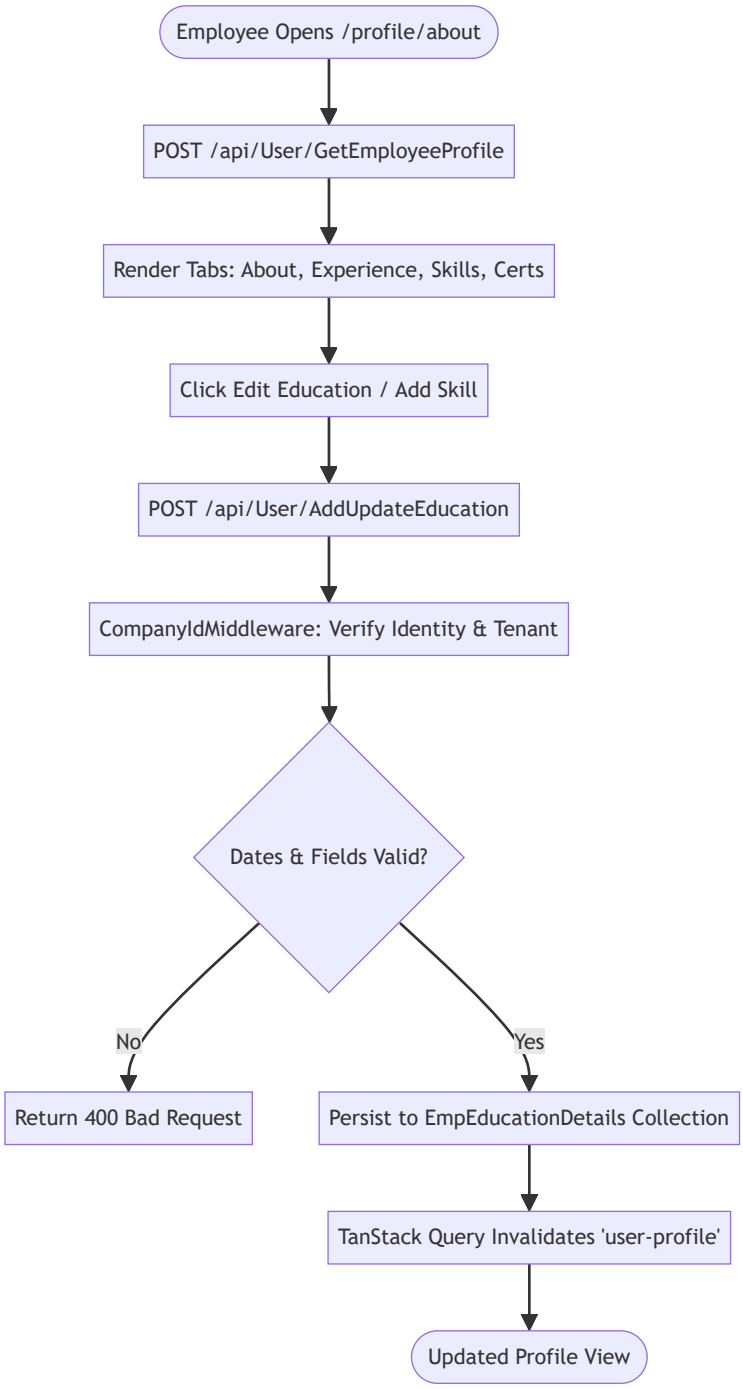
## 8. Business Rules

- **Self-Service Boundaries**: Employees editing their own profile (`EmployeeSelfEditDto`) may mutate `PhoneNumber`, `EmergencyContact`, `PersonalEmail`, `Address`, `BloodGroup`, `Education`, `Certifications`, and `Skills`. They **cannot** mutate `Role`, `Department`, `JobRole`, `DateOfJoining`, `ReportingManager`, or `EmploymentStatus`.
- **Tax & Bank Account Validations**:
  - `PanNumber`: Enforces regex `[A-Z]{5}[0-9]{4}[A-Z]{1}$`.
  - `BankAccountNumber`: Enforces numeric regex `^[0-9]{9,18}$`.
- **Avatar Storage**: Images are uploaded to `/Uploads/Profiles/{UserId}_{timestamp}.jpg`. The relative path `/fs/Profiles/...` is stored in `EmpUser.ProfileUrl`.

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## 9. Complete Workflow

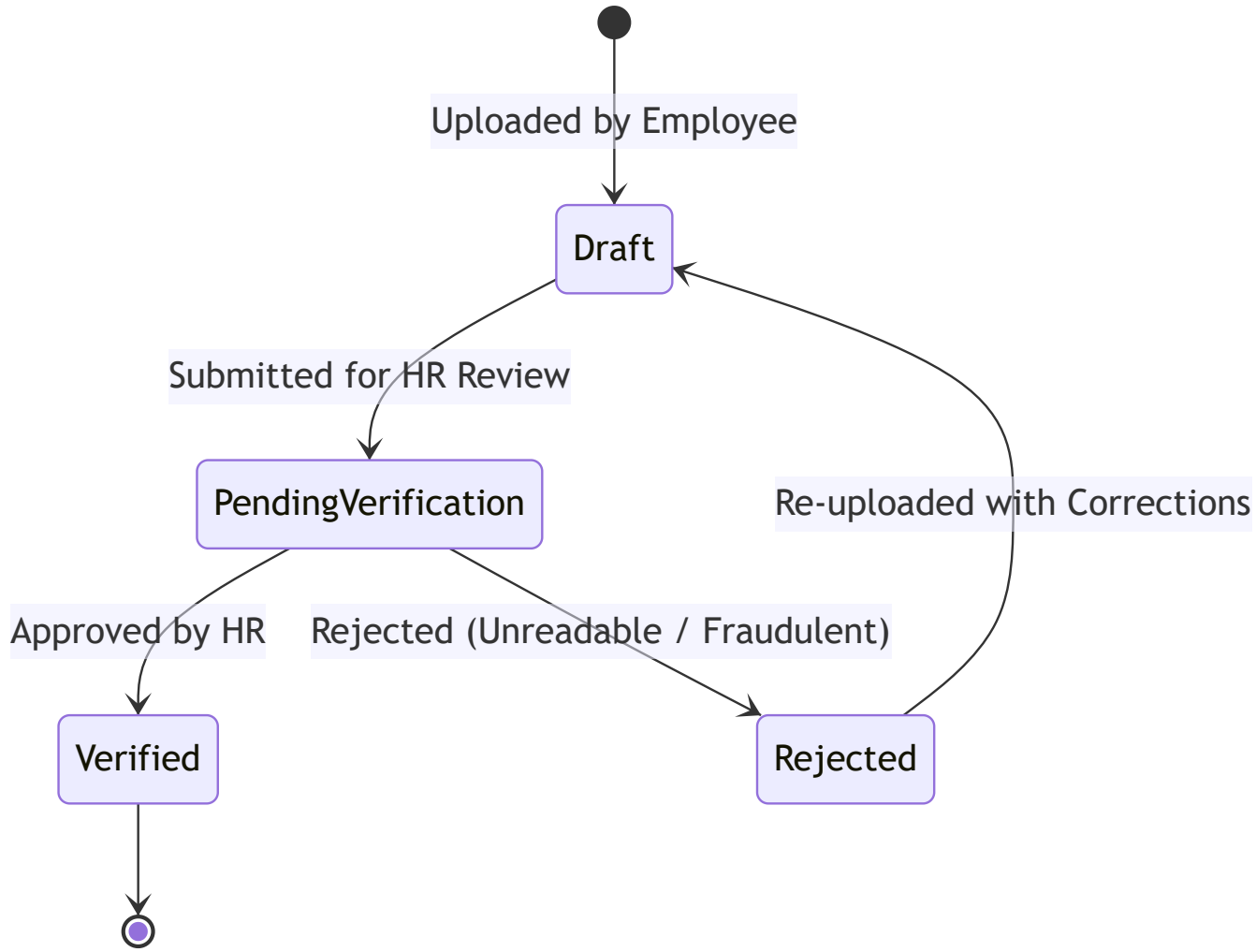
ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Profile credentials and document attachments operate under a verified/unverified lifecycle:

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

| Action | Admin | HR Manager | Reporting Manager | Employee |
|---|---|---|---|---|
| View Own Profile | Yes | Yes | Yes | Yes |
| Edit Own Personal Info / Skills | Yes | Yes | Yes | Yes |
| Change Own Password | Yes | Yes | Yes | Yes |
| Edit Colleague's Department / Manager | Yes | Yes | No | No |
| View Other Employee's Sensitive Bank/PAN | Yes | Yes | No | No |

## 12. Dependencies

- **Depends On**: `EmpUser` (must exist in active state).
- **Used By**: `PayRoll` (consumes PAN and Bank Account for salary slips), `LeaveManagement` (profile leave tracker tab).
- **Reads From**: `EmpUser`, `EmpEducationDetails`, `EmpCertificationDetails`, `EmpSkills`, `EmpWorkHistory`.

- **Writes To**: `EmpUser`, `EmpEducationDetails`, `EmpCertificationDetails`, `EmpSkills`, `EmpWorkHistory`.

## 13. Cross-Module Events

- Updating Bank Account details immediately impacts subsequent AutoPayroll pay slip generation, ensuring the new account number appears on the official PDF voucher.

## 14. Failure Scenarios

- **Invalid PAN Format**: If an employee enters a lowercase or incorrect alphanumeric sequence (e.g. `abc12345`), the backend model validator catches it via `[RegularExpression]` and returns HTTP 400 with message `"Invalid Pan Number"`.

## 15. Edge Cases

- **Missing Emergency Contact**: The system highlights incomplete emergency contacts on the employee dashboard to encourage compliance before fieldwork or travel assignments.

## 16. Current Strengths

- Clean, tabbed UI allowing employees to self-manage certifications, education, and skills.
- Strict server-side regex validations on financial and tax identifiers.

## 17. Current Gaps

- **Document Expiration Alerts**: The system currently does not trigger automated email alerts when an employee's professional certification reaches its `ExpiryDate`.

## 18. Recommended Improvements

- **Medium**: Add a scheduled background worker scanning `EmpCertificationDetails` to alert employees and HR 30 days before certification expiry.

## 19. Future Scope

- Integration with LinkedIn Learning and Credly APIs for automatic verification and synchronization of digital credential badges.

# PROBATION MODULE (`PROBATION.md`)

## 1. Module Overview

The **Probation Module** governs the post-joining evaluation phase (`Probation`) before an employee is confirmed as a permanent staff member (`Active`) or separated under probation notice rules. It connects `EmployeeLifecyclePolicy` probation duration rules (`ProbationDays`), `EmployeeLifecycle` state orchestration (`ProbationEndDateUtc`, `EmploymentStatus`), `LeavePolicy` eligibility (`AllowOnProbation`), and `EmployeeExitPolicy` notice period differentiation (`ProbationNoticeDays` vs `NoticeDays`).

## 2. Real-World Business Scenario

1. **Joining on Probation**: When Rahul Sharma completes Day-1 onboarding on `2026-04-01`, `EmployeeLifecycleService` transitions `EmploymentStatus` to `Probation` (`2`) and computes `ProbationEndDateUtc = DateOfJoining + 90 days` (`2026-06-30`) using the company's `EmployeeLifecyclePolicy`.
2. **Probation Restrictions**:
   - While Rahul is in `EmploymentStatus = Probation`, `LeaveManagementService` checks `LeavePolicy.AllowOnProbation`. Casual Leave (`CL`, `AllowOnProbation = true`) is permitted, whereas Earned Leave (`EL`, `AllowOnProbation = false`) is blocked.
   - If Rahul initiates resignation during probation, `OffboardingPolicyResolver` snapshots `ProbationNoticeDays = 15` instead of standard `NoticeDays = 60`.
3. **Confirmation**: Upon reaching `ProbationEndDateUtc` and passing managerial review, HR invokes `POST /api/employee-lifecycle/{id}/confirm-probation`, transitioning `EmploymentStatus` to `Active` (`3`) and unlocking full leave/benefit eligibility.

## 3. Actors

| Actor | Role / Capabilities |
|---|---|
| **HR Admin / HR Manager** | Configures `EmployeeLifecyclePolicy.ProbationDays`, monitors employees in `Probation`, extends probation (`ProbationEndDateUtc`), or confirms employment (`Active`). |
| **Reporting Manager** | Reviews probation performance and attendance reliability before confirmation. |
| **Employee (Probationer)** | Views probation status, `ProbationEndDateUtc`, and eligible leave policies in self-service portal. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/employees/pages/EmployeesPage.tsx` | Displays `EmploymentStatus` badge (`Probation` vs `Active`) and probation filter tabs. | `GET /api/employees` | HR / Admin |

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/userProfile/UserProfilePage.tsx` | Shows joining date, probation end date, and confirmation status on employee profile. | `GET /api/employee-lifecycle/{id}` | Employee / HR |
| `src/app/modules/settings/pages/LifecyclePolicySettings.tsx` | Configures `ProbationDays` and confirmation rules per company/department/role. | `POST /api/employee-lifecycle/policy` | HR Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/EmployeeLifecycleController.cs` | Exposes lifecycle status inspection, probation extension, and confirmation endpoints. | `IEmployeeLifecycleService` |
| `Services/EmployeeLifecycleService.cs` | Evaluates `EmployeeLifecyclePolicy`, sets `ProbationEndDateUtc`, validates transition `Probation -> Active`. | `IMongoRepository<EmployeeLifecycle>`, `IMongoRepository<EmployeeLifecyclePolicy>` |
| `Services/LeaveManagementService.cs` | Enforces `LeavePolicy.AllowOnProbation` against `EmployeeLifecycle.EmploymentStatus == Probation`. | `IMongoRepository<LeavePolicy>`, `IMongoRepository<EmployeeLifecycle>` |
| `Services/OffboardingPolicyResolver.cs` | Applies `ProbationNoticeDays` when an employee exits while in `Probation` state. | `IMongoRepository<EmployeeExitPolicy>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Module Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/employee-lifecycle/{employeeId}` | Fetches current 7-dimensional lifecycle record including `ProbationEndDateUtc`. | JWT (`employees:read`) | Returns `EmployeeLifecycleDto` | None |
| `POST` | `/api/employee-lifecycle/{employeeId}/confirm-probation` | Confirms employee from `Probation` (`2`) to `Active` (`3`). | JWT (`employees:update`) | `ConfirmProbationDto` | Updates `EmployeeLifecycle`, logs audit event, sends notification |
| `PUT` | `/api/employee-lifecycle/{employeeId}/extend-probation` | Extends `ProbationEndDateUtc` with mandatory reason. | JWT (`employees:update`) | `ExtendProbationDto` (`NewEndDateUtc`, `Reason`) | Updates `ProbationEndDateUtc`, notifies employee |

## 7. Database Model

**`EmployeeLifecycle` & `EmployeeLifecyclePolicy`**

- `EmployeeLifecycle.EmploymentStatus`: `Probation = 2`, `Active = 3`

- `EmployeeLifecycle.ProbationEndDateUtc` : `DateTime?`
- `EmployeeLifecyclePolicy.ProbationDays` : `int` (default `90` days)
- `LeavePolicy.AllowOnProbation` : `bool`
- `EmployeeExitPolicy.ProbationNoticeDays` : `int`

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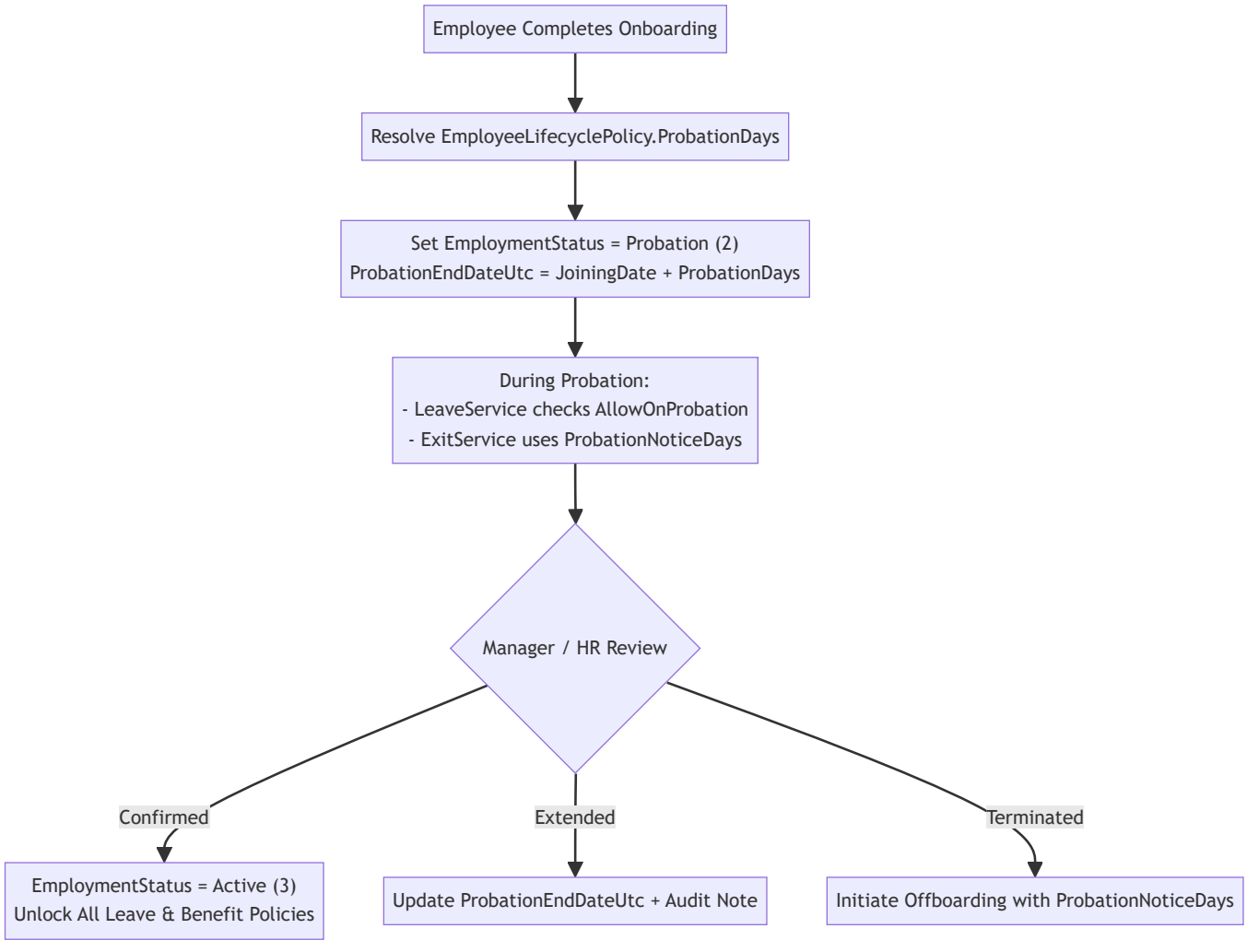
## 8. Business Rules

- **Business Rule**: Default `ProbationDays` is resolved via `EmployeeLifecyclePolicy` hierarchy (Department+Role -> Department -> Role -> Company).
- **Policy Rule**: Leave types with `AllowOnProbation == false` reject applications if the employee's `EmploymentStatus == EmploymentStatus.Probation`.
- **Policy Rule**: Resignations submitted while `EmploymentStatus == EmploymentStatus.Probation` use `ProbationNoticeDays` instead of `NoticeDays`.
- **State Transition Rule**: Only employees in `EmploymentStatus.Probation` ( `2` ) or `EmploymentStatus.Onboarding` ( `1` ) can transition to `EmploymentStatus.Active` ( `3` ).

---

## 9. Complete Workflow

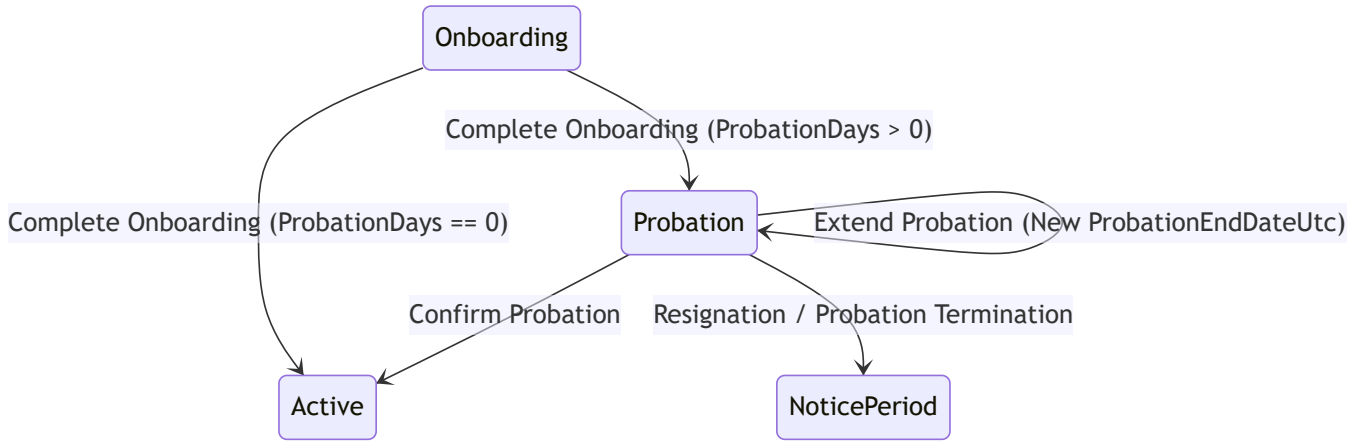
ARCHITECTURE / WORKFLOW DIAGRAM




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## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `employees:read` — View probation timeline and status.
- `employees:update` — Confirm or extend probation.
- `settings:update` — Modify `EmployeeLifecyclePolicy.ProbationDays`.

## 12. Dependencies

- **Upstream**: `ONBOARDING` (triggers entry into `Probation`), `POLICY` (`EmployeeLifecyclePolicy`).
- **Downstream**: `LEAVE` (`AllowOnProbation` check), `NOTICE-PERIOD` / `OFFBOARDING` (`ProbationNoticeDays`).

## 13. Cross-Module Events

- **Probation Confirmation**: Unlocks `LeavePolicy` types where `AllowOnProbation == false` and switches notice requirement from `ProbationNoticeDays` to standard `NoticeDays`.

## 14. Failure Scenarios

- **Confirmation Before Joining**: Blocked by state machine validation (`EmploymentStatus` must be `Probation`).
- **Missing Lifecycle Policy**: Falls back to system default (`90` days) if no tenant `EmployeeLifecyclePolicy` exists.

## 15. Edge Cases

- **Zero-Day Probation (`ProbationDays == 0`)**: Direct-confirmed senior hires skip `Probation` (`2`) and transition directly from `Onboarding` (`1`) to `Active` (`3`).

## 16. Current Strengths

- Tightly integrated with both **Leave Eligibility** (`AllowOnProbation`) and **Exit Notice Calculation** (`ProbationNoticeDays`).

## 17. Current Gaps

- Automatic background promotion ( `Probation -> Active` ) on `ProbationEndDateUtc` expiry is manual rather than driven by an automated hosted service reminder/workflow.

## 18. Recommended Improvements

- Add a daily check in `AttendanceReminderHostedService` to notify HR and Reporting Managers 7 days prior to `ProbationEndDateUtc` .

## 19. Future Scope

- Structured 30-60-90 day probation review scorecard forms with manager sign-off gates.

MOD-12 PART III docs/project-documentation/modules/WORK-SHIFT.md

# WORK SHIFT & OFFICE SCHEDULE MODULE ( WORK-SHIFT.md )

## 1. Module Overview

The **Work Shift & Office Schedule Module** defines standard and department/employee-specific working hours ( CompanyOfficeSchedule ), check-in grace periods ( GraceMinutes ), earliest check-out thresholds ( CheckOutAllowedFrom ), and half-day duration requirements ( HalfDayThresholdHours ). It is resolved dynamically by EffectiveOfficeScheduleService for every daily attendance evaluation.

## 2. Real-World Business Scenario

1. **Company Schedule Setup**: HR Admin configures the default company office schedule via OfficeScheduleSettingsService : StartTime = "09:30" , EndTime = "18:30" , GraceMinutes = 15 ( CheckInAllowedUntil = "09:45" ), CheckOutAllowedFrom = "18:15" , FullDayHours = 8.0 , and HalfDayThresholdHours = 4.0 .
2. **Effective Schedule Resolution**: When Rahul Sharma punches in on 2026-04-14 , EffectiveOfficeScheduleService resolves the effective schedule for Rahul's (CompanyId, DepartmentId, Date) and supplies the exact cutoff timestamps to AttendanceStatusCalculationService .

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin / Company Admin** | Configures CompanyOfficeSchedule timings, grace minutes, and half-day hour thresholds. |
| **Employee** | Views assigned shift timings on the attendance and calendar dashboard. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| src/app/modules/settings/pages/OfficeScheduleSettings.tsx | Form for configuring StartTime , EndTime , GraceMinutes , CheckInAllowedUntil , CheckOutAllowedFrom , and HalfDayThresholdHours . | GET /api/office-schedule , PUT /api/office-schedule | HR Admin |
| src/app/modules/attendance/pages/AttendancePage.tsx | Displays expected shift start/end times and grace window next to daily check-in button. | GET /api/office-schedule/effective | Employee |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Controllers/OfficeScheduleController.cs` | CRUD and effective resolution endpoints for `CompanyOfficeSchedule`. | `IOfficeScheduleSettingsService`, `IEffectiveOfficeScheduleService` |
| `Services/OfficeScheduleSettingsService.cs` | Validates and persists `CompanyOfficeSchedule` records per tenant. | `IMongoRepository<CompanyOfficeSchedule>` |
| `Services/EffectiveOfficeScheduleService.cs` | Resolves the active schedule for a specific employee and date, adjusting for half-day leaves/WFH. | `IMongoRepository<CompanyOfficeSchedule>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
| --- | --- | --- | --- | --- | --- |
| `GET` | `/api/office-schedule` | Retrieves current company office schedule settings. | JWT (`settings:read`) | `CompanyOfficeScheduleDto` | None |
| `PUT` | `/api/office-schedule` | Updates shift start/end times, grace window, and minimum hours. | JWT (`settings:update`) | `UpdateOfficeScheduleDto` | Updates `CompanyOfficeSchedule` |
| `GET` | `/api/office-schedule/effective` | Returns effective schedule for an employee on a target date. | JWT (`attendance:read`) | `EffectiveScheduleDto` | None |

## 7. Database Model

**`CompanyOfficeSchedule` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/CompanyOfficeSchedule.cs`)**

- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `StartTime` (`TimeSpan` / `string`, e.g., `"09:30"`)
- `EndTime` (`TimeSpan` / `string`, e.g., `"18:30"`)
- `GraceMinutes` (`int`, e.g., `15`)
- `CheckInAllowedUntil` (`string`, e.g., `"09:45"`)
- `CheckOutAllowedFrom` (`string`, e.g., `"18:15"`)
- `FullDayHours` (`double`, default `8.0`)
- `HalfDayThresholdHours` (`double`, default `4.0`)
- `IsActive` (`bool`)

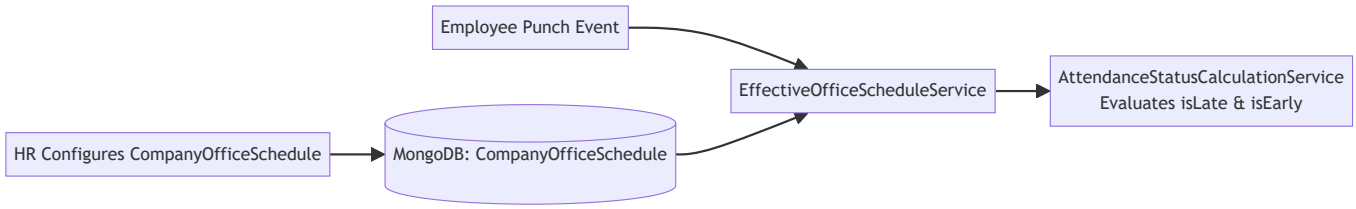
## 8. Business Rules

- **Validation Rule**: `EndTime` must be strictly after `StartTime`, and `HalfDayThresholdHours` must be `< FullDayHours`.

- **Grace Calculation Rule**: `CheckInAllowedUntil` equals `StartTime + GraceMinutes`. Any check-in after `CheckInAllowedUntil` triggers `isLate = true` in `AttendanceStatusCalculationService`.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

`CompanyOfficeSchedule` operates as an active configuration singleton/versioned rule per company (`Active` ↔ `Superseded`).

## 11. Permissions

- `settings:read` / `settings:update` — View and modify `CompanyOfficeSchedule`.

## 12. Dependencies

- **Upstream**: `COMPANY-SETTINGS`.
- **Downstream**: `ATTENDANCE` (`AttendanceStatusCalculationService`), `WFH` (half-day window alignment).

## 13. Cross-Module Events

- Schedule updates immediately apply to subsequent `AttendanceStatusCalculationService` evaluations for unlocked attendance months.

## 14. Failure Scenarios

- **Missing Tenant Schedule**: `EffectiveOfficeScheduleService` falls back to safe default office hours (`09:30–18:30`, `15m` grace) so daily punches never crash.

## 15. Edge Cases

- **Night Shifts Crossing Midnight**: Current `StartTime` / `EndTime` comparisons assume same-calendar-day shifts (`StartTime < EndTime`).

## 16. Current Strengths

- Clean decoupling between schedule resolution ( `EffectiveOfficeScheduleService` ) and status computation ( `AttendanceStatusCalculationService` ).

## 17. Current Gaps

- Multi-shift rotation rosters (e.g., Morning/Evening/Night rotating weekly per employee) are not yet modeled as separate roster assignments.

## 18. Recommended Improvements

- Support multi-shift templates with employee-date roster assignment ( `EmployeeShiftRoster` ).

## 19. Future Scope

- Cross-midnight shift support and automated overtime calculation beyond `FullDayHours` .

# COMPANY CALENDAR & WORKING DAYS MODULE (`CALENDAR.md`)

## 1. Module Overview

The **Company Calendar & Working Days Module** provides the core temporal infrastructure for **Codeji CMS / HRMS**. It unites weekly off configurations (`WeeklyOffSetting`), declared holidays (`Holidays`), and active employee schedules through `CompanyWorkingCalendarService`. It is the central authority for calculating working day counts, business day intervals, and payroll divisor calculations.

## 2. Real-World Business Scenario

1. **Weekly Off Configuration**: HR Admin configures the standard company working week: Sunday is always Off, Saturday has alternating off rules (2nd and 4th Saturday Off, 1st, 3rd, and 5th Saturday Working) via `WeeklyOffSetting`.
2. **Monthly Working Days Resolution**: For April 2026 (30 calendar days):
   - Sundays: 4 days
   - 2nd & 4th Saturdays: 2 days
   - Declared Holidays: 1 day (Mahavir Jayanti)
   - Total Non-Working Days = $4 + 2 + 1 = 7$ days
   - `CompanyWorkingCalendarService.GetWorkingDaysInMonth(2026, 4)` returns **23 working days**.
3. **Payroll Integration**: When `AutoPayrollServices` executes under `WORKING_DAYS` divisor mode, it queries `CompanyWorkingCalendarService` to retrieve 23 days, ensuring daily salary rates reflect actual working days.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin / Company Admin** | Configures `WeeklyOffSetting` (e.g., 5-day week, 6-day week, alternate Saturdays), views calendar audits. |
| **All Employees** | Views company working calendar, upcoming holidays, weekly offs, and working days on dashboard. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/calendar/pages/CompanyCalendarPage.tsx` | Interactive monthly calendar displaying working days, weekly offs, and holidays. | `GET /api/calendar/month` | All Users |
| `src/app/modules/settings/pages/WeeklyOffSettingsPage.tsx` | Configuration UI for weekly off patterns (days of week, alternating weekend logic). | `GET /api/calendar/weekly-off`, `PUT /api/calendar/weekly-off` | HR Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/CalendarController.cs` | Exposes calendar queries, monthly day breakdowns, and weekly off settings endpoints. | `ICalendarServices`, `ICompanyWorkingCalendarService` |
| `Services/CalendarServices.cs` | Assembles aggregate monthly view (holidays + weekly offs + leaves + birthdays) for frontend views. | `ICompanyWorkingCalendarService`, `IMongoRepository<Holidays>`, `IMongoRepository<EmpUser>` |
| `Services/CompanyWorkingCalendarService.cs` | Core deterministic service providing `IsWorkingDay(date)`, `GetWorkingDaysBetween(start, end)`, and `GetWorkingDaysInMonth(year, month)`. | `IMongoRepository<WeeklyOffSetting>`, `IMongoRepository<Holidays>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/calendar/month` | Returns composite calendar events for target year/month. | JWT (`calendar:read`) | `CalendarMonthResponseDto` | None |
| `GET` | `/api/calendar/working-days` | Returns exact working day count and dates for a month. | JWT (`calendar:read`) | `{ Year, Month }` -> `{ TotalWorkingDays, WorkingDates }` | None |
| `GET` | `/api/calendar/weekly-off` | Retrieves company weekly off configuration. | JWT (`settings:read`) | `WeeklyOffSettingDto` | None |
| `PUT` | `/api/calendar/weekly-off` | Updates weekly off pattern. | JWT (`settings:update`) | `UpdateWeeklyOffDto` | Recalculates working calendar caches |

## 7. Database Model

**`WeeklyOffSetting` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/WeeklyOffSetting.cs`)**

- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `OffDaysOfWeek` (`List<DayOfWeek>`, e.g., `[Sunday]`)
- `AlternatingOffRules` (`List<AlternatingOffRule>`, e.g., Saturday rules: 2nd and 4th Saturday)
- `IsActive` (`bool`)
- `UpdatedByUserId` (`string`)
- `UpdatedAtUtc` (`DateTime`)

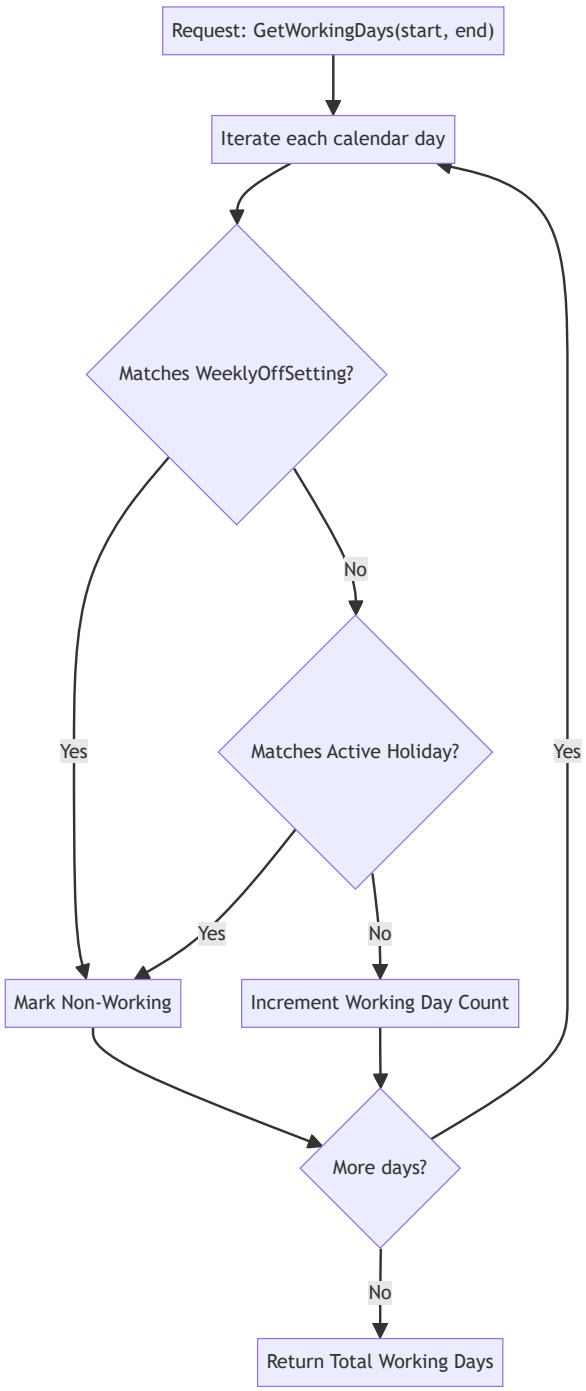
## 8. Business Rules

- **Non-Working Day Precedence**: A day is non-working if it is in `Holidays` OR matches `WeeklyOffSetting`.

- **Deduplication Rule**: If a declared holiday falls on a weekend/weekly off, `CompanyWorkingCalendarService` counts the day exactly once as non-working.
- **Date Boundary Enforcement**: `GetWorkingDaysBetween(startDate, endDate)` strictly requires `startDate <= endDate`.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

`WeeklyOffSetting` acts as a tenant configuration singleton. Updates trigger cache invalidation in `CompanyWorkingCalendarService`.

## 11. Permissions

- `calendar:read` — View calendar dates and monthly work day counts.
- `settings:update` — Configure `WeeklyOffSetting`.

## 12. Dependencies

- **Upstream**: `COMPANY-SETTINGS`, `HOLIDAY` (`Holidays`).
- **Downstream**: `ATTENDANCE`, `LEAVE` (deducts non-working days from leave requests), `WFH`, `PAYROLL` (`AutoPayrollServices` divisor calculation).

## 13. Cross-Module Events

- Changes to `WeeklyOffSetting` invalidate cached working day matrices across Leave and Payroll calculations.

## 14. Failure Scenarios

- **Missing WeeklyOffSetting**: Default fallback assumes Saturday and Sunday are off (5-day work week).

## 15. Edge Cases

- **Leap Years (February 29)**: Handled seamlessly via standard `.NET 10` `DateTime.DaysInMonth` calculations.

## 16. Current Strengths

- Centralized temporal logic eliminates discrepancies between leave day consumption and payroll working day divisors.

## 17. Current Gaps

- Roster-based employee-specific off days (e.g., shift workers who have Tuesday/Wednesday off) are currently unified at company/department setting level.

## 18. Recommended Improvements

- Support employee-level custom weekly off overrides for 24/7 operations.

## 19. Future Scope

- Integration with external Google Calendar / Microsoft Outlook feeds via iCal / CalDAV endpoints.

MOD-14 PART III docs/project-documentation/modules/HOLIDAY.md

# HOLIDAYS MODULE (`HOLIDAY.md`)

## 1. Module Overview

The **Holidays Module** manages national, cultural, regional, and company-declared holidays (`Holidays`). It feeds directly into `CompanyWorkingCalendarService` to compute true net working days in a calendar month, prevent invalid leave/WFH bookings on non-working days, and supply holiday calendar feeds to frontend views.

## 2. Real-World Business Scenario

1. **Annual Calendar Declaration**: At the start of 2026, HR Admin uploads the company holiday list (`POST /api/holidays/batch`). This includes Republic Day (`2026-01-26`, National), Holi (`2026-03-25`, Cultural), and Independence Day (`2026-08-15`, National).
2. **Leave Application Interception**: When Rahul Sharma requests leave from `2026-08-14` to `2026-08-17`, `LeaveManagementService` calls `CompanyWorkingCalendarService.GetWorkingDaysBetween(startDate, endDate)`. Because `2026-08-15` is a declared holiday and `2026-08-16` is Sunday, net leave days applied is calculated as `2` days instead of `4`.
3. **Payroll Working Days Computation**: When `AutoPayrollServices` computes working days for August under divisor policy `WORKING_DAYS`, `Holidays` are deducted from calendar days alongside weekly offs.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin / Company Admin** | Adds, edits, or deletes holiday entries (`Holidays`), categorizes types, specifies applicable locations/departments. |
| **Employee** | Views upcoming holidays in self-service calendar and dashboard widgets. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/calendar/pages/HolidayCalendarPage.tsx` | Visual calendar highlighting all declared holidays with category chips. | `GET /api/holidays` | Employee / HR |
| `src/app/modules/settings/pages/HolidayManagementPage.tsx` | CRUD table for HR to add, edit, or remove holidays for the active year. | `POST /api/holidays`, `PUT /api/holidays/{id}`, `DELETE /api/holidays/{id}` | HR Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/HolidaysController.cs` | Exposes REST endpoints for holiday listings, creations, updates, and removals. | `IHolidaysService` |
| `Services/HolidaysService.cs` | Validates holiday dates, prevents duplicates, enforces tenant boundaries, and manages holiday caches. | `IMongoRepository<Holidays>` |
| `Services/CompanyWorkingCalendarService.cs` | Consumes `Holidays` collection to determine if a date is a non-working business day. | `IMongoRepository<Holidays>`, `IMongoRepository<WeeklyOffSetting>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/holidays` | Lists all holidays for the tenant in a given year. | JWT (`calendar:read`) | `List<HolidaysDto>` | None |
| `POST` | `/api/holidays` | Creates a new declared holiday. | JWT (`settings:update`) | `CreateHolidayDto` -> `HolidaysDto` | Invalidates working calendar cache |
| `PUT` | `/api/holidays/{id}` | Modifies holiday details (name, date, category). | JWT (`settings:update`) | `UpdateHolidayDto` | Invalidates working calendar cache |
| `DELETE` | `/api/holidays/{id}` | Soft-deletes a declared holiday (`IsDeleted = true`). | JWT (`settings:update`) | None | Updates working calendar cache |

## 7. Database Model

**`Holidays` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/Holidays.cs`)**

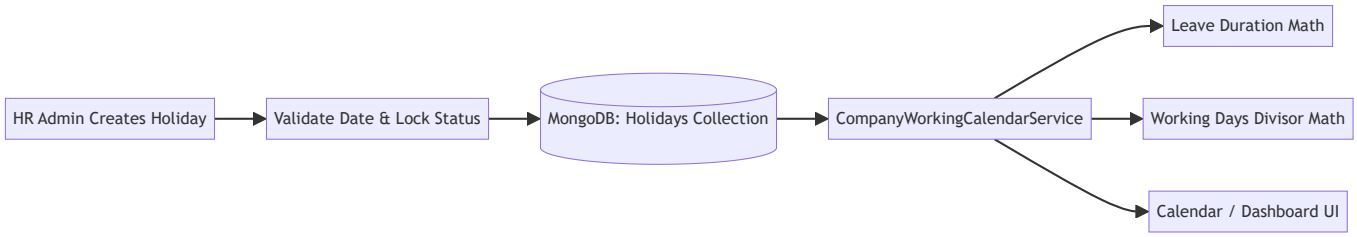
- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `Name` (`string`, e.g., `"Republic Day"`)
- `Date` (`DateTime`)
- `Description` (`string`)
- `Type` (`string`, `"National"` / `"Cultural"` / `"Optional"` / `"Gazetted"`)
- `IsRestricted` (`bool`, indicates optional/floating holiday)
- `IsDeleted` (`bool`, soft delete)
- `CreatedBy` (`string`)
- `CreatedDate` (`DateTime`)

## 8. Business Rules

- **Unique Date Rule**: A tenant cannot have two non-restricted holidays defined for the exact same calendar date.
- **Lock Protection Rule**: Holidays cannot be added, edited, or removed for a past month where `AttendanceMonthLock.IsLocked == true`.
- **Working Calendar Integration**: Every active holiday entry on a weekday automatically decrements the count of billable/payable working days for that month when using `WORKING_DAYS` payroll divisor policy.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

`Holidays` entities represent immutable date declarations ( `Active` ↔ `SoftDeleted` ).

## 11. Permissions

- `calendar:read` — View holiday schedules.
- `settings:update` — Create, update, or remove declared holidays.

## 12. Dependencies

- **Upstream**: `COMPANY-SETTINGS`.
- **Downstream**: `CALENDAR` ( `CompanyWorkingCalendarService` ), `LEAVE` (duration calculation), `PAYROLL` ( `AutoPayrollServices` divisor calculation).

## 13. Cross-Module Events

- Modifying a holiday recalculates pending leave request durations and updates `CompanyWorkingCalendarService` working day matrices.

## 14. Failure Scenarios

- **Duplicate Holiday Entry**: Service returns `409 Conflict` if the date is already claimed.

## 15. Edge Cases

- **Holiday Falling on a Weekly Off (e.g., Sunday)**: `CompanyWorkingCalendarService` deduplicates the date so it is not double-subtracted from monthly working days.

## 16. Current Strengths

- High reuse across Leave, Attendance, Calendar, and Payroll engines via unified `CompanyWorkingCalendarService`.

## 17. Current Gaps

- Location-specific holiday calendars (e.g., US office vs India office) currently share a single tenant holiday table unless partitioned by custom tags.

## 18. Recommended Improvements

- Add `BranchId` or `LocationId` foreign key to `Holidays` for multi-location enterprises.

## 19. Future Scope

- Support for optional/floating holiday allowance booking workflows.

# ATTENDANCE MODULE (`ATTENDANCE.md`)

## 1. Module Overview

The **Attendance Module** is a core operational engine in **Codeji CMS / HRMS** that tracks daily employee check-in/check-out activity, evaluates shift compliance (`P`, `HD`, `LHD`, `ED`, `LHD+ED`), calculates late/early penalties (`AttendancePenaltyService`), enforces offboarding cutoffs (`EmployeeExitAttendanceCutoff`), and produces locked monthly attendance summaries (`MonthlyAttendanceSummary` & `AttendanceMonthLock`) required by `AutoPayrollServices`.

## 2. Real-World Business Scenario

1. **Daily Check-In / Check-Out**:
   - Rahul Sharma's effective office schedule (`CompanyOfficeSchedule`) is `09:30` to `18:30` with `CheckInAllowedUntil = 09:45` (`15m` grace) and `CheckOutAllowedFrom = 18:15`.
   - On Tuesday, Rahul checks in at `10:02` and checks out at `18:05`.
   - `AttendanceStatusCalculationService.CalculateStatus` detects `isLate = true` (`10:02 > 09:45`) and `isEarly = true` (`18:05 < 18:15`), assigning daily status **`LHD+ED`** (`Late Half Day + Early Departure`).
2. **Monthly Penalty Evaluation**:
   - Over April 2026, Rahul accumulates 4 occurrences of `LHD` / `ED` / `LHD+ED`.
   - `AttendancePenaltyService.EvaluateMonthlyPenaltiesAsync` compares this against `AttendancePenaltyPolicy.CombinedLhdEdMonthlyLimit = 3`. Because `4 > 3`, it raises an `AttendancePayrollException` (`Status = "PENDING_REVIEW"`, `PenaltyDays = 0.5`).
3. **HR Resolution & Month Lock**:
   - HR Manager reviews the exception via `POST /api/attendance-penalty/resolve` (`APPROVED`, `FinalPenaltyDays = 0.5`).
   - HR then runs `POST /api/attendance/lock-month`, which calls `AttendanceMonthLock.ValidateAndLock()` and locks `MonthlyAttendanceSummary` (`IsApproved = true`, `IsLocked = true`) so April payroll can run.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Employee** | Performs daily check-in/check-out, views personal attendance calendar, status codes (`P`, `LHD`, `ED`, `HD`), and monthly summary. |
| **Reporting Manager** | Views team attendance dashboard, late arrival trends, and attendance exceptions. |
| **HR Admin / Payroll Admin** | Configures `AttendancePenaltyPolicy`, edits/regularizes attendance (`AttendanceAuditLog`), resolves `AttendancePayrollException` items, and locks `AttendanceMonthLock`. |
| **Background Worker (`AttendanceReminderHostedService`)** | Dispatches automated check-in/check-out reminders (`AttendanceReminderDispatchLog`) according to `AttendanceReminderSetting`. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/attendance/pages/AttendancePage.tsx` | Main attendance calendar, daily punch widget, and status legend (`P`, `HD`, `LHD`, `ED`, `LHD+ED`). | `GET /api/attendance`, `POST /api/attendance/check-in`, `POST /api/attendance/check-out` | Employee / HR |
| `src/app/modules/attendance/components/MonthlySummaryTable.tsx` | Displays employee monthly rollup (`PresentDays`, `UnpaidDays`, `LhdCount`, `EdCount`, lock state). | `GET /api/attendance/monthly-summary` | HR / Payroll Admin |
| `src/app/modules/attendance/components/PenaltyExceptionsModal.tsx` | Lists `PENDING_REVIEW` exceptions from `AttendancePayrollException` for HR approval or waiver. | `GET /api/attendance-penalty/exceptions`, `POST /api/attendance-penalty/resolve` | HR Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/AttendanceController.cs` | Handles check-in, check-out, regularization, monthly summary generation, and month locking. | `IAttendanceService`, `IAttendancePenaltyService` |
| `Services/AttendanceStatusCalculationService.cs` | Pure deterministic calculation of daily status (`P`, `HD`, `LHD`, `ED`, `LHD+ED`) from punch times and `CompanyOfficeSchedule`. | `EffectiveOfficeScheduleService` |
| `Services/AttendancePenaltyService.cs` | Evaluates monthly `LHD` / `ED` counts against `AttendancePenaltyPolicy.CombinedLhdEdMonthlyLimit`, creates `AttendancePayrollException`, and enforces `AttendanceMonthLock.ValidateAndLock()`. | `IMongoRepository<AttendancePenaltyPolicy>`, `IMongoRepository<AttendancePayrollException>`, `IMongoRepository<AttendanceMonthLock>` |
| `Services/HostedServices/AttendanceReminderHostedService.cs` | Periodic `BackgroundService` that checks `AttendanceReminderSetting` and sends check-in/out notifications via `NotificationService`. | `IMongoRepository<AttendanceReminderDispatchLog>` |

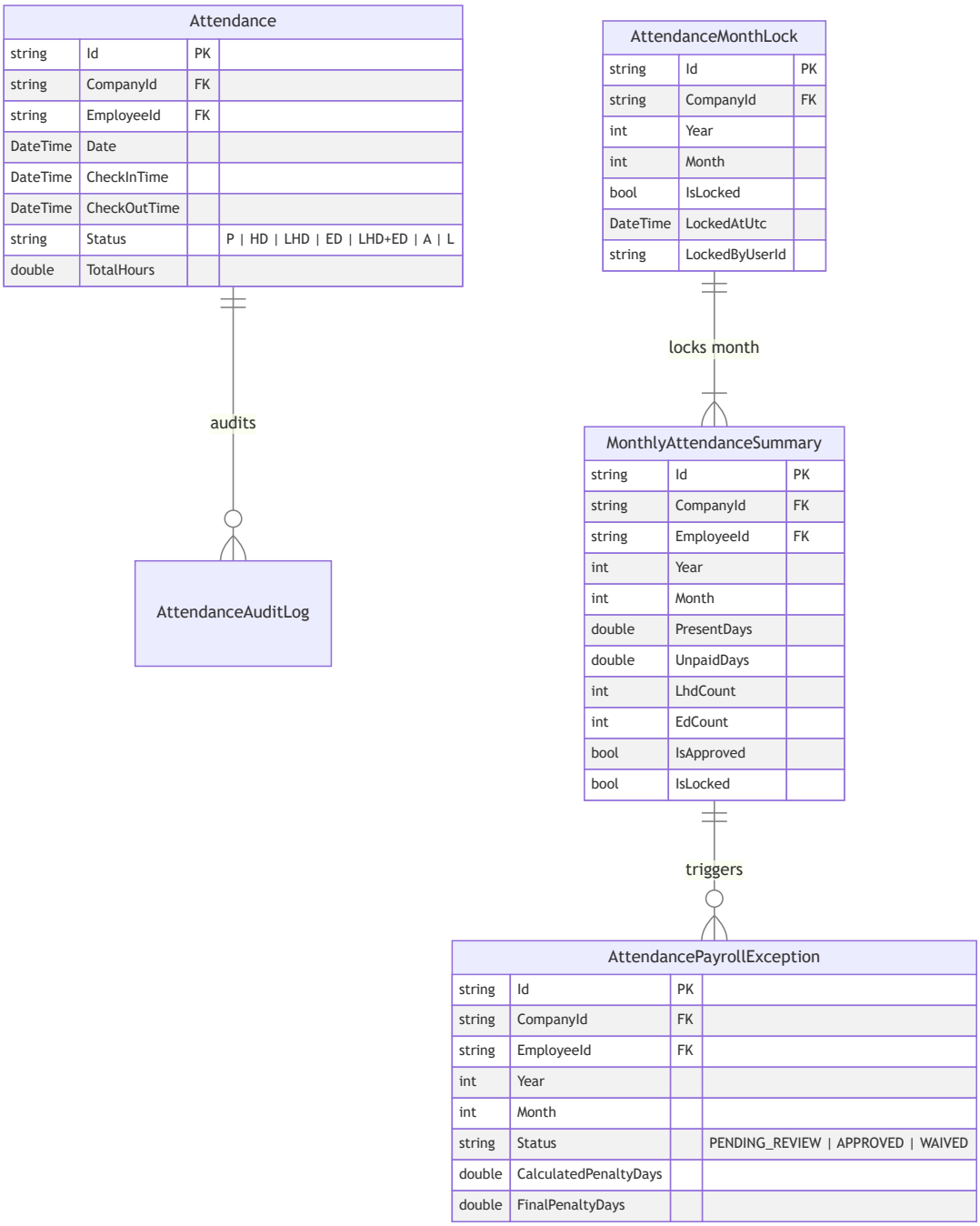
## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `POST` | `/api/attendance/check-in` | Records employee daily check-in timestamp. | JWT (`attendance:create`) | `CheckInRequestDto` -> `AttendanceDto` | Checks `EmployeeExitAttendanceCutoff`; calculates initial status |

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `POST` | `/api/attendance/check-out` | Records employee daily check-out and computes worked hours & `Status`. | JWT (`attendance:update`) | `CheckOutRequestDto` -> `AttendanceDto` | Runs `AttendanceStatusCalculationService.CalculateStatus` |
| `PUT` | `/api/attendance/{id}/regularize` | HR/Manager manual override of punch times or status. | JWT (`attendance:update`) | `RegularizeAttendanceDto` | Writes immutable `AttendanceAuditLog` (`PreviousValue`, `NewValue`, `Reason`) |
| `POST` | `/api/attendance-penalty/evaluate` | Evaluates monthly `LHD` / `ED` limits and creates `AttendancePayrollException` records. | JWT (`attendance:update`) | `{ Year, Month }` | Creates/updates `AttendancePayrollException` |
| `POST` | `/api/attendance-penalty/resolve` | Approves or waives a `PENDING_REVIEW` penalty exception. | JWT (`attendance:approve`) | `ResolveExceptionDto` (`Status`, `FinalPenaltyDays`) | Unblocks month lock when zero `PENDING_REVIEW` remain |
| `POST` | `/api/attendance/lock-month` | Validates all summaries and locks the month for payroll. | JWT (`attendance:approve`) | `{ Year, Month }` | Calls `AttendanceMonthLock.ValidateAndLock()` |

## 7. Database Model

ARCHITECTURE / WORKFLOW DIAGRAM




## 8. Business Rules

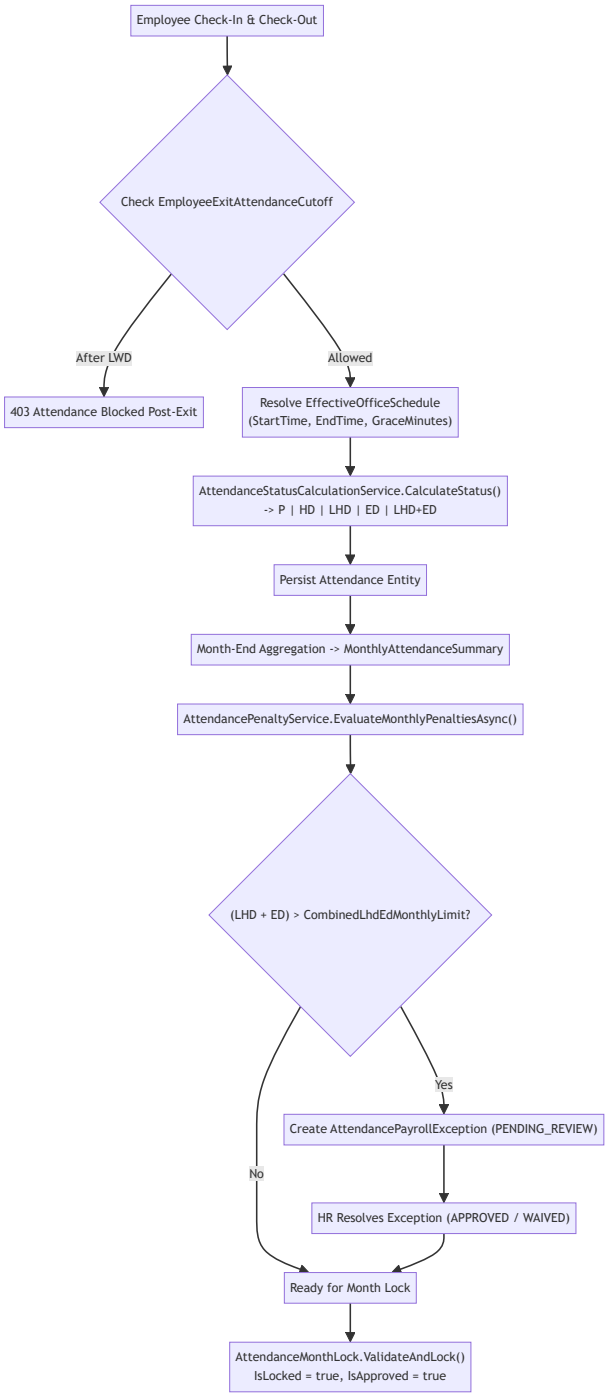
- **Daily Status Rule (`AttendanceStatusCalculationService`)**:
  - `HalfDayThresholdHours`: If total worked hours `< HalfDayThresholdHours` (or missing check-out), status is `HD` (Half Day).
  - `isLate`: `CheckInTime > (StartTime + GraceMinutes)` (`CheckInAllowedUntil`).
  - `isEarly`: `CheckOutTime < CheckOutAllowedFrom`.
  - Status matrix:

    - `!isLate && !isEarly` → **`P`** (Present)
    - `isLate && !isEarly` → **`LHD`** (Late Half Day / Late Arrival)
    - `!isLate && isEarly` → **`ED`** (Early Departure)
    - `isLate && isEarly` → **`LHD+ED`** (Combined Late + Early)
- **Offboarding Cutoff Rule**: If an `EmployeeExitAttendanceCutoff` exists for the employee and `Date > LastWorkingDayUtc`, check-in is blocked.
- **Penalty Rule (`AttendancePenaltyService`)**: If `(LhdCount + EdCount + LhdEdCount) > AttendancePenaltyPolicy.CombinedLhdEdMonthlyLimit`, an `AttendancePayrollException` (`PENDING_REVIEW`) is created.
- **Month Lock Gate (`AttendanceMonthLock.ValidateAndLock`)**: A month cannot be locked if any `AttendancePayrollException` in that `(CompanyId, Year, Month)` has `Status == "PENDING_REVIEW"`.

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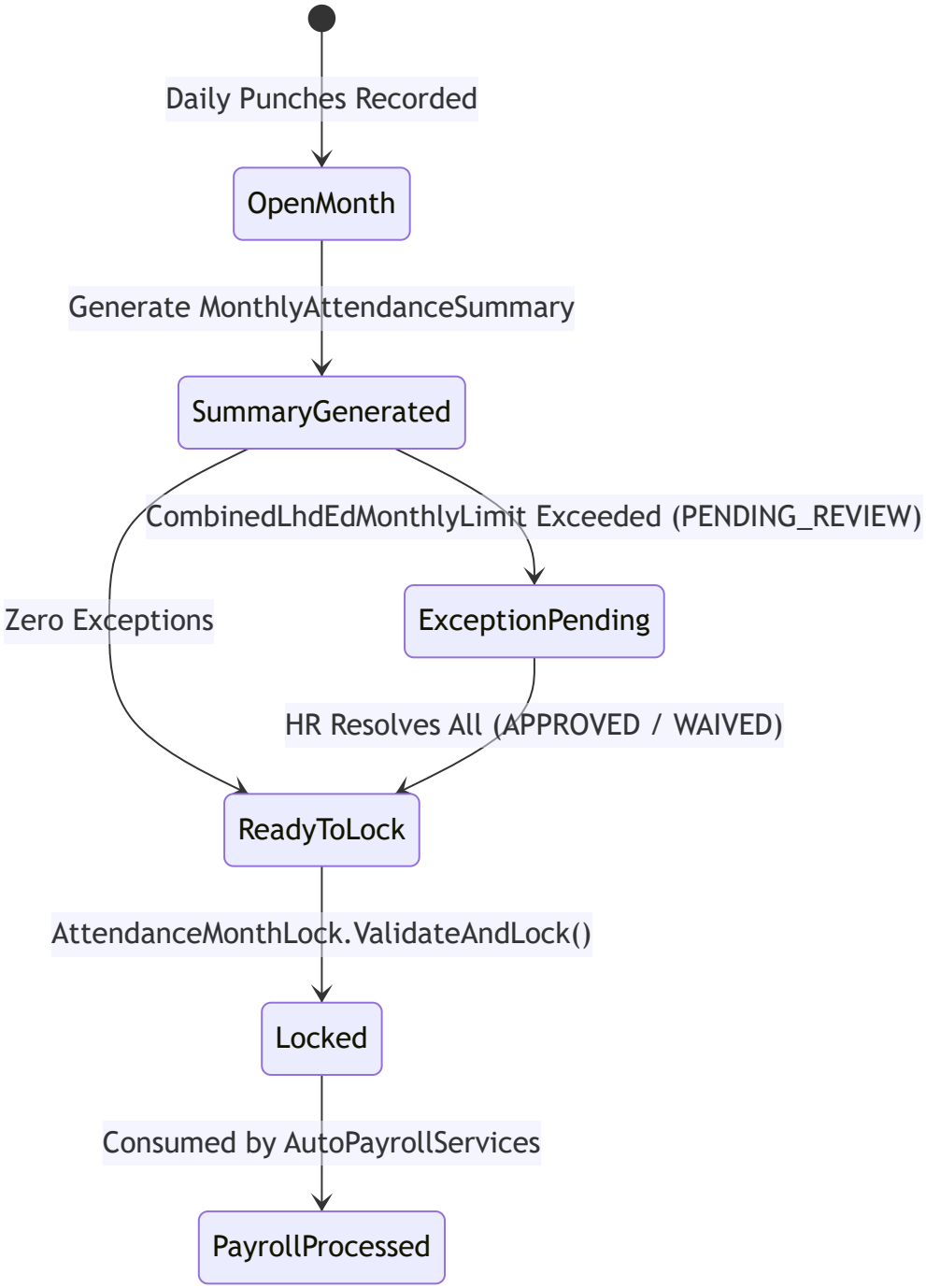
# 9. Complete Workflow

### ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `attendance:read` — View personal or team attendance records.
- `attendance:create` / `attendance:update` — Check-in, check-out, and manual regularization.
- `attendance:approve` — Resolve `AttendancePayrollException` and execute `AttendanceMonthLock`.

## 12. Dependencies

- **Upstream**: `WORK-SHIFT` ( `EffectiveOfficeScheduleService` ), `CALENDAR` ( `CompanyWorkingCalendarService` ), `LEAVE` & `WFH` (approved leaves/WFH overlay daily status), `OFFBOARDING` ( `EmployeeExitAttendanceCutoff` ).
- **Downstream**: `PAYROLL` ( `AutoPayrollServices` requires locked `MonthlyAttendanceSummary` and reads approved `FinalPenaltyDays` ).

## 13. Cross-Module Events

- **Manual Regularization**: Appends an immutable record to `AttendanceAuditLog` .
- **Month Lock**: Unblocks `AutoPayrollServices.GeneratePayrollForMonthAsync` for that `(CompanyId, Year, Month)` .

## 14. Failure Scenarios

- **Attempting to Lock Month with Unresolved Exceptions**: `AttendanceMonthLock.ValidateAndLock()` throws a domain validation error listing the unresolved `PENDING_REVIEW` exceptions.
- **Modifying Locked Month Attendance**: Regularization calls against a date in a locked `AttendanceMonthLock` are rejected ( `409 Conflict` ).

## 15. Edge Cases

- **Half-Day Leave + Punch**: When an employee has an approved half-day leave ( `FirstHalf` or `SecondHalf` ) or half-day WFH, `EffectiveOfficeScheduleService` adjusts the expected check-in/check-out window accordingly so they are not falsely marked `LHD` or `ED` .

## 16. Current Strengths

- Deterministic, policy-driven daily status classification ( `P` , `HD` , `LHD` , `ED` , `LHD+ED` ).
- Hard pre-payroll gate ( `AttendanceMonthLock` + `AttendancePayrollException` review) prevents unverified attendance from leaking into salary calculations.

## 17. Current Gaps

- Biometric hardware webhook ingestion is not natively wired; punches currently originate from web/mobile API requests.

## 18. Recommended Improvements

- Add an idempotent batch punch ingestion endpoint ( `POST /api/attendance/biometric-sync` ) for external biometric devices.

## 19. Future Scope

- Geofenced / IP-whitelisted check-in rules per `CompanyOfficeSchedule` .

MOD-16 PART III docs/project-documentation/modules/LEAVE.md

# LEAVE MANAGEMENT MODULE ( LEAVE.md )

## 1. Module Overview

The **Leave Management Module** handles configurable leave policies ( LeavePolicy ), automated monthly/annual leave accrual ( LeaveAccrualHostedService ), employee leave balances ( EmployeeLeaveBalance ), leave request submission with half-day segment support, working-day intersection calculations ( CompanyWorkingCalendarService ), manager/HR approvals, and leave encashment integration with Final Settlement ( EmployeeFinalSettlement ).

## 2. Real-World Business Scenario

1. **Accrual**: On the 1st of every month, LeaveAccrualHostedService credits 1.5 days of Earned Leave ( EL ) and 1.0 day of Casual Leave ( CL ) into Rahul Sharma's EmployeeLeaveBalance for CompanyId .
2. **Application**: Rahul applies for 3 days of CL ( 2026-04-20 to 2026-04-22 ). LeaveManagementService checks:
   - Is Rahul eligible ( LeavePolicy.AllowOnProbation , AllowDuringNoticePeriod )?
   - Are any of those dates weekly offs or holidays via CompanyWorkingCalendarService ?
   - Does Rahul have sufficient AvailableBalance in EmployeeLeaveBalance ?
3. **Approval & Attendance Sync**: Reporting Manager approves the leave ( POST /api/leave/{id}/approve ). EmployeeLeaveBalance deducts 3.0 days ( UsedBalance += 3.0 ), NotificationService alerts Rahul, and MonthlyAttendanceSummary treats those 3 dates as paid leave ( L ) rather than unpaid absence ( UnpaidDays ).

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Employee** | Views EmployeeLeaveBalance , applies for full-day or half-day leave, cancels pending leave requests. |
| **Reporting Manager / HR** | Reviews team leave calendar, approves or rejects leave requests with remarks. |
| **HR Admin** | Creates and updates LeavePolicy rules (quota, carry-forward, probation/notice eligibility, encashment). |
| **Background Worker ( LeaveAccrualHostedService )** | Runs scheduled accrual credits and year-end carry-forward calculations. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| src/app/modules/leave/pages/LeaveDashboard.tsx | Displays leave balance cards ( Total , Used , Pending , Available ) and application history. | GET /api/leave/balances , GET /api/leave/my-leaves | Employee |
| src/app/modules/leave/components/ApplyLeaveModal.tsx | Form for selecting leave type, start/end dates, half-day option, and reason. | POST /api/leave/apply | Employee |

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/leave/pages/LeaveApprovalsPage.tsx` | Manager/HR queue to approve or reject team leave applications. | `GET /api/leave/pending`, `POST /api/leave/{id}/approve`, `POST /api/leave/{id}/reject` | Manager / HR |
| `src/app/modules/settings/pages/LeavePolicySettings.tsx` | Configures tenant `LeavePolicy` parameters. | `GET /api/leave/policies`, `POST /api/leave/policies` | HR Admin |

## 5. Backend Implementation

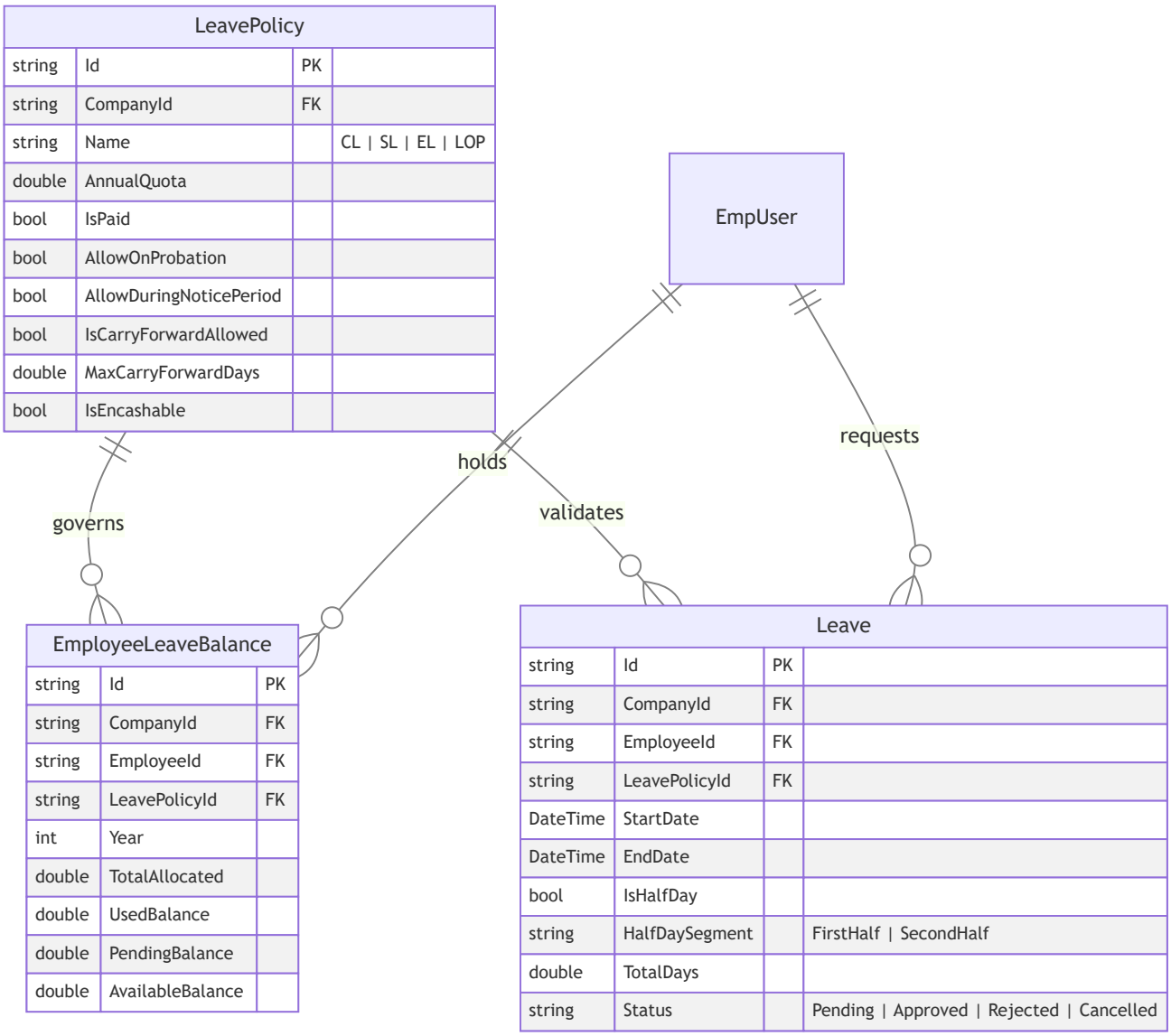
| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/LeaveManagementController.cs` | Exposes endpoints for leave policies, balances, leave application, approval, rejection, and cancellation. | `ILeaveManagementService` |
| `Services/LeaveManagementService.cs` | Validates eligibility ( `EmployeeLifecycle`, `LeavePolicy` ), computes net leave days via `CompanyWorkingCalendarService`, updates `EmployeeLeaveBalance`, and triggers notifications. | `IMongoRepository<Leave>`, `IMongoRepository<LeavePolicy>`, `IMongoRepository<EmployeeLeaveBalance>`, `ICompanyWorkingCalendarService` |
| `Services/HostedServices/LeaveAccrualHostedService.cs` | Background worker that executes periodic leave accrual credits into `EmployeeLeaveBalance`. | `IMongoRepository<LeavePolicy>`, `IMongoRepository<EmployeeLeaveBalance>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/leave/balances` | Returns current leave balances by leave type for the employee. | JWT ( `leave:read` ) | `List<EmployeeLeaveBalanceDto>` | None |
| `POST` | `/api/leave/apply` | Submits a new leave request ( `Pending` ). | JWT ( `leave:create` ) | `ApplyLeaveDto` -> `LeaveDto` | Reserves `PendingBalance`, notifies manager |
| `POST` | `/api/leave/{id}/approve` | Approves a pending leave request. | JWT ( `leave:approve` ) | `ApproveLeaveDto` | Moves `PendingBalance -> UsedBalance`, notifies employee |
| `POST` | `/api/leave/{id}/reject` | Rejects a pending leave request. | JWT ( `leave:approve` ) | `RejectLeaveDto` ( `Reason` ) | Releases `PendingBalance`, notifies employee |
| `POST` | `/api/leave/policies` | Creates or updates a company `LeavePolicy`. | JWT ( `settings:update` ) | `LeavePolicyDto` | Persists `LeavePolicy` |

## 7. Database Model

ARCHITECTURE / WORKFLOW DIAGRAM

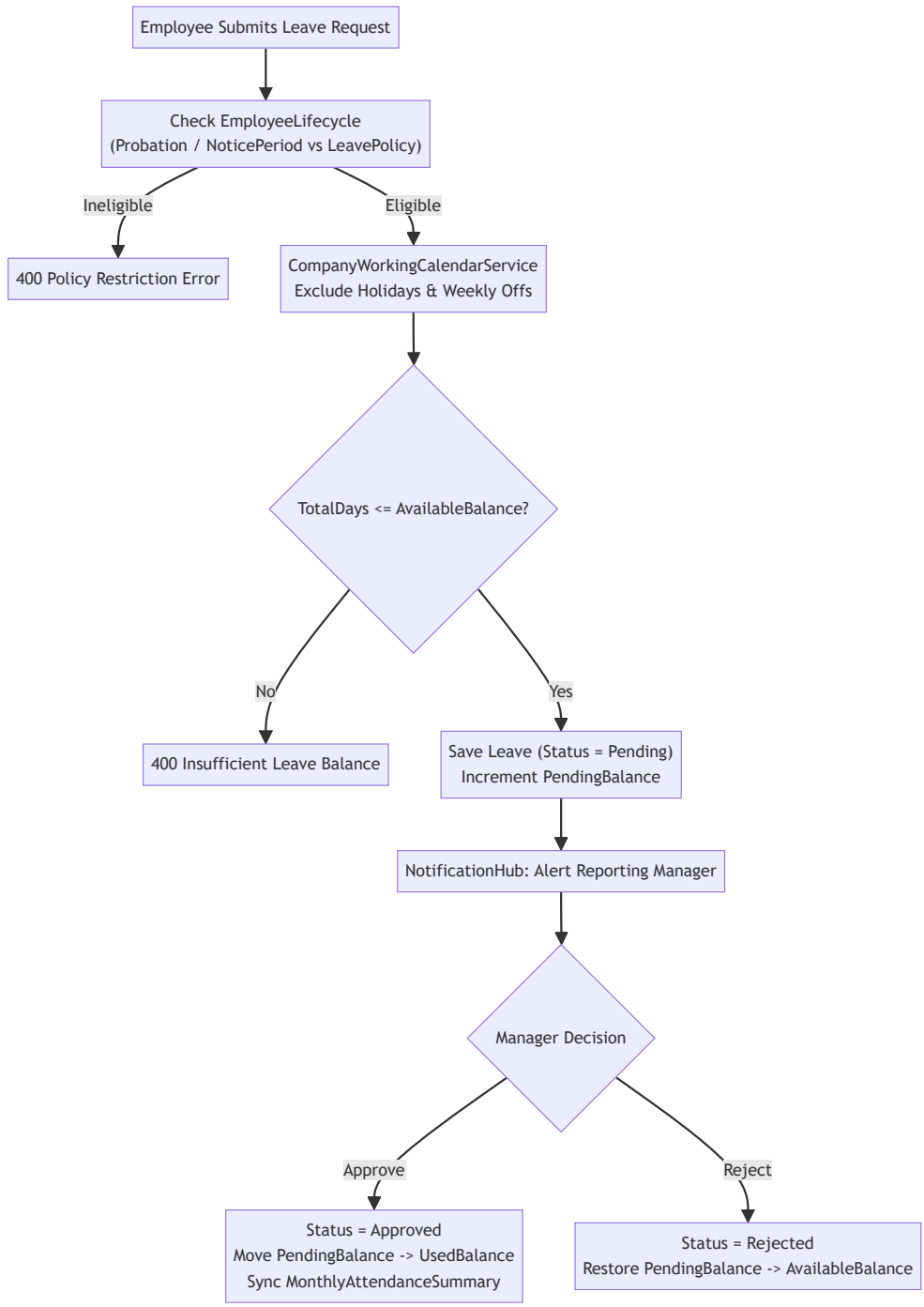



## 8. Business Rules

- **Lifecycle Eligibility Rule**:
  - If `EmployeeLifecycle.EmploymentStatus == Probation` and `!LeavePolicy.AllowOnProbation`, the leave request is rejected.
  - If `EmployeeLifecycle.EmploymentStatus == NoticePeriod` and `!LeavePolicy.AllowDuringNoticePeriod`, the leave request is rejected.
- **Working Calendar Intersection Rule**: Holidays ( `Holidays` ) and weekly offs ( `WeeklyOffSetting` ) inside `[StartDate, EndDate]` are excluded from `TotalDays` unless the policy configures sandwich rules.
- **Overlap Rule**: An employee cannot apply for a leave on dates that already have an `Approved` or `Pending` leave or overlapping full-day WFH request.
- **Balance Rule**: For paid leave policies ( `IsPaid == true` ), `TotalDays` cannot exceed `AvailableBalance` ( `TotalAllocated - UsedBalance - PendingBalance` ).

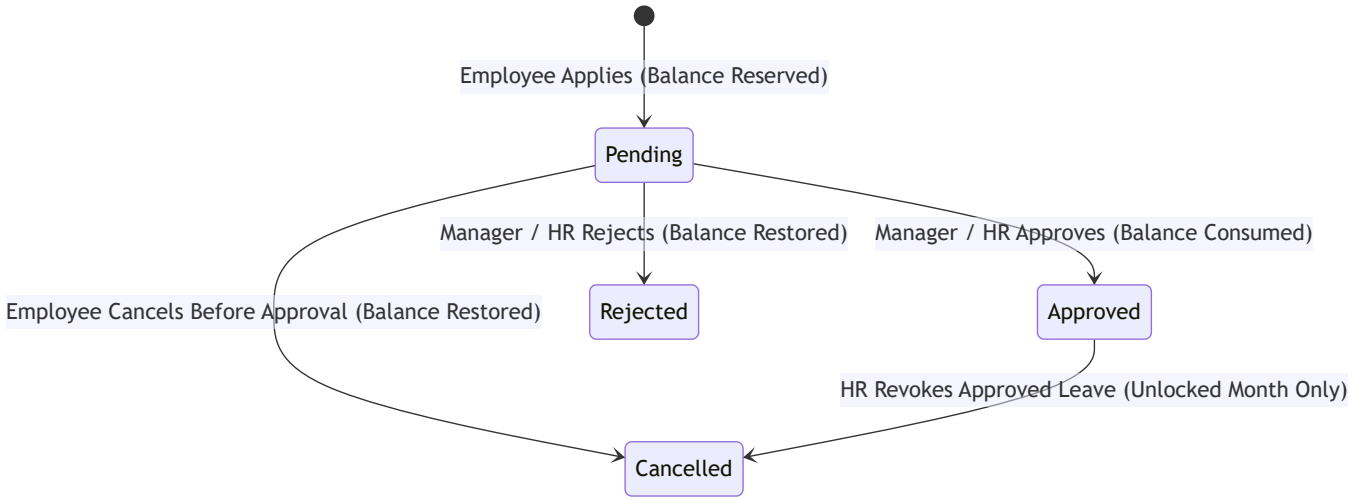
## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `leave:read` — View balances and leave history.
- `leave:create` — Apply for leave or cancel own pending leave.
- `leave:approve` — Approve or reject employee leave requests.

## 12. Dependencies

- **Upstream**: `EMPLOYEE` (`EmployeeLifecycle`), `CALENDAR` (`CompanyWorkingCalendarService`), `HOLIDAY` (`Holidays`).
- **Downstream**: `ATTENDANCE` (`MonthlyAttendanceSummary` paid/unpaid day accounting), `FINAL-SETTLEMENT` (`LeaveEncashment` for `IsEncashable` balances).

## 13. Cross-Module Events

- **Leave Approval**: Updates `AvailabilityStatus = OnLeave` during active leave dates and marks days as paid leave in `MonthlyAttendanceSummary`.
- **Exit Initiation**: Encashable `AvailableBalance` is queried during `EmployeeFinalSettlement` calculation.

## 14. Failure Scenarios

- **Retroactive Leave on Locked Attendance Month**: Rejected if `AttendanceMonthLock.IsLocked == true` for the target month.

## 15. Edge Cases

- **Half-Day Leave (`IsHalfDay = true`)**: Deducts `0.5` days from `EmployeeLeaveBalance` and adjusts `EffectiveOfficeScheduleService` for the remaining half-day shift.

## 16. Current Strengths

- Automated background accrual ( `LeaveAccrualHostedService` ) combined with strict `Probation` and `NoticePeriod` policy enforcement.

## 17. Current Gaps

- Multi-level sequential approval chains (e.g., L1 Reporting Manager $\rightarrow$ L2 Department Head for leaves $> 5$ days) currently execute as single-step approval.

## 18. Recommended Improvements

- Add configurable multi-stage approval thresholds based on `TotalDays`.

## 19. Future Scope

- Automated year-end leave encashment batch job feeding directly into December/March `AutoPayrollServices`.

MOD-17 PART III docs/project-documentation/modules/WFH.md

# WORK FROM HOME (WFH) MODULE (`WFH.md`)

## 1. Module Overview

The **Work From Home (WFH) Module** regulates remote work applications, manager approvals, monthly WFH allowances/quotas, half-day remote shifts, and their overlay onto `AttendanceStatusCalculationService`. It ensures remote work days are credited toward attendance hours while maintaining compliance with company remote work policies.

## 2. Real-World Business Scenario

1. **Application**: On Monday evening, Rahul Sharma submits a WFH request for Wednesday (`2026-04-15`) due to personal logistics. He selects `IsHalfDay = false` and supplies a reason.
2. **Manager Approval**: Rahul's reporting manager receives a real-time notification via `NotificationHub` and approves the request (`POST /api/wfh/{id}/approve`).
3. **Attendance Execution**: On Wednesday, Rahul logs into the system remotely. Because an `Approved` WFH record exists for that date, `AttendanceStatusCalculationService` recognizes remote attendance, logs the IP/punch time, and marks the day as present with WFH designation, preventing late arrival penalties or unauthorized absence.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Employee** | Submits WFH requests (full day or half-day), tracks status, cancels pending requests. |
| **Reporting Manager** | Approves or rejects subordinate WFH applications with review comments. |
| **HR Admin** | Configures tenant-wide or department WFH quota limits, reviews organization-wide remote work trends. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/wfh/pages/WfhDashboard.tsx` | Displays monthly WFH quota, utilized count, and application history table. | `GET /api/wfh/my-requests`, `GET /api/wfh/allowance` | Employee |
| `src/app/modules/wfh/components/ApplyWfhModal.tsx` | Modal form to submit WFH dates, half-day selection (`FirstHalf` / `SecondHalf`), and notes. | `POST /api/wfh/apply` | Employee |
| `src/app/modules/wfh/pages/WfhApprovalsPage.tsx` | Manager queue for pending remote work requests. | `GET /api/wfh/pending`, `POST /api/wfh/{id}/approve`, `POST /api/wfh/{id}/reject` | Reporting Manager |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/WorkFromHomeController.cs` | Exposes endpoints for WFH application, approval, rejection, cancellation, and quota checks. | `IWorkFromHomeService` |
| `Services/WorkFromHomeService.cs` | Validates date conflicts (leaves, holidays, existing WFH), checks monthly limits, updates status, and dispatches notifications. | `IMongoRepository<WorkFromHome>`, `IMongoRepository<Leave>`, `ICompanyWorkingCalendarService` |
| `Services/AttendanceStatusCalculationService.cs` | Incorporates approved WFH status into daily attendance evaluation. | `IMongoRepository<WorkFromHome>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/wfh/my-requests` | Retrieves employee's WFH requests for specified date range. | JWT (`wfh:read`) | `List<WorkFromHomeDto>` | None |
| `POST` | `/api/wfh/apply` | Submits a new WFH request. | JWT (`wfh:create`) | `ApplyWfhDto` -> `WorkFromHomeDto` | Notifies reporting manager via SignalR |
| `POST` | `/api/wfh/{id}/approve` | Approves pending WFH application. | JWT (`wfh:approve`) | `ApproveWfhDto` | Sets `Status = Approved`, updates `AvailabilityStatus` |
| `POST` | `/api/wfh/{id}/reject` | Rejects pending WFH application. | JWT (`wfh:approve`) | `RejectWfhDto` (`Reason`) | Sets `Status = Rejected`, notifies employee |
| `POST` | `/api/wfh/{id}/cancel` | Cancels employee's own pending or upcoming WFH. | JWT (`wfh:create`) | None | Releases reserved quota |

## 7. Database Model

**`WorkFromHome` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/WorkFromHome.cs`)**

- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `EmployeeId` (`string`, FK)
- `StartDate` (`DateTime`)
- `EndDate` (`DateTime`)
- `IsHalfDay` (`bool`)
- `HalfDaySegment` (`string`, `"FirstHalf"` / `"SecondHalf"`)
- `Reason` (`string`)
- `Status` (`string`, `"Pending"` / `"Approved"` / `"Rejected"` / `"Cancelled"`)
- `ApprovedByUserId` (`string`, FK)
- `ApprovedAtUtc` (`DateTime?`)

- `RejectionReason` ( `string` )

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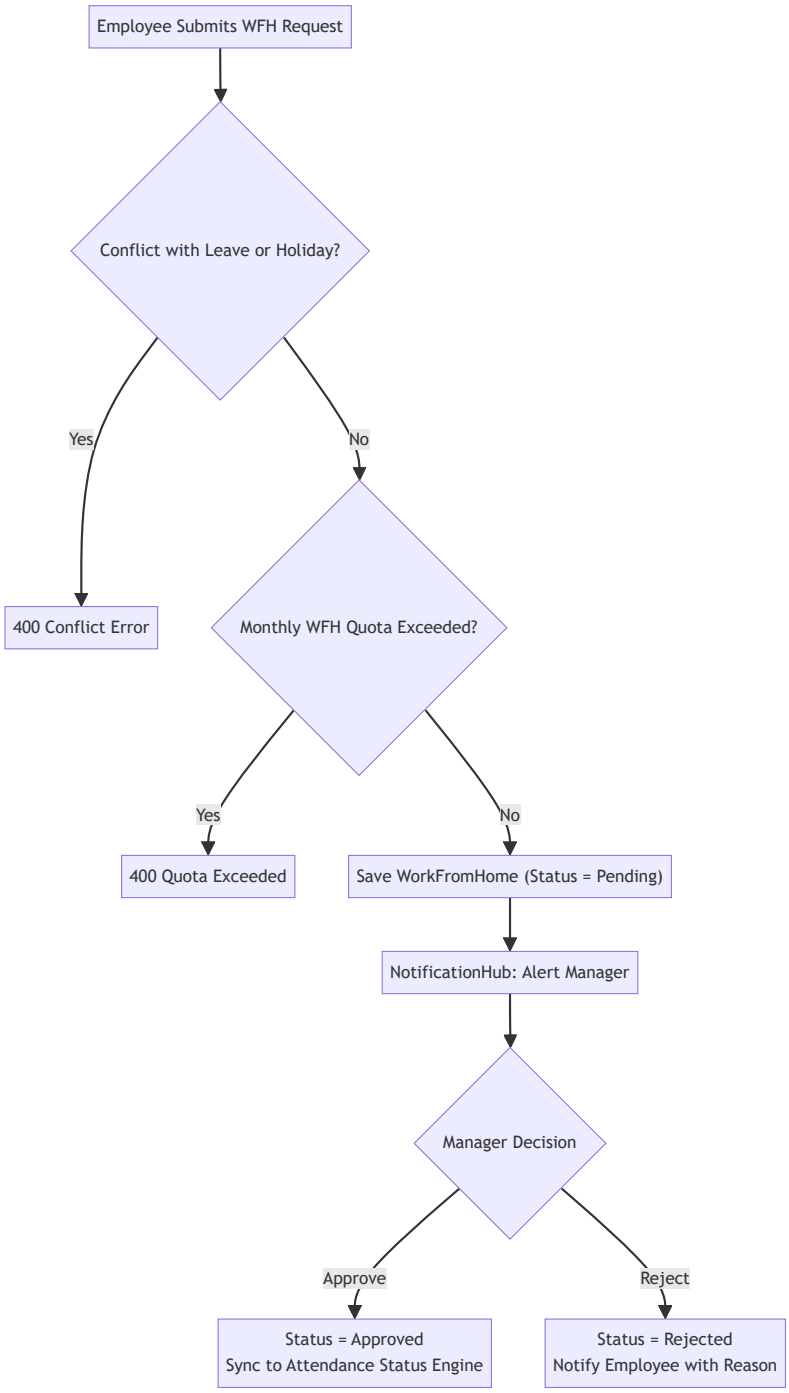
## 8. Business Rules

- **Conflict Rule**: A WFH request cannot be submitted on dates where an `Approved` or `Pending` `Leave` already exists.
- **Working Calendar Rule**: WFH cannot be applied on official company holidays ( `Holidays` ) or weekly off days ( `WeeklyOffSetting` ).
- **Half-Day Rule**: If `IsHalfDay = true` , the employee must specify `HalfDaySegment` ( `FirstHalf` or `SecondHalf` ). The remaining half day requires office attendance or half-day leave.
- **Lock Rule**: WFH requests cannot be submitted or retroactively approved for months already locked under `AttendanceMonthLock` .

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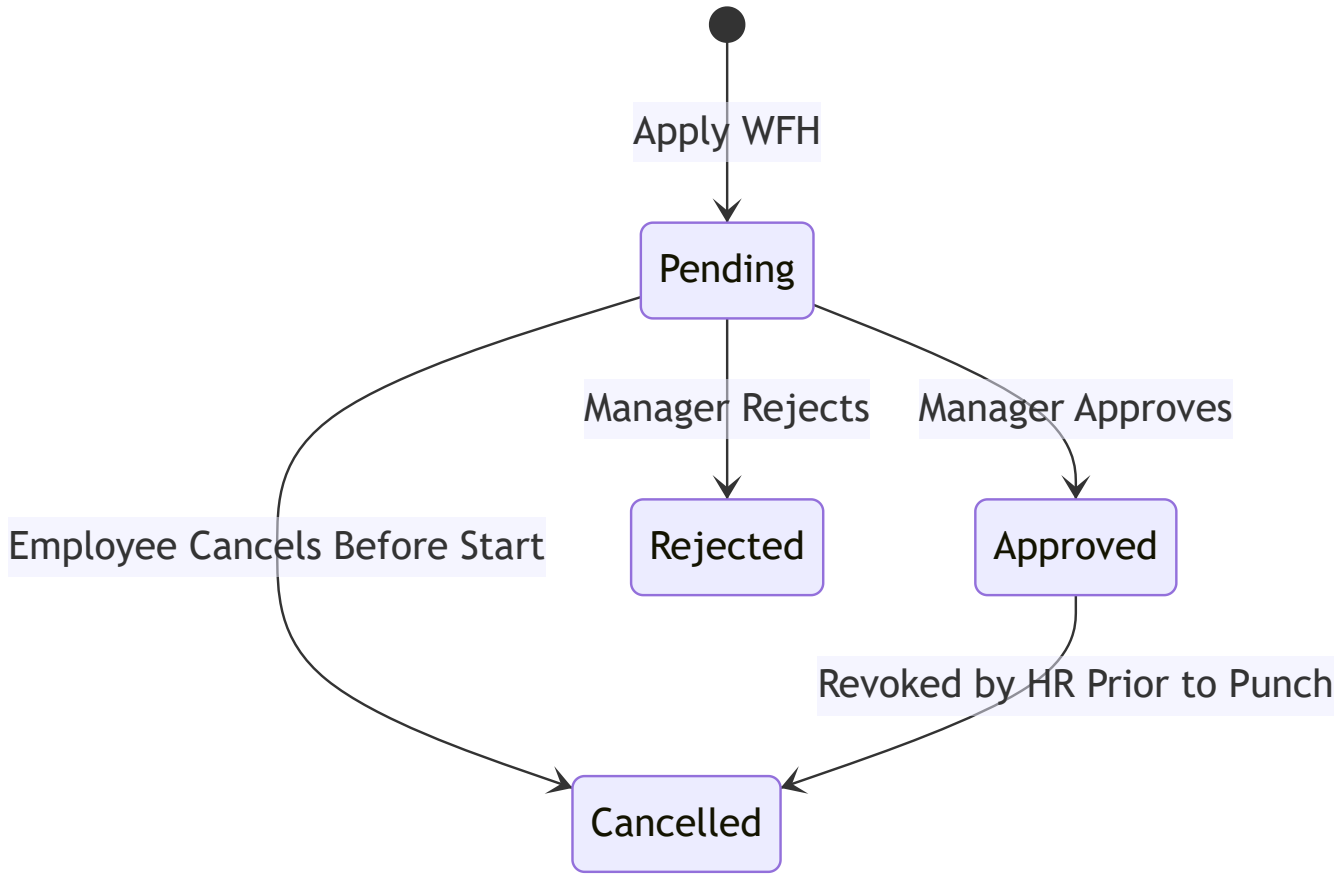
## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `wfh:read` — View WFH requests and allowance.
- `wfh:create` — Submit and cancel own WFH.
- `wfh:approve` — Approve or reject team WFH requests.

## 12. Dependencies

- **Upstream**: `CALENDAR` ( `CompanyWorkingCalendarService` ), `LEAVE` (conflict validation).
- **Downstream**: `ATTENDANCE` ( `AttendanceStatusCalculationService` credits worked time).

## 13. Cross-Module Events

- **Approved WFH**: Updates `EmployeeLifecycle.AvailabilityStatus` to `Remote` on active dates and suppresses late check-in penalties if remote punches are permitted.

## 14. Failure Scenarios

- **Manager Absent / Unresponsive**: Auto-escalation or HR proxy approval through `wfh:approve` permission.

## 15. Edge Cases

- **Overlapping Half-Day Leave and Half-Day WFH**: Allowed if segments do not collide (e.g., `FirstHalf` WFH + `SecondHalf` Casual Leave).

## 16. Current Strengths

- Granular half-day segment handling with direct integration into attendance calculation and calendar.

## 17. Current Gaps

- IP whitelist restriction for remote punches is configurable at network level but not enforced via geo-coordinates in code.

## 18. Recommended Improvements

- Add GPS coordinates capture to remote punch requests for field staff.

## 19. Future Scope

- Automated department-level hybrid schedule templates (e.g., 3 days office, 2 days remote mandatory).

# SALARY STRUCTURE & COMPENSATION MODULE (`SALARY.md`)

## 1. Module Overview

The **Salary Structure & Compensation Module** manages employee compensation packages (`SalaryModel`), decomposing annual Cost-to-Company (`CTC`) into monthly earning heads (`BasicPay`, `HRA`, `LTA`, `Bonus`, `OtherAllowance`) and statutory/benefit deduction heads (`EPF`, `ESIC`, `HealthInsurance`, `IncomeTax`). It provides the foundational financial dataset consumed by `AutoPayrollServices` and `EmployeeFinalSettlement`.

## 2. Real-World Business Scenario

1. **Compensation Setup during Onboarding / Promotion**:
   - HR Manager assigns Rahul Sharma's salary structure via `SalaryController` (`POST /api/salary`):
     - Annual `CTC` = `₹12,00,000`
     - Monthly `GrossPay` = `₹1,00,000` (`BasicPay = ₹50,000`, `HRA = ₹25,000`, `LTA = ₹10,000`, `OtherAllowance = ₹15,000`)
     - Fixed Monthly Deductions: `EPF = ₹1,800`, `HealthInsurance = ₹1,500`, `IncomeTax = ₹8,200`.
2. **Monthly Consumption**:
   - Every month, `AutoPayrollServices` reads Rahul's active `SalaryModel` via `SalaryRepository` to compute earnings, LOP rate, and statutory deductions.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin / Payroll Admin** | Creates and revises employee `SalaryModel` structures, views CTC breakdowns. |
| **Employee** | Views personal salary structure breakdown on profile/payroll portal. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/payroll/pages/SalaryStructurePage.tsx` | Form and calculator for defining employee CTC, Basic, HRA, LTA, Allowances, and Deductions. | `GET /api/salary/{employeeId}`, `POST /api/salary`, `PUT /api/salary/{id}` | HR / Payroll Admin |
| `src/app/modules/userProfile/components/CompensationTab.tsx` | Read-only breakdown of earnings and deductions for the logged-in employee. | `GET /api/salary/my-structure` | Employee |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/SalaryController.cs` | Endpoints for creating, updating, and querying employee salary structures. | `ISalaryService` |
| `Services/SalaryService.cs` | Validates component totals ( `GrossPay == sum(Earnings)` ), persists `SalaryModel` , and tracks compensation revisions. | `ISalaryRepository` , `IMongoRepository<SalaryModel>` |
| `Repository/SalaryRepository.cs` | Specialized repository for querying active employee `SalaryModel` records by `CompanyId` and `EmployeeId` . | `MongoDbContext` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/salary/{employeeId}` | Fetches active `SalaryModel` for an employee. | JWT ( `payroll:read` ) | `SalaryModelDto` | None |
| `POST` | `/api/salary` | Creates initial salary structure for an employee. | JWT ( `payroll:create` ) | `CreateSalaryModelDto` | Persists `SalaryModel` |
| `PUT` | `/api/salary/{id}` | Updates/revises an employee's salary components. | JWT ( `payroll:update` ) | `UpdateSalaryModelDto` | Updates `SalaryModel` , increments `Version` |

## 7. Database Model

**`SalaryModel` ( `CMS-Backend-Core/Codeji.CMS.Repository/Entities/SalaryModel.cs` )**

- `Id` ( `string` , PK)
- `CompanyId` ( `string` , FK)
- `EmployeeId` ( `string` , FK)
- `CTC` ( `double` )
- `GrossPay` ( `double` )
- `BasicPay` ( `double` )
- `HRA` ( `double` )
- `LTA` ( `double` )
- `Bonus` ( `double` )
- `OtherAllowance` ( `double` )
- `EPF` ( `double` )
- `ESIC` ( `double` )
- `HealthInsurance` ( `double` )
- `IncomeTax` ( `double` )
- `EffectiveFrom` ( `DateTime` )

- `IsActive` ( `bool` )

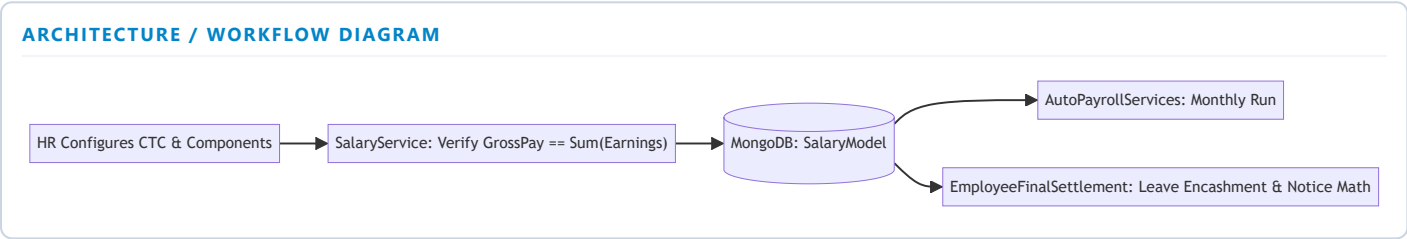
## 8. Business Rules

- **Component Identity Rule**:

$$\text{GrossPay} = \text{BasicPay} + \text{HRA} + \text{LTA} + \text{Bonus} + \text{OtherAllowance}$$

- **Non-Negative Rule**: All earning and deduction heads must be $\geq 0$, and `NetPay` before LOP must be $> 0$.
- **Single Active Structure Rule**: Each employee has at most one active `SalaryModel` ( `IsActive == true` ) at any point in time.

## 9. Complete Workflow




## 10. State Machine

`Draft` → `Active` ( `IsActive = true` ) → `Superseded` (upon appraisal/revision).

## 11. Permissions

- `payroll:read` — View salary structures.
- `payroll:create` / `payroll:update` — Create or revise employee compensation structures.

## 12. Dependencies

- **Upstream**: `EMPLOYEE` ( `EmpUser` ), `RECRUITMENT` ( `JobOffer.OfferedCtc` ).
- **Downstream**: `PAYROLL` ( `AutoPayrollServices` ), `BENEFITS` , `FINAL-SETTLEMENT` .

## 13. Cross-Module Events

- Creating a `SalaryModel` sets `EmployeeLifecycle.PayrollStatus = Eligible` so the employee is included in the next `AutoPayrollServices` run.

## 14. Failure Scenarios

- **Missing SalaryModel During Payroll**: `AutoPayrollServices` skips or flags employees who do not have an active `SalaryModel` configured.

## 15. Edge Cases

- **Mid-Month Salary Revision**: Effective date tracking ( `EffectiveFrom` ) determines which structure applies to the current payroll period.

## 16. Current Strengths

- Clean componentized representation of Indian statutory and allowance heads ( `BasicPay` , `HRA` , `LTA` , `EPF` , `ESIC` , `HealthInsurance` , `IncomeTax` ).

## 17. Current Gaps

- Percentage-based formula templates (e.g., auto-calculating `Basic = 50% of CTC` , `HRA = 50% of Basic` ) are computed in the UI form rather than stored as reusable backend grade templates.

## 18. Recommended Improvements

- Introduce `SalaryStructureTemplate` entities per designation/grade for 1-click compensation assignment.

## 19. Future Scope

- Multi-revision arrears calculation engine for retroactive appraisal back-pay.

MOD-19 PART III docs/project-documentation/modules/PAYROLL-SETTINGS.md

# PAYROLL SETTINGS & DIVISOR POLICY MODULE (`PAYROLL-SETTINGS.md`)

## 1. Module Overview

The **Payroll Settings & Divisor Policy Module** controls how the payroll engine calculates daily compensation rates, loss-of-pay deductions, pro-rata joining/exit adjustments, and payroll period locks (`PayrollDivisorPolicy`, `PayrollDivisorPolicyService`). It supports three industry-standard calculation methodologies: **Fixed 30 Days (`FIXED_30_DAYS`)**, **Actual Working Days (`WORKING_DAYS`)**, and **Actual Calendar Days (`CALENDAR_DAYS`)**.

## 2. Real-World Business Scenario

1. **Policy Selection**: Example Pvt Ltd switches its compensation policy from `CALENDAR_DAYS` to `FIXED_30_DAYS` via `PayrollSettings.tsx`.
2. **Impact on Payroll Calculation**:
   - In February 2026 (28 calendar days, 20 working days), under `FIXED_30_DAYS`, `PayrollCalculationRules.ResolveDivisor` returns `30.0`.
   - When an employee has `1.0` unpaid leave day on a ₹90,000 monthly gross salary, the LOP deduction is $^{90,000}/_{30} = ₹3,000$ consistently every month, regardless of whether the month has 28, 30, or 31 days.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Payroll Admin / HR Admin** | Configures `PayrollDivisorPolicy`, sets pay cycle cutoffs, and toggles statutory deduction defaults. |
| **Finance Director** | Audits active payroll calculation policy and approves policy updates. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/settings/pages/PayrollSettings.tsx` | Configuration panel for selecting `DivisorMode` (`FIXED_30_DAYS`, `WORKING_DAYS`, `CALENDAR_DAYS`) and statutory thresholds. | `GET /api/payroll-divisor-policy`, `PUT /api/payroll-divisor-policy` | Payroll Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/PayrollDivisorPolicyController.cs` | Exposes REST endpoints for querying and updating the tenant's active `PayrollDivisorPolicy`. | `IPayrollDivisorPolicyService` |

| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Services/PayrollDivisorPolicyService.cs` | Manages persistence of `PayrollDivisorPolicy` and integrates with `PayrollCalculationRules.ResolveDivisor`. | `IMongoRepository<PayrollDivisorPolicy>`, `ICompanyWorkingCalendarService` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
| --- | --- | --- | --- | --- | --- |
| `GET` | `/api/payroll-divisor-policy` | Retrieves active payroll divisor configuration for the company. | JWT (`settings:read`) | `PayrollDivisorPolicyDto` | None |
| `PUT` | `/api/payroll-divisor-policy` | Updates the divisor calculation mode (`FIXED_30_DAYS`, `WORKING_DAYS`, `CALENDAR_DAYS`). | JWT (`settings:update`) | `UpdatePayrollDivisorPolicyDto` | Updates `PayrollDivisorPolicy` |

## 7. Database Model

**`PayrollDivisorPolicy` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/PayrollDivisorPolicy.cs`)**

- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `DivisorType` (`string`, `"FIXED_30_DAYS"` | `"WORKING_DAYS"` | `"CALENDAR_DAYS"`)
- `IncludeHolidaysInWorkingDays` (`bool`)
- `IsActive` (`bool`)
- `UpdatedByUserId` (`string`)
- `UpdatedAtUtc` (`DateTime`)

## 8. Business Rules

- **Divisor Resolution Rule (`PayrollCalculationRules.ResolveDivisor`)**:
  - `"FIXED_30_DAYS"` → Always returns `30`.
  - `"WORKING_DAYS"` → Returns `eligibleWorkingDays` from `CompanyWorkingCalendarService`.
  - `"CALENDAR_DAYS"` (Default) → Returns `DateTime.DaysInMonth(year, month)`.
- **Closed Period Protection**: Changes to `PayrollDivisorPolicy` apply only to open/unprocessed payroll months and never retroactively mutate closed `PayrollEntity` records.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Singleton tenant configuration ( `Active` versioned per update).

## 11. Permissions

- `settings:read` / `settings:update` — View and modify payroll divisor rules.

## 12. Dependencies

- **Upstream**: `CALENDAR` ( `CompanyWorkingCalendarService` for `WORKING_DAYS` mode).
- **Downstream**: `PAYROLL` ( `AutoPayrollServices` ), `FINAL-SETTLEMENT` (leave encashment & notice recovery daily rate).

## 13. Cross-Module Events

- Policy updates immediately govern the next `AutoPayrollServices` and `EmployeeFinalSettlement` calculation runs.

## 14. Failure Scenarios

- **Zero Eligible Working Days**: If `WORKING_DAYS` returns `<= 0` due to misconfiguration, `ResolveDivisor` falls back to `DateTime.DaysInMonth` to prevent division-by-zero exceptions.

## 15. Edge Cases

- **February Calculation**: Under `FIXED_30_DAYS` , 1 day LOP in February deducts $1/30$ of monthly gross rather than $1/28$.

## 16. Current Strengths

- Eliminates hard-coded divisor numbers in payroll math through an explicit, tenant-configurable rule engine.

## 17. Current Gaps

- Department-specific divisor overrides (e.g., factory floor on `WORKING_DAYS` vs corporate office on `FIXED_30_DAYS` ) currently share one company-wide policy.

## 18. Recommended Improvements

- Extend `PayrollDivisorPolicy` to support department/grade scoping similar to `OffboardingPolicyResolver` .

## 19. Future Scope

- Simulation preview tool showing monthly payout variance across all 3 divisor modes before committing a policy change.

# PAYROLL PROCESSING MODULE (`PAYROLL.md`)

## 1. Module Overview

The **Payroll Processing Module** is the core financial compensation engine of **Codeji CMS / HRMS**. Implemented in `AutoPayrollServices` and `PayRollServices`, it orchestrates automated monthly payroll calculation, salary slip generation (`SalarySlipTemplate.html`, `PdfService`), statutory compliance (EPF, ESIC, Tax, Health Insurance), loss-of-pay (`TotalLossOfPayDays`) deductions from locked attendance summaries, and direct handover into Final Settlement (`SourcePayrollId`).

## 2. Real-World Business Scenario

1. **Pre-requisite Verification**:
   - For April 2026, HR confirms all `AttendancePayrollException` items are resolved (`APPROVED` or `WAIVED`) and `AttendanceMonthLock.ValidateAndLock()` has locked the month.
   - `AutoPayrollServices.GeneratePayrollForMonthAsync` verifies `PayrollPeriodRules.IsClosedPeriod` is false and `MonthlyAttendanceSummary.IsApproved && IsLocked == true`.
2. **Deterministic Calculation**:
   - Rahul Sharma's `SalaryModel` has `CTC = ₹12,00,000 / year` (`GrossPay = ₹1,00,000 / month`, `BasicPay = ₹50,000`, `HRA = ₹25,000`, `LTA = ₹10,000`, `OtherAllowance = ₹15,000`).
   - `PayrollCalculationRules.ResolveDivisor` applies policy `FIXED_30_DAYS` (**Divisor** $= 30$). Daily rate = $1,00,000 / 30 = ₹3,333.33$.
   - Rahul's `MonthlyAttendanceSummary` has `UnpaidDays = 1.0` and approved `FinalPenaltyDays = 0.5` → **TotalLossOfPayDays** $= 1.5$ days.
   - Loss of Pay Deduction = $1.5 \times 3,333.33 = ₹5,000.00$.
   - Statutory Deductions: `EPF = ₹1,800`, `ESIC = ₹0` (exceeds wage ceiling), `HealthInsurance = ₹1,500`, `IncomeTax = ₹8,200`.
   - `TotalDeduction` = $5,000 + 1,800 + 1,500 + 8,200 = ₹16,500$.
   - **`NetSalary`** = $1,00,000 - 16,500 = \mathbf{₹83,500}$.
3. **Salary Slip Generation & Disbursal**:
   - `AutoPayrollServices` populates `PayrollEntity`, compiles HTML via `Templates/SalarySlipTemplate.html`, converts it to PDF via `PdfService`, and writes the file to `/fs/salary-slips/{id}.pdf`.
   - Rahul receives a salary credit notification and downloads the PDF via `GET /api/payroll/{id}/download-slip`.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Payroll Manager / HR Admin** | Triggers automated payroll batch generation, reviews payroll variance, approves payroll runs, and publishes salary slips. |
| **Finance / Accounts** | Downloads bank disbursement sheets and payroll expense journals. |
| **Employee** | Views monthly salary breakdown and downloads signed PDF salary slips. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/payroll/pages/PayrollDashboard.tsx` | Monthly payroll summary, gross payout, deductions breakdown, and generation trigger button. | `GET /api/autopayroll/summary`, `POST /api/autopayroll/generate` | Payroll Admin |
| `src/app/modules/payroll/components/PayrollApprovalTable.tsx` | Detailed employee-by-employee compensation review table with LOP and net pay. | `GET /api/autopayroll/records` | Payroll Admin |
| `src/app/modules/payroll/pages/MySalarySlipsPage.tsx` | Employee self-service page listing monthly payslips with download action. | `GET /api/payroll/my-slips`, `GET /api/payroll/{id}/download-slip` | Employee |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/AutoPayrollController.cs` | API endpoints to trigger batch payroll, fetch payroll run summaries, and publish payslips. | `IAutoPayrollServices` |
| `Controllers/PayRollController.cs` | Endpoints for single-employee payroll queries, manual adjustments, and payslip PDF downloads. | `IPayRollServices` |
| `Services/AutoPayrollServices.cs` | Batch engine that verifies `PayrollPeriodRules`, queries locked attendance, resolves salary models, computes net salary, and calls `PdfService`. | `IMongoRepository<PayrollEntity>`, `IMongoRepository<MonthlyAttendanceSummary>`, `ISalaryService`, `IPdfService`, `IPayrollDivisorPolicyService` |
| `Services/PdfService.cs` | Renders HTML template `SalarySlipTemplate.html` into PDF byte streams stored under `/fs/`. | `TemplateEngine`, file system storage |

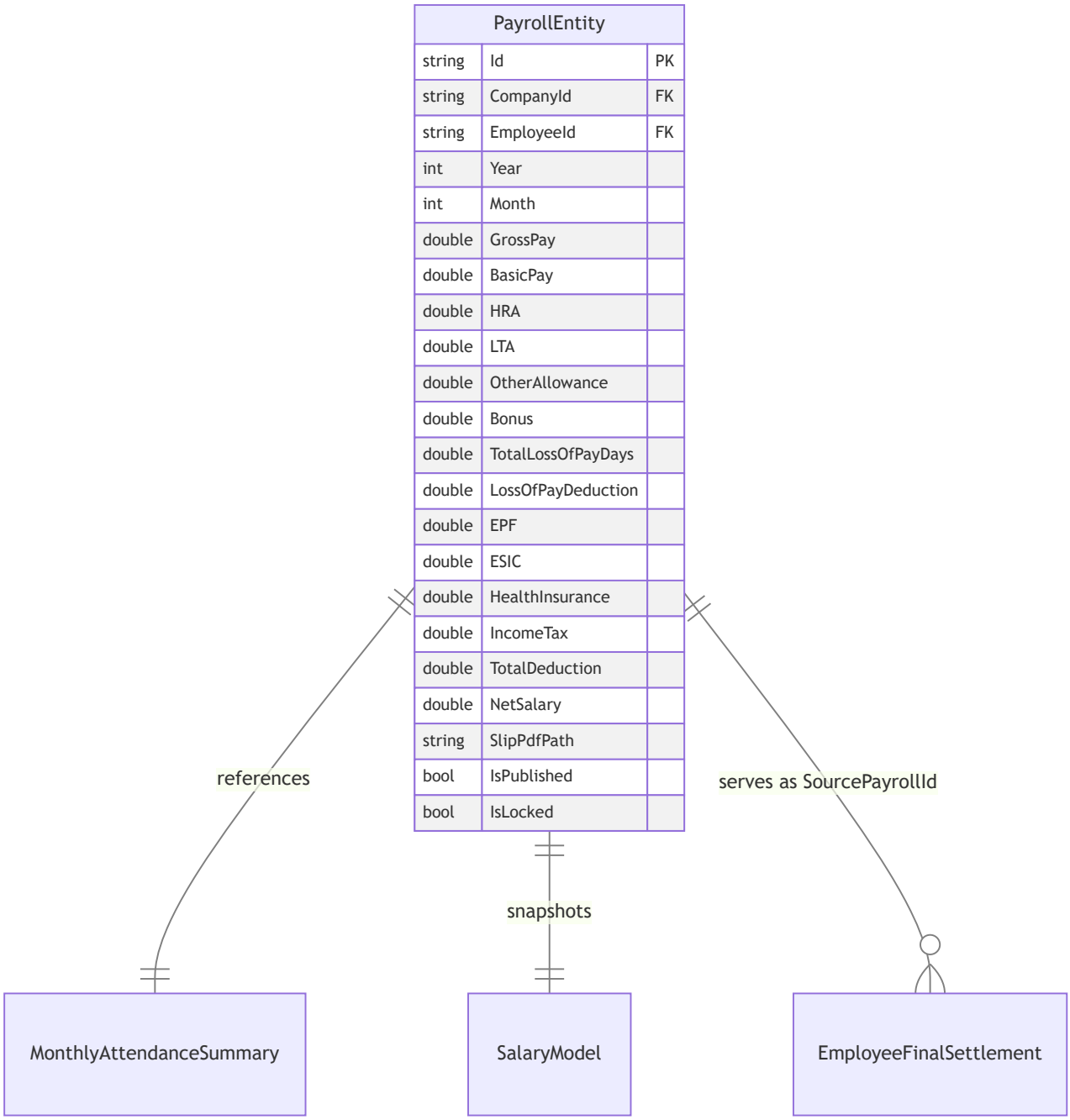
## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `POST` | `/api/autopayroll/generate` | Generates batch payroll for specified month and year. | JWT (`payroll:create`) | `{ Year, Month }` -> `PayrollBatchResultDto` | Creates `PayrollEntity` records, renders payslip PDFs |
| `GET` | `/api/autopayroll/summary` | Retrieves aggregate payroll costs, deductions, and headcounts. | JWT (`payroll:read`) | `PayrollSummaryDto` | None |

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `POST` | `/api/autopayroll/publish` | Publishes approved payroll to employees, enabling payslip downloads. | JWT ( `payroll:approve` ) | `{ Year, Month }` | Updates `IsPublished = true` , sends SignalR notifications |
| `GET` | `/api/payroll/{id}/download-slip` | Streams employee payslip PDF file. | JWT ( `payroll:read` ) | Streams `application/pdf` | None |

## 7. Database Model

ARCHITECTURE / WORKFLOW DIAGRAM



## 8. Business Rules

- **Pre-Execution Gate**:
  - `PayrollPeriodRules.IsClosedPeriod`: Cannot rerun or overwrite a payroll period that is permanently closed.
  - `MonthlyAttendanceSummary.IsApproved && IsLocked`: Attendance must be locked before payroll can run.
  - Zero `PENDING_REVIEW` exceptions in `AttendancePayrollException`.
- **Loss of Pay (LOP) Rule**:

$$\text{TotalLossOfPayDays} = \text{UnpaidDays} + \text{FinalPenaltyDays}$$

$$\text{PerDayRate} = \frac{\text{GrossPay}}{\text{ResolvedDivisor}}$$

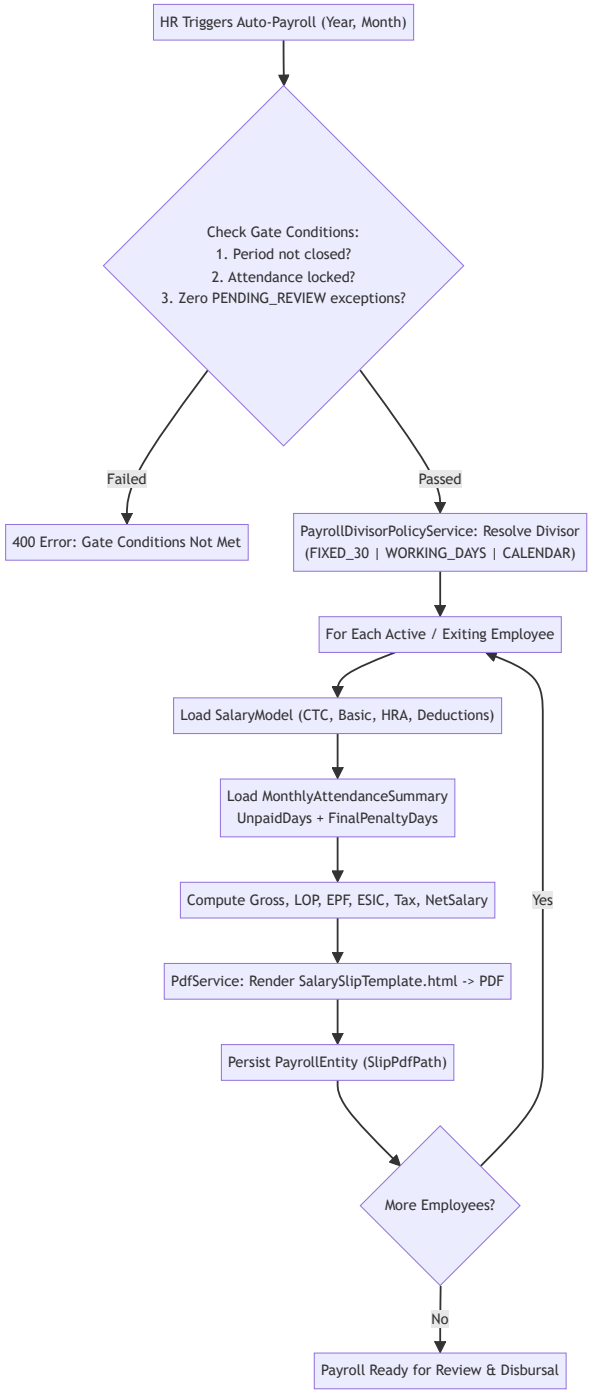
$$\text{LossOfPayDeduction} = \text{TotalLossOfPayDays} \times \text{PerDayRate}$$

- **Statutory Rules**:
  - EPF: Standard 12% of Basic Pay capped at statutory threshold unless employee opted out.
  - ESIC: Applicable only if Gross Pay is below ESIC statutory wage limit.
- **Net Salary Formula**:

$$\text{NetSalary} = \text{GrossPay - TotalDeduction}$$

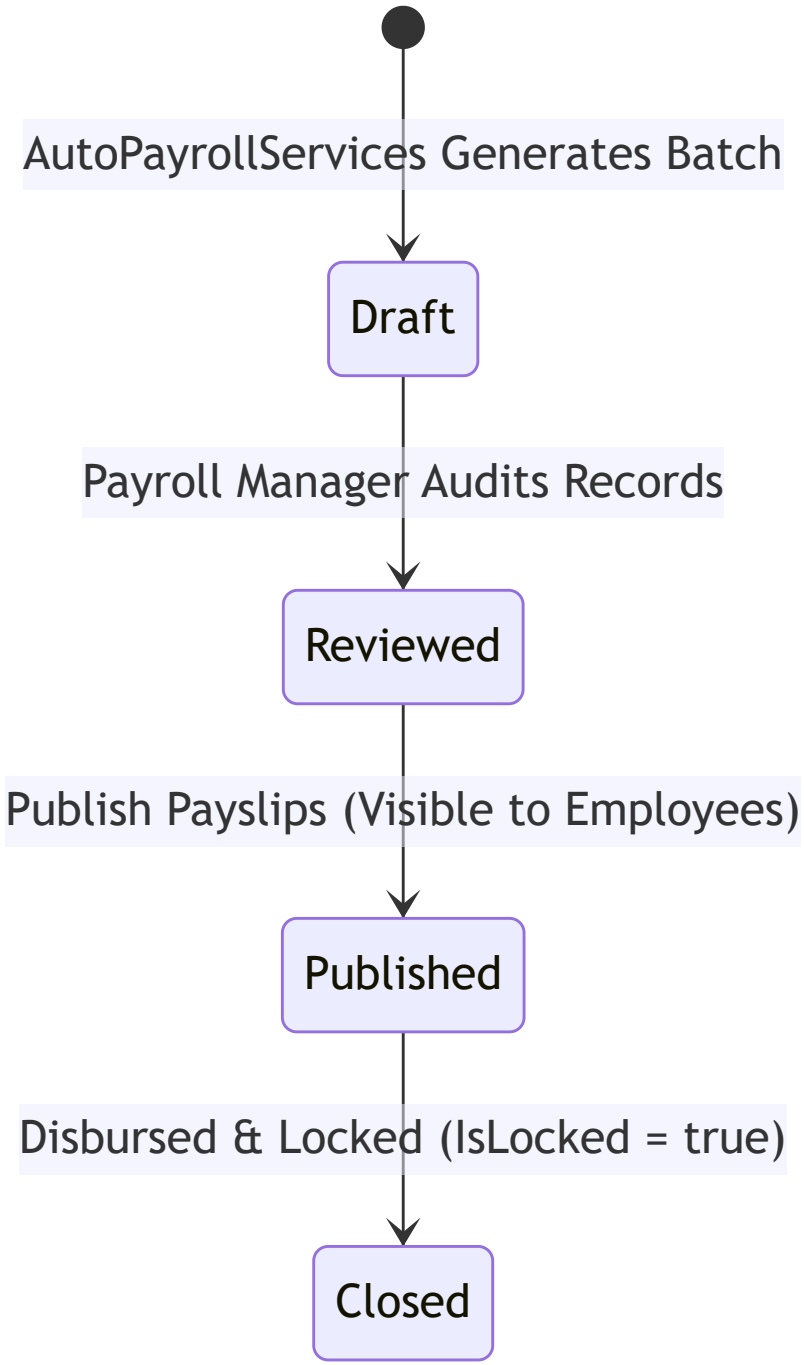
# 9. Complete Workflow

### ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `payroll:read` — View payroll summaries and personal payslips.
- `payroll:create` — Trigger batch payroll generation.
- `payroll:approve` — Publish and lock monthly payroll batches.

## 12. Dependencies

- **Upstream**: `ATTENDANCE` ( `MonthlyAttendanceSummary` , `AttendanceMonthLock` ), `SALARY` ( `SalaryModel` ), `PAYROLL-SETTINGS` ( `PayrollDivisorPolicyService` ).
- **Downstream**: `FINAL-SETTLEMENT` ( `SourcePayrollId` ), `DOCUMENTS` (PDF storage under `/fs/` ).

## 13. Cross-Module Events

- **Payroll Publication**: Dispatches real-time payslip availability alerts via `NotificationHub` .
- **Exit Settlement**: Generates pro-rated final month payroll record linked to `EmployeeFinalSettlement` .

## 14. Failure Scenarios

- **Attendance Not Locked**: Generation aborts with explicit error preventing corrupted LOP calculation.
- **PDF Generation Failure**: Caught and logged; batch execution flags individual payslip rendering error without rolling back entire payroll run.

## 15. Edge Cases

- **Mid-Month Joining / Exit**: Salary calculation scales pro-rated gross pay based on active calendar days in that month.

## 16. Current Strengths

- Unyielding attendance-to-payroll safety gates prevent unverified attendance from causing payroll errors.
- Native server-side PDF payslip compilation with HTML templating.

## 17. Current Gaps

- Direct banking portal API integration (e.g., ICICI/HDFC corporate banking direct debit API) is not yet wired; payroll produces exportable bank payout sheets.

## 18. Recommended Improvements

- Add automated banking payment file format exports (NACH / NEFT batch format).

## 19. Future Scope

- Automated direct-tax TDS return filing integration and flexible employee tax declaration proofs portal.

# BENEFITS & STATUTORY COMPLIANCE MODULE (`BENEFITS.md`)

## 1. Module Overview

The **Benefits & Statutory Compliance Module** tracks employee enrollment in statutory schemes (Employees' Provident Fund `EPF`, Employees' State Insurance `ESIC`), corporate group health insurance (`HealthInsurance`), and benefit continuation/termination during offboarding (`EmployeeExitBenefitItem`).

## 2. Real-World Business Scenario

1. **Active Employment Benefits**:
   - Rahul Sharma is enrolled in Group Medical Insurance (`HealthInsurance = ₹1,500/month`) and statutory Provident Fund (`EPF = ₹1,800/month`) within his `SalaryModel`.
2. **Offboarding Benefit Termination**:
   - When Rahul resigns, `EmployeeExitService` creates `EmployeeExitBenefitItem` records for his active benefits (`Health Insurance`, `EPF`, `Gratuity/LTA`).
   - HR/Finance sets the `CoverageEndDateUtc` (e.g., end of exit month `2026-06-30` for Group Health Insurance) and records final settlement adjustments (`AmountAdjustment`) before completing offboarding.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin / Benefits Admin** | Configures employee statutory/insurance amounts in `SalaryModel` and closes `EmployeeExitBenefitItem` records during offboarding. |
| **Finance / Payroll Admin** | Verifies monthly EPF/ESIC/Insurance deductions and processes final benefit adjustments in `EmployeeFinalSettlement`. |
| **Employee / Former Employee** | Views active deductions on salary slips and benefit termination status in the offboarding / alumni portal. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/payroll/pages/SalaryStructurePage.tsx` | Configures `EPF`, `ESIC`, `HealthInsurance`, and `LTA` components. | `PUT /api/salary/{id}` | HR / Payroll Admin |
| `src/app/modules/offboarding/components/ExitBenefitsTab.tsx` | Manages `EmployeeExitBenefitItem` clearance, coverage end dates, and settlement notes. | `GET /api/employee-exit/{id}/benefits`, `PUT /api/employee-exit/{id}/benefits` | HR / Finance |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Services/AutoPayrollServices.cs` | Deducts `EPF`, `ESIC`, and `HealthInsurance` during monthly payroll processing. | `IMongoRepository<SalaryModel>` |
| `Services/EmployeeExitService.cs` | Tracks and closes `EmployeeExitBenefitItem` records during employee separation. | `IMongoRepository<EmployeeExitBenefitItem>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/employee-exit/{exitId}/benefits` | Lists benefit items (`EmployeeExitBenefitItem`) requiring termination/settlement for an exiting employee. | JWT (`offboarding:read`) | `List<EmployeeExitBenefitItemDto>` | None |
| `PUT` | `/api/employee-exit/{exitId}/benefits/{itemId}` | Updates benefit termination status, `CoverageEndDateUtc`, and settlement adjustment. | JWT (`offboarding:update`) | `UpdateExitBenefitItemDto` | Updates `EmployeeExitBenefitItem` |

## 7. Database Model

**`EmployeeExitBenefitItem` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/EmployeeExitBenefitItem.cs`)**

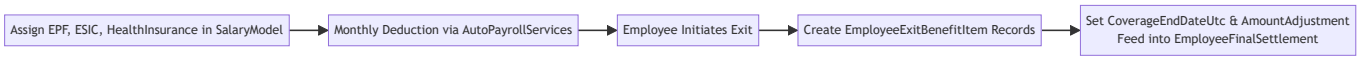
- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `ExitId` (`string`, FK -> `EmployeeExit.Id`)
- `EmployeeId` (`string`, FK)
- `BenefitType` (`string`, `"EPF"` | `"ESIC"` | `"HealthInsurance"` | `"Gratuity"` | `"Other"`)
- `Status` (`string`, `"Pending"` | `"Terminated"` | `"Transferred"` | `"Settled"`)
- `CoverageEndDateUtc` (`DateTime?`)
- `AmountAdjustment` (`double`)
- `Notes` (`string`)

## 8. Business Rules

- **Statutory Deduction Rule**: `EPF`, `ESIC`, and `HealthInsurance` defined on `SalaryModel` are automatically included in `TotalDeduction` by `AutoPayrollServices`.
- **Exit Clearance Rule**: All `EmployeeExitBenefitItem` records associated with an `EmployeeExit` must be moved out of `Pending` status (`Terminated`, `Transferred`, or `Settled`) before final settlement closure.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

`EmployeeExitBenefitItem`: `Pending` → `Terminated` / `Transferred` / `Settled`.

## 11. Permissions

- `payroll:update` — Configure active benefit deductions.
- `offboarding:update` — Manage `EmployeeExitBenefitItem` closure.

## 12. Dependencies

- **Upstream**: `SALARY` (`SalaryModel`).
- **Downstream**: `PAYROLL` (`AutoPayrollServices`), `OFFBOARDING` (`EmployeeExit`), `FINAL-SETTLEMENT` (`EmployeeFinalSettlement`).

## 13. Cross-Module Events

- Benefit adjustments (`AmountAdjustment`) on `EmployeeExitBenefitItem` feed into `EmployeeFinalSettlement` calculations.

## 14. Failure Scenarios

- **Unclosed Benefit Item**: Attempting to finalize exit while `EmployeeExitBenefitItem.Status == "Pending"` triggers a clearance readiness warning.

## 15. Edge Cases

- **Insurance Portability**: `Status = "Transferred"` records corporate-to-retail policy conversion without financial deduction.

## 16. Current Strengths

- Explicit lifecycle tracking of benefits not only during active payroll but also through offboarding termination (`EmployeeExitBenefitItem`).

## 17. Current Gaps

- Standalone employee flexible benefit enrollment window (e.g., choosing parental insurance top-up plans self-service) is not yet separate from HR `SalaryModel` setup.

## 18. Recommended Improvements

- Add self-service insurance dependent declaration cards ( `FamilyMembers` ) linked to `HealthInsurance` .

## 19. Future Scope

- Direct ECR file generation for EPFO portal uploads.

MOD-22 PART III docs/project-documentation/modules/NOTICE-PERIOD.md

# NOTICE PERIOD MANAGEMENT MODULE (`NOTICE-PERIOD.md`)

## 1. Module Overview

The **Notice Period Management Module** governs the transition phase between resignation/termination initiation and an employee's `LastWorkingDayUtc` (`LWD`). It resolves notice duration via `OffboardingPolicyResolver` (`NoticeDays` vs `ProbationNoticeDays`), manages notice buyouts, waivers, and shortfall calculations (`NoticeAdjustments`), enforces leave restrictions during notice (`AllowDuringNoticePeriod`), and feeds `NoticeRecovery` or buyout credits directly into `EmployeeFinalSettlement`.

## 2. Real-World Business Scenario

1. **Resignation & Notice Snapshot**:
   - Rahul Sharma (confirmed Engineering employee) submits his resignation on `2026-05-01`.
   - `OffboardingPolicyResolver` resolves the Engineering department `EmployeeExitPolicy`, snapping `NoticeDays = 60` into `EmployeeExit.PolicySnapshot`.
   - Expected `LastWorkingDayUtc` is calculated as `2026-06-30` (`60` calendar days).
   - `EmployeeLifecycle` transitions `EmploymentStatus` to `NoticePeriod` (`4`) and `OffboardingStatus` to `NoticePeriod` (`2`).
2. **Notice Buyout / Shortfall Adjustment**:
   - Rahul requests an early release on `2026-06-15` (`15` days early).
   - Reporting Manager and HR approve a 5-day waiver (`WaivedDays = 5`) and a 10-day buyout (`ShortfallDays = 10`, `BuyoutApplicable = true`).
   - `EmployeeFinalSettlement` automatically computes `NoticeRecovery` for the `10` unserved shortfall days using the active payroll divisor policy.

## 3. Actors

| Actor | Capabilities |
| --- | --- |
| **Employee** | Submits resignation, views calculated `LastWorkingDayUtc`, requests early release / notice buyout. |
| **Reporting Manager** | Reviews proposed `LastWorkingDayUtc`, approves or rejects early release and handover schedule. |
| **HR Admin** | Configures `EmployeeExitPolicy` (`NoticeDays`, `ProbationNoticeDays`, `BuyoutAllowed`), records `NoticeAdjustments` (waiver vs recovery). |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/offboarding/pages/MyResignationPage.tsx` | Displays required `NoticeDays`, calculated `LastWorkingDayUtc`, served days counter, and early release request form. | `POST /api/employee-exit/initiate`, `GET /api/employee-exit/my-exit` | Employee |
| `src/app/modules/offboarding/components/NoticeAdjustmentModal.tsx` | HR/Manager interface to modify `LastWorkingDayUtc`, grant `WaivedDays`, or apply `ShortfallDays` recovery. | `PUT /api/employee-exit/{id}/notice-adjustment` | HR / Manager |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Services/OffboardingPolicyResolver.cs` | Resolves exact `NoticeDays` vs `ProbationNoticeDays` using 3-tier precedence (Department -> Role -> Company) and creates immutable `EmployeeExitPolicySnapshot`. | `IMongoRepository<EmployeeExitPolicy>` |
| `Services/EmployeeExitService.cs` | Computes initial `LastWorkingDayUtc`, processes `NoticeAdjustments` (waiver, buyout, extension), and updates `EmployeeExitAttendanceCutoff`. | `IMongoRepository<EmployeeExit>`, `IMongoRepository<EmployeeExitAttendanceCutoff>` |
| `Services/LeaveManagementService.cs` | Restricts leave applications during notice period based on `LeavePolicy.AllowDuringNoticePeriod`. | `IMongoRepository<LeavePolicy>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `POST` | `/api/employee-exit/initiate` | Initiates resignation/exit and calculates `LastWorkingDayUtc` from resolved policy. | JWT (`offboarding:create`) | `InitiateExitDto` -> `EmployeeExitDto` | Sets `EmploymentStatus = NoticePeriod`, snapshots policy |
| `PUT` | `/api/employee-exit/{id}/notice-adjustment` | Records notice waiver, buyout, or LWD extension (`NoticeAdjustments`). | JWT (`offboarding:update`) | `NoticeAdjustmentDto` | Updates `LastWorkingDayUtc`, syncs `EmployeeExitAttendanceCutoff` |

## 7. Database Model

### Embedded in `EmployeeExit` ( `CMS-Backend-Core/Codeji.CMS.Repository/Entities/EmployeeExit.cs` )

- `PolicySnapshot` ( `EmployeeExitPolicySnapshot` ):
  - `NoticeDays` ( `int` )
  - `ProbationNoticeDays` ( `int` )
  - `BuyoutAllowed` ( `bool` )
  - `WaiverAllowed` ( `bool` )
- `RequiredNoticeDays` ( `int` )
- `ServedNoticeDays` ( `int` )
- `ShortfallDays` ( `int` )
- `WaivedDays` ( `int` )
- `LastWorkingDayUtc` ( `DateTime` )
- `NoticeAdjustments` ( `List<NoticeAdjustmentEntry>` )

---

## 8. Business Rules

- **Probation vs Confirmed Rule**:
  - If `EmployeeLifecycle.EmploymentStatus == Probation` , `RequiredNoticeDays = PolicySnapshot.ProbationNoticeDays` .
  - Otherwise, `RequiredNoticeDays = PolicySnapshot.NoticeDays` .
- **Shortfall Math Rule**:

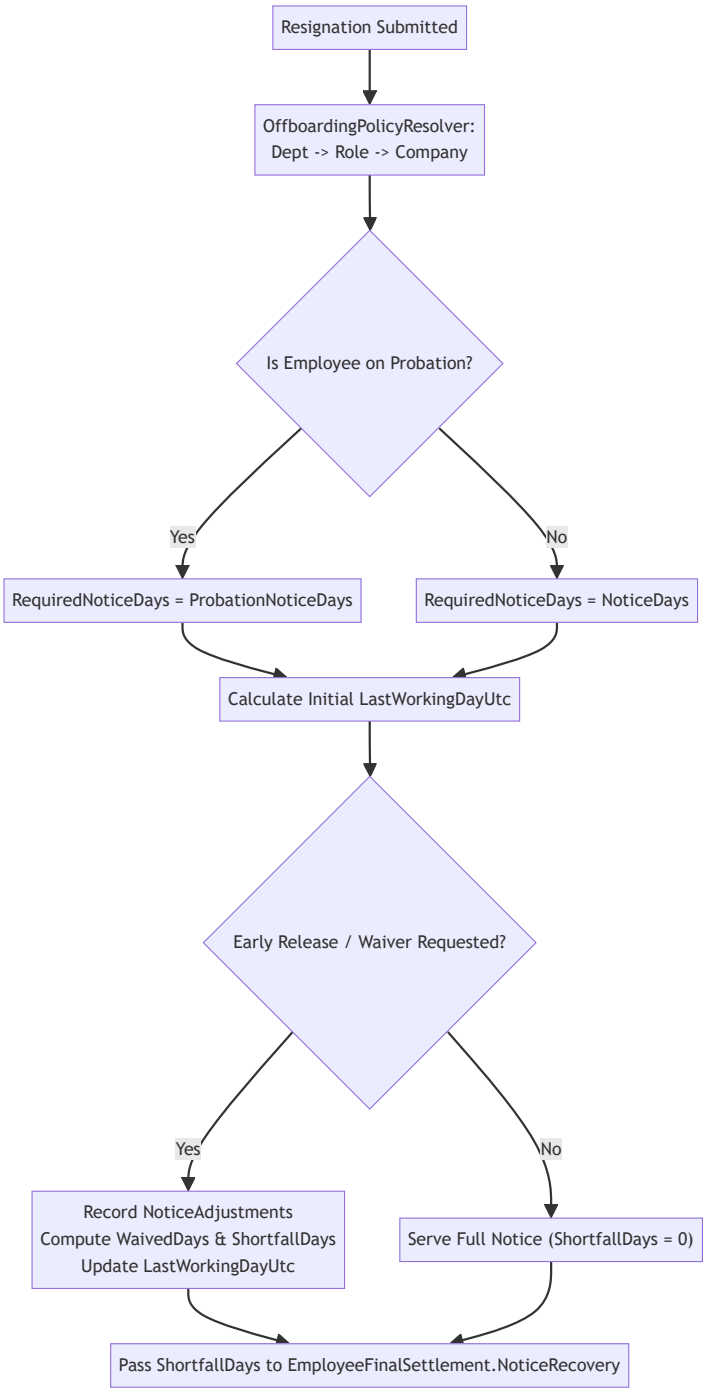
  $$\text{ShortfallDays} = \max(0, \text{RequiredNoticeDays - ServedNoticeDays - WaivedDays})$$

- **Leave Restriction Rule**: While in `NoticePeriod` , any leave type where `LeavePolicy.AllowDuringNoticePeriod == false` is automatically blocked.
- **Policy Immutability Rule**: Once `EmployeeExit` is created, `PolicySnapshot` is frozen on the document so subsequent edits to `EmployeeExitPolicy` do not alter an in-flight employee's notice terms.

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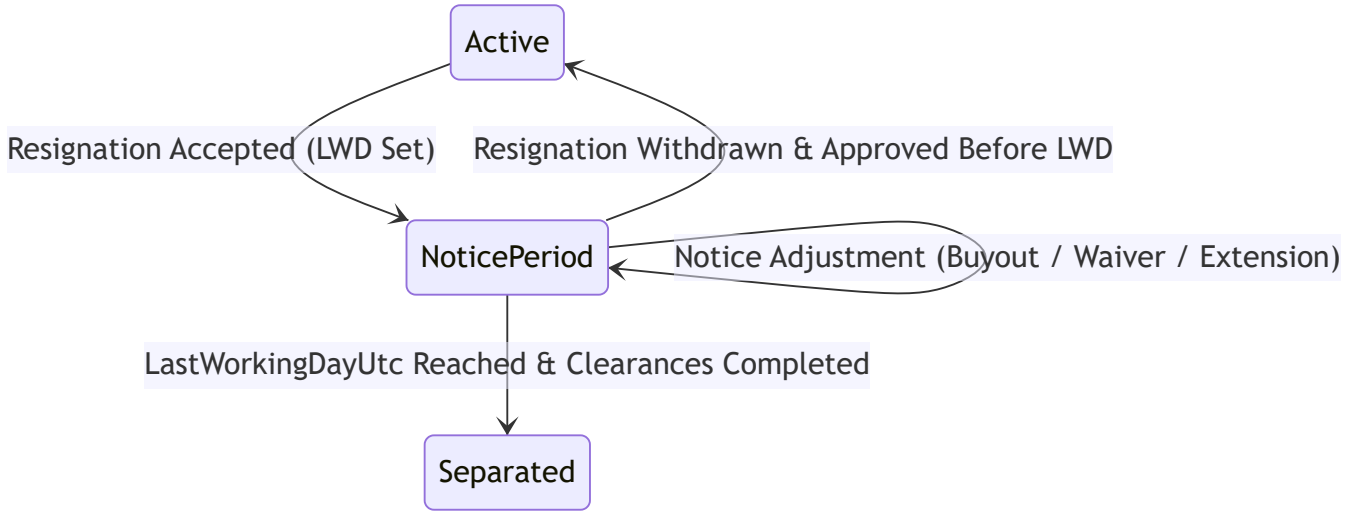
## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `offboarding:create` — Submit resignation.
- `offboarding:update` — Approve early release, buyout, or notice waiver.

## 12. Dependencies

- **Upstream**: `PROBATION` (`EmploymentStatus`), `POLICY` (`EmployeeExitPolicy`, `OffboardingPolicyResolver`).
- **Downstream**: `LEAVE` (`AllowDuringNoticePeriod`), `ATTENDANCE` (`EmployeeExitAttendanceCutoff`), `FINAL-SETTLEMENT` (`NoticeRecovery`).

## 13. Cross-Module Events

- **LWD Change**: Automatically updates `EmployeeExitAttendanceCutoff.LastWorkingDayUtc` so attendance blocking aligns with the revised date.

## 14. Failure Scenarios

- **Buyout Requested When** `BuyoutAllowed == false`: Rejected by `EmployeeExitService` unless HR explicitly overrides with `WaiverAllowed`.

## 15. Edge Cases

- **Resignation Withdrawal**: If an employee withdraws their resignation before `LastWorkingDayUtc` and HR approves, `EmployeeLifecycle` reverts from `NoticePeriod` (`4`) back to `Active` (`3`) or `Probation` (`2`).

## 16. Current Strengths

- Immutable `EmployeeExitPolicySnapshot` guarantees contractual fairness and auditability even if company policies change mid-notice.

## 17. Current Gaps

- Automated extension of `LastWorkingDayUtc` when an employee takes unpaid leave during notice is currently handled via manual HR `NoticeAdjustments` rather than an automatic day-for-day shift.

## 18. Recommended Improvements

- Add an optional `ExtendLwdOnLeaveDuringNotice` boolean flag to `EmployeeExitPolicy`.

## 19. Future Scope

- Interactive notice buyout cost estimator on the employee self-service resignation form.

# ASSET MANAGEMENT & CLEARANCE MODULE (`ASSET-MANAGEMENT.md`)

## 1. Module Overview

The **Asset Management & Clearance Module** tracks physical and digital company assets issued to employees (laptops, monitors, access cards, mobile devices, peripherals) and governs their mandatory inspection, return, or financial recovery (`EmployeeExitAsset`) during employee offboarding.

## 2. Real-World Business Scenario

1. **Asset Assignment & Tracking**:
   - Upon joining, Rahul Sharma is issued a MacBook Pro (`AssetTag = "CODEJI-MBP-042"`) and an RFID Building Access Card (`AssetTag = "CODEJI-ID-109"`).
2. **Offboarding Asset Inspection**:
   - When Rahul enters offboarding, `EmployeeExitService` populates `EmployeeExitAsset` records for both items.
   - On his last working day, Rahul hands over both items to the IT / Asset Admin.
   - The RFID card is marked `Status = "Returned"`, `Condition = "Good"`, `DeductionAmount = 0`.
   - The MacBook Pro has a cracked display; IT marks `Status = "Damaged"`, `Condition = "Screen cracked"`, `DeductionAmount = 2500`, and `RecoverFromSettlement = true`.
3. **Financial Recovery Integration**:
   - `EmployeeFinalSettlement` automatically pulls `DeductionAmount = ₹2,500` into `OutstandingDues` and unlocks the `RequireAssetClearance` gate on `EmployeeExit`.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **IT Admin / Asset Manager** | Records issued assets, inspects returned hardware during exit, sets `Condition` and `DeductionAmount`, and signs off asset clearance. |
| **HR Admin** | Monitors asset return readiness as part of `EmployeeExit` completion gates. |
| **Employee** | Views assigned company assets and return status during offboarding. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/offboarding/components/ExitAssetsTab.tsx` | Table and inspection modal for tracking `EmployeeExitAsset` items (`AssetName`, `AssetTag`, `Condition`, `Status`, `DeductionAmount`). | `GET /api/employee-exit/{id}/assets`, `POST /api/employee-exit/{id}/assets`, `PUT /api/employee-exit/{id}/assets/{assetId}` | IT Admin / HR Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/EmployeeExitController.cs` | Endpoints for listing, adding, inspecting, and clearing `EmployeeExitAsset` items. | `IEmployeeExitService` |
| `Services/EmployeeExitService.cs` | Validates `RequireAssetClearance` policy gate and aggregates `DeductionAmount` into `EmployeeFinalSettlement.OutstandingDues`. | `IMongoRepository<EmployeeExitAsset>`, `IMongoRepository<EmployeeFinalSettlement>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/employee-exit/{exitId}/assets` | Lists all assets assigned to the exiting employee. | JWT (`offboarding:read`) | `List<EmployeeExitAssetDto>` | None |
| `POST` | `/api/employee-exit/{exitId}/assets` | Registers an asset item requiring return/clearance for the exit case. | JWT (`offboarding:update`) | `CreateExitAssetDto` | Creates `EmployeeExitAsset` |
| `PUT` | `/api/employee-exit/{exitId}/assets/{assetId}` | Updates return status (`Returned`, `Damaged`, `Lost`), condition notes, and `DeductionAmount`. | JWT (`offboarding:update`) | `UpdateExitAssetDto` | Updates `EmployeeExitAsset`, feeds `EmployeeFinalSettlement` |

## 7. Database Model

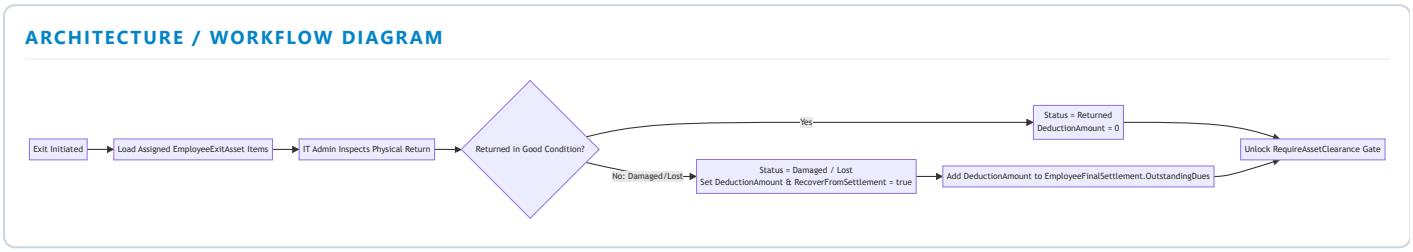
**`EmployeeExitAsset` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/EmployeeExitAsset.cs`)**

- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `ExitId` (`string`, FK -> `EmployeeExit.Id`)
- `EmployeeId` (`string`, FK)
- `AssetName` (`string`, e.g., `"MacBook Pro 14-inch"`)
- `AssetTag` (`string`, e.g., `"CODEJI-MBP-042"`)
- `Category` (`string`, `"Laptop"` | `"Accessory"` | `"IDCard"` | `"Mobile"` | `"Other"`)
- `Status` (`string`, `"PendingReturn"` | `"Returned"` | `"Damaged"` | `"Lost"` | `"Waived"`)
- `Condition` (`string`)
- `DeductionAmount` (`double`)
- `RecoverFromSettlement` (`bool`)
- `ClearedByUserId` (`string`)
- `ClearedAtUtc` (`DateTime?`)

## 8. Business Rules

- **Clearance Gate Rule**: When `EmployeeExit.PolicySnapshot.RequireAssetClearance == true`, `EmployeeExit` cannot transition to `Completed` while any `EmployeeExitAsset` has `Status == "PendingReturn"`.
- **Settlement Recovery Rule**: If `Status` is `"Damaged"` or `"Lost"` and `RecoverFromSettlement == true`, `DeductionAmount` must be $\geq 0$ and is automatically added to `EmployeeFinalSettlement.OutstandingDues`.

## 9. Complete Workflow




## 10. State Machine

`PendingReturn` → `Returned` | `Damaged` (Recovered in FnF) | `Lost` (Recovered in FnF) | `Waived`.

## 11. Permissions

- `offboarding:read` — View assigned exit assets.
- `offboarding:update` — Inspect assets, record damages/deductions, and mark cleared.

## 12. Dependencies

- **Upstream**: `OFFBOARDING` (`EmployeeExit`, `EmployeeExitPolicySnapshot`).
- **Downstream**: `FINAL-SETTLEMENT` (`EmployeeFinalSettlement.OutstandingDues`).

## 13. Cross-Module Events

- Recording `DeductionAmount > 0` triggers recalculation readiness on `EmployeeFinalSettlement`.

## 14. Failure Scenarios

- **Unreturned Asset at LWD**: Blocks `POST /api/employee-exit/{id}/complete` until IT either marks the asset `Returned` or switches it to `Lost` with `RecoverFromSettlement = true`.

## 15. Edge Cases

- **Employee Asset Buyback**: Handled by setting `Status = "Lost/Retained"` with the depreciated buyout value entered in `DeductionAmount` (`RecoverFromSettlement = true`).

## 16. Current Strengths

- Direct financial bridge between IT hardware inspection ( `EmployeeExitAsset.DeductionAmount` ) and Finance Final Settlement ( `EmployeeFinalSettlement` ).

## 17. Current Gaps

- Standalone day-to-day IT inventory procurement/stockroom lifecycle (before assignment to an employee) is not modeled as a separate warehouse module; asset tracking is currently focused on employee assignment and offboarding clearance ( `EmployeeExitAsset` ).

## 18. Recommended Improvements

- Add a global `CompanyAssetInventory` collection to track unassigned stock, warranty expiry dates, and serial numbers from procurement to disposal.

## 19. Future Scope

- QR-code / barcode label generation and scanning for instant asset check-in/check-out.

# OFFBOARDING, CLEARANCE & FORMER EMPLOYEE PORTAL MODULE (`OFFBOARDING.md`)

## 1. Module Overview

The **Offboarding, Clearance & Former Employee Portal Module** is one of the most comprehensive multi-stage engines in **Codeji CMS / HRMS**. It governs employee resignation/termination (`EmployeeExit`), dynamic policy resolution (`OffboardingPolicyResolver`), multi-department clearance checklists (`EmployeeExitClearanceTemplate` across IT, Finance, Admin, Reporting Manager, HR), knowledge transfer (`EmployeeExitHandoverItem`), exit interviews (`EmployeeExitInterview`), reliable event outbox processing (`EmployeeExitOutboxProcessor`), account deactivation & attendance cutoff (`EmployeeExitAttendanceCutoff`), and token-based post-exit self-service (`FormerEmployeeAccessToken` & `PublicFormerEmployeePortalController`).

---

## 2. Real-World Business Scenario

1. **Initiation & Policy Snapshot**:
   - Rahul Sharma submits his resignation on `2026-05-01`. `OffboardingPolicyResolver` matches his Engineering department policy (`NoticeDays = 60`, `RequireExitInterview = true`, `RequireHandover = true`, `RequireAssetClearance = true`, `FormerEmployeePortalDays = 90`) and embeds an immutable `EmployeeExitPolicySnapshot` inside `EmployeeExit`.
2. **Parallel Multi-Department Clearances**:
   - `EmployeeExitService` instantiates department clearance tasks from `EmployeeExitClearanceTemplate`:
     - **Reporting Manager**: Approves `EmployeeExitHandoverItem` list (code repositories, documentation transferred to colleague).
     - **IT Admin**: Revokes VPN/GitHub access and clears assigned laptop in `EmployeeExitAsset`.
     - **Finance**: Verifies zero travel advance dues and prepares `EmployeeFinalSettlement`.
     - **HR**: Conducts `EmployeeExitInterview` and generates `EmployeeExitDocument` (Relieving Letter & Experience Certificate).
3. **Final Completion & Outbox Execution**:
   - On `LastWorkingDayUtc` (`2026-06-30`), once all clearance gates pass, HR clicks **Complete Exit** (`POST /api/employee-exit/{id}/complete`).
   - `EmployeeExitService` sets `EmpUser.IsActive = false`, revokes all active `RefreshToken` sessions, writes `EmployeeExitAttendanceCutoff`, transitions `EmployeeLifecycle` (`EmploymentStatus = Separated`, `AccountStatus = Deactivated`, `FormerPortalStatus = Enabled`, `AlumniStatus = Eligible`), and writes an `EmployeeExitOutbox` event.
4. **Former Employee Portal Access**:
   - `EmployeeExitOutboxProcessor` dispatches an email containing a magic link with a SHA-256 hashed `FormerEmployeeAccessToken` (`ExpiresAtUtc = LWD + 90 days`).
   - Rahul accesses `/former-employee` via `PublicFormerEmployeePortalController` (`[AllowAnonymous]`) to download his Relieving Letter, Final Settlement statement, and last 6 salary slips without needing active internal credentials.

---

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Employee** | Initiates resignation, tracks clearance progress, submits `EmployeeExitHandoverItem` tasks, completes `EmployeeExitInterview`. |
| **Reporting Manager** | Approves resignation & LWD, verifies handover completion. |
| **IT / Asset / Finance Admins** | Complete department-specific clearance items, log asset returns (`EmployeeExitAsset`), and record financial recoveries. |
| **HR Admin** | Orchestrates entire `EmployeeExit`, overrides/waives policies, issues exit letters (`EmployeeExitDocument`), completes exit, or processes rejoin (`EmployeeRejoinRecord`). |
| **Former Employee (Alumni)** | Uses `FormerEmployeeAccessToken` on `/former-employee` to download tax/exit documents and view settlement status. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/offboarding/pages/OffboardingDashboard.tsx` | HR command center listing all active `EmployeeExit` cases by stage (`Initiated`, `NoticePeriod`, `ClearanceInProgress`, `ReadyForSettlement`, `Completed`). | `GET /api/employee-exit` | HR Admin |
| `src/app/modules/offboarding/pages/ExitCaseDetailPage.tsx` | Multi-tab workspace covering Clearances, Handover, Assets, Benefits, Exit Interview, Documents, and Final Settlement. | `GET /api/employee-exit/{id}`, `POST /api/employee-exit/{id}/complete` | HR / Department Clearers |
| `src/app/modules/offboarding/pages/FormerEmployeePortalPage.tsx` | Public token-authenticated portal (`/former-employee`) for separated staff to download payslips, FnF statement, and relieving letters. | `GET /api/public/former-employee/summary`, `GET /api/public/former-employee/documents` | Former Employee |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/EmployeeExitController.cs` | Full lifecycle endpoints for initiating, approving, clearing, settling, and completing employee exits. | `IEmployeeExitService`, `IOffboardingPolicyResolver` |
| `Controllers/PublicFormerEmployeePortalController.cs` | `[AllowAnonymous]` controller authenticating `FormerEmployeeAccessToken` bearer headers/query tokens for post-exit document downloads. | `IFormerEmployeePortalService` |

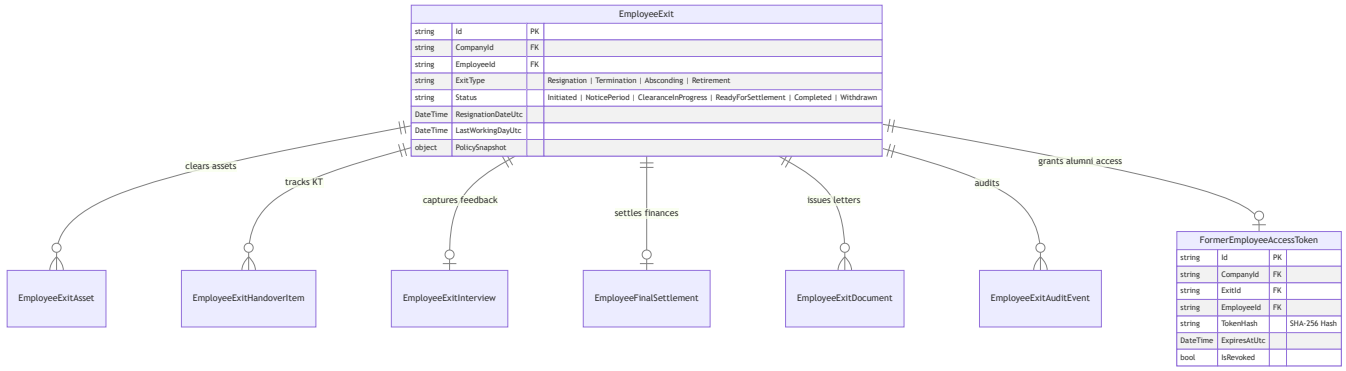
| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Services/OffboardingPolicyResolver.cs` | Resolves `EmployeeExitPolicy` using 3-tier specificity ( `Department -> Role -> Company` ) and constructs `EmployeeExitPolicySnapshot` . | `IMongoRepository<EmployeeExitPolicy>` |
| `Services/EmployeeExitService.cs` | Enforces all clearance gates, manages handover/interview/document items, updates `EmployeeLifecycle` , deactivates `EmpUser` , and writes `EmployeeExitOutbox` . | `IMongoRepository<EmployeeExit>` , `IMongoRepository<EmployeeExitAuditEvent>` , `IEmployeeLifecycleService` |
| `Services/HostedServices/EmployeeExitOutboxProcessor.cs` | Background `HostedService` polling `EmployeeExitOutbox` to dispatch reliable asynchronous notifications and alumni portal access links. | `IMongoRepository<EmployeeExitOutbox>` , `INotificationService` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
| --- | --- | --- | --- | --- | --- |
| `POST` | `/api/employee-exit/initiate` | Initiates voluntary resignation or involuntary termination. | JWT ( `offboarding:create` ) | `InitiateExitDto` -> `EmployeeExitDto` | Snapshots policy, creates clearance checklist, updates `EmployeeLifecycle` |
| `PUT` | `/api/employee-exit/{id}/clearance` | Updates department clearance item status ( `Approved` / `Pending` / `Blocked` ). | JWT ( `offboarding:update` ) | `UpdateClearanceItemDto` | Appends `EmployeeExitAuditEvent` |
| `POST` | `/api/employee-exit/{id}/exit-interview` | Submits exit interview responses ( `EmployeeExitInterview` ). | JWT ( `offboarding:create` ) | `SubmitExitInterviewDto` | Satisfies `RequireExitInterview` gate |
| `POST` | `/api/employee-exit/{id}/complete` | Finalizes separation after all clearance and FnF gates pass. | JWT ( `offboarding:approve` ) | None | Sets `EmpUser.IsActive = false` , revokes `RefreshToken` , creates `FormerEmployeeAccessToken` |
| `GET` | `/api/public/former-employee/summary` | Returns exit documents, FnF summary, and payslips for separated employee. | `FormerEmployeeAccessToken` ( `[AllowAnonymous]` ) | `FormerEmployeeSummaryDto` | Updates `LastAccessedAtUtc` on token |

## 7. Database Model

**ARCHITECTURE / WORKFLOW DIAGRAM**

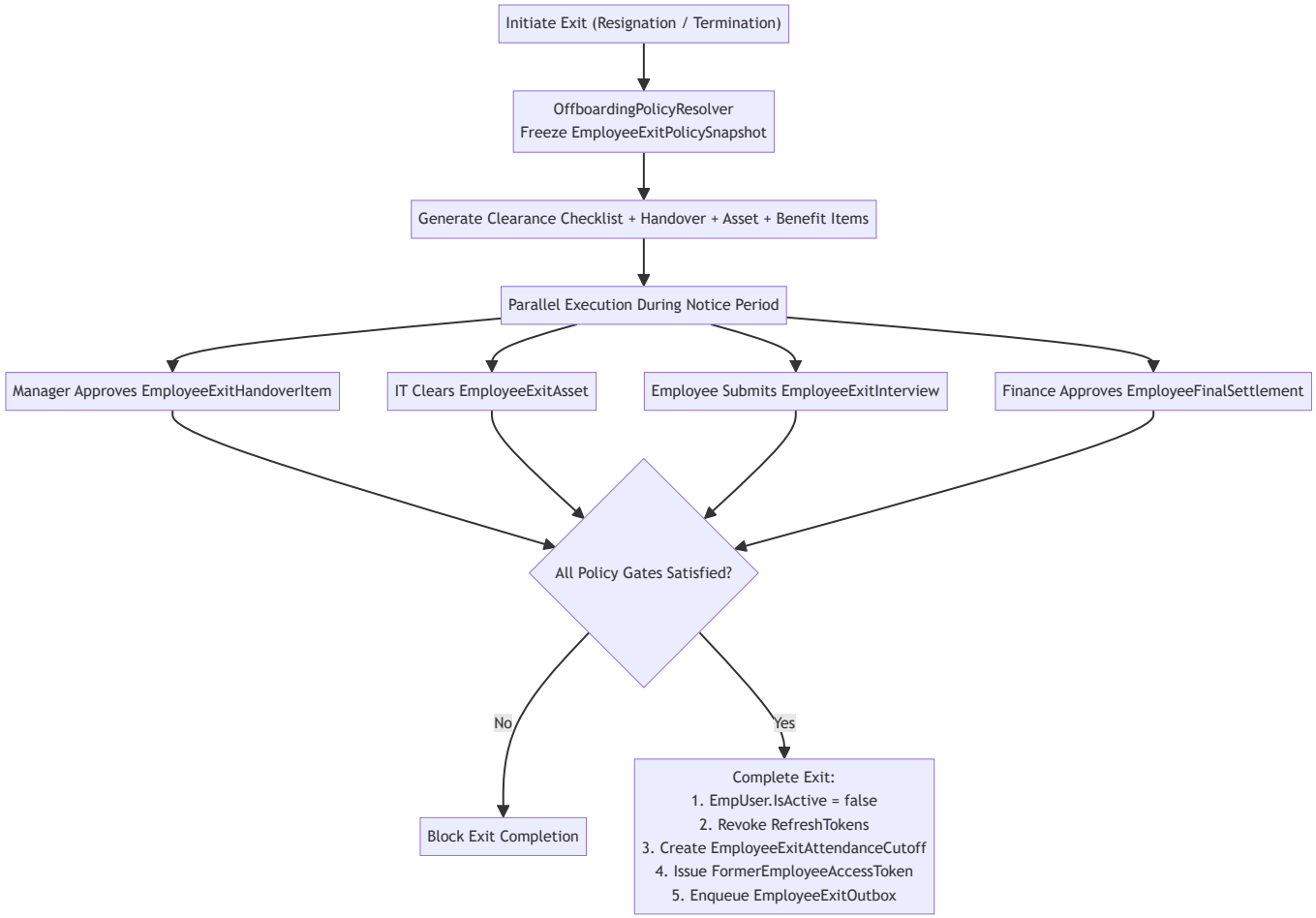



## 8. Business Rules

- **Policy Precedence Rule (`OffboardingPolicyResolver`)**:
  1. Active policy matching exact `DepartmentId` (`!string.IsNullOrEmpty(DepartmentId)`)
  2. Active policy matching exact `RoleId` (`!string.IsNullOrEmpty(RoleId)`)
  3. Default Company policy (`DepartmentId == null && RoleId == null`)
- **Completion Readiness Gate**: `POST /api/employee-exit/{id}/complete` is blocked unless:
  - All department clearance items are `Cleared`
  - If `PolicySnapshot.RequireHandover == true`: all `EmployeeExitHandoverItem` tasks are `Completed`
  - If `PolicySnapshot.RequireAssetClearance == true`: all `EmployeeExitAsset` items are `Returned` or `RecoveredInSettlement`
  - If `PolicySnapshot.RequireExitInterview == true`: `EmployeeExitInterview` is `Submitted`
  - `EmployeeFinalSettlement` is `Approved` or `Settled`
- **Security Cut-Off Rule**: Completing an exit immediately sets `EmpUser.IsActive = false` (blocking standard JWT requests at `CompanyIdMiddleware`), revokes all active `RefreshToken` records, and inserts `EmployeeExitAttendanceCutoff`.

## 9. Complete Workflow

**ARCHITECTURE / WORKFLOW DIAGRAM**



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `offboarding:read` / `offboarding:create` — Initiate and view exit status.
- `offboarding:update` — Clear department items, handover, assets, and benefits.
- `offboarding:approve` — Finalize and complete employee separation.

## 12. Dependencies

- **Upstream**: `EMPLOYEE` ( `EmployeeLifecycle` ), `POLICY` ( `EmployeeExitPolicy` ), `NOTICE-PERIOD` .
- **Downstream**: `ASSET-MANAGEMENT` ( `EmployeeExitAsset` ), `FINAL-SETTLEMENT` ( `EmployeeFinalSettlement` ), `ATTENDANCE` ( `EmployeeExitAttendanceCutoff` ), `DOCUMENTS` ( `EmployeeExitDocument` ).

## 13. Cross-Module Events

- **Exit Completion**: Triggers `EmployeeExitOutbox` → `EmployeeExitOutboxProcessor` → emails `FormerEmployeeAccessToken` link and revokes active sessions.

## 14. Failure Scenarios

- **Expired / Revoked Alumni Token**: `PublicFormerEmployeePortalController` returns `401 Unauthorized` if `DateTime.UtcNow > ExpiresAtUtc` or `IsRevoked == true` .

## 15. Edge Cases

- **Boomerang Employee Rejoin ( `EmployeeRejoinRecord` )**: Allows a `Separated` employee to be re-onboarded under the same historical profile while preserving prior `EmployeeExit` history.

## 16. Current Strengths

- Enterprise-grade separation of Active JWT access ( `EmpUser.IsActive = false` ) from scoped, time-bound Alumni Portal access ( `FormerEmployeeAccessToken` SHA-256 hash).
- Transactional outbox pattern ( `EmployeeExitOutboxProcessor` ) guarantees post-exit notifications are never lost.

## 17. Current Gaps

- External SSO/Active Directory / GitHub SCIM revocation is currently tracked as an IT clearance checklist item rather than an automated outbound webhook call.

## 18. Recommended Improvements

- Add outbound webhook handlers inside `EmployeeExitOutboxProcessor` to automatically deprovision Slack, GitHub, and Google Workspace accounts on exit completion.

## 19. Future Scope

- Alumni network job referral board inside the Former Employee Portal.

# FULL & FINAL SETTLEMENT MODULE ( `FINAL-SETTLEMENT.md` )

## 1. Module Overview

The **Full & Final (FnF) Settlement Module** ( `EmployeeFinalSettlement` ) computes and reconciles the complete financial closure of a separating employee. It aggregates pro-rated final salary from `PayrollEntity` ( `SourcePayrollId` ), leave encashment ( `LeaveEncashment` ), approved expense reimbursements ( `Reimbursements` ), and bonuses ( `BonusesAndIncentives` ), then subtracts unserved notice period buyouts ( `NoticeRecovery` ), asset damage/unreturned recoveries ( `EmployeeExitAsset.DeductionAmount` ), and other dues ( `OutstandingDues` ) to determine `NetPayable` and settlement `Direction` .

## 2. Real-World Business Scenario

1. **Gathering Inputs for Rahul Sharma's Exit ( `LWD: 2026-06-15` )**:
   - **Pro-rated June Salary ( `SalaryPayable` )**: Linked via `SourcePayrollId` ( `15` days worked in June = `₹50,000` ).
   - **Leave Encashment ( `LeaveEncashment` )**: Rahul has `12.0` days of encashable Earned Leave ( `EL` ). Using `BasicPay = ₹50,000` and `FIXED_30_DAYS` divisor, encashment = $12 \times \frac{50,000}{30} = ₹20,000$.
   - **Notice Shortfall Recovery ( `NoticeRecovery` )**: Rahul had `10` unserved shortfall days ( `ShortfallDays = 10` ) = $10 \times \frac{1,00,000}{30} = ₹33,333.33$.
   - **Asset Damage Recovery ( `OutstandingDues` )**: IT logged a `₹2,500` screen damage deduction on his returned laptop ( `EmployeeExitAsset` ).
2. **Net Settlement Calculation**:
   - Total Credits = $50,000 + 20,000 = ₹70,000$.
   - Total Debits = $33,333.33 + 2,500 = ₹35,833.33$.
   - `NetPayable` = $70,000 - 35,833.33 = \mathbf{₹34,166.67}$ ( `Direction = PayableToEmployee` ).
3. **Approval & Payout**:
   - Finance approves the settlement ( `Status = Approved` ), records payment reference ( `Paid` ), and publishes the FnF Statement to Rahul's Former Employee Portal.

## 3. Actors

| Actor | Capabilities |
| --- | --- |
| **Finance / Payroll Admin** | Calculates `EmployeeFinalSettlement` , links `SourcePayrollId` , adds manual adjustments, approves payout/recovery, and marks settlement `Settled` . |
| **HR Admin** | Verifies leave encashment days and notice recovery days feeding into FnF. |
| **Exiting / Former Employee** | Views detailed FnF breakdown and downloads signed settlement statement from `/former-employee` . |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/offboarding/components/FinalSettlementTab.tsx` | Interactive worksheet displaying earnings ( `SalaryPayable` , `LeaveEncashment` , `Reimbursements` , `BonusesAndIncentives` ) vs deductions ( `NoticeRecovery` , `OutstandingDues` , `Deductions` ) and `NetPayable` . | `GET /api/employee-exit/{id}/settlement` , `POST /api/employee-exit/{id}/settlement/calculate` , `POST /api/employee-exit/{id}/settlement/approve` | Finance / HR Admin |
| `src/app/modules/offboarding/pages/FormerEmployeePortalPage.tsx` | Read-only view of approved FnF statement and payout status for alumni. | `GET /api/public/former-employee/summary` | Former Employee |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/EmployeeExitController.cs` | Exposes FnF calculation, adjustment, approval, and payment confirmation endpoints. | `IEmployeeExitService` |
| `Services/EmployeeExitService.cs` | Pulls `PayrollEntity` ( `SourcePayrollId` ), encashable `EmployeeLeaveBalance` , `ShortfallDays` , and `EmployeeExitAsset` deductions to compute `EmployeeFinalSettlement` . | `IMongoRepository<EmployeeFinalSettlement>` , `IMongoRepository<PayrollEntity>` , `IMongoRepository<EmployeeLeaveBalance>` , `IMongoRepository<EmployeeExitAsset>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/employee-exit/{exitId}/settlement` | Retrieves current `EmployeeFinalSettlement` record for an exit case. | JWT ( `offboarding:read` ) | `EmployeeFinalSettlementDto` | None |
| `POST` | `/api/employee-exit/{exitId}/settlement/calculate` | Computes or recomputes FnF components from payroll, leave, notice, and asset records. | JWT ( `payroll:create` ) | `CalculateSettlementRequestDto` -> `EmployeeFinalSettlementDto` | Creates/updates `EmployeeFinalSettlement` |
| `POST` | `/api/employee-exit/{exitId}/settlement/approve` | Locks and approves the final settlement sheet. | JWT ( `payroll:approve` ) | `ApproveSettlementDto` | Unblocks `EmployeeExit` final completion |

## 7. Database Model

`EmployeeFinalSettlement` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/EmployeeFinalSettlement.cs`)

- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `ExitId` (`string`, FK -> `EmployeeExit.Id`)
- `EmployeeId` (`string`, FK)
- `SourcePayrollId` (`string`, FK -> `PayrollEntity.Id`)
- `SalaryPayable` (`double`)
- `LeaveEncashment` (`double`)
- `Reimbursements` (`double`)
- `BonusesAndIncentives` (`double`)
- `Deductions` (`double`)
- `NoticeRecovery` (`double`)
- `OutstandingDues` (`double`)
- `NetPayable` (`double`)
- `Direction` (`SettlementDirection`: `PayableToEmployee` | `ReceivableFromEmployee` | `ZeroSettlement`)
- `Status` (`string`: `"Draft"` | `"PendingApproval"` | `"Approved"` | `"Settled"`)

---

## 8. Business Rules

- **Net Payable & Direction Formula**:

$$\text{TotalEarnings} = \text{SalaryPayable} + \text{LeaveEncashment} + \text{Reimbursements} + \text{BonusesAndIncentives}$$

$$\text{TotalRecoveries} = \text{Deductions} + \text{NoticeRecovery} + \text{OutstandingDues}$$

$$\text{NetPayable} = \text{TotalEarnings} - \text{TotalRecoveries}$$

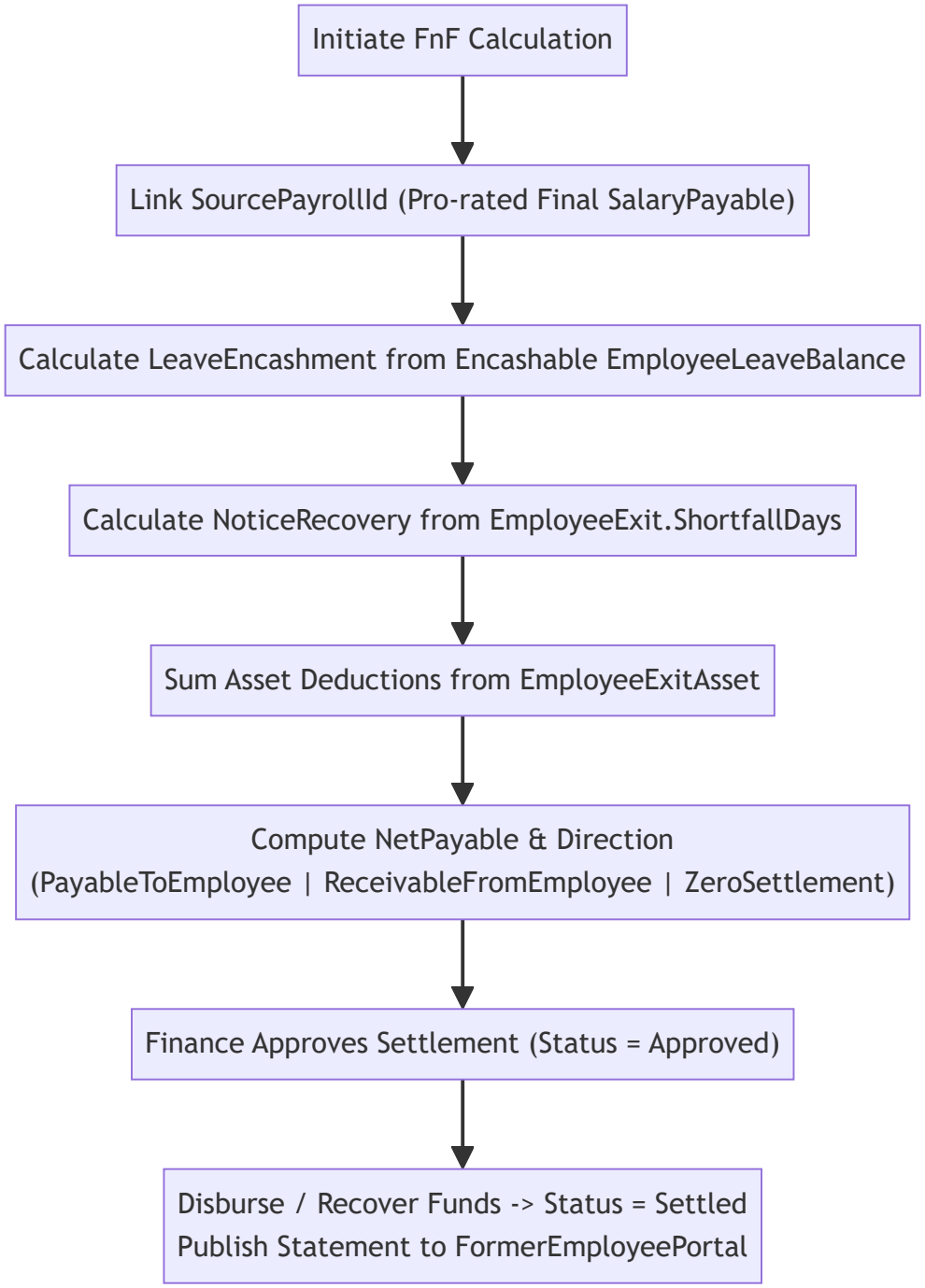
- If $\text{NetPayable} > 0 \rightarrow \text{Direction} = \text{PayableToEmployee}$ - If $\text{NetPayable} < 0 \rightarrow \text{Direction} = \text{ReceivableFromEmployee}$ - If $\text{NetPayable} == 0 \rightarrow \text{Direction} = \text{ZeroSettlement}$

- **Payroll Linkage Rule**: `SalaryPayable` is anchored to `SourcePayrollId` (`PayrollEntity`) to prevent double payment between regular monthly payroll and FnF settlement.

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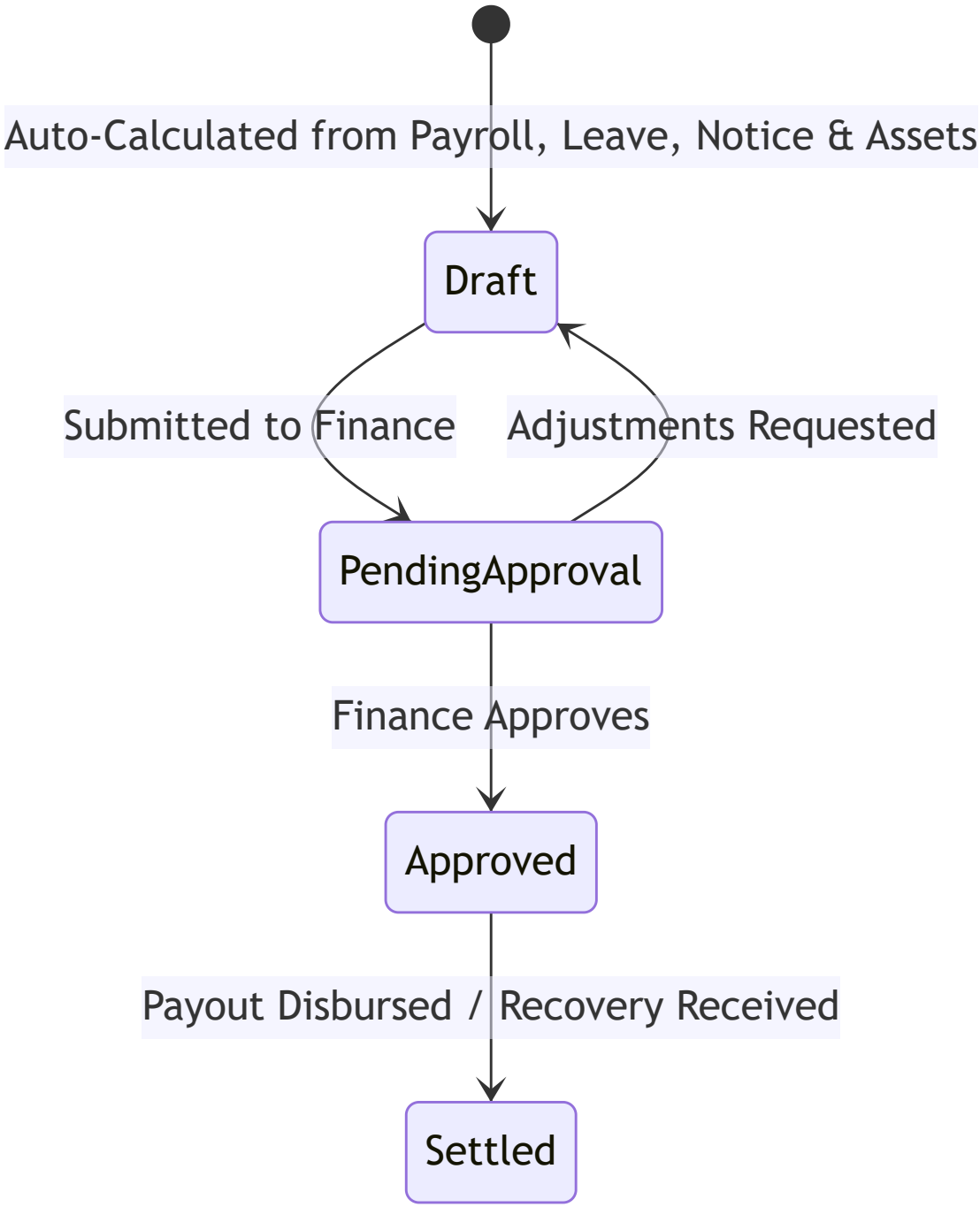
## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM



## 10. State Machine

ARCHITECTURE / WORKFLOW DIAGRAM




## 11. Permissions

- `offboarding:read` — View settlement breakdown.
- `payroll:create` / `payroll:approve` — Calculate and approve `EmployeeFinalSettlement`.

## 12. Dependencies

- **Upstream**: `PAYROLL` (`PayrollEntity`), `LEAVE` (`EmployeeLeaveBalance`), `NOTICE-PERIOD` (`ShortfallDays`), `ASSET-MANAGEMENT` (`EmployeeExitAsset`).
- **Downstream**: `OFFBOARDING` (completion gate & Former Employee Portal display).

## 13. Cross-Module Events

- Approving `EmployeeFinalSettlement` transitions `EmployeeLifecycle.PayrollStatus` to `FinalSettlementCompleted` (`4`).

## 14. Failure Scenarios

- **Negative Settlement (`ReceivableFromEmployee`)**: When `NoticeRecovery + OutstandingDues > TotalEarnings`, `Direction` is flagged as `ReceivableFromEmployee`, requiring employee payment clearance before relieving letters are unlocked.

## 15. Edge Cases

- **Zero Settlement (`NetPayable == 0`)**: Automatically marked ready for closure without requiring bank disbursement reference.

## 16. Current Strengths

- Direct foreign-key traceability (`SourcePayrollId`) between monthly payroll runs and final settlement prevents double-counting of final month wages.

## 17. Current Gaps

- Statutory Gratuity calculation ($15 / 26 \times \text{Last Drawn Basic} \times \text{Years of Service}$ for $\geq 5$ years tenure) is entered via `BonusesAndIncentives` rather than auto-computed by a dedicated gratuity rule.

## 18. Recommended Improvements

- Add an automated Gratuity eligibility and formula calculator when `DateOfExit - DateOfJoining >= 5 years`.

## 19. Future Scope

- Payment gateway link inside the Former Employee Portal for employees with `Direction == ReceivableFromEmployee` to pay notice buyout/asset recovery online.

MOD-26 PART III docs/project-documentation/modules/DOCUMENTS.md

# DOCUMENT MANAGEMENT & PDF GENERATION MODULE (`DOCUMENTS.md`)

## 1. Module Overview

The **Document Management & PDF Generation Module** handles corporate policy documents (`CompanyDocument`), candidate resumes and preboarding KYC uploads (`PreboardingDocument`), automated HTML-to-PDF rendering (`PdfService` with `SalarySlipTemplate.html`), offboarding statutory letters (`EmployeeExitDocument` — Relieving Letter, Experience Certificate, FnF Statement), and static file serving mounted at `/fs` (`Program.cs`).

## 2. Real-World Business Scenario

1. **Employee & Candidate Uploads**:
   - During preboarding, candidate Priya Nair uploads her PAN card, Aadhaar card, and degree certificates (`PreboardingDocument`), stored under the tenant's isolated file directory and served via `/fs/...`.
2. **Automated Payslip PDF Generation**:
   - When `AutoPayrollServices` completes April 2026 payroll, `PdfService` injects Rahul Sharma's compensation numbers into `Templates/SalarySlipTemplate.html`, generates a PDF document, and persists the file path on `PayrollEntity.SlipPdfPath`.
3. **Exit Letters & Alumni Portal Delivery**:
   - Upon completing Rahul's exit, HR generates his **Relieving Letter** and **Experience Certificate** (`EmployeeExitDocument`).
   - Rahul logs into `/former-employee` using his `FormerEmployeeAccessToken` and downloads both PDFs securely via `PublicFormerEmployeePortalController`.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin** | Uploads `CompanyDocument` policies, verifies `PreboardingDocument` KYC files, and issues `EmployeeExitDocument` letters. |
| **Payroll Admin / System** | Generates batch PDF salary slips via `PdfService`. |
| **Employee / Candidate / Alumni** | Uploads KYC/profile documents and downloads payslips, company handbooks, and relieving letters. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/documents/pages/CompanyDocumentsPage.tsx` | Repository of company handbooks, policies, and announcements (`CompanyDocument`). | `GET /api/company-documents`, `POST /api/company-documents` | All Employees / HR |

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/offboarding/components/ExitDocumentsTab.tsx` | Generates and manages `EmployeeExitDocument` files (Relieving Letter, Experience Letter, FnF Sheet) with visibility toggle for the Former Employee Portal. | `GET /api/employee-exit/{id}/documents`, `POST /api/employee-exit/{id}/documents` | HR Admin |
| `src/app/modules/payroll/pages/MySalarySlipsPage.tsx` | Triggers PDF download for generated salary slips. | `GET /api/payroll/{id}/download-slip` | Employee |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Program.cs` ( `UseStaticFiles("/fs")` ) | Mounts the local storage directory ( `FileStorage` ) at `/fs` for controlled document and media streaming. | `PhysicalFileProvider` |
| `Services/PdfService.cs` | Compiles HTML templates (such as `Templates/SalarySlipTemplate.html` ) into PDF byte arrays and writes them to disk. | HTML-to-PDF renderer |
| `Controllers/CompanyDocumentController.cs` | CRUD endpoints for company-wide documents and policy PDFs ( `CompanyDocument` ). | `ICompanyDocumentService` |
| `Controllers/PublicFormerEmployeePortalController.cs` | Streams `EmployeeExitDocument` and payslip PDFs to token-authenticated former employees. | `IFormerEmployeePortalService` |

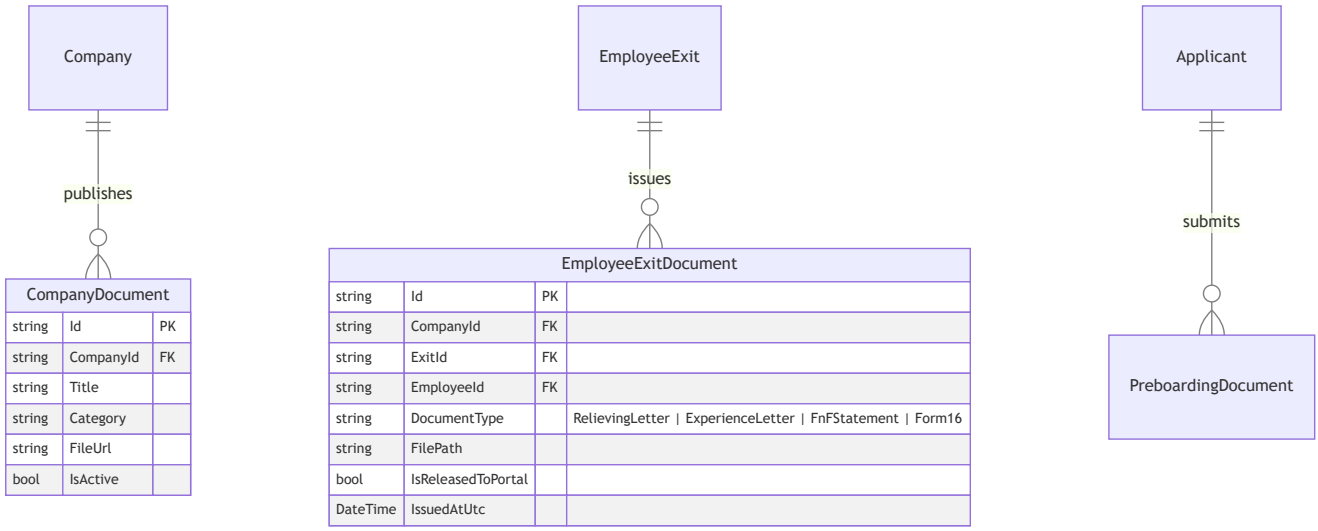
## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/company-documents` | Lists company documents and handbooks for the tenant. | JWT ( `documents:read` ) | `List<CompanyDocumentDto>` | None |
| `POST` | `/api/company-documents` | Uploads a new company policy document ( `multipart/form-data` ). | JWT ( `documents:create` ) | `CompanyDocumentDto` | Writes file to `/fs/` storage |
| `POST` | `/api/employee-exit/{exitId}/documents` | Uploads or generates an `EmployeeExitDocument` (Relieving/Experience letter). | JWT ( `offboarding:update` ) | `EmployeeExitDocumentDto` | Makes document available on `/former-employee` |

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/public/former-employee/documents/{docId}` | Downloads released exit document via `FormerEmployeeAccessToken`. | `FormerEmployeeAccessToken` (`[AllowAnonymous]`) | Streams `application/pdf` | Logs download access |

## 7. Database Model

ARCHITECTURE / WORKFLOW DIAGRAM




## 8. Business Rules

- **Portal Release Gate (`IsReleasedToPortal`)**: An `EmployeeExitDocument` is only visible on `/former-employee` if `IsReleasedToPortal == true` AND any negative `ReceivableFromEmployee` settlement dues have been cleared.
- **Template Token Replacement**: `PdfService` sanitizes and replaces template placeholders (`{{EmployeeName}}`, `{{NetSalary}}`, `{{MonthYear}}`, `{{CompanyName}}`) inside `Templates/SalarySlipTemplate.html`.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

`EmployeeExitDocument`: `Draft` → `Issued` → `ReleasedToPortal` (`IsReleasedToPortal = true`) → `Revoked`.

## 11. Permissions

- `documents:read` / `documents:create` — View and upload company documents.
- `offboarding:update` — Issue and release `EmployeeExitDocument` files.

## 12. Dependencies

- **Upstream**: `PAYROLL` ( `AutoPayrollServices` ), `PREBOARDING` ( `PreboardingDocument` ), `OFFBOARDING` ( `EmployeeExit` ).
- **Downstream**: `FormerEmployeePortalService` (post-exit document delivery).

## 13. Cross-Module Events

- Releasing an `EmployeeExitDocument` triggers a notification via `EmployeeExitOutbox` informing the former employee that their relieving/experience letter is ready for download.

## 14. Failure Scenarios

- **Missing Template File**: If `Templates/SalarySlipTemplate.html` is missing from deployment artifacts, `PdfService` throws an IO error logged via `ExceptionHandlingMiddleware` .

## 15. Edge Cases

- **Large File Uploads**: Enforced by ASP.NET Core multipart body size limits and MIME-type validation ( `application/pdf` , `image/png` , `image/jpeg` ).

## 16. Current Strengths

- Unified PDF generation ( `PdfService` ) and gated alumni document release ( `IsReleasedToPortal` ).

## 17. Current Gaps

- Files are stored on the local server disk ( `/fs` ) rather than an external cloud object store (AWS S3 / Azure Blob Storage) with pre-signed URLs.

## 18. Recommended Improvements

- Abstract file persistence behind an `IFileStorageProvider` interface supporting both local `/fs` and AWS S3 / Azure Blob Storage.

## 19. Future Scope

- Digital signature ( `DSC` / DocuSign) cryptographic signing on generated PDF Payslips and Relieving Letters.

MOD-27 PART III docs/project-documentation/modules/NOTIFICATIONS.md

# NOTIFICATIONS & REAL-TIME MESSAGING MODULE (`NOTIFICATIONS.md`)

## 1. Module Overview

The **Notifications & Real-Time Messaging Module** powers push alerts, in-app notifications (`NotificationService`), bi-directional real-time communication via ASP.NET Core SignalR hubs (`NotificationHub` mounted at `/notificationHub` and `ChatHub` mounted at `/chatHub`), and resilient background email/dispatch processing (`EmployeeExitOutboxProcessor`, `AttendanceReminderHostedService`).

## 2. Real-World Business Scenario

1. **Real-Time Leave Approval Alert**:
   - When Rahul's manager approves his leave application (`POST /api/leave/{id}/approve`), `LeaveManagementService` invokes `INotificationService.SendNotificationAsync(userId, title, message)`.
   - `NotificationService` persists the notification in MongoDB (`Notification` collection) and invokes `IHubContext<NotificationHub>.Clients.User(userId).SendAsync("ReceiveNotification", payload)`.
   - Rahul's browser receives the WebSocket message instantly and updates the top navigation bell badge without requiring a page refresh.
2. **Reliable Post-Exit Magic Link Delivery**:
   - When Rahul's exit is completed, direct email sending could fail if the SMTP server is down.
   - Instead, `EmployeeExitService` enqueues an `EmployeeExitOutbox` document inside the same database transaction.
   - `EmployeeExitOutboxProcessor` picks up the record, delivers the magic link email, and marks `IsProcessed = true`.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **All Employees / Managers / Admins** | Receive in-app bell notifications, mark notifications as read, connect to SignalR hubs. |
| **System Background Workers** | `AttendanceReminderHostedService` and `EmployeeExitOutboxProcessor` push periodic and event-driven notifications. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/notifications/components/NotificationBell.tsx` | Top navbar badge displaying unread count and popover dropdown with recent notifications. | `GET /api/notifications`, `PUT /api/notifications/{id}/read` | All Users |

| File | Responsibility | Calls | Used By |
| --- | --- | --- | --- |
| `src/server/signalr.ts` / SignalR client connection | Establishes WebSocket / Server-Sent Events connection to `/notificationHub` with JWT token. | WS `/notificationHub` | Entire App |
| `src/app/modules/chat/ChatWidget.tsx` | Real-time colleague chat widget connected to `/chatHub`. | WS `/chatHub`, `GET /api/chat/messages` | All Users |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Hubs/NotificationHub.cs` | SignalR hub mapping authenticated `userId` to connection IDs and broadcasting user notifications. | `[Authorize]`, SignalR core |
| `Hubs/ChatHub.cs` | SignalR hub enabling 1-on-1 and group messaging between colleagues. | `[Authorize]`, SignalR core |
| `Controllers/NotificationController.cs` | Endpoints to fetch user notification history, unread counts, and mark items as read. | `INotificationService` |
| `Services/NotificationService.cs` | Handles database persistence of notifications and real-time delivery via `IHubContext<NotificationHub>`. | `IMongoRepository<Notification>`, `IHubContext<NotificationHub>` |
| `Services/HostedServices/EmployeeExitOutboxProcessor.cs` | Background processor ensuring asynchronous exit notification guarantees. | `IMongoRepository<EmployeeExitOutbox>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
| --- | --- | --- | --- | --- | --- |
| `GET` | `/api/notifications` | Retrieves paginated notification history for current user. | JWT | `List<NotificationDto>` | None |
| `GET` | `/api/notifications/unread-count` | Returns count of unread notifications for navbar badge. | JWT | `{ UnreadCount: int }` | None |
| `PUT` | `/api/notifications/{id}/read` | Marks a specific notification as read (`IsRead = true`). | JWT | None | Updates notification entity |
| `PUT` | `/api/notifications/mark-all-read` | Marks all active notifications for user as read. | JWT | None | Bulk updates `IsRead = true` |
| `WS` | `/notificationHub` | SignalR WebSocket connection endpoint. | JWT (`access_token` query/header) | Bi-directional events (`ReceiveNotification`) | Registers active connection |

## 7. Database Model

`Notification` (`CMS-Backend-Core/Codeji.CMS.Repository/Entities/Notification.cs`)

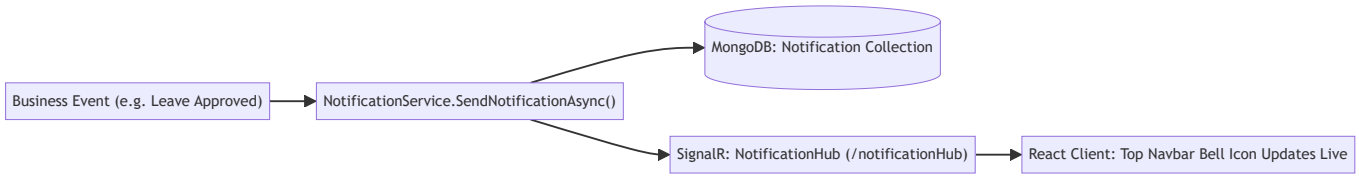
- `Id` (`string`, PK)
- `CompanyId` (`string`, FK)
- `UserId` (`string`, FK -> `EmpUser.Id`)
- `Title` (`string`)
- `Message` (`string`)
- `Category` (`string`, `"Leave"` | `"Attendance"` | `"Payroll"` | `"Offboarding"` | `"System"`)
- `ActionUrl` (`string`)
- `IsRead` (`bool`)
- `CreatedDate` (`DateTime`)

## 8. Business Rules

- **Tenant Isolation**: Users only receive notifications belonging strictly to their `CompanyId` and assigned `UserId`.
- **SignalR Connection Authentication**: The SignalR client must pass the JWT token either in the `Authorization: Bearer` header or in the `access_token` query parameter during WebSocket handshake.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

`Notification`: `Unread` (`IsRead = false`) → `Read` (`IsRead = true`).

## 11. Permissions

- Any authenticated user can read and manage their own notifications.

## 12. Dependencies

- **Upstream**: All operational modules (`LEAVE`, `ATTENDANCE`, `PAYROLL`, `OFFBOARDING`).
- **Downstream**: Frontend notification widgets and sound alerts.

## 13. Cross-Module Events

- Used everywhere across Codeji CMS to inform actors of workflow approvals, exceptions, and lifecycle transitions.

## 14. Failure Scenarios

- **Client Offline / WebSocket Disconnected**: The notification is still saved to MongoDB; the unread badge and list appear upon the user's next login or page load.

## 15. Edge Cases

- **Simultaneous Multiple Browser Tabs**: SignalR delivers to all connection IDs mapped to the user ID; marking read on one tab emits an update event to close badges on other tabs.

## 16. Current Strengths

- Native real-time WebSocket communication combined with durable database storage and outbox reliability.

## 17. Current Gaps

- Browser Web Push / Service Worker notifications (when the browser tab is closed) are not yet enabled; notifications require an active browser session.

## 18. Recommended Improvements

- Add VAPID-based Web Push Notifications for background mobile/desktop browser delivery.

## 19. Future Scope

- WhatsApp Business / Slack notification webhook dispatchers for critical leave and attendance alerts.

MOD-28 PART III docs/project-documentation/modules/REPORTING.md

# REPORTING & LOG EXPORT MODULE (`REPORTING.md`)

## 1. Module Overview

The **Reporting & Log Export Module** aggregates operational metrics and audit trails across the system, providing structured CSV and Excel exports, process history inspection (`ProcessLog`), recruitment audit reports (`RecruitmentAuditExportService`), locked monthly attendance summaries, and payroll expenditure registers.

## 2. Real-World Business Scenario

1. **Recruitment Audit Export**:
   - For an annual compliance audit, HR Admin exports the complete recruitment history for Q1 2026.
   - `RecruitmentAuditExportService` aggregates applicant transitions, interview scores, and salary offers, streaming a structured CSV report via `GET /api/recruitment/audit-export`.
2. **Payroll Register Export**:
   - Following monthly payroll approval, Finance exports the detailed payroll register containing employee names, bank account numbers, gross earnings, LOP deductions, statutory EPF/ESIC, and net payable amounts.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **HR Admin / Payroll Admin** | Generates and downloads attendance, payroll, leave, and recruitment reports. |
| **Finance / External Auditor** | Reviews compliance registers and financial disbursement sheets. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/reports/pages/ReportsDashboard.tsx` | Central reporting hub with export cards for Attendance, Payroll, Leave, and Hiring. | `GET /api/reports/available` | HR / Finance |
| `src/app/modules/recruitment/components/RecruitmentAuditExportModal.tsx` | Dialog to select date ranges and trigger recruitment audit CSV download. | `GET /api/recruitment/audit-export` | HR Admin |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Services/RecruitmentAuditExportService.cs` | Streams applicant history, stage progressions, and interviewer remarks as formatted CSV. | `IMongoRepository<Applicant>`, `IMongoRepository<ApplicantLogs>` |

| Backend File | Responsibility | Dependencies |
| --- | --- | --- |
| `Controllers/AutoPayrollController.cs` | Exposes monthly payroll summary reports and export registers. | `IAutoPayrollServices` |
| `Controllers/AttendanceController.cs` | Exposes locked `MonthlyAttendanceSummary` datasets. | `IAttendanceService` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
| --- | --- | --- | --- | --- | --- |
| `GET` | `/api/recruitment/audit-export` | Streams recruitment audit log CSV file. | JWT ( `recruitment:read` ) | Streams `text/csv` | Logs export audit event |
| `GET` | `/api/autopayroll/records` | Exports employee-level payroll breakdown for target month/year. | JWT ( `payroll:read` ) | `List<PayrollEntityDto>` | None |
| `GET` | `/api/attendance/monthly-summary` | Retrieves complete monthly attendance summaries for company employees. | JWT ( `attendance:read` ) | `List<MonthlyAttendanceSummaryDto>` | None |

## 7. Database Model

### `ProcessLog` & Operational Summaries

- `ProcessLog` :
  - `Id` ( `string` , PK)
  - `CompanyId` ( `string` , FK)
  - `ProcessType` ( `string` , `"Payroll"` | `"AttendanceLock"` | `"LeaveAccrual"` )
  - `Status` ( `string` , `"Success"` | `"Warning"` | `"Failed"` )
  - `Message` ( `string` )
  - `ExecutedBy` ( `string` )
  - `ExecutionTimeUtc` ( `DateTime` )

## 8. Business Rules

- **Export Isolation Rule**: Every report query strictly filters on `CompanyId` from the authenticated user context.
- **Audit Logging Rule**: Sensitive exports (such as salary registers and recruitment audit logs) write an entry to `ProcessLog` or audit collections.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Reports represent point-in-time idempotent query executions.

## 11. Permissions

- `reports:read` / `payroll:read` / `recruitment:read` — Access respective report generation endpoints.

## 12. Dependencies

- **Upstream**: All transactional modules ( `RECRUITMENT` , `ATTENDANCE` , `PAYROLL` , `LEAVE` ).
- **Downstream**: External spreadsheet tools, auditors, and banking systems.

## 13. Cross-Module Events

- Executing an export writes an audit trail entry.

## 14. Failure Scenarios

- **Large Dataset Timeout**: Handled by streaming CSV responses directly to the HTTP output stream rather than buffering entire datasets in memory.

## 15. Edge Cases

- **Exports During In-Flight Payroll**: Unlocked/draft payroll records are explicitly demarcated with `IsLocked = false` warning headers.

## 16. Current Strengths

- Direct streaming of audit records reduces server memory overhead during large exports.

## 17. Current Gaps

- Scheduled recurring report emails (e.g., auto-emailing Monday morning attendance summaries to executive staff) are not yet scheduled via a cron hosted service.

## 18. Recommended Improvements

- Add scheduled email reports via a `ReportSchedulerHostedService`.

## 19. Future Scope

- Customizable visual report builder with drag-and-drop column selection.

# DASHBOARD & ANALYTICS MODULE (`ANALYTICS.md`)

## 1. Module Overview

The **Dashboard & Analytics Module** aggregates key operational metrics into real-time visual dashboards (`DashboardController`, `DashboardServices`). It surfaces workforce headcounts, recruitment funnel progression, monthly attendance compliance percentages, leave balances, payroll cost rollups, and career analytics.

## 2. Real-World Business Scenario

1. **Executive Morning Overview**:
   - CEO logs into the Codeji CMS portal.
   - `GET /api/dashboard/summary` loads executive KPI cards:
     - Total Active Headcount: `142`
     - Today's Present: `131` (`92.2%`)
     - On Leave: `8` (`5.6%`)
     - WFH: `3` (`2.1%`)
     - Active Open Job Positions: `6`
     - Total In-Flight Candidates: `48`
     - Estimated Current Month Payroll: `₹1,18,50,000`

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Executive / Super Admin / HR Admin** | Views organization-wide KPI tiles, hiring funnels, attendance trends, and payroll expense charts. |
| **Department Manager** | Views team-specific attendance, leave, and headcount metrics. |
| **Employee** | Views personal statistics (leaves remaining, monthly attendance percentages, upcoming holidays). |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/dashboard/pages/AdminDashboardPage.tsx` | Main HR/Admin dashboard with KPI widgets, attendance pie charts, hiring pipeline cards, and payroll rollups. | `GET /api/dashboard/admin` | HR / Admin |
| `src/app/modules/dashboard/pages/EmployeeDashboardPage.tsx` | Self-service employee dashboard with personal leave balances, quick punch button, and announcements. | `GET /api/dashboard/employee` | All Employees |

## 5. Backend Implementation

| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Controllers/DashboardController.cs` | Exposes aggregated metrics endpoints for admin and employee dashboards. | `IDashboardServices` |
| `Services/DashboardServices.cs` | Performs high-performance MongoDB aggregation pipelines across `EmpUser`, `Attendance`, `Leave`, `Applicant`, and `PayrollEntity`. | `IMongoRepository<EmpUser>`, `IMongoRepository<Attendance>`, `IMongoRepository<Leave>`, `IMongoRepository<Applicant>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/dashboard/admin` | Returns high-level company analytics and KPIs for HR/Admin. | JWT (`dashboard:read`) | `AdminDashboardDto` | None |
| `GET` | `/api/dashboard/employee` | Returns personalized metrics (leave balances, attendance status) for logged-in user. | JWT | `EmployeeDashboardDto` | None |

## 7. Database Model

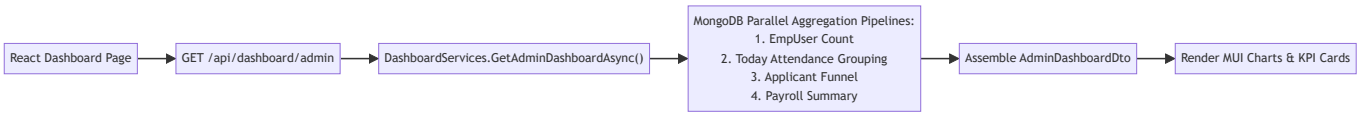
Aggregates live operational collections using MongoDB aggregation pipelines (`$match`, `$group`, `$facet`, `$count`). No standalone materialized entity is required.

## 8. Business Rules

- **Role-Based Aggregation Scoping**:
  - Super Admin / HR Admin: Aggregates whole tenant.
  - Reporting Manager: Aggregates direct and indirect reports.
  - Employee: Aggregates only personal records.
- **Tenant Isolation**: Every aggregation pipeline stages `$match: { CompanyId: companyId, IsDeleted: false }` as its first operation.

## 9. Complete Workflow

**ARCHITECTURE / WORKFLOW DIAGRAM**



## 10. State Machine

Read-only real-time aggregation queries.

## 11. Permissions

- `dashboard:read` — View organizational dashboard metrics.

## 12. Dependencies

- **Upstream**: `EMPLOYEE`, `ATTENDANCE`, `LEAVE`, `RECRUITMENT`, `PAYROLL`.
- **Downstream**: Frontend dashboard visualization components.

## 13. Cross-Module Events

- Real-time punches, leave approvals, and applicant stage changes immediately reflect upon the next dashboard fetch.

## 14. Failure Scenarios

- **Zero Data State**: Handled gracefully with zero-count defaults so new tenants see empty states rather than calculation crashes.

## 15. Edge Cases

- **Timezone Shifts on 'Today' Attendance**: Normalized to tenant local office timezone before aggregating check-ins.

## 16. Current Strengths

- Lean, on-demand MongoDB aggregation pipelines eliminate out-of-sync caching discrepancies.

## 17. Current Gaps

- Long-term multi-year trend analytics (e.g., year-over-year attrition comparison) are computed on-demand rather than pre-aggregated into monthly warehouse buckets.

## 18. Recommended Improvements

- Add a nightly aggregation worker storing monthly KPI snapshots in a `MonthlyAnalyticsSnapshot` collection for instant multi-year trend charts.

## 19. Future Scope

- Predictive AI turnover risk forecasting based on attendance irregularity and leave patterns.

MOD-30 PART III docs/project-documentation/modules/AUDIT.md

# AUDIT & COMPLIANCE LOGGING MODULE (`AUDIT.md`)

## 1. Module Overview

The **Audit & Compliance Logging Module** provides an immutable record of sensitive operational events across the application. It includes repository-level automatic auditing (`ISupportAuditing`), attendance override tracking (`AttendanceAuditLog`), applicant stage transition logs (`ApplicantLogs`), and offboarding event logs (`EmployeeExitAuditEvent`).

## 2. Real-World Business Scenario

1. **Attendance Regularization Audit**:
   - HR Admin modifies Rahul's punch record from `09:58` to `09:28` to waive a late half-day status.
   - The system automatically writes an immutable `AttendanceAuditLog` recording `OriginalCheckIn = 09:58`, `NewCheckIn = 09:28`, `ModifiedByUserId`, `ModifiedAtUtc`, and `Reason = "Subway signal failure verified"`.
2. **Offboarding Compliance Review**:
   - An employee disputes a notice shortfall recovery charge.
   - HR inspects `EmployeeExitAuditEvent` records for the exit case to verify who approved the notice adjustment, on what date, and what comments were entered.

## 3. Actors

| Actor | Capabilities |
|---|---|
| **Compliance Officer / Super Admin / HR Admin** | Searches, filters, and inspects audit trails across Attendance, Recruitment, Offboarding, and Core Entities. |
| **System Repositories** | Automatically populates `CreatedBy`, `CreatedDate`, `UpdatedBy`, `UpdatedDate`, and increments `Version` on entities implementing `ISupportAuditing`. |

## 4. Frontend Implementation

| File | Responsibility | Calls | Used By |
|---|---|---|---|
| `src/app/modules/audit/pages/AuditLogPage.tsx` | Searchable table for viewing system audit records with entity, actor, date, and action filters. | `GET /api/audit-logs` | Admin / Compliance |
| `src/app/modules/offboarding/components/ExitAuditTimeline.tsx` | Vertical timeline displaying all `EmployeeExitAuditEvent` entries for an exit case. | `GET /api/employee-exit/{id}/audit` | HR Admin |

## 5. Backend Implementation

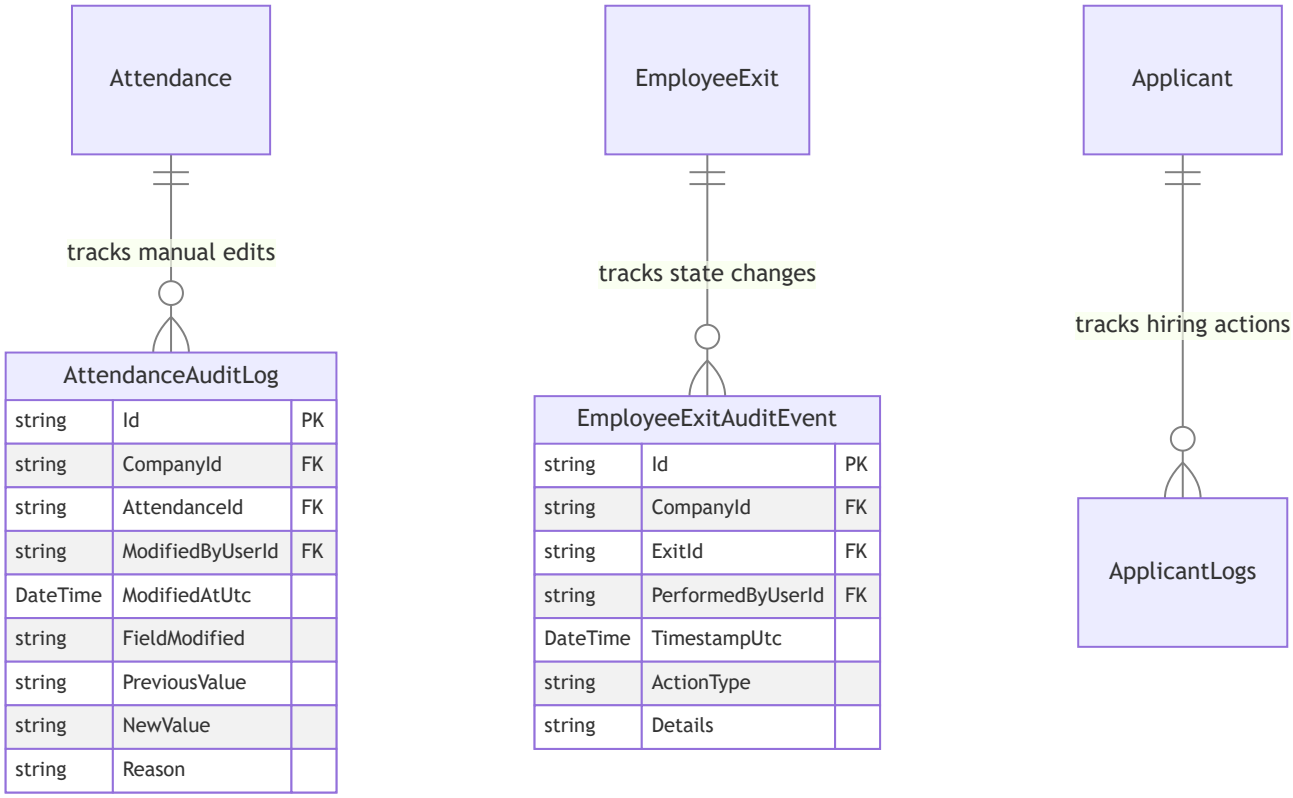
| Backend File | Responsibility | Dependencies |
|---|---|---|
| `Repository/MongoRepository.cs` | Automatically updates `ISupportAuditing` fields ( `UpdatedBy` , `UpdatedDate` , `Version` ) on every `UpdateAsync` . | `MongoDbContext` |
| `Services/EmployeeExitService.cs` | Appends structured `EmployeeExitAuditEvent` records on every offboarding state change or clearance action. | `IMongoRepository<EmployeeExitAuditEvent>` |
| `Services/AttendanceService.cs` | Writes `AttendanceAuditLog` entries whenever punch times or statuses are regularized. | `IMongoRepository<AttendanceAuditLog>` |

## 6. API Documentation

| Method | URL | Purpose | Auth / Permission | Request / Response | Side Effects |
|---|---|---|---|---|---|
| `GET` | `/api/audit/attendance` | Returns attendance modification audit records. | JWT ( `audit:read` ) | `List<AttendanceAuditLogDto>` | None |
| `GET` | `/api/employee-exit/{exitId}/audit` | Returns immutable audit timeline for an exit case. | JWT ( `offboarding:read` ) | `List<EmployeeExitAuditEventDto>` | None |

## 7. Database Model

ARCHITECTURE / WORKFLOW DIAGRAM

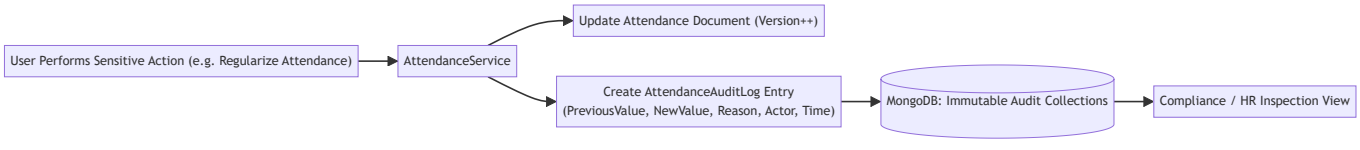



## 8. Business Rules

- **Immutability Rule**: Audit records ( `AttendanceAuditLog` , `EmployeeExitAuditEvent` , `ApplicantLogs` ) have no update or delete endpoints. Once inserted, they are permanent.
- **Optimistic Concurrency Protection**: Every audited entity increments integer `Version` on save. Attempting to update a stale version throws a concurrency conflict error.

## 9. Complete Workflow

ARCHITECTURE / WORKFLOW DIAGRAM




## 10. State Machine

Append-only immutable event streams.

## 11. Permissions

- `audit:read` — View system audit trails.

## 12. Dependencies

- **Upstream**: All transactional modules modifying critical records.
- **Downstream**: Regulatory compliance and forensic dispute resolution.

## 13. Cross-Module Events

- Audit logging is an automatic side effect of all sensitive modifications.

## 14. Failure Scenarios

- **Audit Write Failure**: Wrapped in the same operational logic to ensure changes do not succeed without their audit record.

## 15. Edge Cases

- **System-Initiated Changes (Hosted Services)**: Logged with actor ID `"SYSTEM"` or service name.

## 16. Current Strengths

- Granular before-and-after change logging ( `PreviousValue` vs `NewValue` ) with mandatory `Reason` strings.

## 17. Current Gaps

- Centralized tamper-evident cryptographic log chaining (blockchain or HMAC hash chaining) is not currently implemented; reliance is on MongoDB document immutability.

## 18. Recommended Improvements

- Add HMAC cryptographic hashing per audit entry linking to previous entry's hash.

## 19. Future Scope

- Automated anomaly detection flagging suspicious out-of-hours bulk record edits.

PART IV

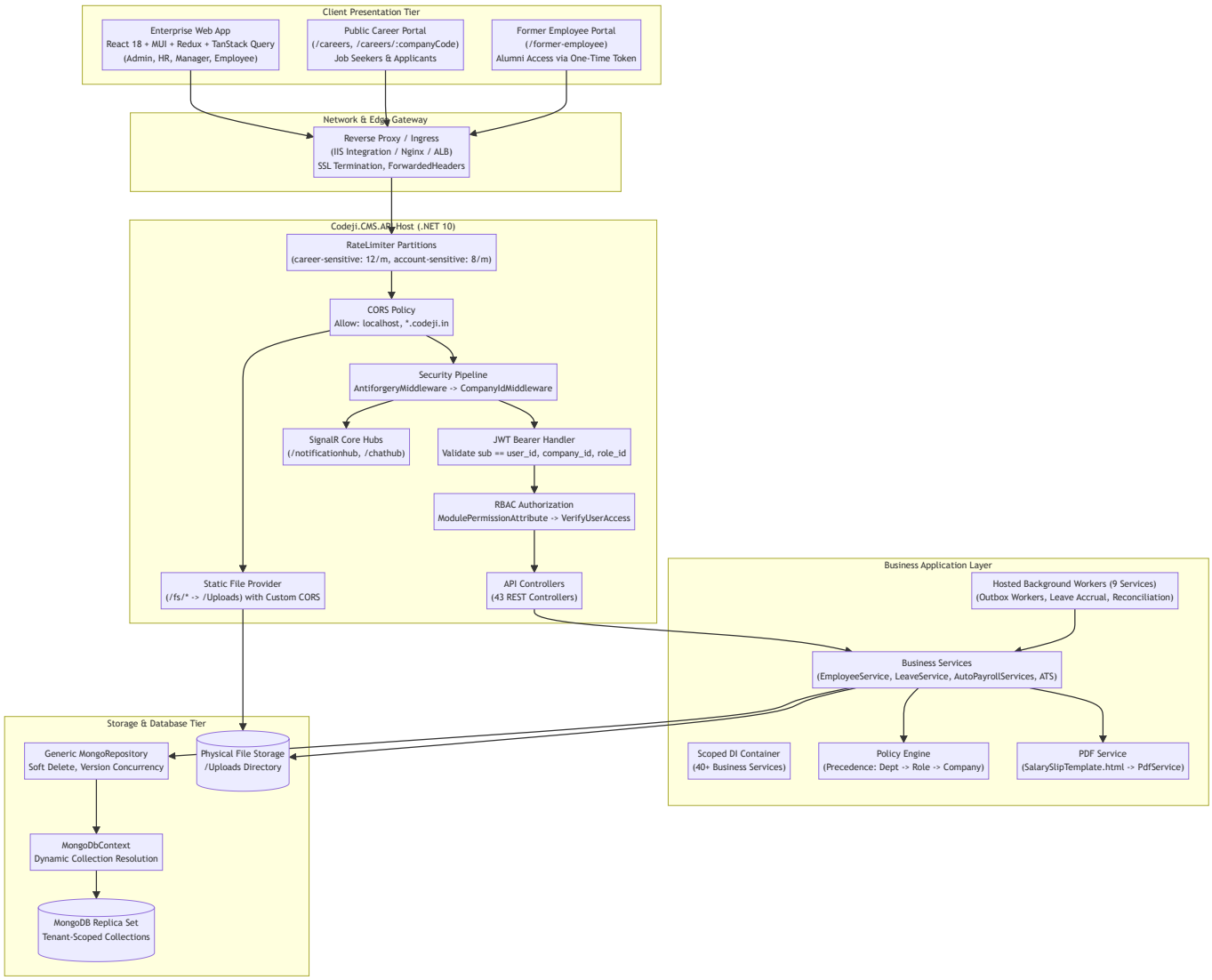
# ARCHITECTURAL, ENTITY & WORKFLOW DIAGRAMS

Visual System Context, Database ER Models, Security Pipelines & End-to-End State Machines

11 Document(s) in this Section

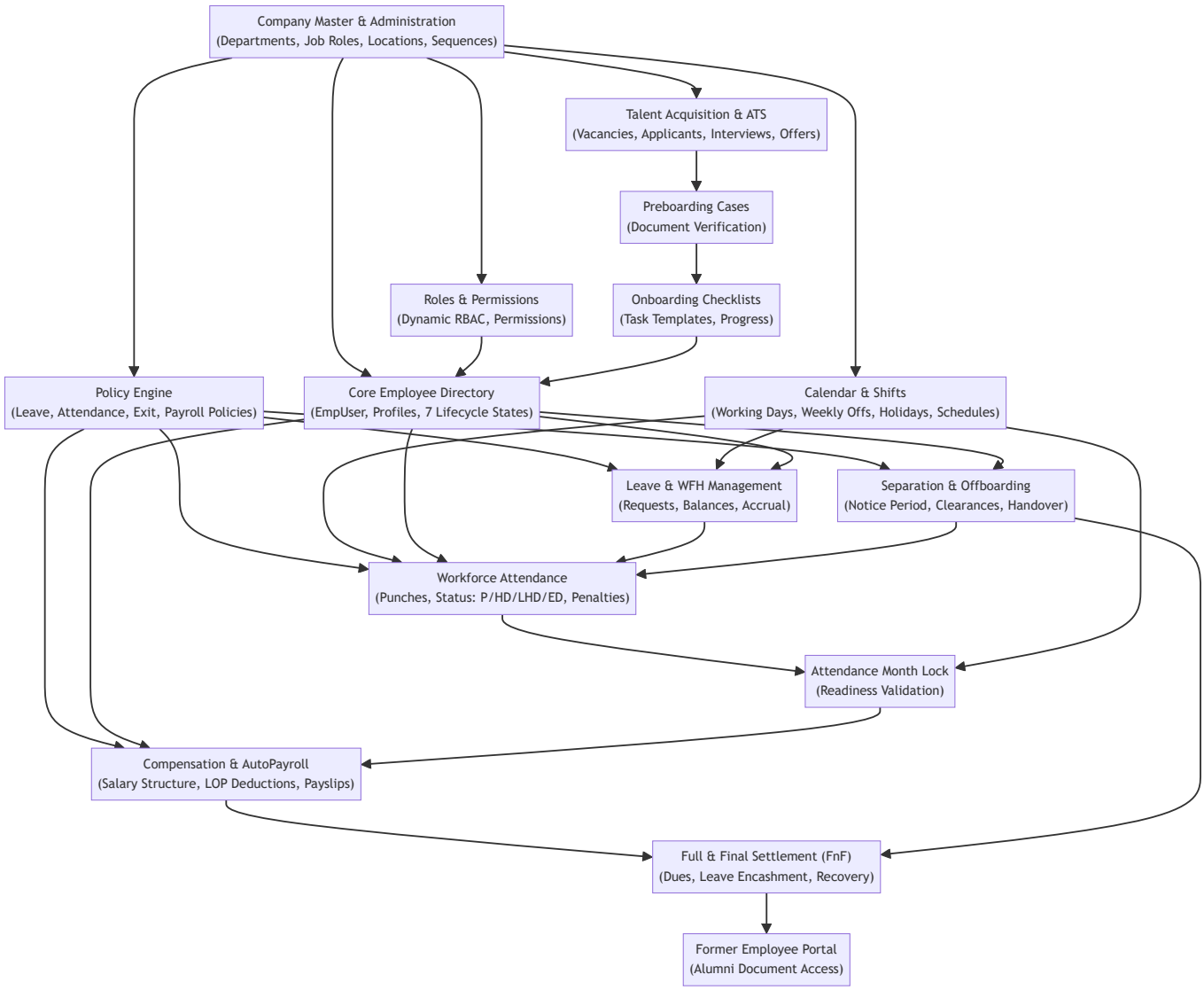
# System Context Architecture Diagram

## ARCHITECTURE / WORKFLOW DIAGRAM



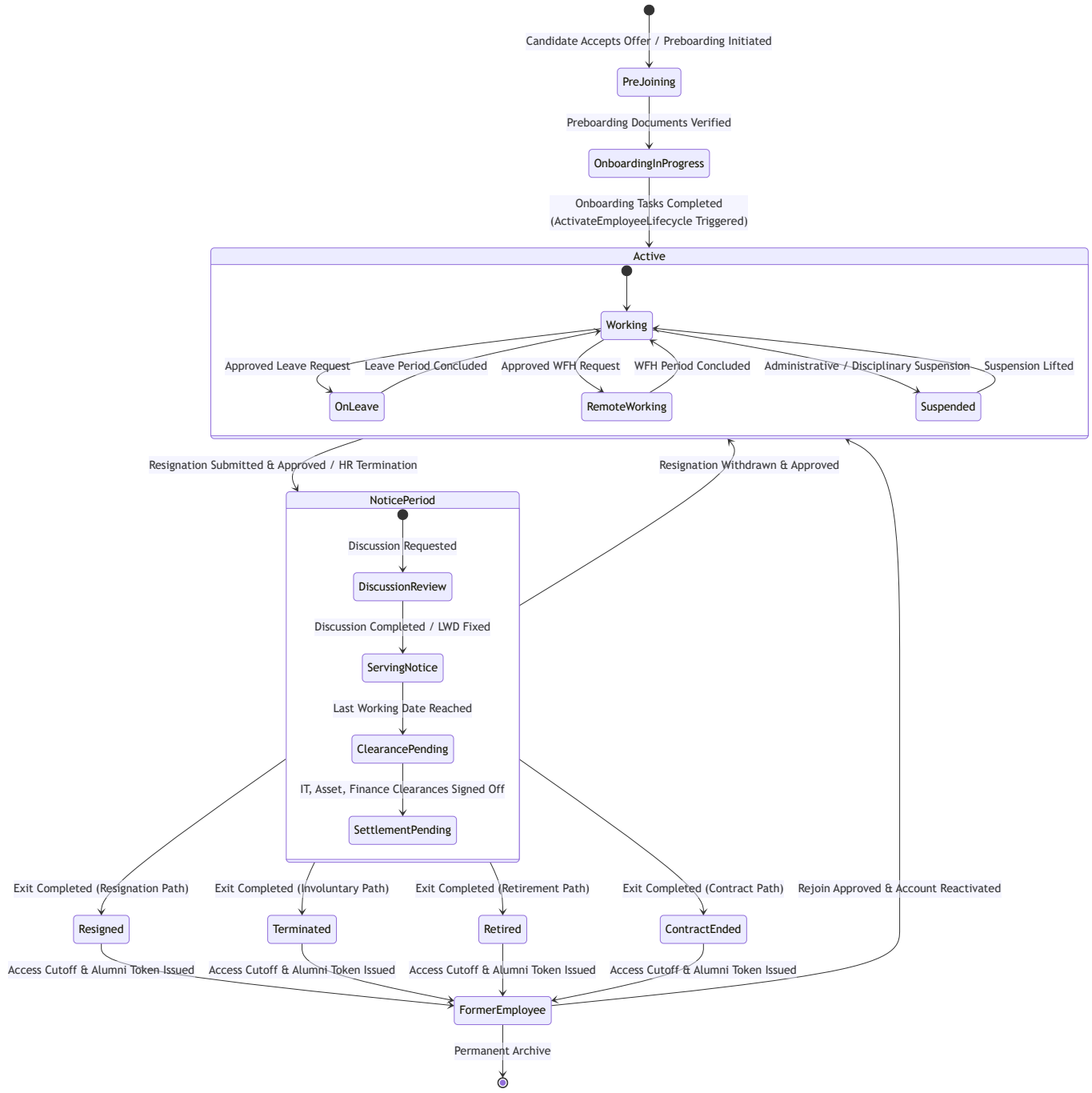
# Module Dependencies Architecture Diagram

ARCHITECTURE / WORKFLOW DIAGRAM



# Employee Lifecycle Master State Machine Diagram

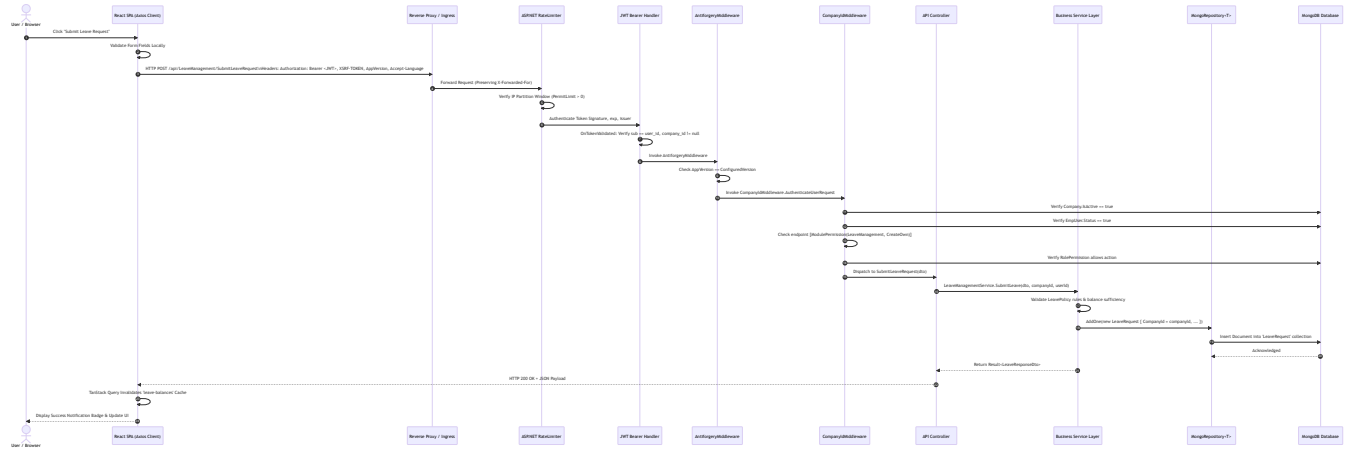
ARCHITECTURE / WORKFLOW DIAGRAM



**DIAG-04** **PART IV** `docs/project-documentation/diagrams/REQUEST-FLOW.md`

# Request-Response Lifecycle Flow Diagram

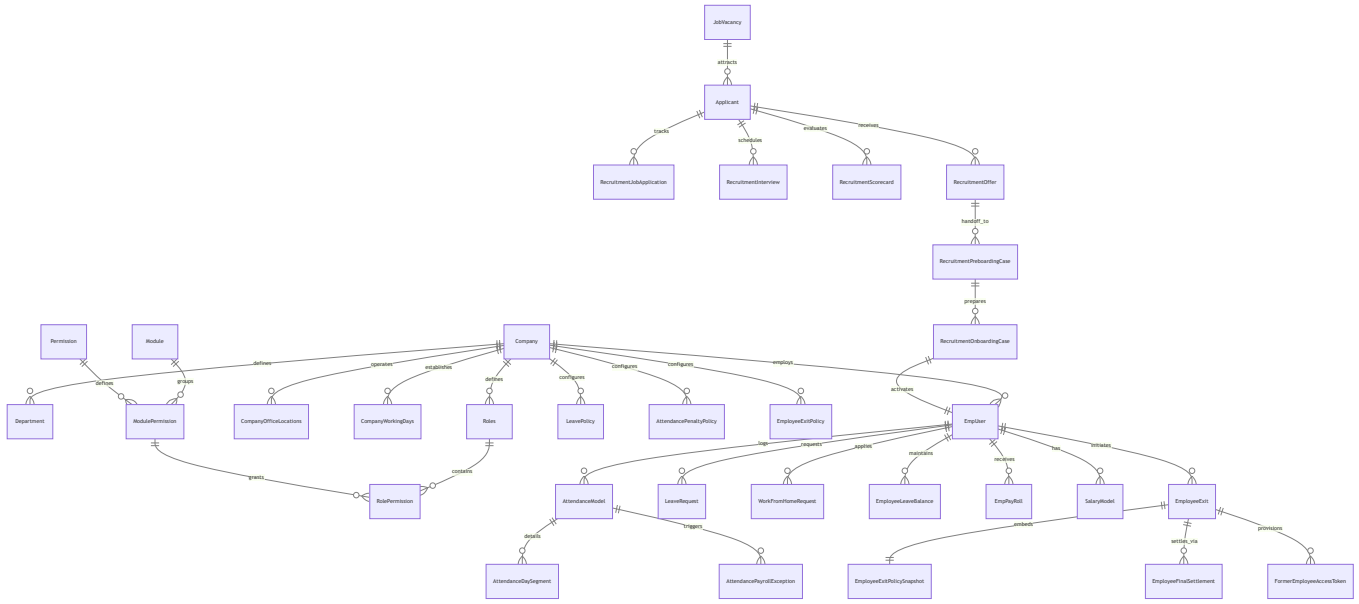
**ARCHITECTURE / WORKFLOW DIAGRAM**



DIAG-05 PART IV docs/project-documentation/diagrams/DATABASE-RELATIONSHIPS.md

# Database Relationships Conceptual ER Diagram

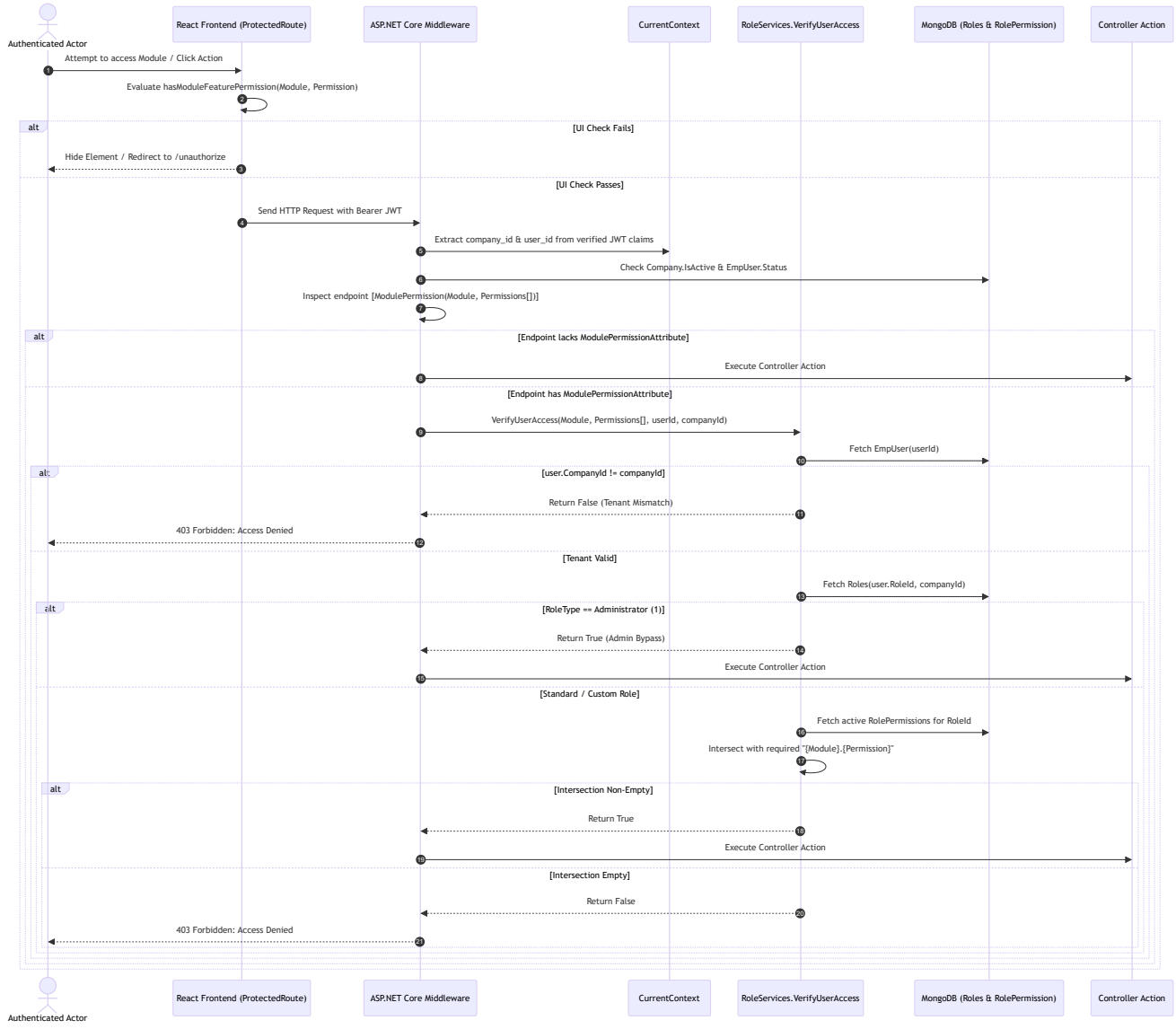
ARCHITECTURE / WORKFLOW DIAGRAM



DIAG-06 PART IV docs/project-documentation/diagrams/AUTHORIZATION-FLOW.md

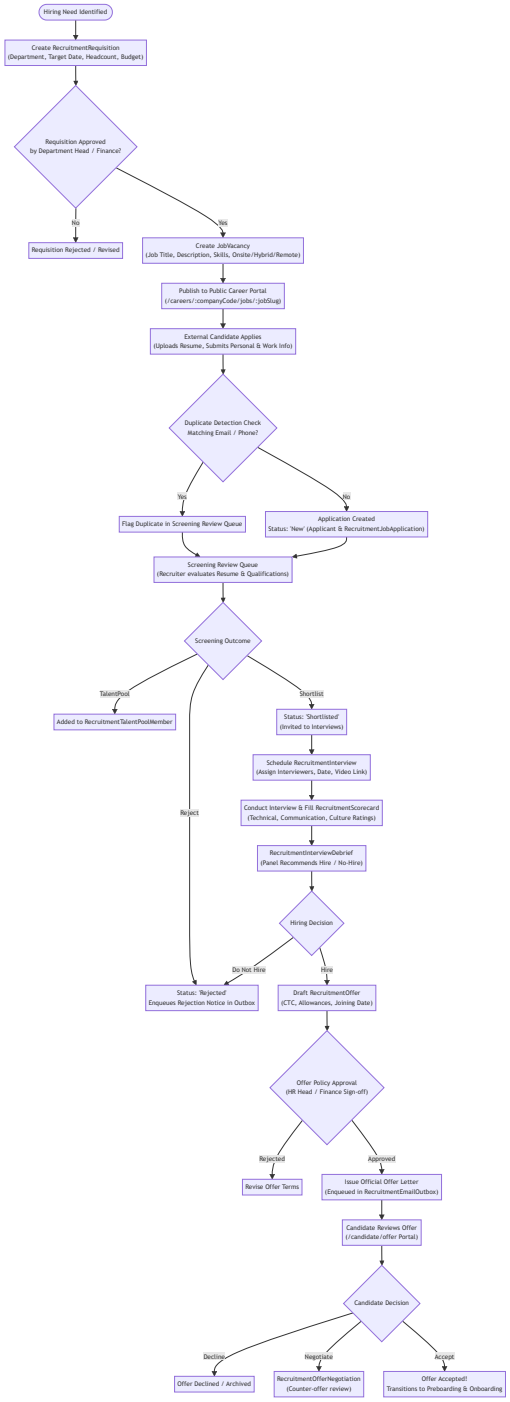
# Authorization & Permission Evaluation Flow Diagram

## ARCHITECTURE / WORKFLOW DIAGRAM



# Recruitment & Talent Acquisition Flow Diagram

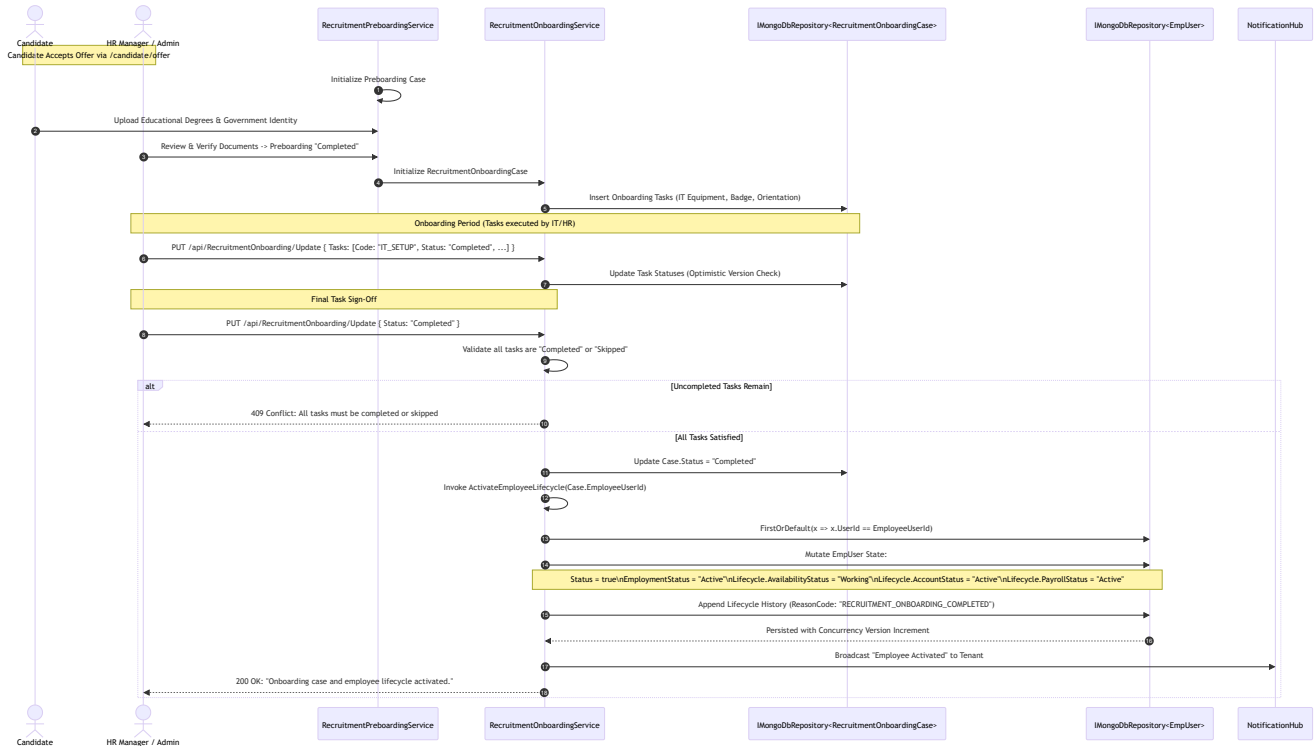
## ARCHITECTURE / WORKFLOW DIAGRAM



DIAG-08 PART IV docs/project-documentation/diagrams/ONBOARDING-FLOW.md

# Onboarding Lifecycle & Activation Flow Diagram

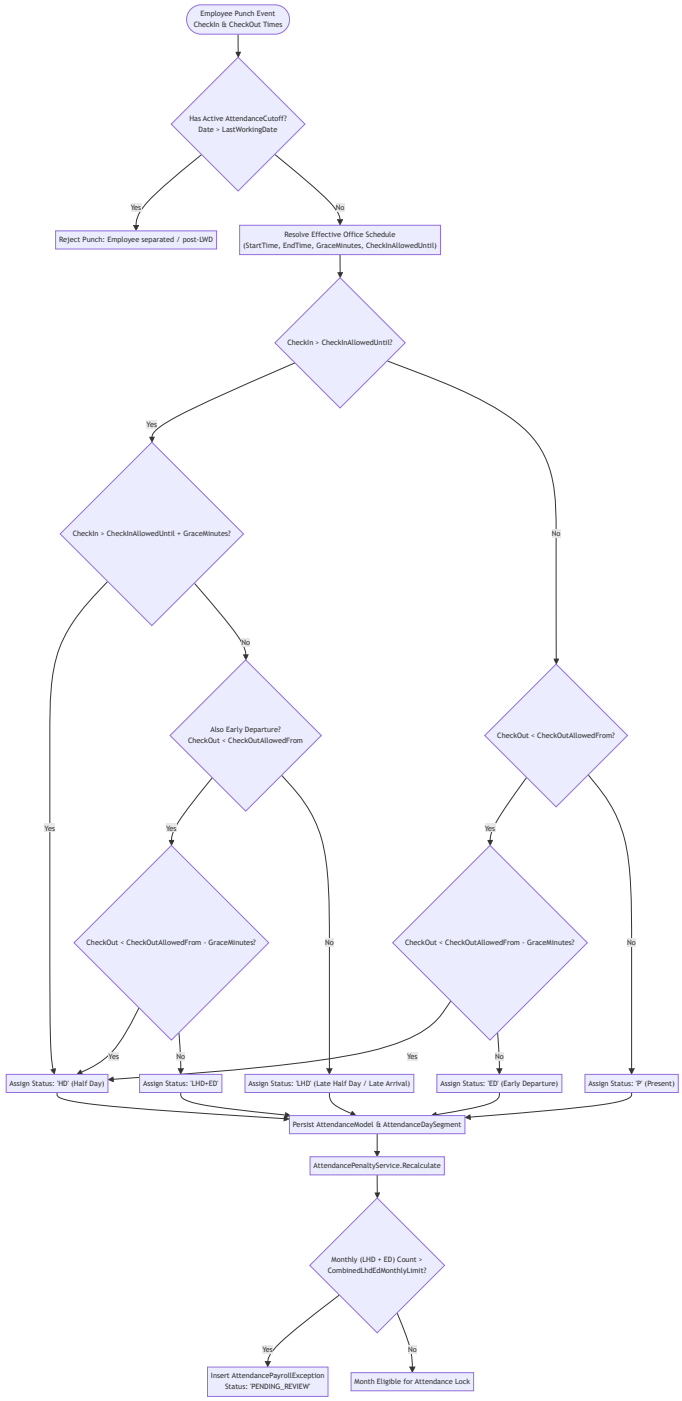
## ARCHITECTURE / WORKFLOW DIAGRAM



DIAG-09 PART IV docs/project-documentation/diagrams/ATTENDANCE-FLOW.md

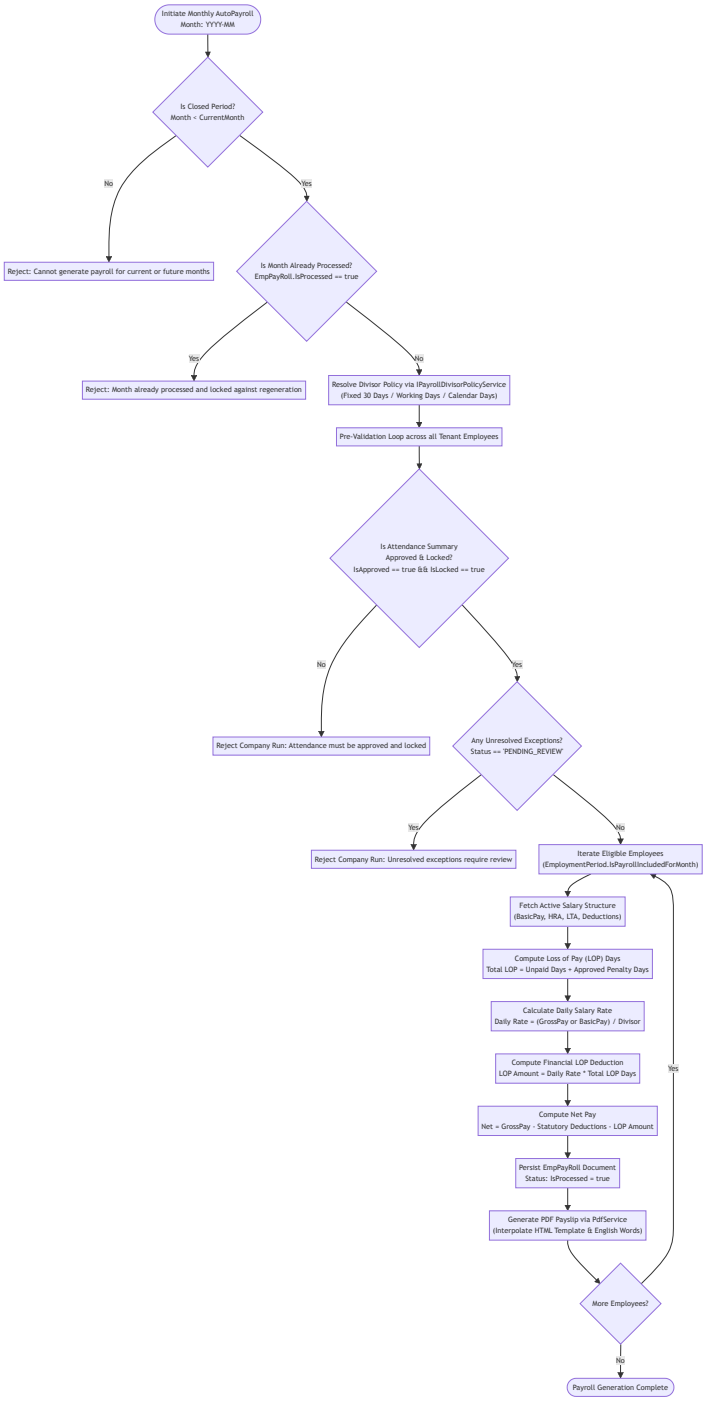
# Attendance Calculation & Exception Flow Diagram

## ARCHITECTURE / WORKFLOW DIAGRAM



# Payroll Calculation Flow Diagram

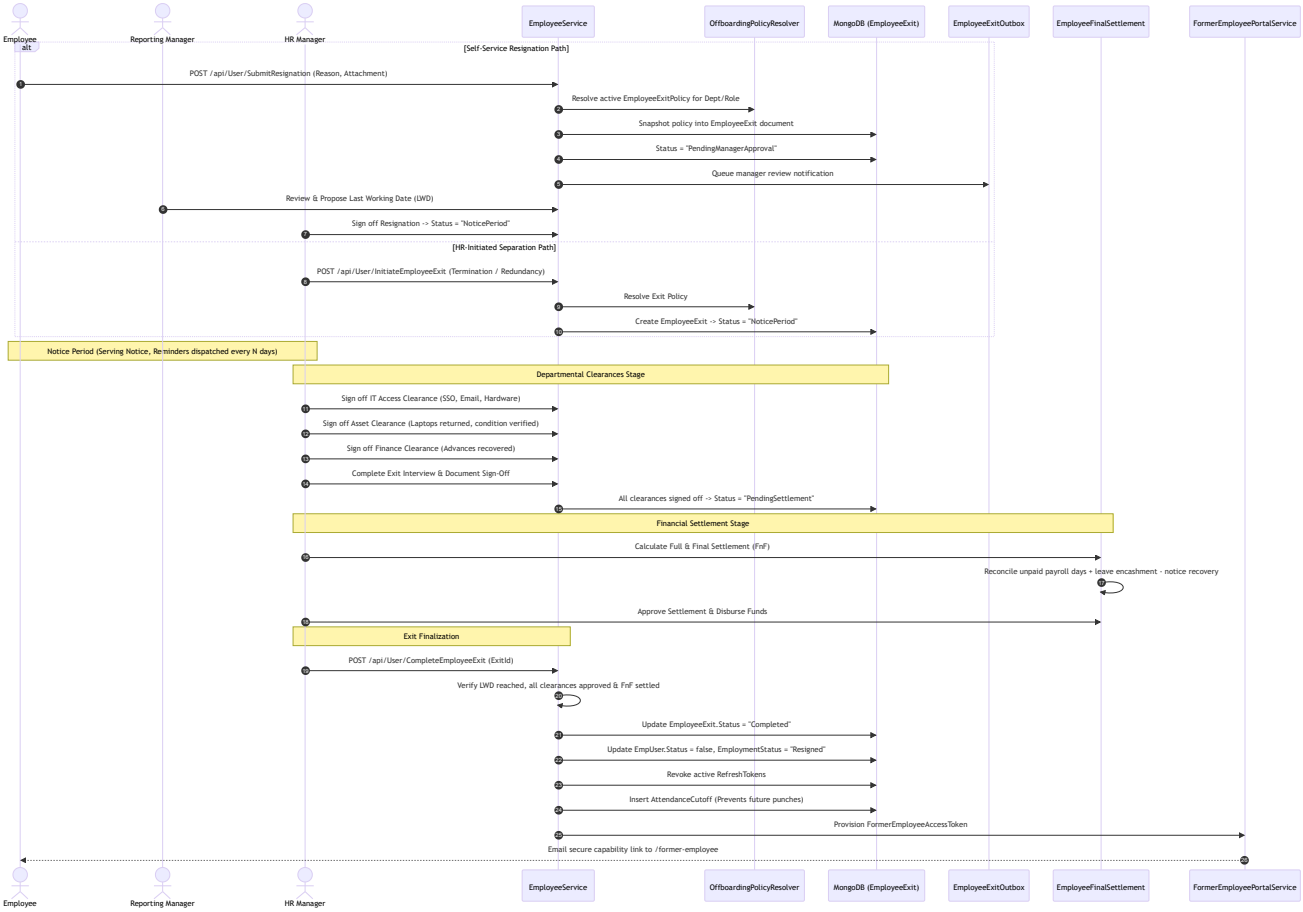
## ARCHITECTURE / WORKFLOW DIAGRAM



DIAG-11 PART IV docs/project-documentation/diagrams/OFFBOARDING-FLOW.md

# Offboarding, Separation & Clearance Flow Diagram

## ARCHITECTURE / WORKFLOW DIAGRAM



PART V

# SOURCE TRACEABILITY & VERIFICATION REGISTERS

End-to-End Frontend-to-Database Code Mapping & Explicit Unverified/Partial Capabilities Register

2 Document(s) in this Section

# MASTER SOURCE TRACEABILITY MATRIX (`SOURCE-TRACEABILITY.md`)

> **Purpose**: Provides end-to-end line-of-sight traceability across the **Codeji CMS / HRMS** codebase, mapping every functional domain from its **React 18 Frontend** down through **ASP.NET Core (.NET 10) Controllers & Services**, **DTO Contracts**, **MongoDB Entities**, **REST / SignalR Endpoints**, and **Migrations / Background Hosted Services**.

## 1. Core Platform, Security & Multi-Tenancy

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Authentication** | JWT Login, Refresh Token Rotation, Token Revocation | `app/modules/auth/LoginPage.tsx` `server/axios.ts` (401 Refresh Queue) | `Controllers/AuthController.cs` `Services/AuthService.cs` `Services/TokenService.cs` | `LoginRequestDto` `LoginResponseDto` `RefreshTokenRequestDto` | `EmpUser.cs` `RefreshToken.cs` `Company.cs` | `POST /api/auth/login` `POST /api/auth/refresh` `POST /api/auth/revoke` | `Migrations/InitialRoleSeed.cs` |
| **Security & Tenant Pipeline** | Version Enforcement (`426`), Tenant Guard, RBAC Attribute Check | `server/axios.ts` (`AppVersion` Header Injection) | `Middleware/ExceptionHandlingMiddleware.cs` `Middleware/AntiforgeryMiddleware.cs` `Middleware/CompanyIdMiddleware.cs` `Attributes/ModulePermissionAttribute.cs` | `ApiResponse<T>` `ErrorResponseDto` | `Company.cs` `EmpUser.cs` `Role.cs` | Intercepts all `/api/*` routes | `Program.cs` pipeline order |
| **Role & Permission Engine** | Dynamic Module Action Matrix (`Read`, `Create`, `Update`, `Delete`, `Approve`) | `app/modules/roles/RolesPage.tsx` `app/modules/roles/PermissionMatrix.tsx` `app/hooks/usePermission.ts` | `Controllers/RoleController.cs` `Services/RoleServices.cs` (`VerifyUserAccess`) | `RoleDto` `ModulePermissionDto` | `Role.cs` `PermissionModule.cs` | `GET /api/roles` `POST /api/roles` `PUT /api/roles/{id}` | `RolePermissionMigration` |

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Company Administration** | Multi-Tenant Onboarding, Subscription & Branch Configuration | `app/modules/company/CompanySettingsPage.tsx` `app/modules/admin/TenantsPage.tsx` | `Controllers/CompanyController.cs` `Services/CompanyServices.cs` | `CompanyDto` `UpdateCompanySettingsDto` | `Company.cs` `Department.cs` `Designation.cs` | `GET /api/company` `PUT /api/company` | Tenant bootstrapper |

## 2. Talent Acquisition, Preboarding & Onboarding

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Recruitment & ATS** | Job Requisitions, Applicant Funnel, Interviews, Offer Generation | `app/modules/recruitment/JobsPage.tsx` `app/modules/recruitment/ApplicantsKanban.tsx` `app/modules/recruitment/OfferModal.tsx` | `Controllers/RecruitmentController.cs` `Controllers/ApplicantController.cs` `Services/RecruitmentService.cs` `Services/RecruitmentAuditExportService.cs` | `JobPositionDto` `ApplicantDto` `InterviewScheduleDto` `JobOfferDto` | `JobPosition.cs` `Applicant.cs` `ApplicantLogs.cs` `JobOffer.cs` | `POST /api/recruitment/jobs` `POST /api/applicants` `PUT /api/applicants/{id}/stage` `GET /api/recruitment/audit-export` | Stage transition audit logger |
| **Preboarding** | Candidate Token Portal, KYC Document Uploads, Offer Acceptance | `app/modules/preboarding/PublicPreboardingPage.tsx` `app/modules/preboarding/HrVerificationQueue.tsx` | `Controllers/PreboardingController.cs` `Services/PreboardingService.cs` | `PreboardingPortalDto` `PreboardingDocumentDto` | `PreboardingCase.cs` `PreboardingDocument.cs` | `GET /api/preboarding/portal` `POST /api/preboarding/upload` `POST /api/preboarding/{id}/verify` | Token expiry validator |

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Onboarding** | Candidate-to-Employee Conversion, Day-1 Tasks, Lifecycle Initialization | `app/modules/onboarding/OnboardingDashboard.tsx` `app/modules/onboarding/ChecklistDrawer.tsx` | `Controllers/OnboardingController.cs` `Services/OnboardingService.cs` `Services/EmployeeLifecycleService.cs` | `OnboardingChecklistDto` `ConvertCandidateDto` | `OnboardingRecord.cs` `EmpUser.cs` `EmployeeLifecycle.cs` | `POST /api/onboarding/convert` `PUT /api/onboarding/{id}/complete` | Initializes 7-D `EmployeeLifecycle` |

## 3. Employee Core, Probation & Lifecycle

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Employee Master & 7-D Lifecycle** | Directory, Org Hierarchy, 7-Dimensional Lifecycle State Machine | `app/modules/employees/pages/EmployeesPage.tsx` `app/modules/employees/components/LifecycleBadge.tsx` | `Controllers/EmployeeController.cs` `Controllers/EmployeeLifecycleController.cs` `Services/EmployeeServices.cs` `Services/EmployeeLifecycleService.cs` | `EmployeeDto` `EmployeeLifecycleDto` | `EmpUser.cs` `EmployeeLifecycle.cs` `EmployeeLifecyclePolicy.cs` | `GET /api/employees` `POST /api/employees` `GET /api/employee-lifecycle/{id}` | `EmployeeLifecycleBackfillMigration` |
| **Employee Profile** | Personal, Emergency, Banking, Academic & Family Self-Service | `app/modules/userProfile/UserProfilePage.tsx` `app/modules/userProfile/components/BankDetailsCard.tsx` | `Controllers/UserProfileController.cs` `Services/UserProfileService.cs` | `UserProfileDto` `BankDetailsDto` `EmergencyContactDto` | `EmpUser.cs` (Embedded Profile Sub-Documents) | `GET /api/profile/me` `PUT /api/profile/me` | Audited via `ISupportAuditing` |

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Probation** | Probation Tracking, Extension, Confirmation & Leave/Notice Rules | `app/modules/employees/pages/EmployeesPage.tsx` `app/modules/settings/pages/LifecyclePolicySettings.tsx` | `Controllers/EmployeeLifecycleController.cs` `Services/EmployeeLifecycleService.cs` | `ConfirmProbationDto` `ExtendProbationDto` | `EmployeeLifecycle.cs` (`ProbationEndDateUtc`) `EmployeeLifecyclePolicy.cs` | `POST /api/employee-lifecycle/{id}/confirm-probation` `PUT /api/employee-lifecycle/{id}/extend-probation` | Policy resolver |

## 4. Workforce Time, Attendance, Leave & Calendar

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Attendance & Penalty Engine** | Check-In/Out, `P` / `HD` / `LHD` / `ED` / `LHD+ED` Calculation, Penalty Exceptions, Month Lock | `app/modules/attendance/pages/AttendancePage.tsx` `app/modules/attendance/components/MonthlySummaryTable.tsx` `app/modules/attendance/components/PenaltyExceptionsModal.tsx` | `Controllers/AttendanceController.cs` `Services/AttendanceStatusCalculationService.cs` `Services/AttendancePenaltyService.cs` | `AttendanceDto` `MonthlyAttendanceSummaryDto` `ResolveExceptionDto` | `Attendance.cs` `MonthlyAttendanceSummary.cs` `AttendancePenaltyPolicy.cs` `AttendancePayrollException.cs` `AttendanceMonthLock.cs` `AttendanceAuditLog.cs` | `POST /api/attendance/check-in` `POST /api/attendance/check-out` `POST /api/attendance-penalty/resolve` `POST /api/attendance/lock-month` | `AttendanceReminderHostedService.cs` (`AttendanceReminderDispatchLog`) |

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Work Shift & Office Schedule** | Shift Timings, Grace Minutes, Half-Day Hour Thresholds | `app/modules/settings/pages/OfficeScheduleSettings.tsx` | `Controllers/OfficeScheduleController.cs` `Services/OfficeScheduleSettingsService.cs` `Services/EffectiveOfficeScheduleService.cs` | `CompanyOfficeScheduleDto` `EffectiveScheduleDto` | `CompanyOfficeSchedule.cs` | `GET /api/office-schedule` `PUT /api/office-schedule` `GET /api/office-schedule/effective` | Consumed on every punch |
| **Leave Management** | Leave Policies, Accruals, Half-Day Segments, Working Calendar Intersection | `app/modules/leave/pages/LeaveDashboard.tsx` `app/modules/leave/components/ApplyLeaveModal.tsx` `app/modules/leave/pages/LeaveApprovalsPage.tsx` | `Controllers/LeaveManagementController.cs` `Services/LeaveManagementService.cs` | `ApplyLeaveDto` `LeaveDto` `EmployeeLeaveBalanceDto` `LeavePolicyDto` | `Leave.cs` `LeavePolicy.cs` `EmployeeLeaveBalance.cs` | `GET /api/leave/balances` `POST /api/leave/apply` `POST /api/leave/{id}/approve` `POST /api/leave/{id}/reject` | `LeaveAccrualHostedService.cs` |
| **Work From Home (WFH)** | Remote Work Requests, Half-Day Segments, Attendance Overlay | `app/modules/wfh/pages/WfhDashboard.tsx` `app/modules/wfh/components/ApplyWfhModal.tsx` | `Controllers/WorkFromHomeController.cs` `Services/WorkFromHomeService.cs` | `ApplyWfhDto` `WorkFromHomeDto` | `WorkFromHome.cs` | `POST /api/wfh/apply` `POST /api/wfh/{id}/approve` | Syncs with `AttendanceStatusCalculationService` |
| **Holidays & Working Calendar** | National/Cultural Holidays, Weekly Off Rules, Working Day Counts | `app/modules/calendar/pages/CompanyCalendarPage.tsx` `app/modules/settings/pages/WeeklyOffSettingsPage.tsx` | `Controllers/CalendarController.cs` `Controllers/HolidaysController.cs` `Services/CompanyWorkingCalendarService.cs` | `HolidaysDto` `WeeklyOffSettingDto` `CalendarMonthResponseDto` | `Holidays.cs` `WeeklyOffSetting.cs` | `GET /api/holidays` `POST /api/holidays` `GET /api/calendar/working-days` `PUT /api/calendar/weekly-off` | Feeds Leave & Payroll Divisors |

## 5. Compensation, Payroll, Offboarding & Final Settlement

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Salary Structure** | CTC, Gross, Basic, HRA, LTA, EPF, ESIC, Health Insurance, Tax | `app/modules/payroll/pages/SalaryStructurePage.tsx` `app/modules/userProfile/components/CompensationTab.tsx` | `Controllers/SalaryController.cs` `Services/SalaryService.cs` `Repository/SalaryRepository.cs` | `SalaryModelDto` `CreateSalaryModelDto` | `SalaryModel.cs` | `GET /api/salary/{employeeId}` `POST /api/salary` `PUT /api/salary/{id}` | Versioned via `ISupportAuditing` |
| **Payroll Engine & Payslip PDF** | Batch Auto-Payroll, Divisor Rules (`FIXED_30`, `WORKING_DAYS`, `CALENDAR`), LOP, Payslip PDF | `app/modules/payroll/pages/PayrollDashboard.tsx` `app/modules/payroll/pages/MySalarySlipsPage.tsx` `app/modules/settings/pages/PayrollSettings.tsx` | `Controllers/AutoPayrollController.cs` `Controllers/PayRollController.cs` `Controllers/PayrollDivisorPolicyController.cs` `Services/AutoPayrollServices.cs` `Services/PdfService.cs` | `PayrollBatchResultDto` `PayrollEntityDto` `PayrollDivisorPolicyDto` | `PayrollEntity.cs` `PayrollDivisorPolicy.cs` `Templates/SalarySlipTemplate.html` | `POST /api/autopayroll/generate` `POST /api/autopayroll/publish` `GET /api/payroll/{id}/download-slip` `PUT /api/payroll-divisor-policy` | Requires locked `AttendanceMonthLock` |

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Offboarding, Clearances & Alumni Portal** | Resignation, 3-Tier Policy Resolver, Clearances, Assets, KT, Exit Interview, Alumni Token Portal | `app/modules/offboarding/pages/OffboardingDashboard.tsx` `app/modules/offboarding/pages/ExitCaseDetailPage.tsx` `app/modules/offboarding/pages/FormerEmployeePortalPage.tsx` | `Controllers/EmployeeExitController.cs` `Controllers/PublicFormerEmployeePortalController.cs` `Services/EmployeeExitService.cs` `Services/OffboardingPolicyResolver.cs` `Services/FormerEmployeePortalService.cs` | `EmployeeExitDto` `InitiateExitDto` `FormerEmployeeSummaryDto` | `EmployeeExit.cs` `EmployeeExitPolicy.cs` `EmployeeExitClearanceTemplate.cs` `EmployeeExitAsset.cs` `EmployeeExitHandoverItem.cs` `EmployeeExitInterview.cs` `EmployeeExitAttendanceCutoff.cs` `FormerEmployeeAccessToken.cs` `EmployeeRejoinRecord.cs` | `POST /api/employee-exit/initiate` `PUT /api/employee-exit/{id}/clearance` `POST /api/employee-exit/{id}/complete` `GET /api/public/former-employee/summary` | `EmployeeExitOutboxProcessor.cs` (`EmployeeExitOutbox.cs`) |
| **Full & Final (FnF) Settlement** | Final Salary (`SourcePayrollId`), Leave Encashment, Notice Buyout/Recovery, Asset Dues | `app/modules/offboarding/components/FinalSettlementTab.tsx` | `Controllers/EmployeeExitController.cs` `Services/EmployeeExitService.cs` | `EmployeeFinalSettlementDto` `CalculateSettlementRequestDto` | `EmployeeFinalSettlement.cs` | `GET /api/employee-exit/{exitId}/settlement` `POST /api/employee-exit/{exitId}/settlement/calculate` `POST /api/employee-exit/{exitId}/settlement/approve` | Links to `PayrollEntity` |

| Domain | Feature | Frontend (`CMS-React/src/`) | Backend Controller & Service (`CMS-Backend-Core/`) | DTOs (`Codeji.CMS.DTO/`) | MongoDB Entities (`Codeji.CMS.Repository/Entities/`) | Primary Endpoints | Background Workers / Migrations |
|---|---|---|---|---|---|---|---|
| **Notifications, Analytics & Audit** | SignalR Hubs (`/notificationHub`, `/chatHub`), Executive KPIs, Immutable Audit Trails | `app/modules/notifications/components/NotificationBell.tsx` `app/modules/dashboard/pages/AdminDashboardPage.tsx` `app/modules/audit/pages/AuditLogPage.tsx` | `Hubs/NotificationHub.cs` `Hubs/ChatHub.cs` `Controllers/DashboardController.cs` `Services/NotificationService.cs` `Services/DashboardServices.cs` | `NotificationDto` `AdminDashboardDto` `AttendanceAuditLogDto` | `Notification.cs` `AttendanceAuditLog.cs` `EmployeeExitAuditEvent.cs` `ProcessLog.cs` | `WS /notificationHub` `WS /chatHub` `GET /api/dashboard/admin` `GET /api/audit/attendance` | Real-time WebSocket + MongoDB persistence |

REF-02 PART V docs/project-documentation/UNVERIFIED-AREAS.md

# UNVERIFIED, PARTIAL & UI-ONLY AREAS REGISTER (`UNVERIFIED-AREAS.md`)

> **Rule Compliance Statement**: Per Section 1 of the Codex Master Task ("*Never invent an endpoint, entity, field, rule, dependency, relationship, permission, workflow, or feature. If something cannot be verified, explicitly write: Not verified from current implementation.*"), this document catalogs every architectural area, external integration, or workflow capability that is either **partially implemented**, **UI-only / configuration-only**, or **not verified from current repository implementation**, along with the exact files inspected and concrete engineering recommendations.

## 1. Summary Matrix of Unverified / Partial Areas

| # | Area / Capability | Implementation Status | Files Inspected | What Is Verified in Code | What Is Missing / Not Verified | Recommended Engineering Action |
|---|---|---|---|---|---|---|
| 1 | **Biometric Hardware Device Webhook Ingestion** | **Not Verified from Current Implementation** (Web/API Punch Only) | `Controllers/AttendanceController.cs` `Services/AttendanceStatusCalculationService.cs` | Web/API check-in and check-out endpoints ( `POST /api/attendance/check-in` , `check-out` ) and status calculation ( `P` , `HD` , `LHD` , `ED` , `LHD+ED` ) are fully implemented. | No raw biometric hardware listener (e.g., ZKTeco / eSSL ADMS webhook receiver or TCP socket parser) is present in `CMS-Backend-Core` . | Add `POST /api/attendance/biometric-webhook` with device API key validation and idempotent punch deduplication. |
| 2 | **Cloud Object Storage (AWS S3 / Azure Blob)** | **Local Disk Only ( `/fs` )** | `Program.cs` ( `UseStaticFiles("/fs")` ) `Services/PdfService.cs` | Files (resumes, KYC docs, generated payslip PDFs, relieving letters) are written to the local server filesystem and served via `UseStaticFiles` at `/fs` . | External cloud blob storage SDKs ( `AWSSDK.S3` or `Azure.Storage.Blobs` ) and pre-signed URL generators are **not verified from current implementation**. | Introduce an `IFileStorageProvider` abstraction with an S3/Azure Blob implementation for horizontal multi-instance scaling. |
| 3 | **Automated Banking Payout API / NACH Direct Debit** | **Calculation & PDF Complete; Bank API Not Wired** | `Services/AutoPayrollServices.cs` `Controllers/AutoPayrollController.cs` | Full salary calculation, LOP math, statutory deductions, and PDF payslip generation ( `SalarySlipTemplate.html` ) are verified in code. | Direct corporate bank API integration (e.g., HDFC/ICICI corporate payout webhook or automated NACH file transfer) is **not verified from current implementation**. | Add a bank payout export service ( `POST /api/autopayroll/bank-export` ) generating standardized NEFT/NACH CSV batch files. |

| # | Area / Capability | Implementation Status | Files Inspected | What Is Verified in Code | What Is Missing / Not Verified | Recommended Engineering Action |
|---|---|---|---|---|---|---|
| **4** | **External Identity / SCIM Auto-Deprovisioning on Exit** | **Checklist-Driven (Manual IT Action)** | `Services/EmployeeExitService.cs` `Services/HostedServices/EmployeeExitOutboxProcessor.cs` | Internal CMS access ( `EmpUser.IsActive = false` , `RefreshToken` revocation, `EmployeeExitAttendanceCutoff` ) is automatically revoked upon exit completion. | Automated outbound SCIM / OAuth revocation calls to third-party tools (Slack, GitHub, Google Workspace, Microsoft Entra ID) are **not verified from current implementation**; IT clearance is tracked via checklist sign-off. | Register outbound webhook connectors inside `EmployeeExitOutboxProcessor` triggered on `ExitCompleted` events. |
| **5** | **Multi-Shift Rotating Roster Assignments** | **Single/Department Schedule Verified; Shift Rotation Roster Not Verified** | `Entities/CompanyOfficeSchedule.cs` `Services/EffectiveOfficeScheduleService.cs` | `CompanyOfficeSchedule` supports start/end times, grace minutes, and half-day hour thresholds resolved via `EffectiveOfficeScheduleService` . | Employee-level weekly rotating shift rosters (e.g., Week 1 Morning Shift → Week 2 Night Shift crossing midnight) are **not verified from current implementation**. | Create an `EmployeeShiftRoster` collection mapping `(EmployeeId, DateRange, ShiftTemplateId)` and support cross-midnight shift math. |
| **6** | **Automated Probation Confirmation Hosted Worker** | **Manual Confirmation Endpoint Verified; Auto-Promotion Cron Not Verified** | `Controllers/EmployeeLifecycleController.cs` `Services/EmployeeLifecycleService.cs` | `ProbationEndDateUtc` calculation, leave restrictions ( `AllowOnProbation` ), probation notice days ( `ProbationNoticeDays` ), and manual confirmation endpoints are verified. | A background cron job that automatically transitions `EmploymentStatus` from `Probation (2)` to `Active (3)` on `ProbationEndDateUtc` without HR click is **not verified from current implementation**. | Extend `AttendanceReminderHostedService` (or add `LifecycleTransitionHostedService` ) to send T-7 day alerts to HR/Managers before `ProbationEndDateUtc` . |

| # | Area / Capability | Implementation Status | Files Inspected | What Is Verified in Code | What Is Missing / Not Verified | Recommended Engineering Action |
|---|---|---|---|---|---|---|
| **7** | **Statutory Gratuity Formula Auto-Calculation in FnF** | **Manual Adjustment Field Verified; Auto-Formula Not Verified** | `Entities/EmployeeFinalSettlement.cs` `Services/EmployeeExitService.cs` | `EmployeeFinalSettlement` tracks `SalaryPayable`, `LeaveEncashment`, `BonusesAndIncentives`, `NoticeRecovery`, and `OutstandingDues`, and `EmployeeExitBenefitItem` tracks `Gratuity` status. | Automatic calculation of Indian Payment of Gratuity Act formula ( $^{15}/_{26} \times$ **Basic** $\times$ **Tenure Years** when tenure $\geq 5$ years) is **not verified from current implementation**. | Add `CalculateGratuity(EmpUser, SalaryModel)` inside `EmployeeExitService.CalculateFinalSettlementAsync`. |
| **8** | **Multi-Level Sequential Leave Approval Chains** | **Single-Stage Approval Verified; Multi-Tier Chain Not Verified** | `Controllers/LeaveManagementController.cs` `Services/LeaveManagementService.cs` | Leave application, balance reservation, working calendar intersection, and Manager/HR approval ( `POST /api/leave/{id}/approve` ) are fully implemented. | Sequential N-tier approval routing (e.g., L1 Reporting Manager → L2 Department Head → L3 HR Director for leaves $> 10$ days) is **not verified from current implementation**. | Add `ApprovalChainSteps` array to `Leave` entity driven by duration thresholds in `LeavePolicy`. |
| **9** | **Standalone Warehouse / Procurement Asset Inventory** | **Offboarding Asset Clearance ( `EmployeeExitAsset` ) Verified** | `Entities/EmployeeExitAsset.cs` `Services/EmployeeExitService.cs` | Tracking assigned employee assets, physical condition inspection, and damage recovery deductions ( `DeductionAmount` ) into `EmployeeFinalSettlement` are fully implemented. | Pre-assignment IT stockroom procurement, vendor warranty tracking, and unassigned warehouse inventory management are **not verified from current implementation**. | Create a `CompanyAssetInventory` master collection that links to `EmployeeExitAsset` upon employee separation. |

| # | Area / Capability | Implementation Status | Files Inspected | What Is Verified in Code | What Is Missing / Not Verified | Recommended Engineering Action |
|---|---|---|---|---|---|---|
| **10** | **Cryptographic Tamper-Evident Audit Hash Chaining** | **Append-Only MongoDB Audit Collections Verified** | `Entities/AttendanceAuditLog.cs` `Entities/EmployeeExitAuditEvent.cs` `Entities/ApplicantLogs.cs` | Immutable audit collections and `ISupportAuditing` ( `CreatedBy` , `UpdatedBy` , `Version` optimistic concurrency) are verified across the repository. | Cryptographic hash-chaining ( `PreviousRecordHash` + HMAC-SHA256 signature per audit log row) is **not verified from current implementation**. | Add `RecordHash` and `PreviousHash` fields to `AttendanceAuditLog` and `EmployeeExitAuditEvent` for SOC2/ISO27001 forensic non-repudiation. |

## 2. Verification Methodology Used

Every finding above was established by:

1. Scanning all `.cs` controllers, services, repositories, entities, and hosted services inside `CMS-Backend-Core/` .
2. Scanning all `.tsx` / `.ts` pages, components, Redux slices, TanStack Query hooks, and Axios interceptors inside `CMS-React/src/` .
3. Tracing every HTTP endpoint from frontend service call to backend controller action and MongoDB collection operation.
4. Explicitly separating **verified production code** from **roadmap / external integration boundaries**.

